1198

**Make Limited**

The Crown Estate

# St James's Market Phase 2

## Architectural Specification

Rev 001

04-09-2026

| Title | Code | Clause | Description |
|---|---|---|---|
| Design Compliance | | F10/100 | |
| External Glazed Balustrade & Planting (Visual Intent C | BAL-101 | L30/561 | Purpose made external glass balustrade system with planting to weave parapet and listed cinema parapet interface, as shown on the Design |
| Metal Balustrade System | BAL-102 | L30/562 | Metal balustrade system to historic mansard, modern mansard and roof, as shown on the |
| External Glazed Inclined Balustrade & Planting (Visual | BAL-103 | L30/563 | Purpose made external glass inclined balustrade system with planting to modern mansard, as shown on the Design Drawings. |
| Metal Balustrade and Timber Handrail | BAL-141 | L30/551 | Metal balustrade and timber handrail as shown on the Design Drawings |
| Metal Balustrade and Leather Finish Handrail | BAL-142 | L30/552 | Metal balustrade and leather finish handrail as shown on the Design Drawings |
| External Glass Balustrade | BAL-171 | L30/552 | External glass balustrade as shown on the |
| External Modular Guardrails | BAL-191 | L30/553 | External modular guardrails as shown on the Design Drawings |
| Leather Finish Handrail - Wall Mounted | BAL-341 | L30/554 | Leather finish wall mounted handrail as shown on the Design Drawings. |
| Leather Finish Handrail - Floor Mounted | BAL-342 | L30/555 | Leather finish floor mounted handrail as shown on the Design Drawings. |
| Fluted Glass Balustrade | BAL-401 | L30/556 | Fluted glass balustrade as shown on the Design |
| UKPN Brickwork | BRK-101 | F10/110 | 240min fire rated brickwork configured and to locations indicated on the Design Drawings. |
| Prefabricated Metal Canopy System (Visual Intent Onl | CAN-101 | L30/591 | Bespoke-made cantilevered aluminium sheet canopy system, configured and to locations indicated as Design Drawings. |
| Prefabricated Metal Canopy System (Visual Intent Onl | CAN-102 | L30/592 | Bespoke-made cantilevered aluminium sheet canopy system, configured and to locations indicated as Design Drawings. |
| Mass Barrier Ceiling | CLG-151 | K10/220 | Suspended monolithic plasterboard ceiling lining fixed to suspended metal support system. |
| Acoustic Raft Ceiling System | CLG-501 | K40/110 | Acoustic raft ceiling system, as shown on the Design Drawings. |
| Acoustic Raft Ceiling System | CLG-503 | K40/113 | Acoustic raft ceiling system, as shown on the Design Drawings. |
| Metal Mesh Raft Ceiling System | CLG-521 | K40/115 | Metal mesh raft ceiling system. |
| Suspended Metal Ceiling Tile System | CLG-601 | K40/151 | Suspended metal ceiling tile system, as shown on the Design Drawings. |
| Feature Acoustic Ceiling | CLG-701 | K40/175 | Feature acoustic ceiling, as shown on the Design |
| Durasteel Ceiling | CLG-901 | K40/201 | Proprietary fire rated suspended metal ceiling tile |
| Plain Finish To In-Situ Concrete | CON-201 | E20/620 | 'Plain formed' in-situ concrete surface finish, as defined in the National Structural Concrete Specification (NSCS) 4th edition. |
| External Single Leaf Metal Framed Glazed Hinged Do | DRS-101 | L20/501 | External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and |
| External Single Leaf Metal Framed Glazed Hinged Do | DRS-102 | L20/502 | External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and |
| External Single Leaf Metal Hinged Doorset (Visual Int | DRS-103 | L20/503 | External single leaf metal hinged doorset, as shown on the Design Drawings and Door |
| External Double Leaf Metal Framed Glazed Hinged D | DRS-104 | L20/504 | External double leaf metal framed glazed hinged doorset, as shown on the Design Drawings and |
| External Double Leaf Metal Framed Glazed Hinged D | DRS-105 | L20/505 | External double leaf metal framed opaque glazed hinged doorset, as shown on the Design Drawings and Door Schedule. |
| External Double Leaf Metal Screen (Visual Intent Only | DRS-106 | L20/506 | External double leaf metal screen, as shown on the Design Drawings and Door Schedule. |
| External Double Leaf Metal Hinged Doorset (Visual In | DRS-107 | L20/507 | External double leaf metal hinged doorset, as shown on the Design Drawings and Door |
| External Bi-fold Doorset - Vehicle Lift (Visual Intent On | DRS-108 | L20/508 | External bi-fold doorset to vehicle lift, as shown on the Design Drawings and Door Schedule. |
| External Double Leaf Metal Hinged Doorset (Visual In | DRS-109 | L20/509 | External double leaf metal hinged doorset, as shown on the Design Drawings and Door |

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| Circle Slide Entrance Door (Visual Intent Only) | DRS-110 | L20/510 | Circle slide fully automated entrance doorset integrated within the curtain wall system where indicated on the Design Drawings. |
| External Double Leaf Metal Framed Glazed Hinged D | DRS-111 | L20/511 | External double leaf metal framed glazed hinged doorset - curved, as shown on the Design Drawings and Door Schedule. |
| External Bi-fold Metal Framed Glazed Doorset (Visual | DRS-112 | L20/512 | External bi-fold metal framed doorset, as shown on the Design Drawings and Door Schedule. |
| External Single Leaf Metal Framed Glazed Hinged Do | DRS-113 | L20/513 | External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and |
| External Single Leaf Metal Framed Hinged Doorset (V | DRS-114 | L20/514 | External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door |
| External Single Leaf Metal Framed Hinged Doorset (V | DRS-115 | L20/515 | External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door |
| External Double Leaf Metal Framed Glazed Hinged D | DRS-116 | L20/516 | External double leaf metal framed glazed hinged doorset, as shown on the Design Drawings and |
| External Bi-fold Metal Framed Glazed Doorset (Visual | DRS-117 | L20/517 | External bi-fold metal framed doorset, as shown on the Design Drawings and Door Schedule. |
| External Metal Framed Glazed Hinged Doorset (Visual | DRS-118 | L20/518 | External metal framed glazed hinged doorset, as shown on the Design Drawings and Door |
| External Integrated Green Wall Doorset (Visual Intent | DRS-119 | L20/519 | External integrated green wall doorset, as shown on the Design Drawings and Door Schedule. |
| Louvre Doorset (Visual Intent Only) | DRS-120 | L20/520 | Proprietary external metal louvred doorset to louvred screening. |
| External Metal Framed Hinged Doorset (Visual Intent ( | DRS-121 | L20/521 | External metal framed hinged doorset, as shown on the Design Drawings and Door Schedule. |
| External Single Leaf Metal Framed Hinged Doorset (Vi | DRS-122 | L20/522 | External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door |
| External Double Leaf Metal Framed Hinged Doorset (V | DRS-123 | L20/523 | External double leaf metal framed hinged doorset, as shown on the Design Drawings and |
| External Single Riser Doorset (Visual Intent Only) | DRS-124 | L20/524 | External single riser doorset, as shown on the Design Drawings and Door Schedule. |
| External Double Riser Doorset (Visual Intent Only) | DRS-125 | L20/525 | External double riser doorset, as shown on the Design Drawings and Door Schedule. |
| Louvre Doorset (Visual Intent Only) | DRS-126 | L20/526 | Proprietary external metal louvred doorset to louvred screening. |
| Glazed Access Gate (Visual Intent Only) | DRS-127 | L20/527 | Glazed access gate to roof glazed balustrading. |
| External Top Hinged Delivery Hatch (Visual Intent Only | DRS-128 | L20/528 | External top hinged delivery hatch, as shown on the Design Drawings and Door Schedule. |
| External Dry Riser Cabinet (Visual Intent Only) | DRS-129 | L20/529 | External dry riser cabinet, as shown on the Design Drawings and Door Schedule. |
| External Dry Riser/ Sprinkler Fill Cabinet (Visual Intent | DRS-130 | L20/530 | External dry riser/ sprinkler fill cabinet, as shown on the Design Drawings and Door Schedule. |
| Gatic Hatch | DRS-951 | L35/631 | Waterproof gatic hatch for plant replacement |
| Portland Stone Soffit Panel System (Visual Intent Only | ESS-101 | H51/450 | Portland stone soffit panel system, as the Design |
| Woven Facade (Visual Intent Only) | EWS-01A | H92/461 | Woven façade cladding, as the Design |
| Woven Facade Corner Setback (Visual Intent Only) | EWS-01B | H92/462 | Woven façade corner setback cladding, as the Design Drawings. |
| Barrel Curve Facade (Visual Intent Only) | EWS-02A | H92/463 | Barrel curve façade cladding, as the Design |
| Modern Mansard Facade (Visual Intent Only) | EWS-02B | H92/464 | Modern mansard façade cladding, as the Design |
| Side Hung Window (Visual Intent Only) | EWS-04A | L10/311 | Side hung window system. |
| Curved Fixed Glazing - North West Setback Corner (V | EWS-04B | L10/312 | Curved fixed pane glazing to North West setback |
| Incline Fixed Glazing - Modern Mansard (Visual Intent | EWS-04C | L10/313 | Incline fixed glazing to modern mansard. |
| Side Hung Window - Historic Façade (Visual Intent On | EWS-04D | L10/314 | Side hung window to historic façade. |
| Side Hung Windows - Historic Mansard Dormers (Visu | EWS-04E | L10/315 | Side hung windows to historic mansard dormers. |
| Bi-fold Windows - Sky Lobby (Visual Intent Only) | EWS-04F | L10/316 | Bi-fold windows to sky lobby. |
| Rebuilt Historic Façade (Visual Intent Only) | EWS-05 | H51/110 | Rebuilt historic façade, as the Design Drawings. |
| Slate Mansard With Dormers (Visual Intent Only) | EWS-06 | H62/105 | Slate tile mansard system with dormers configured as indicated on the Design Drawings. |
| Curtain Wall System - Ground Floor (Visual Intent Only | EWS-08 | H11/110 | Capped stick built curtain walling system with integrated doorsets, as the Design Drawings. |
| Architectural Metal Patterned Screening (Visual Intent | EWS-09 | L10/666 | Architectural metal patterned screening to north core roof plant. |

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| Curtain Wall System - Terrace (Visual Intent Only) | EWS-10 | H11/111 | Capped stick built curtain walling system with integrated doorsets, as the Design Drawings. |
| Insulated Wall System | EWS-106 | H43/120 | Insulated wall system, as the Design Drawings. |
| Insulated Wall System | EWS-107 | H43/122 | Insulated wall system, as the Design Drawings. |
| Blockwork Wall | EWS-107 | F10/202 | Concrete fair-faced blockwork configured and to locations indicated on the Design Drawings. |
| Roof Plant Screen (Visual Intent Only) | EWS-11 | L10/661 | Roof plant screening |
| Core Cladding (Visual Intent Only) | EWS-12 | H92/465 | Core cladding, as the Design Drawings. |
| Steel Framing System - Insulated (Visual Intent Only) | EWS-13A | G12/150 | Load-bearing metal section external stud wall system, forming the backing wall between or across the external face of perimeter elements of the primary structure, to receive green wall |
| Steel Framing System - Uninsulated (Visual Intent Onl | EWS-13B | G12/151 | Load-bearing metal section external stud wall system, forming the backing wall between or across the external face of perimeter elements of the primary structure, to receive green wall |
| Acoustic Seperation Wall (Visual Intent Only) | EWS-14 | F10/301 | Acoustic separation wall to locations indicated on the Design Drawings. |
| Insulated Panel System | EWS-151 | H43/122 | Insulated panel cladding system, as the Design |
| Self-Smoothing Levelling Compound | FBS-131 | M10/110 | Self smoothing levelling compound. |
| Raised Access Flooring System - Medium Duty | FBS-201 | K41/111 | Fully accessible raised access flooring system. |
| Battened Timber Sprung Flooring | FLS-131 | K20/110 | Hardwood sprung dance floor system. |
| Stone Large Format Tile | FLS-201 | M40/111 | Stone large format tile as shown on the Design |
| Ceramic Tile Flooring - Large Format | FLS-202 | M40/112 | Ceramic tile floor finish as shown on the Design |
| Porcelain Tile Flooring | FLS-204 | M40/114 | Porcelain tile floor finish as shown on the Design |
| Precast Terrazzo Tile Floor Finish | FLS-205 | M41/110 | Slip resistant precast terrazzo tile floor finish |
| Resin Floor Finish | FLS-401 | M12/111 | External grade, resin anti-slip floor finish, as shown on the Design Drawings. |
| Fire and Smoke Curtain Systems | FRP-301 | L20/611 | Fire curtain with smoke seals, as shown on the Design Drawings. Generally integrated within lift architraves or adjacent wall finishes. |
| Fixed Vertical Ladder | FUS-102 | L35/131 | Fixed vertical ladder, as shown on the Design |
| External Maintenance Metal Staircase | FUS-111 | L30/501 | External maintenance metal staircase, as shown on the Design Drawings. |
| Internal Metal Grate Platform | FUS-141 | L35/141 | Non-slip metal maintenance walkways. |
| External Metal Grate Platform | FUS-142 | L35/142 | Non-slip metal maintenance walkways. |
| Internal Timber Hinged Doorset - Painted | IDS-101 | L20/251 | Internal timber doorset with paint finish. |
| Internal Timber Hinged Doorset - Veneered | IDS-102 | L20/252 | Internal timber doorset with veneer finish. |
| Internal Timber Hinged Doorset - WC Cubicle | IDS-103 | L20/253 | Internal timber doorset with veneer finish to WC |
| Internal Timber Hinged Doorset - Shower Cubicle | IDS-104 | L20/254 | Internal timber doorset with veneer finish to |
| Internal Metal Hinged Doorset | IDS-201 | L20/401 | Internal metal hinged doorset. |
| Internal Insulated Metal Hinged Doorset | IDS-202 | L20/402 | Internal insulated metal hinged doorset. |
| Fire Rated Glazed Doors | IDS-302 | L20/302 | Internal fire rated glazed doors, as shown on the Design Drawings and Door Schedule. |
| Glazed Sliding Doors - Fire Rated | IDS-303 | L20/303 | Glazed sliding door system |
| Internal Riser Doorset | IDS-401 | L20/481 | Internal riser doorset, as shown on the Design |
| Internal Insulated Metal Hinged Riser Doorset | IDS-402 | L20/403 | Internal insulated metal hinged doorset. |
| Internal Timber Hinged Doorset - Feature Timber Linin | IDS-406 | L20/255 | Internal timber doorset with feature timber lining |
| Internal Wall Tiled Access Panel | IDS-512 | L20/480 | Proprietary non-rated visually discreet metal wall access panel for ceramic/ marble tiled walls. |
| Internal Ceiling Access Panel | IDS-521 | K10/265 | Plasterboard ceiling access panel. |
| Internal Sliding/ Folding Partition System | IDS-701 | K30/121 | Top hung, sliding/ folding partition system. |
| Internal Folding Partition Door System | IDS-702 | K30/122 | Bi-fold partition door system, as shown on the Design Drawings. |
| Durasteel Wall | IWL-901 | K13/251 | Proprietary fire rated wall system. |
| Gypsum Board Partition/ Linings/ Ceiling Systems (Ge | 01/ LIN-101/ CL | K10/111 | Metal stud plasterboard partitions/ wall linings/ ceiling types with acoustic slab insulation. |
| Metal Stud Partition System - Shaftwall | IWS-181 | K10/119 | Metal stud plasterboard partition. |
| Aluminium Framed Glazed Screen System | IWS-201 | K30/110 | Internal metal-framed glazed screen with |
| Internal Green Wall | IWS-901 | K13/191 | Internal preserved moss feature wall panels |
| Bronze Cladding - Lift Doors | LIN-106 | K13/141 | Internal bronze cladding |
| Bronze Cladding - Lift Opening | LIN-107 | K13/142 | Internal bronze cladding |
| Wall Protection - Metal | LIN-171 | K13/146 | Internal metal wall protection |

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| Timber Veneered Feature Walls | LIN-236 | K13/151 | Timber veneered feature wall panels featuring curved blocks/ vertical timber battens and |
| Timber Veneered Wall Lining | LIN-237 | K13/152 | Solid hardwood timber panel wall lining, as shown on the Design Drawings. |
| Natural Stone Tile Wall Finish | LIN-301 | M40/141 | Natural stone tile wall finish, as the Design |
| Porcelain Tile Wall Finish | LIN-411 | M40/151 | Large format porcelain tile wall finish, as the Design Drawings. |
| Porcelain Tile Wall Finish | LIN-412 | M40/152 | Proprietary porcelain medium format wall tile finish, as the Design Drawings. |
| Porcelain Tile Wall Finish | LIN-413 | M40/153 | Proprietary porcelain medium format wall tile finish, as the Design Drawings. |
| Accessible Duct Panel Lining Systems (IPS) | LIN-501 | K32/151 | Accessible duct panel lining systems (IPS) with metal support frame. |
| GRG Panel Wall Lining | LIN-646 | K13/236 | GRG wall lining, as shown on the Design |
| Leather Panel Wall Lining | LIN-651 | K13/241 | Leather panel wall lining, as shown on the |
| Shopfront Louvre - Ground Floor (Visual Intent Only) | LVR-101 | L10/651 | Shopfront louvre banding to ground floor. |
| Louvre - Barrel Curve Façade (Visual Intent Only) | LVR-102 | L10/652 | Louvre to barrel curve façade. |
| Incline Architectural Louvres - Modern Mansard Façad | LVR-103 | L10/653 | Incline architectural louvres to modern mansard |
| Acoustic Louvres - Modern Mansard Façade (Visual In | LVR-104 | L10/654 | Acoustic performance louvres behind incline architectural louvres to the modern mansard |
| Louvre Screening - Level 7 Plant (Visual Intent Only) | LVR-105 | L10/655 | Louvre screening to level 7 external plant. |
| Louvre Screening - Adiabatic Cooler (Visual Intent Onl | LVR-106 | L10/656 | Louvre screening to adiabatic cooler. |
| Louvre Screening - North Core Roof Plant (Visual Inter | LVR-107 | L10/657 | Louvre screening to north core roof plant. |
| UKPN Louvre Screening | LVR-108 | L10/658 | UKPN louvre screening. |
| Accessories/ Sundry Items Scope (Generally) | MAS-101 | F30/105 | All necessary masonry accessories required to complete the works to be proposed by the Contractor for the acceptance of the Architect and will achieve the required performance of |
| Blockwork Wall | MAS-206 | F10/201 | Concrete fair-faced blockwork configured and to locations indicated on the Design Drawings. |
| Blockwork Wall - WI System | MAS-401 | F10/270 | Hollow, WI blockwork wall. |
| Automatic Opening Vent | RFL-101 | L20/636 | Automatic opening vent. |
| Hot Melt Warm Roof System | RFS-101 | J31/130 | Hot melt warm roof system. |
| Hot Melt Warm Roof System - Concrete Pavers | RFS-102 | J31/131 | Hot melt warm roof system. |
| Overslab Roof System | RFS-103 | J31/132 | Overslab roof system |
| Hot Melt Warm Roof System - Bio Solar | RFS-104 | J31/133 | Warm roof system, hybrid system that has waterproofing below and above. |
| Overrun Roof System | RFS-105 | J31/134 | Overrun warm roof system |
| Floating Screed | SCR-101 | M10/106 | Proprietary modified floating screed |
| Bonded Screed | SCR-201 | M10/111 | Modified bonded screed |
| Internal Metal Staircase | STR-301 | L30/231 | Internal metal staircase and landings, as the Design Drawings |
| Internal Feature Metal Stair | STR-302 | L30/236 | Internal feature metal staircase and landings, as the Design Drawings |
| Internal Feature Metal Stair | STR-303 | L30/240 | Internal feature metal staircase and landings, as the Design Drawings |
| Internal In-Situ Concrete Stair | STR-341 | L30/251 | Internal in-situ concrete staircase |
| Internal In-Situ Concrete Stair | STR-501 | L30/256 | Internal in-situ concrete staircase |
| Internal In-Situ Concrete Stair | STR-506 | L30/260 | Internal in-situ concrete staircase |
| Steel Framed Glazed Window System | WIN-111 | K30/111 | Internal metal-framed glazed window |
| Waterproofing Systems (Generally) | WPS-101 | J30/110 | Waterproofing system providing a fully waterproof, damp-proof and water vapour impermeable barrier resistant to gases and contamination from the ground, to locations |

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# A10
# Project Particulars

## Clauses

### 110 Project Particulars

1. Project is located St James Marketplace, London
2. The Project measures 13,436m² office NIA, 26,467m² GEA

### 120 Two Stage Procurement of JCT Design & Build Contract

1. The Main Contract Works are being procured through a Two Stage Design & Build process comprising:
    1.1. Procurement is based on RIBA Stage 3 level of design information. During the First D&B Stage, certain packages will be jointly procured by the preferred Main Contractor and Client Team to infirm the agreement of the Contract Sum.
    1.2. Contract will be awarded based upon RIBA Stage 4 and 5 level of design information, at the end of the second D&B Stage, with novation of the Design Team as depicted in the Employer's Requirements.
2. The Contractor shall examine the Design, accept full responsibility for its entirety and complete the Detailed Design, etc. and deliver the Works, such that they maintain or exceed the quality, performance criteria and design/ visual intent of the Design.
3. The Design will generally be of a descriptive nature, requiring the appointment of suitable specialist subcontractors, as necessary, to complete the Detailed Design. Refer to Clause A21.140 herein for further details.
4. Specialist subcontractors are to complete the Detailed Design, manufacture, supply and installation of fully warranted and guaranteed systems, etc. as described in the Specification. The Detailed Design submissions are to be reviewed for performance and compliance within the Design by the Employer's Agent and/or the novated Design Team. Any proposed alternatives are to be fully visually compliant and be equal or better, in terms of robustness, performance, guarantee, suitability and quality to that specified or implied.
5. The Design may include a limited number of Prescriptive Elements, which are for incorporation in the Works, strictly in accordance with the Employer's Requirements and manufacturer's recommendations.

Ω End of Section

# A21
# Definitions and Compliance

## Clauses

### 110 Definitions

1. **Defined Terms**: The following definitions appear from time to time throughout the Specification. Where reference is made using slightly different terms they are to be interpreted as one of the below by the Architect. If unsure, request clarification at time of Tender.
2. **Design Stages**: Design to be carried out as follows:
    2.1. **Stage 3**: Design Requirements carried out prior to award of the PCSA.
    2.2. **Stage 4**: Detailed Design carried out by the design team or the Contractor.
    2.3. **Stage 5**: Detailed Design, Technical Design and Working Drawings carried out by the Contractor or Specialist Subcontractors during the second D&B procurement stage.
3. **Employer's Agent**: The company or person representing the Client which may also be referred to as Employer's Representative or Employer herein.
4. **Specification/ Architectural Specification/ this specification**: This Specification, comprising sections A to Z inclusive, representing the architectural design requirements, which links the products, systems and materials shown on the Design Drawings to the Specification via a Material and System Sheet (MSS).
5. **Design**: All information (drawings, schedules, specifications and other supplemental information) forming part of the Employer's Requirements (the Contract information and the Contractor's Proposals) and agreed for Contract purposes, which sets out Descriptive Design, the visual intent, any Prescriptive Elements, performance criteria and minimum quality standards to be maintained in the completion of the Detailed Design by the Contractor.
6. **D(d)etailed D(d)esign**: Working Drawings, schedules, detailed specifications and Technical Proposals to complete the Design and confirm the final agreed solution to meet the fundamental intrinsic requirements of the Design completed by, or on behalf of the Contractor.
7. **Working Drawings**: Construction drawings and supporting information produced to an appropriate scale, showing final co-ordinated layouts, configuration, details, interfaces and finishes of the Works.
8. **Technical Proposals**: Detailed specifications, calculations, assembly drawings, technical submittals, data sheets, etc. provided by a specialist subcontractor confirming the final agreed solution to the Design.
9. **Technical Submittal(s)**: Product literature, Calculations, Certificates, Technical Data Sheets and manufacturers details supplied to confirm the compliance with the specification. All submittals to be provided with an executive summary confirming compliance and/or any deviations proposed.
10. **Descriptive Design**: Design information, performance criteria and Visual Reference that describe the visual intent of the Works, parts of the Works or elements within the Works where the specialist subcontractors shall be responsible for meeting the visual design intent and the requirements of the Specification, including any Visual References, including the performance criteria.
11. **Visual Reference**: Where the Design is of a descriptive nature, proprietary products or materials may be identified to indicate the visual intent, appearance aspirations, performance requirements, product quality, etc. only of the Design. The Contractor may use the stated products, having first satisfied themselves as to their full compliance with the Specification,

performance criteria, as well as contractual and Statutory requirements. Alternatively, the Contractor may submit similar products for consideration as set out in clause A21.120.4.

12. **Prescriptive Elements**: Named elements of the Works that are proprietary systems, products or materials, which are to be incorporated in the Works strictly in accordance with the manufacturer's recommendations and the Employer's Requirements. If alternative systems, products or materials are proposed, the Contractor shall remain responsible for meeting the requirements of the Design and such elements should be treated as Descriptive Design.
13. **Review Status**: The Contractor (including specialist subcontractor) submissions are to be reviewed using the A-B-C and D system as described below. The Contractor shall not instruct manufacture or installation until A status is achieved for each element of the Works. For clarity the status definitions are:
    13.1. A = Reviewed. No comments.
    13.2. B = Reviewed. Incorporate comments and resubmit.
    13.3. C = Reviewed. Rejected for reasons noted, rework and resubmit.
    13.4. D = Not Reviewed. Not applicable to the architectural design scope.
    13.5. If the Review Status is marked 'A' or 'B' it does not relieve the Contractor/ subcontractor of their obligations to comply with the design intent and performance criteria, in any way, unless explicitly stated otherwise. Where the submittal is marked 'D' it is incumbent for the Contractor to ensure that the correct, qualified and appropriate reviewer is utilised.
14. **Samples**: Labelled physical products/ materials along with supportive technical information, representing elements to be incorporated in the Works, subject to acceptance.
15. **Visual Mock-ups**: Full size, three dimensional indicative model demonstrating size, orientation, shape, etc. manufactured from temporary materials.
16. **Wider Mock-ups**: As 'Visual Mock-ups' but covering multiple elements and sections of the Specification and also to include, where required, elements designed by others, including but not limited to the Structural Engineer and Services Engineer. Wider Mock-ups to also be used for demonstrating buildability and co-ordination/ interface between different Specialist contractor elements.
17. **Prototypes**: Full size elements representing the final Detailed Design, incorporating actual materials and finishes, but not forming part of the permanent works. Prototypes may be tested to confirm compliance with performance requirements.
18. **Benchmarks**: First section of a pre-identified installation on site for inspection and acceptance.
19. **A(a)cceptance/ A(a)pproval/ A(a)pproved**: Confirmation that the proposed drawings, submittals etc. appear to meet the requirements of the Design and the performance criteria as set out in the Specification. It does not relieve the Contractor/subcontractor of their obligations to comply with the Specification, relevant Codes and Standards, or any other requirement, unless explicitly stated otherwise.
20. **Additional Information**: Other information necessary to adequately demonstrate Detailed Design compliance with the Design, including, but not limited to certification, manufacturers' information, warranties/ guarantees, programme/ delivery information, detailed QA/ QC systems, relevant reports and data/ calculations.
21. **Reviewed**: The review of a Technical Proposal for non-compliance with the Design.

## 120 Specification Compliance

1. The Contractor /Subcontractor Shall comply with the Specification which sets out minimum standards of quality and performance to be maintained or exceeded in the Works.
2. For sections/ elements where the Design is of a descriptive nature, the Contractor is to appoint a specialist subcontractor to complete the Detailed Design for review and Acceptance.

3. All named systems, products and materials shall be installed where specified in accordance with the manufacturer's recommendations.
4. Any proposed alternatives shall be visually compliant and of the same or better in terms of robustness, performance, guarantee, suitability and quality. Technical submittals are required with full manufacturer's details to substantiate acceptability of alternative systems, products and materials.
5. The Contractor / Subcontractor shall not commence any part of the Works without acceptance of all submittals.
6. Provide all information to substantiate Specification compliance and adherence to safety requirements, including as necessary, any additional structural work.

## 130 Contractor and Specialist Subcontractor Design Responsibility

1. For sections/ elements where the Design is of a descriptive nature, the Contractor / Subcontractor shall complete the Detailed Design by adopting the Design in compliance with all relevant Statutory Regulations, Codes of Practice, National Standards, Local Authority Bye-Laws/ Regulations, Fire/ Building Codes and any other applicable rules and regulations, and shall:
2. Check buildability, performance requirements, regulations, codes and manufacturers' recommendations to confirm suitability. Report any concerns at time of Tender.
3. Provide Working Drawings, as necessary, and detailed specifications for Detailed Design elements, together with all necessary additional information for review and acceptance before placing orders or commencing manufacture.
4. Provide detailed specifications for all elements as necessary, including any Technical Proposals, together with relevant additional information for review and acceptance, before placing orders or commencing manufacture.
5. Demonstrate and maintain buildability and structural integrity .
6. Maintain function, performance and visual intent.
7. Ensure that all interface conditions are acceptable and comply with performance and design intent requirements.
8. Comply with all performance requirements which are to be considered as the minimum acceptable.
9. 1. If the facade or any other works are split into multiple packages, completed by multiple parties, the Main Contractor must produce or appoint one of the specialist subcontractors to produce a single coordinated set of drawings incorporating the details produced by the relevant subcontractors. These coordinated drawings are to be those made available for review and ultimately form the final as-built set of drawings.
10. Provide a design risk assessment.
11. Confirm structural principles in terms of performance and how loads are transferred through the structure in a safe manner whilst taking account of all tolerance requirements.
12. Minimise modification of proprietary products and maintain suitability at all times.
13. Comply with CDM Regulation Designer Duties.
14. Comply with The Building Safety Act 2022 Designer Duties.
15. Comply with manufacturers' recommendations and ensure suitability for use in the Works.
16. Carry out and satisfy all testing requirements.
17. Provide a complete and fully co-ordinated Detailed Design.
18. Working Drawings to be complete and fully co-ordinated to finalise all details, assemblies, junctions, tolerance allowances, fixings, interfaces, etc.

19. Comply with all planning and Local Authority requirements and provide them with any information/ submissions requested.
20. Provide necessary supplemental information to demonstrate full compliance, including structural information/ calculations, etc.
21. Provide summary of deviations for consideration. These may not be acceptable. Take full responsibility for those that are deemed acceptable.
22. All alternative materials, products, systems, details, and the like to be submitted for review:

    22.1. In good time so as to have no adverse effect on the project programme.

    22.2. To meet all criteria of the Design, including quality, design/ visual intent, performance, etc.

    22.3. No orders to be placed until acceptance is confirmed in writing.

## 140 Use of Specialist Subcontractors

1. The Contractor shall engage specialist subcontractors on a design and build basis where stipulated in the Specification, to ensure that the Detailed Design, including Technical Proposals comply with the requirements of the Design and provide adequate warranties and guarantees. The Design of such elements will be of a descriptive nature, will be completed by the Contractor's in-house, outsourced, or novated design team through the production of Working Drawings and Specifications or other methods as agreed.
2. Subcontractors will be required to fully comply with the Design. Any variation or alternatives must be submitted as technical proposals to demonstrate the alternatives meet or exceed the requirements of the Design, only agreed and equivalent or better (technically, performance and visual) design alternatives will be acceptable. If alternative designs are put forward, the specialist subcontractors shall remain responsible for meeting the requirements of the Design, Specification, and the performance criteria, as set out above.
3. All specialist subcontractors are to provide collateral warranties in favour of the Employer, to the effect that the specialist subcontractor will have the same obligations to the Employer as the Contractor has to the Employer with respect to the specialist subcontractors' work.
4. The Contractor shall provide a list of proposed specialist subcontractors with the Tender return.
5. Specialist subcontractors to be engaged to complete the Detailed Design (including Technical Proposals), for the following elements as described in the Design Responsibility Matrix.

## 141 Architectural Specification Design Type

1. Specialist subcontractors shall be engaged to complete the Detailed Design (including the provision Technical Proposals), in accordance with the specification categories P, P/D or D as defined below, for the elements of the works as described in the Design Responsibility Matrix.
2. **P - Performance Specification (with Prescriptive elements) and Detailed Design intent.**
   The Designer/Specifier outlines performance requirements and may reference specific products or systems. The specialist contractor can propose equivalent alternatives. Sub-contractors select proprietary products that meet specifications, providing technical submissions for approval and following manufacturer recommendations during installation.
3. **P/D - Performance Specification with Descriptive Design intent requiring Contractor design input/assistance).**
   The Designer/Specifier requires technical submissions and proposals to ensure the contractor's final solution meets requirements without significant deviation from the original intent. The Designer coordinates design details and general arrangements, while the subcontractor provides input, selects suitable products, adapts them as needed, and ensures compliance with performance and design objectives.

4. **D - Descriptive Design intent and Performance specification (Visual and Functional requirements only with full or complex in context contractor design and warranty).**
The Designer/Specifier outlines visual and functional intent but does not provide detailed design or production information. The subcontractor is responsible for creating a full design proposal, selecting products, and constructing and warranting the work to meet the specified requirements.

## 142 Subcontractor responsibility

1. Specialist subcontractors shall be engaged to complete the Detailed Design (including the provision Technical Proposals), in accordance with the sub-contractor responsibility as defined below, for the elements of the works as described in the Design Responsibility Matrix.

    1.1. **1: No design by Contractor.**
    Sub-Contractor Constructs work in accordance with production information and prescriptive specification. No design responsibility. Contractor responsible for materials and workmanship only. May obtain product manufacturer's guarantees and warranties to Client.

    1.2. **2: No "in context" design by Contractor. Limited to product selection.**
    Sub-Contractors Design input limited to selecting 'off the shelf' products that satisfy performance specification. Constructs work using selected components. Contractor responsible for materials and workmanship and for design input in selecting products. May obtain product manufacturer's guarantees and warranties for Client.

    1.3. **3: Simple "in context" design by Contractor. Limited to product selection and adapted details.**
    Specialist Sub-Contractor input limited to selecting 'off the shelf' products that satisfy the performance specification and adapting them as necessary to suit the Specifier's design. Constructs work using selected components as adapted.
    Sub-Contractor responsible for materials and workmanship and for design input in selecting and adapting products. May obtain product manufacturer's guarantees and warranties for Client.

    1.4. **4: Complex "in context" design by Contractor. Provides production information (shop drawings).**
    Specialist Sub-Contractor provides detail design and production information, including confirming prescriptive specification of products. Constructs work in accordance with that design, using selected components adapted as necessary. Specialist's Subcontractor responsible for materials and workmanship and for design compliance with design intent. May obtain product manufacturers' guarantees and warranties for Client.

    1.5. **5: Full Contractor Design. Complete the full package including drawings and specification.**
    Contractor Completes design for the project and all production information, including prescriptive specifications. Constructs complete package in accordance with that design. Contractor's responsibilities for design of package and for materials and workmanship. May obtain product manufacturer's guarantees and warranties for Client.

## 150 Satisfying Performance Criteria and Design Intent

1. Generally:

    1.1. Maintain design and visual intent through to completion of the Works.

    1.2. Stated performance criteria to be considered as minimum requirements.

    1.3. Provide the necessary warranties and guarantees for all products, systems and components.

    1.4. Demonstrate compliance with the Specification at all times.

2. Design Life: Comply with:
    - BS ISO 15686: Part 1, Table B.1
    - BS 7543 table D1.
3. Service Life - Comply with:
    3.1. BS ISO 15686: Buildings and construction assets - Service Life Planning.
    3.2. BS 7543 Guide to durability of buildings and building elements, products, and components. Table D 1 - Service Life.
    3.3. Primary components are those generally inaccessible with a Service Life equal to Design Life, whereas Secondary components are generally accessible and require maintenance or replacement according to manufacturers' recommendations.
    3.4. Identify and confirm Secondary components and confirm Service Life expectations.
    3.5. Refer to Table D. 1 for Service Life of secondary components with the exception of the following which are to take precedence:

| Item | Years | Status |
|---|---|---|
| Anodising | 50 | Maintainable |
| Applied coatings to metalwork | 30 | Maintainable |
| Blinds / Blind motors | 15 | Replaceable |
| Coated glass / Fritted glass | 25 | Replaceable |
| Double skin blinds | 25 | Replaceable |
| Exposed sealant / Structural silicone sealant | 25 | Replaceable |
| Fire stops at slab edges | 60 | Permanent |
| Flashings / EPDM Membranes | 25 | Maintainable |
| Galvanised steel | 40 | Maintainable |
| Gaskets - external | 25 | Replaceable |
| Gaskets - internal | 30 | Replaceable |
| Glass units / Single laminated glass | 25 | Replaceable |
| Green roofing systems | 20 | Replaceable |
| Insulating material | 60 | Replaceable |
| Ironmongery (excluding those given above) | 15 | Replaceable |
| Metal brackets | 60 | Replaceable |
| Metal frames | 60 | Replaceable |
| Paint coating: PPC / Paint coating: epoxy, acrylic | 25 | Maintainable |
| Single membrane waterproofing | 20 | Maintainable |
| Stainless steel | 30 | Maintainable |
| Stone | 40 | Maintainable |
| Structural steel elements and brackets, Aluminium frames and profiles / | 50 | Permanent |
| The cladding system | 60 | Permanent |

4. Warranties and Guarantees: Provide certified Warranties and Guarantees for the Detailed Design, materials, products, goods and systems incorporated in the Works.

## 160 Contractor's and Subcontractor's Proposals

1. Contractor to submit Technical Proposals and Technical Submittals, as required and set out within the Project Vista 'sub-contractor design specification matrix, demonstrating compliance with the Employer's Requirements, which includes the Design.

2. Contractor's proposals to include plans, elevations, key details, layouts, risk analysis, build methodology, technical specifications, etc. which show a complete understanding of the Design in context with the project and interfacing elements.
3. The Detailed Design is not to change the design/ visual intent, quality, performance, function or design/ service life of the Works.
4. Any variation or alternatives must be submitted as technical proposals to demonstrate the alternatives meet the performance requirements of the Design. Only agreed and equivalent design alternatives will be acceptable. If alternative designs are put forward by the Contractor, the Works shall be treated as Descriptive, and a specialist subcontractor shall become responsible for meeting the requirements of the Design, the requirements of the Specification, and the performance criteria, as set out above, and provide detailed specifications, full details, calculations and Working Drawings as required to complete the Works.
5. Provide a detailed Risk Assessment.

Ω End of Section



make

# A33
# Quality standards/ control

## Clauses

### 110 Incomplete documentation

1. General: Where and to the extent that products or work are not fully documented, they are to be:
   1.1. Of a kind and standard appropriate to the nature and character of that part of the Works where they will be used.
   1.2. Suitable for the purposes stated or reasonably to be inferred from the project documents.
   1.3. Contract documents: Omissions or errors in description and/ or quantity shall not vitiate the Contract nor release the Contractor from any obligations or liabilities under the Contract.

### 120 Workmanship skills

1. Operatives: Appropriately skilled and experienced for the type and quality of work.
2. Registration: With Construction Skills Certification Scheme.
3. Evidence: Operatives must produce evidence of skills/ qualifications when requested.

### 130 Quality of products

1. Standards: Comply with the Standards stated in the Specification. Where stated Standards have been superseded/ replaced/ enhanced the version published at the time of Contract award shall prevail. Advise at time of Tender any such occurrences.
2. Generally: All products, materials, components, systems, etc. to be new (unless otherwise agreed by the Employer) and compliant with appropriate/ applicable National Standards, Codes of Practice and best practice.
3. National Standards/ Codes of Practice: Comply with current Standards, Codes, Planning Requirements, etc. Any such documents referred to herein which have been updated or superseded, advise at the time of Tender and accommodate the latest publication.
4. Supply of each product: From the same source or manufacturer.
   Whole quantity of each product required to complete the Works: Consistent kind, size, quality and overall appearance.
5. Tolerances: Where critical, measure a sufficient quantity to determine compliance. Such measurement is to be cumulative in nature to ensure completeness.
6. Deterioration: Prevent. Order in suitable quantities to the programme of the Works and use in an appropriate sequence.

### 131 Quality Control

1. Generally: Comply with BS EN ISO 9001 and ensure compliance with the Specification.
2. Certification: In accordance with BS EN ISO 9001.
3. Compatibility: Materials, systems, components and products to be compatible.
4. Codes and Standards: Comply with current British Standards/ Codes, European Norms and relevant By-Laws. Notify changes immediately.
5. Sustainability: Comply with all sustainability requirements to achieve Client targets.
6. Storage and Handling Materials:

6.1. Compliance: With BS 8000.

6.2. Protection: Protect all items for inclusion in the Works at all times from point of manufacture to installation and ensure pristine condition of all elements at handover.

6.3. Pest Infestations: Implement recommendations of BRE Digest 415 - DIG 415 to avoid pest infestations.

6.4. Cleaning: Clean all elements thoroughly prior to handover.

6.5. Defects: Scratches, blemishes, dents, mould, or any other visual defects will not be accepted.

7. Safety: Comply with and demonstrate compliance with CDM and applicable Health and Safety Regulations at all times. Do not incorporate any material, product or component that may be a potential health hazard. Maintain a specific/ residual risk analysis document.
8. Maintenance Manual: Provide and maintain a detailed Maintenance Manual, to include training requirements, test/ inspection results, replacement processes, manufacturers/ suppliers, recommendations for spares, etc, covering all elements and components of the Works.
9. Material Quality:

9.1. QA/QC: Implement QA/ QC procedures to guarantee compliance with the qualitative, performance and visual intent described in the Specification.

9.2. Certification: Products and systems to be certified by the British Board of Agrement (BBA).

9.3. Installation: Final installations to be to a standard of accepted samples and prototypes.

10. Deleterious Materials: Implement the "Good Practice in the Selection of Construction Materials" published by the British Council of Offices. Do not incorporate any of the following in the Works:
    - Asbestos of any kind.
    - Wood wool slabs.
    - Lead, lead paint or other liquid applications. (Note: Some roofing and rainwater goods containing minimal amounts of lead may be acceptable in certain circumstances. Seek acceptance prior to ordering).
    - Urea formaldehyde foam and any product that may release formaldehyde.
    - Polyurethane based insulation.
    - Expanded polystyrene and polyurethane.
    - Composite metal panels containing an interstitial polyethelene insulated core.
    - Chloroflourocarbons and hydrochloroflourocarbons of any kind.
    - High alumina cement.
    - Tributyltin (TBT).
    - Chromated Copper Arsenate (CCA).
    - Pentachlorophenol PCP).
    - Contaminated gypsum board.
    - Pitch polymer DPC.
    - Tropical hardwoods.
    - Man made or natural mineral fibres that are 3 microns diameter and 200 microns in length, to include those that are encapsulated, unsealed and stabilised preventing migration.
    - Admixtures containing calcium chloride.
    - Polychlorinated biphenyis (PCB) and polychlorinated terphenyis (PCT).
    - Lindane wood treatment/ insecticidal spray.
11. Timber: Procure only from proven and demonstrable sustainable sources, including but not limited to:

11.1. Avoid the use of tropical hardwoods, including veneers, lippings and boards.

11.2. Avoid timber species noted on the IUNC REDLIST listed other than "least concern".

11.3. Timber and timber products to be labeled with the forestry steward ship council (FSC) trade mark, programme for the Endorsement of Forest Certification or other acceptable label from a recognised independent certification provider.

11.4. UK Government Timber Procurement Policy.

## 132 Sustainability

1. Comply with the Employer's sustainability and environmental objectives to achieve the BREEAM, WELL, NABERS and EPC ratings required confirmed in a Final Assessment demonstrating achievement of the target rating score. The Final Assessment is to confirm and compare where overarching sustainability criteria have been met, exceeded or not met.
2. Comply with the project Sustainability Report(s) to achieve target ratings and the Crown Estates's DSP V4 Performance requirements.
3. Follow the principles of the relevant Standards set out in clause A33.132.1 so that the Detailed Design maximises the use of materials, products, systems, techniques, methods etc. which are consistent with those Standards and good sustainability practice.
4. With regard material selection and use pay particular attention to the following aspects to minimise the detrimental impact of the project on the health and wellbeing of the planet and occupants:
    4.1. Environmental Product Declarations (EPD's) including compliance with BS EN ISO 14025, BS EN 15804 and ISO 21930. EPDs to be provided for all systems and materials covered within this Specification to inform product selection and whole life carbon calculations to be checked against the design stage LCA.
        4.1.1. Where EDPs, or an EPD without embodied carbon data is not available, a complete EPD should be requested from the manufacture/supplier.
        4.1.2. Supply Health Product Declaration (HDP) for interior finishes and finish materials, furnishings, and built-in furniture.
    4.2. Durability: All key exposed building elements shall align with the relevant industry durability or quality standards as required by BREEAM.
    4.3. Sourcing of materials: All construction materials should be sourced in accordance with the project's Sustainable Procurement Plan. Sourcing should be in line with guidance provided by BS ISO 20400 and requirements set out in clause A33.131.11.
    4.4. Material Efficiency Principles should follow BS8895: Parts 1 to 3.
    4.5. Transportation confirmed by a travel plan that minimises delivery miles.
    4.6. Waste should be minimised and properly disposed of. Refer to project's Sustainability Report for waste targets.
    4.7. Prioritise low VOC products, natural materials and minimum standards for product selection based on British Standards, and EU standards.
    4.8. Recycled Content of materials and/or products: Refer to project's Sustainability Report for recycled content targets.
    4.9. Circular economy principles including but not limited to, design for adaptability, design for construction, circular materials selection and recourse efficiency. Refer to project's Sustainability Report for circular economy targets.
    4.10. Climate change risk as defined by the IPCC Assessment Reports.
5. Comply with the requirements outlined in the table below. For further information on targeted credits/conditions of each accreditation scheme refer to relevant trackers included in the Employer's Requirements.

| Accreditation | Type | Version | Aspirational |
|---|---|---|---|
| BREEAM | Commercial OfficesShell & Core | 6 | Outstanding |
| BREEAM | Commercial RetailShell Only | 6 | Excellent |
| WELL | Core | 2 | Gold (core building)/ Platinum 'ready'. |
| NABERS | Design for performance / Energy for offices | N/A | 5.5 star |
| EPC | EPC A | 1 | 675kgCO2c/m²/<19,000t/CO2 |
| Wired Score | Commercial | N/A | Platinum |
| SMART Score | N/A | N/A | Platinum |

## 135 Quality of Execution

1. Generally: Fix, apply, install or lay products securely, accurately, plumb, neatly and in alignment.
2. All of the Works to be completed to the satisfaction of the Architect.
3. Compliance: Execute the Works to a very high acceptable standard and comply with:
    - BS 8000
    - BS EN ISO 8501: Part 1:
    - BS 5493:
    - BS EN ISO 12944: Parts 1-8:
    - BS EN ISO 14713:
4. Visible elements: Angles, junctions, edges, corners, arises, component interfaces and all visible elements to be tight, smooth, correctly positioned, neat, properly aligned, and ready for use without remedial work.
5. Tolerances: Maintain acceptable tolerances at all times.
6. Preceding works:
    6.1. Check and ensure accuracy of preceding elements before commencement of installation to guarantee accuracy.
    6.2. Do not cut, drill or alter the preceding work or interfacing installations without prior authorisation. Any such authorised work shall not change the visual intent.
7. Sequencing:
    7.1. Implement proper sequencing of elements and components to guarantee quality of final installation.
    7.2. Ensure relevant work areas/building zones are suitably watertight before commencing internal finishes.
8. Preparation: Surfaces to receive treatment or attachments to be suitably prepared to ensure accurate installation without causing any damage or reduction in performance, quality or visual intent.
9. Tools: Minimise use of power activated tools.
10. Cleaning:
    10.1. Blast cleaning: Blast cleaning to be grade Sa 2.5 after removal of all debris and deposits to achieve required surface finish quality. Primer to be applied immediately after blast cleaning.

10.2. Prior to installation: Remove efflorescence, dirt, dust, traces and any other contaminant prior to installation or application of finishes.

10.3. Handover: Professionally clean the Works prior to handover.

11. Manufacture: Maximise off site manufacture where specified and appropriate.

12. Installations:

12.1. General: Installations to be permanently stable and remain in position without twisting, warping, loosening, vibrating, knocking, rattling, whistling, etc.

12.2. Weather: Do not carry out installations during inclement weather, unless measures have been taken to mitigate the effects of low temperature, frost, high rainfall, snow, high humidity, etc.

13. Joints:

13.1. Joints to be as shown on the Working Drawings, be perfectly formed, tight, properly aligned, without lipping and visually acceptable, all within stipulated tolerances.

13.2. Movement joints to be installed as shown on the Working Drawings.

13.3. Minimise the number of joints and maximise alignment.

14. Fire Stopping: Provide permanent fire stopping solutions maintaining compliance with the visual intent. Fire stopping to meet all fire codes and regulations, including spread of flame and smoke. Material to be installed strictly in accordance with manufacturers' recommendations, ensuring that all gaps are properly sealed, and remain permanently in place for the Design Life of the Building.

15. Membranes: Fix membranes to adequately cleaned surfaces.

16. Insulation: Install dry sheet insulation using recommended mechanical or bonded methods, horizontally, vertically or sloping, as appropriate, without lipping, gaps or uneven joints. Stagger joints when installing filter fabric.

17. Protection: Protect the work at all times, including all sides at junctions, joints, perimeters, etc. when installing high performance, reinforced metal or plastic vapour control and air leakage barriers.

18. Welding: When working on site use manual inert gas tungsten arc welding techniques in accordance with BS EN 1011. Welding to be smooth, continuous and visually acceptable.

19. Colour batching: Do not use different colour batches where they can be seen together.

20. Dimensions: Check on-site dimensions.

21. Finished work: Not defective, e.g. not damaged, disfigured, dirty, faulty, or out of tolerance.

22. Location and fixing of products: Adjust joints open to view so they are even and regular.

## 140 Evidence of Compliance

1. Proprietary products: Retain on site evidence that the proprietary product specified has been supplied.

2. Performance specification: Submit evidence of compliance, including test reports indicating:

- Properties tested.
- Pass/ fail criteria.
- Test methods and procedures.
- Test results.
- Identity of testing agency.
- Test dates and times.
- Identities of witnesses.
- Analysis of results.

## 150 Inspections

1. Products and executions: Inspection or any other action must not be taken as approval unless confirmed in writing referring to:
    - Date of inspection.
    - Part of the work inspected.
    - Respects or characteristics which are approved.
    - Extent and purpose of the acceptance.
    - Any associated conditions.

## 160 Related Work

1. Details: Provide all trades with necessary details of related types of work. Before starting each new type or section of work to ensure previous related work is:
    - Appropriately complete.
    - In accordance with the project documents.
    - To a suitable standard.
    - In a suitable condition to receive the new work.
2. Preparatory work: Ensure all necessary preparatory work has been carried out.

## 170 Manufacturer's recommendations/ instructions

1. General: Comply with all manufacturer's printed recommendations and instructions.
2. Exceptions: Submit details of changes to recommendations or instructions.
3. Execution: Use ancillary products and accessories supplied or recommended by main product manufacturer.
4. Products: Comply with limitations, recommendations and requirements of relevant valid certificates.

## 210 Samples

1. Scope: Provide the samples listed in the Specification plus any others which are deemed necessary to demonstrate the Contractor's proposed alternatives.
2. Products or executions: Comply with all other specification requirements and in respect of the stated or implied characteristics either:
    - To an acceptable level.
    - To match a sample previously accepted as a standard for the purpose.

## 211 Mock-ups

1. Scope: Provide the Mock-ups listed in the Specification and/or shown on Design Drawings.

## 212 Wider mock-ups

1. Scope: Provide the Wider Mock-ups listed in the Specification and/or shown on Design Drawings.
    1.1. Scope:
        1.1.1. Refer to the Design Drawings
2. Detailed requirements/ configurations: Refer to the Design Drawings and individual work Sections of the Specification



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## 213 Prototypes

1. Scope: Provide the prototypes listed in the Specification and/or shown on Design Drawings.
2. Use: Finalising details, appearance, connections, finish, etc. and testing where necessary to prove compliance with performance requirements.

## 214 Benchmarks

1. Scope: Provide the Benchmarks listed in the Specification.

## 220 Acceptance of products

1. Submissions, samples, inspections and tests: Undertake or arrange to suit the Works programme.
2. Acceptance: Relates to a sample of the product and not to the product as used in the Works. Do not confirm orders or use the product until acceptance of the sample has been obtained.
3. Complying/control sample: Retain in good, clean condition on site until Practical Completion of the Works. Remove when no longer required.

## 230 Acceptance of execution

1. Submissions, samples, inspections and tests: Undertake or arrange to suit the Works programme.
2. Acceptance: Relates to the stated characteristics of the sample. (If approval of the finished work as a whole is required this is specified separately). Do not conceal, or proceed with affected work until compliance with requirements is confirmed.
3. Complying sample: Retain in good, clean condition on site. Remove when no longer required.

## 320 Setting out

1. General: Submit details of methods and equipment to be used in setting out the Works.
2. Levels and dimensions: Set out the Works accurately in terms of position, level and alignment. Check and record the results on a copy of drawings. Notify discrepancies and obtain instructions before proceeding.
3. Inform: When complete and before commencing construction.
4. Accuracy: Set out plan positions of principal elements of the Works within a tolerance from primary grid positions of +/-2mm unless otherwise agreed in advance. Check accuracy of previous installations to ensure acceptable installation of following work.

## 330 Appearance and fit

1. System Tolerances:
    1.1. Comply with BS EN ISO 286: Part 1.
    1.2. Maintain dimensional tolerances included in the Detailed Design.
    1.3. Immediately notify any tolerance omissions, inconsistencies, incompatibilities and/ or deviations from tolerance requirements when discovered.
    1.4. Working Drawings to show how tolerances are to be achieved and demonstrate adherence to tolerance requirements throughout manufacture and installation.
    1.5. Check completed work on site prior to commencing manufacture where possible to check existing installation tolerances.
    1.6. Accommodate anticipated structural tolerances and movement.
    1.7. Continually monitor tolerances achieved and keep detailed records.

1.8. Where tolerance clashes/ discrepancies occur, the most stringent is to be met.

1.9. Keep records and report deviations/ discrepancies of all tolerances from manufacture to installation.

1.10. Apply the most stringent tolerance requirement when different solutions are determined for the same condition.

1.11. Do not compound tolerances for adjoining elements, each of which is to be treated individually.

1.12. Prepare written methodology adopted to maintain and check tolerances as the work proceeds.

2. General tolerances (maximum): To BS 5606, tables 1 and 2.

## 340 Critical dimensions

1. Critical dimensions: Set out and construct the Works to ensure compliance with the tolerances stated.

## 350 Levels of structural floors

1. Maximum tolerances for designed levels to be +/- 10mm for:
   1.1. Floors to be self-finished, and floors to receive sheet or tile finishes directly bedded in adhesive
   1.2. Floors to receive dry board/ panel construction with little or no tolerance on thickness
   1.3. Floors to receive mastic asphalt flooring/ underlays directly
   1.4. Floors to receive mastic asphalt flooring/ underlays laid on mastic asphalt levelling coat(s)
   1.5. Floors to receive fully bonded screeds/ toppings/ beds
   1.6. Floors to receive unbonded or floating screeds/ beds
   1.7. Floors to be self-finished, and floors to receive sheet or tile finishes directly bedded in adhesive
   1.8. Floors to receive dry board/ panel construction with little or no tolerance on thickness

## 351 General structural requirements

1. Take account of Structural Engineer's documents and specifications, etc. related to loadings, tolerances and design and safely accommodate the worst combination of structural conditions, including pressure coefficients.
2. Take account of BS EN 1991: Part 1.
3. Ensure rigidity of materials and components so that they are not adversely affected by any external or internal vibration and safely withstand all forces and loads imposed at all times.
4. Install and align all parts of the Work to accommodate deviations of any part of the structure.
5. Safely accommodate all movement and stresses imposed by the structure as well as all dead/ live loads imposed by self weight, attachments, environmental conditions, temporary loads during construction, without affecting visual intent.
6. All movement joints to be visually acceptable.
7. Components, materials, systems and products to accommodate expansion/ contraction when conditions range between 5°C and 28°C air temperature, as well as surface differential temperature of between five and eighty degrees centigrade solar radiation.
8. Working Drawings to show methods by which tolerances of elements, including interfaces, are accommodated.
9. Take into account the following:

- BS 6180.
- BS EN 1991 - Eurocode 1.
- BS EN 1991 Parts 1 - 4 (as applicable to location). 9.4. PD 6688.

10. All elements to resist all static and dynamic loads, without deformation or changes to visual intent, including conditions of extreme wind loads, moving elements/ components in close proximity and unexpected water absorption.
11. Take account of all loads imposed during cleaning and maintenance operations including operatives, machinery, products, temporary structures and the like to a minimum of 500 Newtons applied through 100mm square.
12. All imposed loads to be safely transferred to the adjoining structure, without detrimental effect to visual intent, performance or functionality.
13. Accommodate all deflections, including those of a temporary nature, without detrimental effect to visual intent, performance or functionality.
14. Maximum allowable deflection, when fully loaded, to be the lower of either 15mm or 1/175 of clear span.
15. Accommodate all thermal movements caused by climatic conditions externally and internally. Provide necessary permanent/ temporary structural support, as necessary, and take measures to mitigate against noise arising.
16. All components, products, materials, fixtures, fittings and systems are to remain rigid under all conditions unless purposefully designed otherwise.
17. Maximum permitted deflection to be span/400 (span maximum 1500mm).

## 352 General environmental requirements

1. Take account of Services Engineer's documents, specifications, Approved Document Part L of the Building Regulations, etc. related to local conditions, performance requirements, eradication of cold bridging etc.
2. When selecting materials, components, products, etc. take account of prevailing local conditions to ensure suitability for the Design and Service Life of the Works.
3. All products, materials, components etc. to be moisture and water resistant, including resisting changes in moisture content of air or humidity, the effect of freezing and material moisture content.
4. Protect against frost, precipitation of all kinds, humidity, etc. during manufacture, delivery, storage and installation up to handover.
5. Maintain weatherproof and watertight barriers between external and internal conditions, including prevention of water ingress, condensation and capillarity at all times and under all conditions.
6. External, internal or interstitial condensation is not acceptable. Comply with BS 5250 and other applicable codes or standards with regard condensation calculation methods and demonstrate compliance through calculations, testing and adequate detailing.

## 353 General acoustic requirements

1. Refer to the Acoustic Report and ensure compliance, including BS EN ISO 10848 related to flanking transmission and the provision to maintain no less than DnTf, w 55db across transoms, floor edges and mullions.
2. Ensure that the acoustic performance criteria of Sections B to Y of the Specification are achieved, including at penetrations, junctions, and interfaces to maintain acoustic integrity.
3. Audible noise arising from the finished building, including rattles, creaks, whistling, movement etc. is not acceptable. Airborne noise caused by wind, precipitation, air infiltration, etc. may not

exceed 35db (A) at a distance of 1000mm from any external wall, or less if so specified in Sections B to Y of the Specification.

## 354 General security requirements

1. Take account of the Security Report and comply with the Secured By Design Initiative (Association of Chief Police Officers) and LPCB standard LPS 1175 and classification levels stated in Sections B to Y of the Specification.
2. Comply with BS EN 1627 with regard burglar resistance requirements (doorsets, windows, grilles, shutters etc.) and resistance class requirements stated in Sections B to Y of the Specification.
3. Comply with BS EN 356 with regard to manual impact resistance to glazing.

## 355 General fire requirements

1. All products, materials, components, systems and the like to be non combustible and properly certified as such. Internal surfaces and linings to be appropriately rated in accordance with the current Building Regulations and Approved Document.
2. Comply with the following:
    - Project Fire Strategy Report and all relevant Approved Documents of the Building Regulations (including Parts A and B), including maintaining very low flame spread characteristics and the non production of smoke or toxic gases during combustion.
    - BS EN 13501: All relevant parts.
    - BS EN 13501-2 to 4, Building Regulations related to fire resistance and performance.
    - BS7974 - Code of Practice, fire safety engineering.
    - BS 9991 - Code of Practice (design/ management/ use).
    - BS 9999 - Code of Practice (design/ management/ use).
    - BS EN 12101 - Smoke/ heat control systems.
    - LPS 1207 - LPCB Approval and listing of internal temporary covering materials.
    - LPS 1215 - LPCB Approval and listing of internal temporary covering external containment net/ sheet materials.
    - TRADA - Timber Research and Development Association.
    - Requests received from and recommendations made by the Fire Services and Statutory Authorities.
3. Provide test certificates to confirm compliance of all products, materials, components and systems.
4. Provide all necessary smoke and fire resistant cavity barriers, seals and fire/ smoke stops to maintain integrity and insulation, which continue to be effective and in place during a fire.
5. Prevent fire and smoke spreading between fire separation compartments in both floor to floor and area to area on individual floors.
6. Provide designated fire fighting enclosures, lift shafts and lobbies that are adequately ventilated, incorporating sufficient operable vents being a minimum of 25% of the horizontal cross sectional area of each enclosure.

## 356 Resisting damage

1. Refer to and comply with BS 6180 to ensure that all materials, goods, products, systems, finishes, etc. are durable and maintain original appearance and performance throughout the required Design Life and Service Life, provided adequate maintenance and cleaning are carried out in accordance with the manufacturer's requirements.

2. Barriers and guardrails to be adequate and able to resist loads to which they are subjected in the use for which they are designed, without deformation or loss of integrity, including impact from hand held objects and tools.
3. Maintenance and cleaning activities to have no detrimental effect on performance or appearance.
4. No reduction in performance or visual appearance to emanate from exposure to sunlight or weather conditions.

## 357 Resisting corrosion

1. Refer to and comply with:
    - BS 4479 - Material to receive coatings.
    - BS 5493 - Code of Practice for protective coating to steel and iron.
    - BS EN ISO 12944: Parts 1-8 - Protective paint systems to steel structures.
    - BS EN ISO 14713 - Zinc coatings.
    - PD 6484 - Bimetallic contact corrosion avoidance.
    - BS EN ISO 9223 - Urban marine environmental conditions.
    - BS EN ISO 14713: Part 1, Table 2 - Zinc coatings Life to First maintenance to be “Very High”.
    - For buildings adjacent to water, take special precautions to avoid corrosion of metalwork including effect of salinity: Categories to be:
    - External Corrosion Category 4.
    - Interior corrosion Categories: Interior spaces - C1 for insulated/ heated areas and C2 for uninsulated/ unheated spaces.
2. Take all necessary protective measures to avoid corrosion, including electrochemical corrosion/ staining, from material sourcing through manufacture, delivery, storage and installation to handover.
3. Provide guarantees and certification.
4. Prevent corrosion as a result of electrolytic and galvanic action, in particular where dissimilar metals connect and other points of connection where the use of bolts, welding, screwing, etc. are used.
5. All cut edges of metal components to be treated to avoid corrosion.

## 358 Lightning protection

1. Conceal, where possible, earth bonding and lightning protection systems.
2. Provide testing certificates.
3. Refer to and comply with:
    - Services Engineer’s Specification and associated reports.
    - BS 7651 - Electrical installations.
    - BS EN 62305: Parts 1-4 - Lightning protection.
    - BS 7430 - Code of Practice for protective earthing.

## 359 Safe access

1. Provide safe access to any part of the Works for independent maintenance, component replacement, or any such other purpose such that disturbance to adjacent/ adjoining elements are not damaged or disturbed.
2. Show all required access panels on the Working Drawings.

## 360 Services

1. Services such as wiring and pipework shall be concealed, unless otherwise noted or agreed. Refer to Services Engineer's drawings.
2. Ensure allowance for all necessary service penetrations has been made.
3. At points of penetration, adequate and suitable seals, gaskets, grommets, support frames, etc. are to be provided without reduction in performance.

## 370 Record drawings

1. Site setting out drawing: Record details of all grid lines, setting-out stations, benchmarks and profiles. Retain on site throughout the Contract and hand over on completion.

## 540 Defects in existing work

1. Undocumented defects: When discovered, immediately give notice. Do not proceed with affected related work until response has been received.
2. Documented remedial work. Do not execute work which may:
    - Hinder access to defective products or work; or
    - Be rendered abortive by remedial work.

## 550 Access for inspection and testing

1. Timing: Agree and record dates and times of tests and inspections to enable all affected parties to be represented.
2. Confirmation: One working day prior to each such test or inspection. If sample or test is not ready, agree a new date and time.
3. Records: Submit a copy of test certificates and retain copies on site.
4. Testing generally:
    4.1. Testing to be carried out in accordance with British Standards (including where applicable European Norms) and UKAS requirements to demonstrate compliance with the Specification by independent and accredited establishments, undertaken by qualified staff using regularly calibrated equipment. In house testing facilities and prior certification may only be used when proven to be at least the same level of competence, relevant and agreed in advance.
    4.2. Testing samples to be new, properly prepared and tested to reflect all conditions (edges, interfaces, penetrations, perimeters, etc.) and location in the final product or systems incorporated in the Works.
    4.3. Successful testing does not negate responsibility for compliance with the Specification.
    4.4. Any item which fails a test or is deemed to be inadequate in any way is not to be used in the Works.
    4.5. Tests to be recorded and notified in advance such that the Employer's Agent may attend. A post test detailed report is to be prepared and issued.
    4.6. Maintain test records for inspection at any time.
5. Prototype testing:
    5.1. When testing Prototypes ensure that the Employer's Agent is present unless instructed otherwise in advance.
    5.2. Ensure that the Prototype is complete and fully prepared reflecting its site location, including interfaces.
    5.3. Prepare separate Working Drawings for each Prototype.

5.4. Under testing the prototype will show no signs of deformation, leakage, reduction in performance or changes to visual intent.

5.5. Testing to be carried out at the manufacturer's specialist facility or a pre agreed independent facility.

6. Proprietary Products: Retain on site evidence that the proprietary product specified has been supplied.
7. Acoustic Testing:
   7.1. Ensure and demonstrate that the acoustic performance requirements of the Specification are met.
   7.2. Comply with the following:
      - BS EN ISO 10140 - Sound insulation.
      - Approved Document Part E of the Building Regulations. 7.2.3.The project Acoustic Report.
   7.3. The Project Acoustic Report.
8. Weatherproofing/ Watertightness:
   8.1. Comply with the requirements of BS EN 1027.
   8.2. Tests of facades to confirm that the greater of class 600 Pa or 0.25 of design wind pressure has been achieved in respect of water penetration.
   8.3. All external walls/ roofs are to be tested or certified to demonstrate that the building is weather and watertight, without leakage or ponding. Any such occurrence to be immediately notified, repaired and re-tested.
9. Flood Testing:
   9.1. Supervised and recorded site flood tests to be actioned, as necessary. Such tests are to be carried out by filling the designated area with water to a depth of at least 50mm and 12mm below kerb level, after sealing all outlets, joints, junctions, etc. for a minimum period of 48 hours.
   9.2. Depths to be checked every four hours and if lower water levels are recorded, immediately locate the leak point, carry out a satisfactory repair and re-test until no drop in levels is detected.
10. Hose testing: Site hose tests to comply with CWCT TN41 - "Site Testing for Watertightness" and be carried out on benchmarks and designated areas of the final installation as instructed.
11. Fire testing:
   11.1. Examine carefully all fire/ smoke seals prior to concealment and test a minimum of one of each type to confirm adequacy.
   11.2. Comply with the following:
      - BS EN 1364 - Fire testing no load bearing elements.
      - BS EN 13501-2 to 4.
      - BE EN 13381
      - All recommendations and regulations arising from the Grenfell Enquiry and subsequent Fire Authorities' requirements.
12. Fixings: Carry out tests on items for which acceptable manufacturers' test results/ data is not available. Relevant tests to include proof loading, pull out and ultimate loading as a minimum for each fixing type.
13. Slip Resistance:
   13.1. Test finished floor surfaces regularly as the work proceeds. For floors where surface roughness is expressed as Rz, carry out tests with a roughness meter.

13.2. Pendulum Test to be carried out in wet and dry conditions to demonstrate the achievement of the specified value (PTV).

13.3. Comply with the following:

- BS 7976 - Pendulum testers.
- HSE document - "Assessing the Slip Resistance of Flooring". 13.3.3. UK Slip Resistance Group recommendations.

14. Coatings Generally:

14.1. Comply with the following:

- BS EN ISO 2064 - Metallic/ inorganic coatings.
- AAMA 2605 - Organic coating performance/ testing on aluminium panels/ extrusions. 14.1.3. BS 4842: Clause 10.2.

14.2. To ensure no flaking, cracking or colour change occurs during the Service Life of any coating, carry out both artificial and natural weather tests. For clarification, colour change means less than one step in hue or chroma from original colour, in accordance with the Munsell Atlas.

14.3. Visible defects from a distance of 1000mm, such as dust, blistering, entrapped particles, scratches, indents, chips etc. are not acceptable.

15. PVDC/ stoved Acrylic Coatings:

15.1. Comply with the following:

- BS EN ISO 2064.
- BS 6001: Parts 1 and 2.

15.2. Selected areas and components to be inspected on site by an independent qualified inspector to confirm colour and quality compliance. LQ (Limited Quality) (Pa=10%) = 5%.

15.3. Test 300mm x 200mm flat samples of at least four selected areas for a level 2 inspection to ensure that an Acceptable Quality Level of 1% for each colour is being maintained.

16. Performance Specified Elements: Submit evidence of compliance, including test reports indicating:

- Properties tested.
- Pass/ fail criteria.
- Test methods and procedures. 16.4. Test results.
- Identity of testing agency. 16.6. Test dates and times.
- Identities of witnesses.
- Analysis of results.

## 570 Air permeability

1. Airtightness/ Wind Resistance:

1.1. Comply with the following:

- BS EN 1026 - Infiltration.
- BS EN 12211 - Window/ door wind loads.
- BS EN 12179 - Curtain walling wind loads.
- BS EN ISO 9972 - Thermal glazing/ air permeability.
- CWCT recommendations and guidance.
- Approved Document Part L Building Regulations, CIBSE TM23 - Air Leakage.

1.2. Air leakage tests to comprise at least five positive pressure cycles to represent maximum wind gusts, followed by five cycles negative pressure to show maximum deflection values are not exceeded.



make

2. Testing organization: UKAS accredited and registered with the Air Tightness Testing and Measurement Association (ATTMA) or the Independent Air Tightness Testing Scheme (iATS).
3. Infiltration: Air infiltration through external walls is to be controlled such that maximum air infiltration does not exceed 3.0m3/ hr/ m² at 50Pa at any point or condition when tested to 600Pa (max.) as per CWCT guidance. Control and spread evenly over each surface exfiltration and infiltration such that it is not concentrated in a single location. Comply with the Services Engineer's recommendations and specification including provision of test results to confirm performance.
4. Results:
    - Content: Include test results and all supporting data.
    - Copies: Required for building control inspection and inclusion in Building Manual.
    - Electronic deposit: Through the ATTMA lodgement database
    - Additional copies: Provide on request.

## 580 Continuity of thermal insulation

1. Record and report: Confirm that work to new, renovated or upgraded thermal elements has been carried out to confirm that the requirements of the Specification have been met. Include:
    - The address of the premises.
    - The Contractor's name and address.
    - The name, qualification and signature of the competent person responsible for checking compliance.
    - The date on which the installation was checked.
2. Submit: Before completion of the Works.
3. Copy: To be lodged in the building manual.

## 610 Proposals for rectification of defective products/ executions

1. Proposals: Immediately any execution or product is known, or appears, to be not in accordance with the Contract, submit proposals for opening up, inspection, testing, making good, adjustment of the Contract Sum, or removal and re-execution.
2. Acceptability: Such proposals may be unacceptable and contrary instructions may be issued.

## 620 Measures to establish acceptability

1. General: Wherever inspection or testing shows that the work, materials or goods are not in accordance with the contract and measures (e.g. testing, opening up, experimental making good) are taken to help in establishing whether or not the work is acceptable, such measures:
    - Will be at the expense of the Contractor.
    - Will not be considered as grounds for revision of the completion date.

## 630 Quality Control

1. Procedures: Establish and maintain to ensure that the Works, including the work of subcontractors, comply with specified requirements.
2. Records: Maintain full records, keep copies on site for inspection, and submit copies on request.
3. Content of records
    - Identification of the element, item, batch or lot including location in the Works.
    - Nature and dates of inspections, tests and approvals.
    - Nature and extent of nonconforming work found.

- Details of corrective action.

## 710 Work before completion

1. General: Make good all damage consequent upon the Works.
2. Temporary markings, coverings and protective wrappings: Remove unless otherwise instructed.
3. Cleaning: Clean the Works thoroughly inside and out, including all accessible ducts and voids. Remove all splashes, deposits, efflorescence, rubbish and surplus materials.
4. Cleaning materials and methods: As recommended by manufacturers of products being cleaned, and must not damage or disfigure other materials or construction.
5. COSHH dated data sheets: Obtain for all materials used for cleaning and ensure they are used only as recommended by their manufacturers.
6. Minor faults: Touch up in newly painted work, carefully matching colour and brushing out edges. Repaint badly marked areas back to suitable breaks or junctions.
7. Moving parts of new work: Adjust, ease and lubricate as necessary to ensure easy and efficient operation, including doors, windows, drawers, ironmongery, appliances, valves and controls.

## 721 Handover

1. General: Leave the Works secure with, where appropriate, all accesses closed and locked.
2. Keys: Account for and adequately label all keys, and hand over together with an itemised schedule, retaining duplicate schedule signed as a receipt.
3. Information: Provide information and records of all materials, products, goods, components, systems, etc. to identify source, manufacturers, suppliers, subcontractors, registration codes, warranties and guarantees.
4. Notification: Provide sufficient time prior to handover to allow proper examination and snagging prior to handover.
5. Completion: Handover will not be achieved until the Architect is satisfied that all installations have been completed to the quality and standards required.

Ω End of Section

# E10
# Mixing/casting/curing in situ concrete

## Concrete

### 101 Specification

1. Concrete generally: To BS 8500-2.
2. Exchange of information: Provide concrete producer with information required by BS 8500-1, clauses 4 and 5.
3. Refer to the Structural Engineer's documentation

## Materials, batching and mixing

### 201 Generally

1. Refer to the Structural Engineer's Information

### 418 Proprietary admixture

1. Type: As per the manufacturers recommendations to suit the conditions of service.

## Project testing/ certification

### 501 Generally

1. Refer to the Structural Engineer's Information

## Placing/ compacting/ curing and protecting

### 601 Generally

1. Refer to the Structural Engineer's Information

Ω End of Section



make

# E20
# Formwork for In-Situ Concrete

## Generally/ preparation

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 620 CON-201 Plain Finish To In-Situ Concrete

1. Description: 'Plain formed' in-situ concrete surface finish, as defined in the National Structural Concrete Specification (NSCS) 4th edition.
2. Surfaces:
    2.1. Free from discoloration caused by contamination from release agent, grout leakage, rust staining or other sources.
    2.2. Smooth even finish with even aggregate distribution and colour consistency.
3. Finish: Even and dense.
    3.1. Formwork: Arrange formwork panels in a regular pattern as a feature of the surface.
4. Crack widths:
    4.1. Shall not exceed 0.3 mm generally in the exposed surfaces of finished concrete.
    4.2. Water resisting concrete: Shall not exceed 0.2 mm in the exposed surfaces of finished concrete.
5. Permissible deviation of surfaces
    5.1. Sudden irregularities (maximum): 3 mm.
    5.2. Gradual irregularities (maximum): 3 mm, when measured from the underside of a 1000 mm straightedge, placed anywhere on surface.
6. Variations in colour
    6.1. Permitted: Those caused by impermeable formwork linings.
    6.2. Not permitted: Those caused by contamination or grout leakage.
7. Surface blemishes
    7.1. Permitted: Blowholes less than 10 mm in diameter and at an agreed frequency.
    7.2. Not permitted: Voids, honeycombing, segregation and other large defects.
8. Formwork tie holes: In a regular pattern and filled with matching mortar.

## Construction

### 310 Accuracy

1. General requirement for formwork: Accurately and robustly constructed to produce finished concrete in the required positions and to the required dimensions.
2. Formed surfaces: Free from twist and bow (other than any required cambers).
3. Intersections, lines and angles: Square, plumb and true.

## 315 Substructure formwork and underslab insulation

1. Cutting: Neat and accurate to edges, and around penetrations and downstands.
2. Laying: Tightly butted and fully supported on firm, even substrate.
3. Vertical faces: Stiffen as necessary to act as shutter.
4. Formwork/ insulation surfaces: Protect from indentation by spacers and other items.
5. Joints in formwork/ insulation and with edge structure and penetrations: Seal to prevent penetration of concrete.
6. Concrete placement: Restrain formwork/ insulation against movement.

## 320 Joints in forms

1. Requirements including joints in form linings and between forms and completed work
   1.1. Prevent loss of grout, using seals where necessary.
   1.2. Prevent formation of steps. Secure formwork tight against adjacent concrete.

## 330 Inserts, holes and chases

1. Positions and details
   1.1. Dimensioned on drawings provided on behalf of the Employer: Do not change without consent.
   1.2. Undimensioned or from other sources: Submit proposals.
2. Positioning relative to reinforcement: Give notice of any conflicts well in advance of placing concrete.
3. Method of forming: Fix inserts or box out as required. Do not cut hardened concrete without approval.

## 340 Kickers

1. Method statement: Submit proposals including means of achieving quality of concrete consistent with that specified for the column or wall.
   1.1. Kicker height: To be confirmed

## 350 Form ties

1. Metal associated with form ties/ devices: Prohibited within cover to reinforcement. Compatible with reinforcement metal.

## Striking - Not Used

## Formed finishes

## 600 Samples of finishes available for inspection

1. Quality of finish: To be agreed with the Architect.
2. Location: To be agreed with the Architect.
3. Variations: None

## 605 Control samples (part of the finished work)

1. Quality of finish: To be agreed with the Architect.
2. Location: Obtain instructions.

3. Area (minimum): To be agreed with the Architect.
4. Features to be included: To be agreed with the Architect.
5. Viewing distance (maximum): To be agreed with the Architect.
6. Approval of appearance: Obtain before proceeding with remainder of the work.

## 610 Basic finish

1. Location: To be agreed with the Architect.
2. Finish: Faces fully compacted and cover to reinforcement provided.

## 613 Ordinary finish

1. Location: To be agreed with the Architect.
2. Finish: Faces fully compacted. Formed surfaces free from major blemishes and honeycombing. Steps at joints to be less than 5 mm.

## 615 Finish to receive asphalt tanking

1. Finish: Even and suitable to receive asphalt.
2. Permissible deviation of surfaces
   2.1. Sudden irregularities (maximum): 3 mm.
   2.2. Gradual irregularities (maximum): 3 mm, when measured from underside of a 1 m straightedge, placed anywhere on surface.
3. Surface blemishes
   3.1. Permitted: Blowholes less than 10 mm in diameter.
   3.2. Not permitted: Voids, honeycombing, segregation and other large defects.
4. Projecting fins: Remove.
5. Formwork tie holes: Filled with mortar.

## 620 Plain finish

1. Location: To be agreed with the Architect.
2. Finish: Even and dense. Arrange formwork panels in a regular pattern as a feature of the surface.
3. Permissible deviation of surfaces
   3.1. Sudden irregularities (maximum): 3 mm.
   3.2. Gradual irregularities (maximum): 3 mm, when measured from the underside of a 1 m straightedge, placed anywhere on surface.
4. Variations in colour
   4.1. Permitted: Those caused by impermeable formwork linings.
   4.2. Not permitted: Those caused by contamination or grout leakage.
5. Surface blemishes
   5.1. Permitted: Blowholes less than 10 mm in diameter and at an agreed frequency.
   5.2. Not permitted: Voids, honeycombing, segregation and other large defects.
6. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 630 Special finish

1. Location: To be agreed with the Architect.

2. Finish: Smooth and even. Formwork panels as large as practicable. Arrange in a regular pattern as a feature of the surface.
    2.1. Special requirements: None
3. Permissible deviation of surfaces
    3.1. Sudden irregularities (maximum): 3 mm.
    3.2. Gradual irregularities (maximum): 3 mm when measured from the underside of a 1 m straightedge, placed anywhere on surface.
4. Variations in colour
    4.1. Permitted: Those caused by impermeable formwork linings.
    4.2. Not permitted: Those caused by contamination, grout leakage and replacement of formwork panels.
5. Cover spacers: Do not use
6. Surface blemishes
    6.1. Permitted: Blowholes less than 5 mm in diameter and at an agreed frequency.
    6.2. Not permitted: Voids, honeycombing, segregation and other defects.
7. Formwork tie holes: In a regular pattern and filled with matching mortar

## 640 Rough board finish

1. Forms or form linings: Rough textured softwood boards with smooth planed joint faces.
    1.1. Width of boards: 100 mm
    1.2. Moisture content at time of casting: 18-25%.
2. Arrangement of boards: Vary textures to give uniform overall effect.
    2.1. Orientation of boards: Diagonal/ Horizontal/ Vertical - to be agreed with the Architect.
    2.2. End joints: Not permitted
    2.3. Nail heads: Flush with surface.
3. Surface blemishes: Penetration of grout between boards not permitted. Wet pre-assembled forms with clean water before erecting and keep damp until concrete is placed.
4. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 650 Ribbed finish

1. Forms or form linings: Splayed timber battens secured to plywood backing.
    1.1. Finished size of battens: To be agreed with the Architect.
    1.2. Centres of battens: To be agreed with the Architect.
    1.3. Fixing to plywood backing: Adhesive and screws.
    1.4. Finish: Smooth with clean, sharp arrises at junctions with backing. Prevent penetration of grout behind battens.
2. Cover spacers: Do not use
3. Surface blemishes
    3.1. Permitted: Blowholes less than 5 mm in diameter and at an agreed frequency.
    3.2. Not permitted: Voids, honeycombing, segregation and other defects.
4. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 660 Proprietary form lining finish

1. Installation of linings: Securely to support formwork and jointed neatly.
2. Cover spacers: Do not use
3. Surface blemishes
   3.1. Permitted: Blowholes less than 5 mm in diameter and at an agreed frequency.
   3.2. Not permitted: Voids, honeycombing, segregation and other defects.
   3.3. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 665 Controlled permeability form lining finish

1. Installation of linings: As part of a controlled permeability formwork (CPF) system. Prevent any restrictions to drainage of water.
2. Cover spacers: Do not use
3. Surface blemishes
   3.1. Permitted: Blowholes less than 5 mm in diameter and at an agreed frequency.
   3.2. Not permitted: Voids, honeycombing, segregation and other defects.
4. Formwork tie holes: Regular pattern and filled with matching mortar.

## 700 Finish for abrasive blasting

1. Finish: Even with accurate, true and even joints.
2. Permissible deviation of surfaces
   2.1. Gradual irregularities when measured from the underside of a 1 m straightedge, placed anywhere on surface (maximum): 3 mm.
3. Cover spacers: Do not use
4. Surface blemishes
   4.1. Permitted: Blowholes less than 10 mm in diameter and at an agreed frequency.
   4.2. Not permitted: Voids, honeycombing, segregation and other large defects.
5. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 710 Finish for tooling

1. Finish: Even with accurate, true and even joints.
2. Permissible deviation of surfaces
   2.1. Gradual irregularities when measured from the underside of a 1 m straightedge, placed anywhere on surface (maximum): 5 mm.
3. Cover spacers: Do not use
4. Surface blemishes
   4.1. Permitted: Blowholes less than 10 mm in diameter and at an agreed frequency.
   4.2. Not permitted: Voids, honeycombing, segregation and other large defects.
5. Formwork tie holes: In a regular pattern and filled with matching mortar.

## 750 Arrises, margins and junctions

1. Requirements: Refer to the Design Drawings

Ω End of Section



make

# F10
# Brick/ Block Walling

## Types of walling

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 BRK-101 UKPN Brickwork

1. Description: 240min fire rated brickwork configured and to locations indicated on the Design Drawings.
2. Bricks: To BS EN 771-1.
    2.1. Arrangment: Refer to the Design Drawings
    2.2. Appearance: To match accepted sample.
    2.3. Compressive strength (minimum): 20N/mm².
    2.4. Freeze / thaw Resistance: F1.
    2.5. Active soluble salts content: : S1.
    2.6. Water absorption: 23% maximum.
    2.7. Recycled content: To be agreed with the Architect.
    2.8. Work size (length x width x height): 215 x 102 x 65 mm.
3. Tolerances:
    3.1. Mean value: Tm.
    3.2. Range: Rm.
4. Durability designation: F2.
5. Reaction to Fire: Class A1 to BS EN 13501-1.
6. Mortar: As section Z21.
    6.1. Standard: To BS EN 998-2
    6.2. Type: Cement: Lime: Sand.
    6.3. Colour: Natural. To match blockwork.
7. Bond: English.
8. Joints: Neat struck joints and walls to provide a flush finish internally.
9. Other requirements: Common bricks should be 'frogged' and laid with 'frogs' facing upward. Refer to Appendix C.1 for typical details.

### 201 MAS-206 Blockwork Wall

1. Description: Concrete fair-faced blockwork configured and to locations indicated on the Design Drawings.
2. Blocks: To BS EN 771-3.
    2.1. Manufacturer: Lignacite, or acceptable equivalent
        2.1.1. Product reference: SP Fair Faced.
    2.2. Configuration: Group 1



make

2.3. Compressive strength
2.3.1. Mean value: 7.3 N/mm²
2.3.2. Category: I
2.4. Freeze/ thaw resistance: Frost-resistant
2.5. Work sizes (length x width x height): 440 x 100 x 215 mm
2.5.1. Tolerance category: D1.
3. Reaction to fire: Class A1 to BS EN 13501-1.
4. Mortar: As section Z21.
4.1. Standard: To BS EN 998-2
4.2. Type: Cement: Lime: Sand.
4.3. Colour: Natural. To match block colour.
5. Bond: Stretcher.

## 202 EWS-107 Blockwork Wall

1. Description: Concrete fair-faced blockwork configured and to locations indicated on the Design Drawings.
2. Specific performance: Performance and detail interfaces to be validated by Façade Consultant.
2.1. Fire rating: Refer to the Fire Strategy Report
2.2. Acoustic rating: Refer to the Acoustic Strategy Report
2.3. Duty rating: Severe.
3. Types: Refer to the Design Drawings for partitions/ wall linings/ ceiling types and locations.
4. Build-ups, construction, finishes, sizes and zone: Refer to the Design Drawings
5. Support framing: Varies, as per the manufacturer's recommendations to achieve the stipulated performance requirements.
6. Blocks: To BS EN 771-3.
6.1. Manufacturer: Lignacite, or acceptable equivalent
6.1.1. Product reference: SP Fair Faced.
6.2. Configuration: Group 1
6.3. Compressive strength
6.3.1. Mean value: 7.3 N/mm²
6.3.2. Category: I
6.4. Freeze/ thaw resistance: Frost-resistant
6.5. Work sizes (length x width x height): 440 x 100 x 215 mm
6.5.1. Tolerance category: D1.
7. Insulation: Non-combustible.
8. Linings: Refer to the Design Drawings
9. Pattressing: Plywood pattress, timber noggings and fixing plates to be provided where necessary to support surface-mounted fixtures and fittings.
10. Finishing:
10.1. Unless specified, or indicated otherwise, outer lining plasterboard shall be square edged, scrim taped and finished with plaster skim suitable to receive applied finishes.
10.2. Generally, skim plaster shall not be applied to moisture resistant plasterboard in areas subject to high humidity.

10.3. Flush jointing (tape and fill) needs to be undertaken in line with the manufacturer's recommendations and shall only be used if samples of the joints have been accepted by the Employer's Agent.

11. Mortar: As section Z21.
    11.1. Standard: To BS EN 998-2
    11.2. Type: Cement: Lime: Sand.
    11.3. Colour: Natural. To match block colour.
12. Bond: Stretcher.
13. Other requirements: System shall include all studs/ framing, insulation, linings, fixings and accessories necessary to complete the installation.

## 270 MAS-401 Blockwork Wall - WI System

1. Description: Hollow, WI blockwork wall.
2. Manufacturer: Plasmor/ S Morris, or acceptable equivalent.
    2.1. Product reference: WI Hollow Block Wall
3. Performance:
    3.1. Fire rating: Refer to the Fire Strategy Report
    3.2. Acoustic rating: Refer to the Acoustic Report
4. Structural support: The quantities and positioning of the structural support columns is to be determined by the Specialist Sub-Contractor to suit the service conditions, loading requirements and co-ordination with the Design Drawings and all interfacing components, systems, services and associated BWIC as appropriate to each required instance.
5. Loading criteria:
    5.1. Internal wind pressure: Refer to the Structural Engineer's documentation
    5.2. Barrier loads: Refer to the Structural Engineer's documentation
6. Air tightness: Walls to be painted to both sides where required to achieve air tightness requirements. Extent of air tightness requirements to be proposed by accredited testing company and accepted by Building Control.
7. Blocks:
    7.1. Type: Paint grade hollow, WI column block units.
    7.2. Special shapes: Hollow block with side slit.
    7.3. Nominal face size: 440mm x 215mm.
    7.4. Thickness: To suit corresponding wall thickness, Refer to the Design Drawings
    7.5. Average compressive strength: 7.3 N/mm² (excluding concrete infill).
        7.5.1. Including concrete infill: 40 N/mm² - To be confirmed by Wembley Innovation.
    7.6. Freeze/ Thaw resistance: Not to be left exposed.
    7.7. Thermal resistance (internal): 0.3 K/W $m^2$.
    7.8. Density: 1450 kg/$m^3$ (excluding concrete infill).
        7.8.1. Including concrete infill: Approx. 2100 kg/$m^3$.
    7.9. Weight of blocks to be less than 20kg, to comply with manual handling requirements.
8. Bond: Stack bonded 440mm wide full blocks or Stretcher bond.
9. Mortar:
    9.1. Type: Cement: Lime: Sand.

9.2. Compressive strength class: M4.

9.3. Joint profile: 10mm, to match the wall the element is set in.

9.4. Colour: Natural to match the block.

10. Masonry accessories: In accordance with Section F30, to be determined by the Contractor.
11. Movement joints:
    11.1. At nominated design lengths.
    11.2. Adjacent to or between full block WI walls.
12. Steel reinforcing:
    12.1. H16 bar to EN 10083-1.
    12.2. Minimum overlap of 640mm tied with tie wire at 300mm centres.
13. Concrete Infill: To EN1996-1-1.
    13.1. C40 mixture strength with a maximum aggregate size of 6mm.
14. Head Cleats:
    14.1. Manufacturer: Wembley Innovation, or acceptable equivalent.
    14.2. Connected to structural frame as per detailed drawings.
15. Other:
    15.1. WI Beam metal components: Zinc plated to BS 1706.

## 301 EWS-14 Acoustic Seperation Wall (Visual Intent Only)

1. Description: Acoustic separation wall to locations indicated on the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Blockwork: Refer to the Façade Performance Specification
4. Insulation: Refer to the Façade Performance Specification
5. Mortar:
    5.1. Type: Refer to the Façade Performance Specification
    5.2. Colour: Refer to the Façade Performance Specification

# Testing

## 400 Hard landscaping materials specification

1. Minimum BRE 'Green Guide to Specification' online rating: Refer to Section A.

## 410 Compressive strength of mortar for each walling type

1. Testing authority: A UKAS Accredited laboratory
2. Test method: To BS EN 1015-11.
3. Preliminary tests procedure: As follows:
    3.1. Specimens
        3.1.1. Number of specimens: 6.
        3.1.2. Type: 40 x 40 x 160 mm prism.
        3.1.3. Preparation: At least six weeks before walling commences.

3.2. Specimen testing: Half of specimens at 7 days. Remainder at 28 days.

3.2.1. Retarded mixes: Extend curing periods to include retardation period.

3.3. Response to result: If mean compressive strength at 28 days is not within the range given below repeat tests with more suitable sand or next higher designation of mortar.

4. Site tests procedure: As follows.

4.1. Number of specimens: Six per 150m² of walling or per storey whichever the more frequent.

4.2. Specimen types: As preliminary test, but prepared during construction.

4.3. Specimen testing: Half of specimens at 7 days. Remainder at 28 days.

4.3.1. Retarded mixes: Extend curing periods to include retardation period.

5. Required test mean compressive strength at 28 days (N/mm²): To be agreed with the Architect.

6. Results: Submit.

## Workmanship generally

### 421 Prototypes

1. Provide reference panels representing each masonry wall comprising;

1.1. Size (nominal): 2000mm x 2000mm.

1.2. Mortar: As specified.

1.3. Location of build: On-Site.

1.4. Masonry units: Randomly selected.

1.5. Rejected panels: In the event that a sample panel is rejected, additional sample panels of the rejected element to be constructed until acceptance is gained.

### 430 Conditioning of clay bricks and blocks

1. Bricks and blocks delivered warm from manufacturing process: Do not use until cold.

2. Absorbent bricks in warm weather: Wet to reduce suction. Do not soak.

### 440 Conditioning of concrete blocks

1. Autoclaved concrete bricks/ blocks delivered warm from manufacturing process: Do not use.

2. Age of nonautoclaved concrete bricks/ blocks: Do not use until at least four weeks old.

3. Avoidance of suction in concrete bricks/ blocks: Do not wet.

3.1. Use of water retaining mortar admixture: Submit details.

### 460 Mortar designations

1. Mix proportions: For a specified designation select a mix from the following:

1.1. Designation (i) (BS EN 998-2 M12 equivalent)

1.1.1. 1:0-¼:3 (Portland cement:lime:sand with or without air entraining additive).

1.1.2. 1:3 (Portland cement:sand and air entraining additive).

1.2. Designation (ii) (BS EN 998-2 class M6 equivalent)

1.2.1. 1:½:4-5 (Portland cement:lime:sand with or without air entraining additive).

1.2.2. 1:3 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).



make

1.2.3. 1:2½-3½ (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).

1.2.4. 1:3-4 (Portland cement:sand and air entraining additive).

1.3. Designation (iii) (BS EN 998-2 class M4 equivalent)

1.3.1. 1:1:5-6 (Portland cement:lime:sand with or without air entraining additive).

1.3.2. 1:3½-4 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).

1.3.3. 1:4-5 (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).

1.3.4. 1:5-6 (Portland cement:sand and air entraining additive).

1.4. Designation (iv) (BS EN 998-2 class M2 equivalent)

1.4.1. 1:2:8-9 (Portland cement:lime:sand with or without air entraining additive).

1.4.2. 1:4½ (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).

1.4.3. 1:5½-6½ (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).

1.4.4. 1:7-8 (Portland cement:sand and air entraining additive).

2. Batching: Mix proportions by volume.

3. Mortar type: Continuous throughout any one type of masonry work.

## 500 Laying generally

1. Mortar joints: Fill vertical joints. Lay bricks, solid and cellular blocks on a full bed.

2. AAC block thin mortar adhesive and gypsum block adhesive joints: Fill vertical joints. Lay blocks on a full bed.

3. Clay block joints

3.1. Thin-layer mortar: Lay blocks on a full bed.

3.2. Interlocking perpends: Butted.

4. Bond where not specified: Half-lap stretcher.

5. Vertical joints in brick and concrete block facework: Even widths. Plumb at every fifth cross joint.

## 520 Accuracy

1. Courses: Level and true to line.

2. Faces, angles and features: Plumb.

3. Permissible deviations

3.1. Position in plan of any point in relation to the specified building reference line and/ or point at the same level ± 10 mm.

3.1.1. Straightness in any 5 m length ± 5 mm.

3.1.2. Verticality up to 3 m height ± 10 mm.

3.1.3. Verticality up to 7 m height ± 14 mm.

3.1.4. Overall thickness of walls ± 10 mm.

3.1.5. Level of bed joints up to 5 m

3.1.6. (brick masonry) ± 11 mm.

3.1.6.1. Level of bed joints up to 5 m



make

3.1.7. (block masonry) ± 13 mm.

## 535 Height of lifts in walling using cement gauged or hydraulic lime mortar

1. Quoins and advance work: Rack back.
2. Lift height (maximum): 1.2 m above any other part of work at any time.
3. Daily lift height (maximum): 1.5 m for any one leaf.

## 540 Height of lifts in walling using thin layer mortar

1. Quoins and advance work: Rack back.
2. Lift height (maximum): 1.3 m above any other part of work at any time.

## 545 Levelling of separate leaves

1. Locations for equal levelling of cavity wall leaves: As follows:
   1.1. Every course containing vertical twist type ties or other rigid ties.
   1.2. Every third tie course for double triangle/ butterfly ties.
   1.3. Courses in which lintels are to be bedded.

## 560 Coursing brickwork

1. Gauge: Four brick courses including bed joints to 300 mm.

## 561 Coursing brickwork with existing

1. Gauge: Line up with existing brick courses.

## 580 Laying frogged bricks

1. Single frogged bricks: Frog uppermost.
2. Double frogged bricks: Larger frog uppermost.
3. Frog cavity: Fill with mortar.

## 595 Lintels

1. Bearing: Ensure full length masonry units occur immediately under lintel ends.

## 610 Support of existing work

1. Joint above inserted lintel or masonry: Fully consolidated with semidry mortar to support existing structure.

## 615 Brickwork to receive asphalt dpc

1. Substrate: Mortar bed finished flush, smooth and level.

## 620 Block bonding new walls to existing

1. Pocket requirements: Formed as follows
   1.1. Width: Full thickness of new wall.
   1.2. Depth (minimum): 100mm.
   1.3. Vertical spacing
      1.3.1. Brick to brick: 4 courses high at 8 course centres.

1.3.2. Block to block: Every other course.

2. Pocket joints: Fully filled with mortar.

## 635 Jointing

1. Profile: Consistent in appearance.

## 645 Accessible joints not exposed to view

1. Jointing: Struck flush as work proceeds.

## 665 Pointing

1. Description: Generally where required.
2. Joint preparation: Remove debris. Dampen surface.
3. Mortar: As Section Z21.
   3.1. Standard: To BS EN 998-2
   3.2. Additional requirements: Coloured mortar to match bricks
4. Profile: Flush

## 671 Fire stopping

1. Avoidance of fire and smoke penetration: Fit tightly between cavity barriers and masonry. Leave no gaps.

## 690 Adverse weather

1. General: Do not use frozen materials or lay on frozen surfaces.
2. Air temperature requirements: Do not lay bricks/ blocks:
   2.1. In cement gauged mortars when at or below 3°C and falling or unless it is at least 1°C and rising.
   2.2. In hydraulic lime:sand mortars when at or below 5°C and falling or below 3°C and rising, or as manufacturer's/ supplier's recommendations.
   2.3. In thin layer mortars when outside the limits set by the mortar manufacturer.
3. Temperature of walling during curing: Above freezing until hardened.
4. Newly erected walling: Protect at all times from:
   4.1. Rain and snow.
   4.2. Drying out too rapidly in hot conditions and in drying winds.

# Additional requirements for facework

## 710 The term facework

1. Definition: Applicable in this specification to block walling finished fair.
   1.1. Painted facework: The only requirement to be waived is that relating to colour.

## 730 Concrete block samples

1. General: Before placing orders with suppliers submit for approval of appearance labelled samples of the following: All types specified .
2. Selection of samples: Representative of the range in variation of appearance.



make

## 740 Finished masonry work reference panels

1. General: Before proceeding to construct the following walling types, construct panels as specified. Give notice when panels are dry.
2. Selection of masonry units: Reasonably representative of the average quality of the whole order to be delivered
3. Panel types
    3.1. Walling type: All
        3.1.1. Size: 1.5 x 1.5 m
        3.1.2. Other requirements: As shown on drawing

## 745 Masonry sample panels

1. Sampling frequency: A panel for each type Specified and delivery of masonry unit.
2. Selection of masonry units: Reasonably representative of the average quality of the whole order to be delivered
3. Panel types
    3.1. Brickwork and blockwork: 5 No. of each.
    3.2. Walling type: All
        3.2.1. Size: 1.5 x 1.5 m
        3.2.2. Other requirements: As shown on drawing

## 750 Colour consistency of masonry units

1. Colour range: Submit proposals of methods taken to ensure that units are of consistent and even appearance within deliveries.
2. Conformity: Check each delivery for consistency of appearance with previous deliveries and with approved reference panels; do not use if variation is excessive.
3. Facing bricks should be blended on site from a minimum of three packs to ensure an even distribution of colour and texture variation.
4. Finished work: Free from patches, horizontal stripes and racking back marks.

## 760 Appearance

1. Brick/ block selection: Do not use units with damaged faces or arrises.
2. Cut masonry units: Where cut faces or edges are exposed cut with table masonry saw.
3. Quality control: Lay masonry units to match relevant reference panels.
    3.1. Setting out: To produce satisfactory junctions and joints with built-in features and components.
    3.2. Coursing: Evenly spaced using gauge rods.
4. Lifts: Complete in one operation.
5. Methods of protecting facework: Submit proposals.

## 770 Textured bricks

1. Units with directional texturing
    1.1. Masonry units:
        1.1.1. Laying orientation: Refer to the Design Drawings



make

## 780 Ground level

1. Commencement of facework: Not less than 150 mm below finished level of adjoining ground or external works level.

## 790 Putlog scaffolding

1. Use: Not permitted in facework.

## 800 Toothed bond

1. New and existing facework in same plane: Bond together at every course to achieve continuity.

## 810 Fixing of brick slips

1. Adhesive system: As required to achieve the required performance.
2. Concrete substrate
   2.1. Preparation: Remove laitance and expose aggregate.
   2.2. Substrate surface: Clean and free of loose material.
3. Slips
   3.1. Preparation of slips: Clean and free of loose material.
   3.2. Slips: Fully bonded.
4. Movement joints in substrate: Do not bridge.
5. Pointing: Allow adhesive bedding to set. Use same mortar as adjacent facework and finish to same profile.

## 830 Cleanliness

1. Facework: Keep clean.
2. Mortar on facework: Allow to dry before removing with stiff bristled brush.
3. Removal of marks and stains: Rubbing not permitted.

Ω End of Section

# F30
# Accessories/ Sundry Items for Brick/ Block/ Stone Walling

## Cavities

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 105 MAS-101 Accessories/ Sundry Items Scope (Generally)

1. Description: All necessary masonry accessories required to complete the works to be proposed by the Contractor for the acceptance of the Architect and will achieve the required performance of interfacing masonry works and suit the conditions of service.
2. Provide samples of all exposed accessories to the Architect for acceptance in regard to the finish and colour requirements.
3. Proposed accessories to achieve the performance of the interfacing masonry works and to suit the conditions of service.

### 110 Concrete fill to base of cavity

1. Concrete generally: To BS EN 206 and BS 8500-2.
2. Concrete type:
   2.1. Workability: High.
3. Extent: Maintain 75 mm between top of fill and external ground level and a minimum of 225 mm between top of fill and ground-level dpc.
4. Placement: Compact to eliminate voids.

### 120 Cleanliness

1. Cavity base and faces, ties, insulation and exposed dpcs: Free from mortar and debris.

### 130 Perpend joint weepholes

1. Form: Open perpend joint.
2. Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
3. Provision: At not greater than 900 mm centres and not less than two over each opening at maximum 450 mm centres.

### 131 Bed joint weepholes

1. Form: Open 10 mm-diameter hole.
2. Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
3. Provision: At not greater than 900 mm centres and not less than two over each opening at maximum 450 mm centres.

## 132 Perpend joint plastics weep holes

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
3. Provision: At not greater than 900 mm centres and not less than two over each opening at maximum 450 mm centres.

## 150 Full fill cavity insulation

1. Insulation:
   1.1. Product certification: British Board of Agrément (BBA) Certificate number
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Face size (nominal length x width): In accordance with system manufacturers details/ recommendations and the requirements of the testing/ certification criteria applicable to each system to achieve the specific performance requirements and to suit the service conditions of each individual instance.
4. Thickness (nominal): Refer to the Design Drawings
5. Thermal conductivity: Refer to the A Sections.
6. Reaction to fire class: A1.
7. Additional requirements: None
8. Placement: Continuous and free of mortar and debris.

## 155 Partial fill cavity insulation

1. Insulation:
   1.1. Product certification: British Board of Agrément (BBA) Certificate
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Face size (length x width): In accordance with system manufacturers details/ recommendations and the requirements of the testing/ certification criteria applicable to each system to achieve the specific performance requirements and to suit the service conditions of each individual instance.
4. Thickness (nominal): Refer to the Design Drawings
5. Thermal conductivity: Refer to the A Sections.
6. Reaction to fire class: A1
7. Placement: Secure against face of inner leaf.
   7.1. Residual cavity: Clear and unobstructed.
8. Joints between boards, at closures and penetrations: No gaps and free from mortar and debris.

## 160 Air Bricks in External Walling

1. Description: Air brick to external walls, as shown on the Design Drawings.
2. Standard: To BS 493, Class 1.
3. Manufacturer: Submit proposals
   3.1. Product reference: Submit proposals

4. Apertures: Refer to the Design Drawings.
5. Work sizes: To suit brickwork coursing and size, refer to the Design Drawings.
6. Material and colour: To match adjoining masonry.
7. Placement: Built in with no gaps at joints.

## 165 Gratings/ ventilators in internal walling

1. Standard: To BS 493, Class 2.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Apertures: Refer to the Design Drawings
4. Work sizes: Refer to the Design Drawings
5. Material and colour: To match adjoining masonry.
6. Other requirements: Refer to the Design Drawings
7. Placement: Built in with no gaps at joints.

## 171 Ventilation ducts in external walling

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Placement: Across cavity, sloping away from inner leaf. Full mortar joints to seal cavity.
3. Protection from water penetration to inner leaf: Where barrier is not integral to duct, form stepped dpc cavity tray with stop ends above duct, extending 150 mm on each side.

## 175 Cavity ventilators

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Material and colour: Plastics/ to match adjoining masonry
3. Number and location: Refer to the Design Drawings

## 180 Cavity closers

1. Description: Cavity closers
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Accessories: To include integral dpc/ To include integral fixing cramp/ To include integral insulation - to be agreed with the Architect.

## 185 Transmitted noise reduction systems

1. Description: Noise reduction system
2. Type: Continuous ground floor raft system/ Flanking wall system/ Party wall system
3. Manufacturer: Submit proposals
   3.1. Product reference: Submit proposals.
4. Components: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

## Reinforcing/ fixing accessories

### 214 Cavity wall ties

1. Standard: To BS EN 845-1.
   1.1. Type: 2 (Masonry general purpose).
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
6. Embedment length (minimum): 50 mm.
7. Movement: Tolerant.
8. Additional requirements: Resistance to water crossing cavity: resistant.

### 215 Cavity wall ties used with partial fill insulation

1. Standard: To BS EN 845-1.
   1.1. Type: 2 (Masonry general purpose)
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
6. Embedment length (minimum): 50 mm.
7. Movement: Tolerant.
8. Additional requirements: Resistance to water crossing cavity: resistant.
9. Tie-mounted insulation retaining clips: As recommended by the manufacturer.

### 218 Shear wall ties

1. Standard: To BS EN 845-1.
   1.1. Type: 2 (Masonry general purpose).
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
6. Embedment length (minimum): 50 mm.

## 220 Wall ties

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Material/ finish: Stainless steel.
3. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.

## 224 Individual brackets

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1
   3.1. Thickness (minimum): In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. Vertical load capacity: Refer to the Structural Engineer's Information

## 225 Fixing ties in masonry cavity walls

1. Embedment in mortar beds (minimum): 50 mm.
2. Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
3. Spacing: Staggered in alternate courses.
   3.1. Horizontal centres: Refer to the Design Drawings
   3.2. Vertical centres: Refer to the Design Drawings
4. Provision of additional ties: Within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
   4.1. Spacing: Refer to the Design Drawings

## 228 Fixing ties in masonry cavity walls with full fill cavity insulation

1. Embedment in mortar beds (minimum): 50 mm.
2. Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
3. Spacing: Staggered in alternate courses.
   3.1. Horizontal centres: Refer to the Design Drawings
   3.2. Vertical centres: Refer to the Design Drawings
4. Provision of additional ties
   4.1. One row to support lowest row of insulation batts.
   4.2. Within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
   4.3. Spacing: Refer to the Design Drawings

## 233 Fixing ties in masonry cavity walls with partial fill cavity insulation

1. Embedment in mortar beds (minimum): 50 mm.
2. Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
3. Spacing: Evenly space in non-staggered horizontal and vertical rows.
   3.1. Horizontal centres: Refer to the Design Drawings
   3.2. Vertical centres: Refer to the Design Drawings
4. Provision of additional ties: Within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
   4.1. Spacing: Refer to the Design Drawings

## 236 Fixing ties in masonry cladding to timber frames

1. Fixing ties to timber frame studs through sheathing: Securely with 50 mm x 11 gauge stainless steel annular ringed shank nails.
2. Embedment in mortar beds (minimum): 50 mm.
3. Placement: Slope downwards away from timber frame, without bending.
4. Spacing
   4.1. Horizontal centres: To suit stud centres.
   4.2. Vertical centres: To suit stud centres.
5. Provision of additional ties: Within 225 mm of reveals of openings and at no more than 300 mm centres vertically.

## 241 Wall starters/ connectors

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – BS EN 845-1 material/ coating reference 1
4. Sizes: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
5. End type: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.

## 242 Surface mounted channels/ slots

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Material/ finish: Austenitic stainless steel – BS EN 845-1 material/ coating reference 1.
3. Type: Plain back.
4. Profile: Manufacturer's standard.
5. Size (width x depth): Refer to the Design Drawings and manufacturers information to suit the required cavity width.
6. Length: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
7. Fixing: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.

## 245 Channel/ slot ties

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1.
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. Channel/ slot fixing: Embedded.
6. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
7. Embedment length (minimum): 50 mm.
8. Additional requirements: Resistance to water crossing cavity: resistant.

## 248 Channel/ slot ties used with partial fill insulation

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. Channel/ slot fixing: Embedded.
6. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
7. Embedment length (minimum): 50 mm.
8. Additional requirements: Resistance to water crossing cavity: resistant.
9. Tie-mounted insulation retaining clips: As recommended by the manufacturer.

## 251 Wall head restraint slip ties

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1.
4. Sizes: Refer to the Design Drawings and manufacturers information to suit the required cavity width.
5. Shear load capacity: Refer to the Structural Engineer's Information
6. End types: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
7. Stem length: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
8. Embedment length (minimum): 50 mm.

## 255 Wind posts

1. Standard: To BS EN 1090-1.

2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material: Aluminium to European standards indicated in BS EN 1090-3.
4. Sizes: Refer to the Structural Engineer's Information
5. Fixings (complete with washers and shims): Resistance to water crossing cavity: resistant.
6. Additional requirements: Refer to the Structural Engineer's Information

## 260 Angle supports

1. Standard: To BS EN 1090-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material: Aluminium to European standards indicated in BS EN 1090-3.
4. Size: Refer to the Structural Engineer's Information
5. Fixings (complete with washers and shims): Refer to the Structural Engineer's Information
6. Additional requirements: Refer to the Structural Engineer's Information

## 265 Support systems

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Material: Austenitic stainless steel - material/ coating reference 1 to BS EN 845-1.
3. Components, arrangement and dimensions: Refer to the Design Drawings

## 270 Meshwork joint reinforcement

1. Standard: To BS EN 845-3.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Type: Refer to the Structural Engineer's Information
4. Material: Austenitic stainless steel – material/ coating reference R1 to BS EN 845-3.
5. Width: Approximately 40–50 mm less in width than wall or leaf.
6. Placement: Lay on an even bed of mortar in a continuous strip with full laps at angles. Keep back 20 mm from face of external work, 12 mm back from face of internal work and finish joint to normal thickness.
   6.1. Lap length (minimum): In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.

## 282 Slip ties

1. Standard: To BS EN 845-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Austenitic stainless steel – material/ coating reference 1.
4. Sizes: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
5. Shear load capacity: Refer to the Structural Engineer's Information

6. End types: Symmetrical flat plate with slip case.
7. Embedment length (minimum): 50 mm.
8. Fixing centres: Alternate courses.
9. Additional requirements: Resistance to water crossing cavity: resistant.

### 290 Proprietary special fixings

1. Standard: To BS EN 1090-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material/ finish: Aluminium to European standard indicated in BS EN 1090-3.
4. Sizes: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
5. Placement: Fill pockets with bedding mortar, finish flush where exposed to view.

### 291 Special fixings

1. Standard: To BS EN 1090-1.
2. Type: Submit proposals.
3. Material/ finish: Aluminium to European standard indicated in BS EN 1090-3.
4. Performance: Of type, size, strength and number necessary to resist loads likely to occur during the life of the building, and prevent lateral displacement or pulling apart of construction.
5. Placement: Fill pockets with bedding mortar finished flush where exposed to view.

## Flexible damp proof courses/ cavity trays

### 310 Damp proof courses – bitumen based

1. Standard: To BS EN 14967, BS 6398 and BS 743
   1.1. Type: Class A (hessian-based bitumen) to BS 6398/ Class B (fibre-based bitumen) to BS 6398/ Class D (laminated hessian and lead-based bitumen) to BS 6398/ Class E (laminated lead and fibre-based bitumen) to BS 6398/ Bitumen polymer
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Additional requirements: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

### 320 Damp proof courses – plastics

1. Standard: To BS EN 14909 and BS 6515.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Material: Ethylene propylene diene monomer (EPDM).
4. Additional requirements: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

### 330 Damp-proof courses

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.

## 340 Site formed flexible sheet cavity trays – bitumen based

1. Standard: To BS EN 14967, BS 6398 and BS 743
2. Type: Class A (hessian-based bitumen) to BS 6398/ Class B (fibre-based bitumen) to BS 6398/ Class D (laminated hessian and lead-based bitumen) to BS 6398/ Class E (laminated lead and fibre-based bitumen) to BS 6398/ Bitumen polymer
3. Manufacturer: Submit proposals
   3.1. Product reference: Submit proposals.
4. Additional requirements: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

## 345 Site-formed flexible sheet cavity trays – plastics

1. Standard: To BS EN 14909 and BS 6515
2. Material: Bitumen polymer/ Ethylene polypropylene/ Polypropylene/ Lead
3. Manufacturer: Submit proposals
   3.1. Product reference: Submit proposals.
4. Additional requirements: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

## 360 Gas-resistant damp-proof courses/ cavity trays

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.

## 370 Preformed cavity trays

1. Manufacturer: Submit proposals
   1.1. Product references and locations: Submit proposals.
2. Placement: To provide a free draining and watertight installation.

## 380 Junction cloaks/ stop ends for preformed dpcs/ cavity trays

1. Manufacturer: Submit proposals
   1.1. Product references and locations: Submit proposals.
2. Placement: To provide a free draining and watertight installation. Seal laps with dpcs and/ or cavity trays.

## 385 Junction cloaks/ stop ends for preformed dpcs/ cavity trays

1. Manufacturer: Submit proposals
   1.1. Product references and locations: Submit proposals.
2. Placement: Seal laps with dpcs and/ or cavity trays.

## 390 Junction cloaks/ stop ends for site-formed dpcs/ cavity trays

1. Three dimensional changes in shape: Form to provide a free draining and watertight installation. Seal laps.
2. Alternative use of preformed junction cloaks/ stop ends: Submit proposals.

# Installation of dpcs/ cavity trays

## 415 Installation of horizontal dpcs

1. Placement: In continuous lengths on full even bed of fresh mortar, with 100 mm laps at joints and full laps at angles.
2. Width: At least full width of leaf unless otherwise specified. Edges of dpc not covered with mortar or projecting into cavity.
3. Overlying construction: Immediately cover with full even bed of mortar to receive next masonry course.
4. Overall finished joint thickness: As close to normal as practicable.

## 425 Installation of ground-level dpcs

1. Joint with damp-proof membrane: Continuous and effectively sealed.

## 435 Installation of stepped dpcs in external walls

1. External walls on sloping ground: Install dpcs not less than 150 mm above adjoining finished ground level.

## 445 Installation of sill dpcs

1. Form and placement: In one piece and turned up at back when sill is in contact with inner leaf.

## 455 Installation of coping/ capping dpcs

1. Placement: Bed in one operation to ensure maximum bond between masonry units, mortar and dpc.
2. Dpcs crossing cavity: Provide rigid support to prevent sagging.

## 465 Sealing of dpcs

1. Description: Generally.
2. Overlaps and junctions: Seal with Adhesive recommended by dpc manufacturer.

## 475 Installation of site-formed cavity trays

1. Requirements to prevent downward ingress of water
    1.1. Profiles: To match those shown on drawings. Firmly secured.
    1.2. Joint treatment: Use continuous length wherever possible, otherwise lap at least 100 mm and seal to produce a free draining and watertight installation.
    1.3. Horizontal cavity trays: Support using cavity closer.
    1.4. Sloping cavity trays: Prevent sagging.
    1.5. Cleanliness: Free from debris and mortar droppings.

## 485 Installation of cavity trays over openings and other cavity bridgings

1. Length: To extend not less than 150 mm beyond ends of lintels/ bridgings.

## 495 Installation of gas-resistant dpcs/ cavity trays

1. Joint treatment: Use continuous length wherever possible, otherwise lap at least 150 mm and seal to form a gas and watertight installation.

2. Joint with damp-proof membrane: Overlap dpc/ cavity tray not less than 150 mm.

## 515 Dpc/ cavity tray leading edge in facework - flush

1. Treatment at face of masonry: Finish flush and clear of mortar at the following locations: Generally.

## 525 Dpc/ cavity tray leading edge in facework – set back

1. Treatment at face of masonry: Set back 5 mm from face of wall with recessed mortar joint to expose edge at the following locations: Generally.

## 535 Dpc/ cavity tray leading edge in facework - projecting

1. Treatment at face of masonry: Projecting 5 mm from face of wall at the following locations: Generally.

## 560 Installation of vertical dpcs

1. Form: In one piece wherever possible.
    1.1. Joints: Upper part overlapping lower not less than 100 mm.

## 570 Installation of jamb dpcs at openings

1. Joint with cavity tray/ lintel at head: Full underlap.
2. Joint with sill/ horizontal dpc at base: Full overlap.
3. Projection into cavity: Not less than 25 mm.
4. Relationship with frame: In full contact.

## 580 Installation of jamb dpcs to built-in timber frames

1. Fixing: Securely fastened to back of frame.
    1.1. Fasteners: Galvanized clout nails or staples.

# Joints

## 610 Movement joints with sealant

1. Joint preparation and sealant application: As section Z22.
2. Filler:
    2.1. Thickness: To match design width of joint.
    2.2. Manufacturer: Submit proposals
        2.2.1. Product reference: Submit proposals.
    2.3. Placement: Build in as work proceeds with no projections into cavities and to correct depth to receive sealant system.
3. Sealant
    3.1. Designation: ISO 11600
    3.2. Manufacturer: Submit proposals.
        3.2.1. Product reference: Submit proposals.
    3.3. Colour: To match adjoining masonry.

## 615 Movement joints without sealant

1. Filler:
    1.1. Thickness: To match design width of joint.
    1.2. Manufacturer: Submit proposals
        1.2.1. Product reference: Submit proposals.
    1.3. Placement: Build in as work proceeds with no projections into cavities and to correct depth to receive sealant system.

## 616 Fire resisting movement joints without sealant

1. Fire-resistant filler
    1.1. Thickness: Refer to the Design Drawings
    1.2. Fire resistance period: Refer to the Fire Strategy Report
    1.3. Manufacturer: Submit proposals
    1.4. Product reference: Submit proposals.
    1.5. Placement: Compress and insert into place in open joint.
    1.6. Adhesives and accessories: Types recommended by filler manufacturer.

## 630 Unexposed contraction joints

1. Formation: Close butt as work proceeds.

## 650 Pointing in flashings

1. Joint preparation: Free of debris and lightly wetted.
2. Pointing mortar: As for adjacent walling.
3. Placement: Fill joint and finish flush.

## 655 Pointing in asphalt skirtings

1. Joint preparation: Free of debris and lightly wetted.
2. Pointing mortar: 1:4 cement: sand incorporating a bonding agent.
    2.1. Colour: Match adjacent work.
3. Placement: Fill joint and finish flush.

## 660 Pinning up to soffits

1. Top joint of loadbearing walls: Fill and consolidate with mortar.

## 670 Head of non-loadbearing walls

1. Restraints: As clause 251.
    1.1. Fixing: Secure to soffit.
2. Joint filler: Refer to the Fire Strategy Report andRefer to the Acoustic Strategy Report
    2.1. Placement: Full, no gaps.

# Proprietary sills/ lintels/ copings/ dressings

## 720 Sills

1. Standard: To BS 5642-1.

2. Material: Clayware.
3. Manufacturer: Submit proposals
   3.1. Product reference: Submit proposals.
4. Finish: To match sample.
5. Mortar for bedding/ jointing: Cement-gauged, as section Z21.
   5.1. Standard: To BS EN 998-2.
   5.2. Additional requirements: Coloured mortar to match bricks.
6. Joints: Flush.
7. Bedding one-piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

## 721 Natural stone sills laid in hydraulic lime:sand mortar

1. Standard: To BS 5642-1.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish: To match sample.
5. Mortar for bedding/ jointing: Hydraulic lime:sand, as section Z21.
   5.1. Hydraulic lime: Eminently hydraulic lime.
   5.2. Sand source/ type: Sharp, well graded sand to approval
6. Joints: Flush.
7. Bedding one-piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

## 725 Cast stone sills laid in cement gauged mortar

1. Standard: To BS 1217.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish/ colour: To match sample.
5. Mortar for bedding/ jointing: Cement-gauged, as section Z21.
   5.1. Standard: To BS EN 998-2.
   5.2. Additional requirements: Coloured mortar to match bricks.
6. Joints: Flush.
7. Bedding one-piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

## 726 Cast stone sills laid in hydraulic lime:sand mortar

1. Standard: To BS 1217.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish/ colour: To match sample.

5. Mortar for bedding/ jointing: Hydraulic lime:sand, as section Z21.
    5.1. Hydraulic lime: Eminently hydraulic lime
    5.2. Sand: Graded crushed stone.
        5.2.1. Colour: To match sill.
        5.2.2. Source/ type: Sharp, well-graded sand to approval.
6. Joints: Flush.
7. Bedding one-piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

## 735 Precast concrete lintels

1. Standard: To BS EN 845-2.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Types: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
4. Sizes: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
5. Additional requirements: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
6. Placement: Bed on mortar used for adjacent work.
    6.1. Bearing length (minimum): Refer to the Structural Engineer's Information

## 745 Prestressed concrete lintels

1. Standard: To BS EN 845-2.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Types: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
4. Sizes: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
5. Additional requirements: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
6. Placement: Bed on mortar used for adjacent work. Prop at not more than 1.2 m centres to prevent displacement during construction. Retain props in position for not less than 14 days or until mortar has matured, whichever is longer.
    6.1. Bearing length (minimum): Refer to the Structural Engineer's Information

## 755 Prefabricated steel lintels

1. Standard: To BS EN 845-2.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Types: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
4. Material/ finish: Austenitic stainless steel – material/ coating reference L1.

5. Sizes: Refer to the Design Schedule Information and Refer to the Structural Engineer's Information
6. Additional requirements: Refer to the Structural Engineer's Information
7. Placement: Bed on mortar used for adjacent work.
    7.1. Bearing length (minimum): Refer to the Structural Engineer's Information

## 760 Coping units

1. Standard: To BS 5642-2.
2. Material: Clayware.
3. Manufacturer: Submit proposals
    3.1. Product reference: Submit proposals.
4. Dimensions: Refer to the Design Drawings
5. Finish: To match sample.
6. Mortar for bedding/ jointing: Cement-gauged, as section Z21.
    6.1. Standard: To BS EN 998-2.
7. Joints: Full and finished flush.
8. Placement: Lay on a full bed of mortar to line and level.

## 761 Natural stone coping units laid in hydraulic lime:sand mortar

1. Standard: To BS 5642-2.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish: To match sample.
5. Mortar for bedding/ jointing: Hydraulic lime:sand, as section Z21.
    5.1. Hydraulic lime: Eminently hydraulic lime.
    5.2. Sand source/ type: Sharp, well-graded sand to approval.
6. Joints: Full and finished flush.
7. Placement: Lay on a full bed of mortar to line and level.

## 765 Cast stone coping units laid in cement gauged mortar

1. Standard: To BS 1217.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish/ colour: To match sample.
5. Mortar for bedding/ jointing: Cement gauged as section Z21.
    5.1. Standard: To BS EN 998-2.
6. Joints: Full and finished flush.
7. Placement: Lay on a full bed of mortar to line and level.

## 766 Cast stone coping units laid in hydraulic lime:sand mortar

1. Standard: To BS 1217.

2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Dimensions: Refer to the Design Drawings
4. Finish/ colour: To match sample.
5. Mortar for bedding/ jointing: Hydraulic lime:sand, as section Z21.
    5.1. Hydraulic lime: Eminently hydraulic lime.
    5.2. Sand: Graded crushed stone.
        5.2.1. Colour: To match coping.
        5.2.2. Source/ type: Sharp, well-graded sand to approval.
6. Joints: Full and finished flush.
7. Placement: Lay on a full bed of mortar to line and level.

## 770 Cast stone dressings laid in cement gauged mortar

1. Standard: To BS 1217.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Finish/ colour: To match sample.
4. Mortar for bedding/ jointing: Cement gauged as section Z21.
    4.1. Standard: To BS EN 998-2.
5. Joints: Full and finished flush.
6. Placement: Lay on a full bed of mortar.
7. Bed joints under one-piece sills/ thresholds: Leave open except under end bearings and masonry mullions. On completion, point with mortar to match adjacent work.

## 771 Cast stone dressings laid in hydraulic lime:sand mortar

1. Standard: To BS 1217.
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Finish/ colour: To match sample.
4. Mortar for bedding/ jointing: Hydraulic lime:sand, as section Z21.
    4.1. Hydraulic lime: Eminently hydraulic lime.
    4.2. Sand: Graded crushed stone.
        4.2.1. Colour: To match component.
        4.2.2. Source/ type: Sharp, well-graded sand to approval.
5. Joints: Full and finished flush.
6. Placement: Lay on a full bed of mortar.
7. Bed joints under one-piece sills/ thresholds: Leave open except under end bearings and masonry mullions. On completion, point with mortar to match adjacent work.

## 780 Coping systems

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.
2. Material/ finish: Aluminium.

# Miscellaneous items

## 810 Tile sills

1. Tiles: Plain clay to BS EN 1304.
    1.1. Manufacturer: Submit proposals
        1.1.1. Product reference: Submit proposals.
    1.2. Size: Refer to the Design Drawings
2. Placement: Two courses, broken jointed, true to line and level on full bed of 1:¼:3 cement:lime:sand mortar, as section Z21.
3. Joints: Full and finished flush.

## 820 Tile creasing

1. Tiles: Plain clay to BS EN 1304.
    1.1. Manufacturer: Submit proposals
        1.1.1. Product reference: Submit proposals.
    1.2. Size: Refer to the Design Drawings
2. Placement: Two courses, broken jointed, on full bed of mortar as used for adjacent work.
3. Joints: Full and finished flush.

## 830 Building in frames

1. Preparation: Remove horns and provide support.
2. Fixing cramps: Fully bed in mortar.

## 840 Openings for frames

1. Formation: Use accurate, rigid templates to required size.

## 850 Wall plates

1. Placement: On full bed of mortar to correct horizontal level.

## 910 Clay/ ceramic flue lining system

1. Lining standard: To BS EN 1457-1 for flue liners operating under dry conditions, and BS EN 1457-2 for wet conditions.
    1.1. Section/ internal dimensions: Refer to the Design Drawings andRefer to the Services Engineer's Information
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
4. Jointing material: As recommended by flue manufacturer.
5. Linings placement: Fully bed with socket or rebate uppermost using correct starters, adaptors, bends, etc.
    5.1. Joints: Flush to provide an unrestricted flueway with smooth interior surfaces.
6. Void between lining and surrounding work: Fill with 1:20 ordinary Portland cement:lightweight expanded clay aggregate, minimally wetted.

7. Core ball test and smoke test to BS EN 15287-1: On completion, carry out each test. If obstructions or leaks are revealed, submit proposals for making good.

## 915 Concrete flue lining system

1. Lining standard: To BS EN 1857.
    1.1. Section/ internal dimensions: Refer to the Design Drawings and Refer to the Services Engineer's Information
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
4. Jointing material: As recommended by flue manufacturer.
5. Linings placement: Fully bed with socket or rebate uppermost using correct starters, adaptors, bends, etc.
    5.1. Joints: Flush to provide an unrestricted flueway with smooth interior surfaces.
6. Void between lining and surrounding work: Fill with 1:20 ordinary Portland cement:lightweight expanded clay aggregate, minimally wetted.
7. Core ball test and smoke test to BS EN 15287-1: On completion carry out each test. If obstructions or leaks are revealed, submit proposals for making good.

## 923 Concrete flue block system

1. Block standard: To BS EN 1858.
    1.1. Chimney designation to BS EN 1443: As required by appliance manufacturer.
    1.2. Flue form: Refer to the Services Engineer's Information
    1.3. Section/ internal dimensions: Refer to the Design Drawings and Refer to the Services Engineer's Information
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.
3. Components/ accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
4. Jointing material: As recommended by flue manufacturer.
5. Blocks placement: Finish joints flush to ensure a smooth, unrestricted flueway. Use correct starter, offset and transfer blocks. Course bonded blocks with adjacent masonry. Support non-bonded blocks. Seal joints between blocks.
6. Coring ball test and smoke test to BS EN 15287-1: On completion, carry out each test. If obstructions or leaks are revealed, submit proposals for making good.

## 925 Clay/ ceramic flue block system

1. Block standard: To BS EN 1806.
    1.1. Flue form: Refer to the Services Engineer's Information
    1.2. Class: Refer to the Services Engineer's Information
    1.3. Section/ internal dimensions: Refer to the Design Drawings and Refer to the Services Engineer's Information
2. Manufacturer: Submit proposals
    2.1. Product reference: Submit proposals.

3. Components/ accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
4. Jointing material: As recommended by flue manufacturer.
5. Blocks placement: Finish joints flush to ensure a smooth, unrestricted flueway. Use correct starter, offset and transfer blocks. Course bonded blocks with adjacent masonry. Support non-bonded blocks. Seal joints between blocks.
6. Core ball test and smoke test to BS EN 15287-1: On completion carry out each test. If obstructions or leaks are revealed, submit proposals for making good.

## 930 Fireplace components

1. Standard: To BS 1251.
2. Manufacturer: Submit proposals
   2.1. Product reference: Submit proposals.
3. Components: Refer to the Design Drawings

## 940 Chimney pots

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Coping: Refer to the Design Drawings
3. Placement: Bed solid in mortar specified for chimney stack.

## 970 Refuse chutes and accessories

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. General: Build in to provide a smooth and even interior surface free from voids and restrictions. Fill void between chute and surrounding work with concrete.

Ω End of Section



make

# G12
# Steel Framing System (SFS)

## Products

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 150 EWS-13A Steel Framing System - Insulated (Visual Intent Only)

1. Description: Load-bearing metal section external stud wall system, forming the backing wall between or across the external face of perimeter elements of the primary structure, to receive green wall externally and drylining internally.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Loadbearing metal section stud walling: Refer to the Façade Performance Specification
5. Insulation: Refer to the Façade Performance Specification
6. Fixings: Refer to the Façade Performance Specification
7. Green wall: Refer to the Façade Performance Specification

### 151 EWS-13B Steel Framing System - Uninsulated (Visual Intent Only)

1. Description: Load-bearing metal section external stud wall system, forming the backing wall between or across the external face of perimeter elements of the primary structure, to receive green wall externally and drylining internally.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Loadbearing metal section stud walling: Refer to the Façade Performance Specification
5. Fixings: Refer to the Façade Performance Specification
6. Green wall: Refer to the Façade Performance Specification

## Fabrication

### 510 Fabrication of steel members

1. Cuts and holes: Accurate and neat.
2. Welding: Metal arc method to BS EN 1011-2.
   2.1. Welded joints: Fully fused, with mechanical properties not less than those of the parent metal.
   2.2. Site welding: Obtain approval

3. Joints: Location and layout of fastenings to suit fabrication drawing package.

### 520 Samples

1. Provide the following samples:
   1.1. 300mm - Colour finish types.
   1.2. 500mm - Metal elements.
   1.3. Visible fixings.

## Execution

### 610 Installation

1. Accuracy: Members positioned true to line and level using, if necessary, steel packs of sufficient area to allow full transfer of loads to bearing surfaces.
2. Fixing: Use washers under bolt heads and nuts.
   2.1. Tapered washers: Provide under bolt heads and nuts bearing on sloping surfaces. Match taper to slope angle and align correctly.
3. Install metalwork systems within the following maximum tolerances:
   3.1. Length/ width/ squareness/ alignment: ±2mm
   3.2. Joints: ±1mm
   3.3. Twist over complete length: 1°

### 620 Bonded anchors

1. Holes: Clean and free from dust at time of installing anchor.
2. Permeable sleeves: Use in conditions where otherwise the loss of bonding agent would be unacceptably high.

### 630 Pull-out testing generally

1. Standard: Refer to the Structural Engineer's Information

### 640 Preparation and priming

1. Sequence of working: Fabricate, prepare, prime.
2. Surfaces inaccessible after assembly: Apply full treatment and coating system, including, if necessary, local application of site coatings.
3. Galvanized/ sherardized fasteners: After steelwork erection and before applying site coatings, thoroughly degrease and clean. Etch prime.

### 650 Shop priming

1. Description: Refer to the Structural Engineer's Information

## Completion

### 910 Steel-to-timber joints

1. General: Inspect accessible bolted and coach-screwed joints and tighten fasteners if necessary.
   1.1. Timing: Immediately prior to installation of finishes, on completion, and at end of the defects rectification or liability period.

1.2. Confirmation: Give notice when inspections and adjustments have been made.

Ω End of Section

# H11
# Curtain Walling

## Tendering

### 1 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 10 Information to be provided with tender

1. Submit the following curtain walling particulars
    1.1. Evidence of compliance: All curtain walling to be CE marked to BS EN 13830. Submit Declaration of Performance (DoP). CE marking and Declaration of Performance (DoP) requirements for structural sealant glazing to be submitted on the basis of a manufacturer's European Technical assessment (ETA) prepared in accordance with EOTA technical guide ETAG 002 (Parts 1, 2 and 3).
    1.2. Typical plan, section and elevation drawings at suitable scales.
    1.3. Typical detailed drawings at large scales, including to be agreed.
    1.4. Technical information and certification demonstrating compliance with specification of proposed incorporated products and finishes, including to be agreed.
    1.5. Certification, reports and calculations demonstrating compliance with specification of proposed curtain walling.
    1.6. Proposals for connections to and support from the building structure and building components.
    1.7. Proposals for amendments to primary supporting structure and for secondary supporting structure additional to that shown on preliminary design drawings.
    1.8. Schedule of builder's work, special provisions and special attendance by others.
    1.9. Examples of standard documentation from which project quality plan will be prepared.
    1.10. Preliminary fabrication and installation method statements and programme.
    1.11. Schedule of products and finishes with a design life expectancy less than that specified in clause 440, with proposals for frequencies and methods of replacement.
    1.12. Proposals for replacing damaged or failed products.
    1.13. Areas of non-compliance with the specification.

## Types of curtain walling

### 110 EWS-08 Curtain Wall System - Ground Floor (Visual Intent Only)

1. Description: Capped stick built curtain walling system with integrated doorsets, as the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification

4. Sizes and configurations: Refer to the Design Drawings
5. Specific performance: Refer to the Façade Performance Specification
6. Supporting structure/ framing/ external cover cap: Refer to the Façade Performance Specification
7. Finish/ colour/ samples: Refer to the Façade Performance Specification
8. Fire, smoke and acoustic stopping: Refer to the Façade Performance Specification

### 111 EWS-10 Curtain Wall System - Terrace (Visual Intent Only)

1. Description: Capped stick built curtain walling system with integrated doorsets, as the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings
5. Specific performance: Refer to the Façade Performance Specification
6. Colour/ finish/ sampling: Refer to the Façade Performance Specification
7. Fire, smoke and acoustic stopping: Refer to the Façade Performance Specification

### 150 Materials specification

1. Minimum BRE 'Green Guide to Specification' online rating: Submit proposals
2. Specification of glazing materials to be in accordance with guidance in CIRIA document ‘Guidance on glazing at height’.

## General requirements/ preparatory work

### 210 Design

1. Curtain walling and associated features: Complete the detailed design. Submit before commencement of fabrication.
2. Related works: Coordinate in the detailed design.

### 215 Design proposals

1. Submit detailed proposals related to the following with the Tender:
   1.1. Response to the Design depicting technical solutions whilst maintaining the design intent.
   1.2. QA/QC procedures.
   1.3. List of proposed Working Drawings
   1.4. List of tests included together with details of Testing Authority and timing.
   1.5. List of Samples, Mock-Ups and Prototypes proposed.
   1.6. Proposed deviations from the Design.
   1.7. Outline specifications for proposed technical solution to the Design.
   1.8. Thermal stress calculations.
   1.9. Risk assessment.
   1.10. Light and solar gain calculations.

1.11. Movement/ tolerance expectations and parameters for dead and live loads.

1.12. Methods to be adopted to counter the occurrence of condensation.

1.13. Methods to be adopted to ensure acoustic requirements are met.

1.14. Methods to be adopted to meet environmental performance requirements.

## 220 Specification

1. Compliance standards: BS EN 13830 and The Centre for Window and Cladding Technology (CWCT) 'Standard for systemised building envelopes'.
2. Reference information: For the duration of the contract, keep available at the design office, workshop and on site copies of:
   2.1. The CWCT 'Standard for systemised building envelopes'.
   2.2. Publications invoked by the CWCT 'Standard for systemised building envelopes'.

## 230 Information to be provided during detailed design stage

1. Submit the following curtain walling particulars
   1.1. A schedule of detailed drawings and dates for submission for comment.
   1.2. A schedule of loads that will be transmitted from the curtain walling to the structure.
   1.3. Proposed fixing anchor details relevant to structural design and construction.
   1.4. A detailed testing programme in compliance with the main contract master programme.
   1.5. A detailed fabrication and installation programme in compliance with the main contract master programme.
   1.6. Proposals to support outstanding applications for Building Regulation consents or relaxations.

## 232 Quality plan

1. Requirement: Submit during detailed design.
2. Content: In accordance with BS EN ISO 9001 and including the following:
   2.1. Name of the quality manager.
   2.2. Quality assessment procedures.
   2.3. Inspection procedures to be adopted in checking the work.
   2.4. Stages at which check lists will be used and samples of the lists.
   2.5. List of work procedures on the correct use of materials or components, both off site and on site.
   2.6. List of product information with latest revisions.
   2.7. Subcontractors involved in the work.
   2.8. Subcontractors' quality plans.
   2.9. Storage, handling, transport and protection procedures.
   2.10. Procedure for registering and reporting non compliances.
   2.11. Maintenance procedures and calibration records.
   2.12. Certification that completed work complies with specification.
   2.13. Check list register to ensure all items have been inspected and non compliances discharged.

## 235 Information to be provided before commencement testing or fabrication of curtain walling

1. Submit the following curtain walling particulars
    1.1. Detailed drawings to fully describe fabrication and installation.
    1.2. Detailed calculations to prove compliance with design/ performance requirements.
    1.3. Project specific fabrication, handling and installation method statements.
2. Certification for incorporated components manufactured by others confirming their suitability for proposed locations in the curtain walling.
    2.1. Recommendations for spare parts for future repairs or replacements.
3. Recommendations for safe dismantling and recycling or disposal of products.

## 240 Information to be provided before commencement of structural sealant/ bonded glazing

1. Submit structural bonding sealant manufacturer's project specific approval for
    1.1. Compatibility and adhesion of products and finishes.
    1.2. Full details of structural sealant/ bonded glazing design.
    1.3. Structural sealant dimensions.
    1.4. Project specific sealant application method statement.

## 250 Product samples

1. General: Before commencing detailed design, submit labelled samples of:
    1.1. 300mm length of each type of framing.
    1.2. All proposed gasket types, extrusions, etc. including at least one joint.
    1.3. Glass samples 300mm x 300mm of all proposed glass types.
    1.4. All fixings.
    1.5. Structural silicone.
    1.6. Proposed types of finish.
    1.7. Ironmongery.

## 260 Samples of fixings

1. General: During detailed design, submit labelled samples of each type of fixing anchor, including casting-in restraints and shims, together with manufacturers' recommended torque figures.

## 270 Fabrication samples

1. General: During detailed design, submit samples of: to be agreed .
    1.1. Obtain acceptance of appearance before proceeding.

## 280 Prototypes

1. General: Construct during detailed design work in an approved location. Obtain approval of appearance before proceeding. Retain undisturbed until completion of curtain walling installation.
2. Extent: Refer to the Design Drawings
3. Purpose: Testing and visual review.

## Design/ performance requirements - Not Used

## Breeam performance requirements - Not Used

## Testing

### 510 Comparison (type) testing

1. Requirement: To CWCT 'Standard for systemised building envelopes', part 8.
2. Test results and reports: Before commencement of curtain walling fabrication and installation, submit proof of compliance with this specification.

### 515 Project testing (laboratory)

1. Test results and reports: Before commencement of curtain walling fabrication and installation, submit proof of compliance with this specification.

### 520 Project testing (site)

1. Test results and reports: Before installation of general areas of curtain walling, submit proof of compliance with this specification.

### 530 Testing authority

1. Requirement: Project testing must be carried out by a United Kingdom Accreditation Service (UKAS) approved independent laboratory.

### 555 Test sequence – standard laboratory tests

1. Requirement: To CWCT 'Standard for systemised building envelopes', Standard sequence B, clause 8.12.2, and as follows:
2. Air permeability (infiltration): To CWCT clause 8.6.
   - 2.1. Peak test pressure: As clause 340.
   - 2.2. Method of identifying areas of concentrated air leakage: Hot wire anemometers
3. Air permeability (exfiltration): To CWCT clause 8.6.
   - 3.1. Peak test pressure: As clause 340.
   - 3.2. Allowable exfiltration rates: As clause 345.
   - 3.3. Method of identifying areas of concentrated air leakage: To be agreed.
4. Water penetration resistance - static: To CWCT clause 8.7.1.
   - 4.1. Peak test pressure: As clause 350.
5. Wind resistance - serviceability: To CWCT clause 8.8.
   - 5.1. Peak test pressure: Equal to design wind pressure specified in clause 311 and 313.
   - 5.2. Position of displacement measuring devices: To be agreed.
   - 5.3. Allowable elastic and residual deformation: To be agreed.
6. Air permeability (infiltration): Repeat as above.
7. Air permeability (exfiltration): Repeat as above.
8. Water penetration resistance - static: Repeat as above.
9. Water penetration resistance - dynamic: To CWCT clause 8.7.2.
   - 9.1. Test method: Dynamic fan
   - 9.2. Test pressure: As clause 350.

10. Hose test: To CWCT clause 8.16.2.
    10.1. Joints to be tested: To be agreed with the Architect.
11. Wind resistance - safety: To CWCT clause 8.8.
    11.1. Peak test pressure: Equal to 1.5 times design wind pressure specified in clause To be agreed..
    11.2. Position of displacement measuring devices: To be agreed with the Architect.
    11.3. Allowable residual deformation: To be agreed with the Architect.
12. Impact - safety: To CWCT clause 8.10
    12.1. Location, sequence and number of impacts to be applied: To be agreed with the Architect.
    12.2. Type of soft body impactor: To be agreed with the Architect.
    12.3. Energy of soft body impactor (E): To be agreed with the Architect.
    12.4. Required hard body impact load category: To be agreed with the Architect.
    12.5. Required soft body impact classification: To be agreed with the Architect.
13. Dismantle inspect and record.

### 630 Site spray bar test

1. Requirement: To CWCT 'Standard for systemised building envelopes', 'Standard test methods for building envelopes' Section 10.
    1.1. Area to be tested: To be agreed with the Architect.
    1.2. Pressure difference across curtain wall: To be agreed with the Architect.

### 635 Site hose test

1. Requirement: To CWCT 'Standard for systemised building envelopes' Section 8.16.2, 'Standard test methods for building envelopes' Section 9.
    1.1. Joints to be tested: To be agreed with the Architect.

### 655 Wind load fatigue test, small specimen

1. Requirement: To CWCT 'Standard for systemised building envelopes', 'Standard test methods for building envelopes' Section 14.
    1.1. Test sequence: To be agreed with the Architect.

### 660 Structural sealant/ bonded glazing tests

1. Product samples: Provide the structural bonding sealant manufacturer with framing profiles, glass, gaskets, assembly/ weathering sealants and other curtain walling products that are proposed for contact with structural bonding sealant.
2. Testing: By sealant manufacturer to determine compatibility and adhesion of structural bonding sealant under specified design loadings.
3. Modification of product to enable compliance with test criteria: Details must be recorded in the sealant manufacturer's project specific approval.

## Products

### 710 Aluminium alloy framing sections

1. Standard: To relevant parts of BS EN 515, BS EN 573, BS EN 755 and BS EN 12020.

2. Alloy, temper and thickness: Suitable for the application and specified finish.
3. Structural members: To BS EN 1999-1-1.

## 712 Aluminium alloy sheet

1. Standards: To relevant parts of BS EN 485, BS EN 515 and BS EN 573.
2. Alloy, temper and thickness: Suitable for the application and specified finish.

## 730 Mechanical fixings

1. Stainless steel: To BS EN ISO 3506, grade A2 generally, grade A4 when used in severely corrosive environments.
2. Carbon steel: To BS 4190 and suitable for galvanizing or other protective coating.
3. Aluminium brackets, rivets and shear pins: To relevant parts of BS EN 755.

## 732 Adhesives

1. General: Not degradable by moisture or water vapour, or exposure to UV light.

## 735 Fixing anchors

1. Type and use: Reviewed and approved by fixing manufacturers. Submit confirmatory information on request.
2. Dimensions: Not less than recommended by their manufacturers.
3. Adjustment capability: Sufficient in three dimensions to accommodate building structure and curtain walling fabrication/ installation tolerances.

## 737 Glass generally

1. Standards: To BS 952 and relevant parts of:
    1.1. BS EN 572 for basic soda lime silicate glass.
    1.2. BS EN 1096 for coated glass.
    1.3. BS EN 1748 for borosilicate glass.
    1.4. BS EN 1863 for heat-strengthened soda lime silicate glass.
    1.5. BS EN 12150 for thermally toughened soda lime silicate glass.
    1.6. BS EN 13024 for thermally toughened borosilicate glass.
    1.7. BS EN ISO 12543 for laminated glass.
2. Selection of glass type and thickness in accordance with recommendations of CIRIA publication 'Guidance on glazing at height'.
3. Glass quality: Clean and free from obvious scratches, bubbles, cracks, ripplings, dimples and other defects.
4. Glass edges: Generally undamaged. Shells and chips not more than 2 mm deep and extending not more than 5 mm across the surface are acceptable if ground out.

## 739 Dimensional tolerances on glass

1. Measurement of tolerances: Before any thermal toughening/ heat-strengthening.
2. Pane dimensions less than 1500 mm
    2.1. For 3 to 6 mm-thick glass: ± 1.0 mm.
    2.2. For 8 to 12 mm-thick glass: ± 1.5 mm.

2.3. For 15 mm-thick glass: ± 2.0 mm.

2.4. For 19 mm and 25 mm-thick glass: ± 2.5 mm.

3. Pane dimensions more than 1500 mm

3.1. For 3 to 6 mm-thick glass: ± 1.5 mm.

3.2. For 8 to 12 mm-thick glass: ± 2.0 mm.

3.3. For 15 mm-thick glass: ± 2.5 mm.

3.4. For 19 mm and 25 mm-thick glass: ± 3.0 mm.

4. Pane squareness: Not more than 4 mm difference in diagonal measurements.

## 741 Distortional tolerances on glass

1. Measurement of tolerances: After any thermal toughening/ heat-strengthening.
2. Maximum bow: 0.2% of pane dimension.
3. Maximum roller wave

3.1. For 3 to 5 mm-thick glass: 0.5 mm.

3.2. For 6 to 10 mm-thick glass: 0.3 mm.

3.3. For 12 mm and thicker glass: 0.15 mm.

4. Maximum edge dip

4.1. For 3 to 5 mm-thick glass: 0.8 mm.

4.2. For 6 to 10 mm-thick glass: 0.5 mm.

4.3. For 12 mm and thicker glass: 0.25 mm.

## 742 Heat-soaked thermally toughened glass

1. Standard: To BS EN 14179.

1.1. Holding period: To be agreed with the Architect.

2. Locations of heat-soaked glass: To be agreed with the Architect.

## 745 Insulating glass units

1. Standard and labels for hermetically sealed units: To BS EN 1279.
2. Label: Each pane.
3. Colour of aluminium perimeter spacers: To be agreed with the Architect.
4. Perimeter taping: Not to be used.
5. Perimeter seals

5.1. Resistant to UV light degradation on exposed edges.

5.2. Compatible with structural, assembly and weather sealants.

## 750 Infill panels/ Facings

1. Tolerances

1.1. Deviation in size (maximum): ± 1 mm.

1.2. Deviation in flatness from plane per 2 m length (maximum): ± 1 mm.

2. Rigidity: Adequate to comply with design/ performance requirements.
3. Fire performance: BS EN 13501-1, Class A1

## 760 Gaskets

1. Material
    1.1. Noncellular rubber to BS 4255-1.
    1.2. Cellular rubber to ASTM-C509-06.
2. Continuity: Outer gaskets of single front sealed curtain walling systems and inner gaskets of drained and ventilated or pressure equalized curtain walling systems must be formed in a complete frame with sealed joints. Vulcanized rubber gaskets must have factory moulded corner joints.
3. Durability: Resistant to oxidation, ozone and UV degradation.

## 765 Weatherstripping of opening units

1. Material
    1.1. Noncellular rubber to BS 4255-1.
    1.2. Cellular rubber to ASTM-C509-06.
    1.3. Polypropylene woven pile, silicone treated.
2. Attachment: Fixed in undercut grooves in framing sections using preformed corners, with any joints in the length.

## 770 General sealants

1. Selection: In accordance with BS 6213 from:
    1.1. Silicone.
    1.2. One part polysulfide.
    1.3. Two parts polysulfide.
    1.4. One or two parts polyurethane.
2. Classification and requirements: To BS EN ISO 11600.
3. Reaction to contact products and finishes: Stable and compatible.

## 772 Curtain walling joint assembly sealants

1. Material: One part, low modulus silicone to BS EN ISO 11600, type F or G. Neutral curing where in contact with or close proximity to other products that may be adversely affected by acetoxy curing.

## 780 Thermal insulation

1. Material: Mineral wool boards to BS EN 13162
    1.1. Recycled content: Submit proposals
    1.2. Properties: Durable, rot and vermin proof and not degradable by moisture or water vapour.
2. Fixing: Attached to or supported within the curtain walling so as not to bulge, sag, delaminate or detach during installation or in situ during the life of the curtain walling.

## 785 Air and vapour control layer

1. Acceptable materials
    1.1. Aluminium alloy.
    1.2. Carbon steel, galvanized or protective-coated.

1.3. Stainless steel.

1.4. Reinforced membranes: Foil, plastics or rubbers, protected both sides by rigid facings/ linings.

2. Location: Warm side of thermal insulation.
3. Integrity: Continuous, free from gaps and sealed at joints.

## Finishes

### 810 Protective coating of carbon steel framing sections/ Reinforcement

1. Treatment: One of the following to all surfaces:
   1.1. Hot dip galvanized to BS EN ISO 1461.
   1.2. An appropriate equivalent coating to BS EN ISO 12944-5 or BS EN ISO 14713-1, -2 and -3.

### 820 Protective coating of carbon steel mechanical fixings

1. Treatment: One of the following to all surfaces:
   1.1. Hot dip galvanized to BS EN ISO 1461.
   1.2. Sherardized to BS 7371-8, Class 30 coating thickness and passivated.
   1.3. Zinc plated to BS EN ISO 2081, coating designation Fe//Zn//C for an iridescent (yellow passivate) chromate conversion coating or Fe//Zn//D for an opaque (olive green) chromate conversion coating.

### 830 Powder-coating

1. Requirement: As Section Z31.

### 840 Anodizing

1. Requirement: As Section Z33.

### 850 Polyvinylidene fluoride (PVDF)-coating of aluminium alloy

1. Standard: To BS 4842 or AAMA 2604-20, subject to minimum coating thicknesses recommended by the manufacturer on significant surfaces.
2. PVDF resin content of coatings: Not less than 70%.
3. Process: Prepare base metals, prime, PVDF-coat, test samples, protect components and repair damage in accordance with manufacturer's recommendations.
4. Sequence: Wherever possible, apply coatings after fabrication is complete.
   4.1. Fabrication of prefinished lengths: Submit proposals beforehand.
   4.2. Uncoated edges: Invisible in completed assemblies.

## Fabrication and installation

### 910 Generally

1. Electrolytic corrosion: Prevent. Submit proposed methods.
2. Fixings: Concealed unless indicated on detailed drawings. Where exposed they must match material and finish of the products fixed. Where fixings will be visible, stainless steel hex socket machine (Allen) screws to be used and countersunk where possible.
3. Fabrication: Machine cut and drill products in the workshop wherever possible.

4. Identification of products: Mark or tag to facilitate identification during assembly, handling, storage and installation. Do not mark surfaces visible in the completed installation.

## 912 Metalwork

1. Requirement: As Section Z11, unless specified otherwise in this section.

## 915 Glazing

1. Requirement: As Section L40, unless specified otherwise in this section.
2. Directional patterned/ wired glass: Generally fix parallel to surround and align adjacent panes where seen together at close quarters.

## 917 Fixings/ Adhesives application

1. Requirement: As Section Z20, unless specified otherwise in this section.

## 920 Sealant application

1. Requirement: As Section Z22, unless specified otherwise in this section.

## 930 Assembly

1. General: Carry out as much assembly as possible in the workshop.
2. Joints (other than movement joints): Rigidly secured, reinforced where necessary and fixed with hairline abutments.
3. Displacement of components in assembled units: Submit proposals for reassembly on site.

## 935 Openable windows in naturally ventilated buildings

1. Location: Over 10 m from sources of external pollution

## 955 Fixing anchor installation

1. Site drilling or cutting into structure: Submit proposals for positions other than shown on detailed drawings.
2. Concrete supporting structure
    2.1. Cast-in inserts: Provide detailed locational information. Protect cavities in inserts from entry of concrete.
    2.2. Edge fixing distances: Not less than recommended by fixing anchor manufacturers.
3. Corrective fabrication: Minimize. Where necessary, submit proposals.

## 965 Preliminary curtain walling installation

1. Requirement: Complete an area for inspection and approval of appearance as follows: To be agreed.

## 970 Curtain walling installation

1. Securing to fixing anchors: Through holes formed during fabrication only.
2. Tightening mechanical fasteners: To manufacturer's recommended torque figures. Do not overtighten fasteners intended to permit differential movement.
3. Protective coverings: Remove only where necessary to facilitate installation and from surfaces that will be inaccessible on completion.

## 975 Welding

1. In situ welding: Permitted, subject to completion of and compliance with a 'hot work permit' form

## 978 Installing fire and smoke stops

1. Fire and smoke stops: To be located at all junctions between the curtain wall and compartment or separating walls and floors. To be installed strictly in accordance with manufacturers' guidelines and as recommended in CWCT Technical Note TN 98.
2. Installer qualification: To be a member of a UKAS-accredited installer scheme.

## 980 Interfaces

1. Flashings, closers, etc: Locate and form correctly to provide weathertight junctions with the curtain walling.

## 982 Ironmongery

1. Assembly and fixing: Accurately, using fasteners with matching finish supplied by ironmongery manufacturer.
2. Completion: Check, adjust and lubricate as necessary to ensure correct functioning.

## 985 Maintenance

1. Maintenance manual: Incorporate details within the Building Manual in accordance with CWCT 'Standard for systemised building envelopes' clause 7.6.1.
    1.1. Materials certification and test reports to be included: To be agreed with the Architect.

Ω End of Section

# H43
# Metal Insulating Sandwich Panel Cladding/ Roofing

## Types of cladding/ roofing system

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 120 EWS-106 Insulated Wall System

1. Description: Insulated wall system, as the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Performance requirements: Performance and detail interfaces to be validated by Façade Consultant.
    3.1. Fire rating: Refer to the Fire Strategy Report
    3.2. Acoustic rating: Refer to the Acoustic Report
    3.3. U-value: 0.26 W/m2K.
4. Support structure: Corrosion resistant support components, mechanically secured to the loadbearing elements of the backing wall/ primary structure and concealed in the finished works.
5. Panels:
    5.1. External facing material: Anodized aluminium to BS EN ISO 7599:2018.
    5.2. Composition:
        5.2.1. Profile: Flat
        5.2.2. Finish: Anodized.
        5.2.3. Colour: Standard colour to be confirmed.
    5.3. Internal facing material: Anodized aluminium to BS EN ISO 7599:2018.
        5.3.1. Profile: Flat
        5.3.2. Finish: Anodized.
        5.3.3. Colour: Standard colour to be confirmed.
    5.4. Core insulation: Non-combustible core insulation to suit composite panel construction, fully bonded to inner and outer sheets and to achieve the performance requirements
6. Joint type
    6.1. Type: Lapped, interlocking, watertight joints consistent in size
    6.2. Side/ vertical: Butt joints with cover flashing
    6.3. End/ horizontal: To include inconspicuous factory applied black gasket extrusions/ weather seals.
7. Primary fasteners: Concealed or inconspicuous fixings. Exposed fixings to be colour co-ordinated with the cladding panels and uniformly configured.
8. Flashings, trims, drips, cappings and other such elements: Visually to match the cladding.
9. Fire, smoke and acoustic stopping: As required to be concealed in the finished works and to meet the current requirements of the Building Regulations.

## 122 EWS-107 Insulated Wall System

1. Description: Insulated wall system, as the Design Drawings.
2. Specific performance: Performance and detail interfaces to be validated by Façade Consultant.
    2.1. Fire rating: Refer to the Fire Strategy Report
    2.2. Acoustic rating: Refer to the Acoustic Strategy Report
    2.3. Duty rating: Severe.
3. Types: Refer to the Design Drawings for partitions/ wall linings/ ceiling types and locations.
4. Build-ups, construction, finishes, sizes and zone: Refer to the Design Drawings
5. Support framing: Varies, as per the manufacturer's recommendations to achieve the stipulated performance requirements.
6. Blocks: To BS EN 771-3.
    6.1. Manufacturer: Lignacite, or acceptable equivalent
        6.1.1. Product reference: SP Fair Faced.
    6.2. Configuration: Group 1
    6.3. Compressive strength
        6.3.1. Mean value: 7.3 N/mm²
        6.3.2. Category: I
    6.4. Freeze/ thaw resistance: Frost-resistant
    6.5. Work sizes (length x width x height): 440 x 100 x 215 mm
        6.5.1. Tolerance category: D1.
7. Insulation: Non-combustible.
8. Linings: Refer to the Design Drawings
9. Pattressing: Plywood pattress, timber noggings and fixing plates to be provided where necessary to support surface-mounted fixtures and fittings.
10. Finishing:
    10.1. Unless specified, or indicated otherwise, outer lining plasterboard shall be square edged, scrim taped and finished with plaster skim suitable to receive applied finishes.
    10.2. Generally, skim plaster shall not be applied to moisture resistant plasterboard in areas subject to high humidity.
    10.3. Flush jointing (tape and fill) needs to be undertaken in line with the manufacturer's recommendations and shall only be used if samples of the joints have been accepted by the Employer's Agent.
11. Mortar: As section Z21.
    11.1. Standard: To BS EN 998-2
    11.2. Type: Cement: Lime: Sand.
    11.3. Colour: Natural. To match block colour.
12. Bond: Stretcher.
13. Other requirements: System shall include all studs/ framing, insulation, linings, fixings and accessories necessary to complete the installation.

## 122 EWS-151 Insulated Panel System

1. Description: Insulated panel cladding system, as the Design Drawings.
2. Size and configuration: Refer to the Design Drawings

3. Performance requirements:
    3.1. Fire rating: Refer to the Fire Strategy Report
    3.2. Acoustic rating: Refer to the Acoustic Report
4. Support structure: Corrosion resistant support components, mechanically secured to the loadbearing elements of the backing wall/ primary structure and concealed in the finished works.
5. Panels:
    5.1. Manufacturer: Rockwool Ltd, or acceptable equivalent
        5.1.1. Product reference: To be agreed with the Architect
    5.2. External facing material: Zinc-coated steel to BS EN 10346
    5.3. Composition:
        5.3.1. Profile: Flat
        5.3.2. Finish: Polyester powder coated
        5.3.3. Colour: Standard RAL colour to be confirmed.
    5.4. Internal facing material: Zinc-coated steel to BS EN 10346
        5.4.1. Profile: Flat
        5.4.2. Finish: Polyester powder coated
        5.4.3. Colour: White.
    5.5. Core insulation: Non-combustible core insulation to suit composite panel construction, fully bonded to inner and outer sheets and to achieve the performance requirements
    5.6. System to be hermetically sealed to prevent the ingress of moisture, to avoid delamination.
6. Joint type
    6.1. Type: Lapped, interlocking, watertight joints consistent in size
    6.2. Side/ vertical: Butt joints with cover flashing
    6.3. End/ horizontal: To include inconspicuous factory applied black gasket extrusions/ weather seals.
7. Primary fasteners: Concealed or inconspicuous fixings. Exposed fixings to be colour co-ordinated with the cladding panels and uniformly configured.
8. Flashings, trims, drips, cappings and other such elements: Visually to match the cladding.
9. Fire, smoke and acoustic stopping: As required to be concealed in the finished works and to meet the current requirements of the Building Regulations.

## General requirements

### 165 Contractor's design

1. Description: Refer to A Section.
2. Design responsibility: Determine depth and thickness of profile and type, sizes and number of fixings to suit the layout and details of supporting steelwork.
3. Design standard: In accordance with BS 5427.
4. Product specification and requirements: To BS EN 14509.
5. Structural and fire requirements
    5.1. Generally: Refer to the Fire Strategy Report and Refer to the Structural Engineer's Information

5.2. Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.

6. Functional requirements: As specified in this section and Section A.
7. Design and production information: As Preliminaries section A31
8. Timing of submissions: As Preliminaries section A31

## 167 Completion of design

1. Requirement: Complete the detailed design to satisfy specified performance criteria and coordinate with the detailed design of related and adjacent work.
   1.1. Design standard: In accordance with BS 5427.
   1.2. Product specification and requirements: To BS EN 14509.
2. Structural requirements: Refer to the Structural Engineer's Information
3. Design and production information: As Preliminaries section A31
4. Timing of submissions: As Preliminaries section A31

## 172 Thermal bridging

1. Requirement:
   1.1. Standard and process, in accordance with: BS EN ISO 6946 and BS EN ISO 10211

## 175 Product samples

1. General:: During initial design, submit labelled samples of the panel finish and colour.

## 176 Fastener samples

1. General: During detailed design, submit labelled samples of each type of fastener. Retain on site as a reference standard until instructed to dispose of.

# Design/ performance requirements

## 185 Performance compliance

1. Verification: Before commencing fabrication, submit evidence based on laboratory testing or computer modelling.

## 187 Deflection of metal cladding/ roofing

1. Standard: Calculation or test in accordance with BS EN 14509.
2. Roofing: Maximum permitted deflection under distributed loads as a multiple of span and taking account of:
   2.1. Permanent load: Refer to the Structural Engineer's Information
   2.2. Permanent load plus imposed roof load or undrifted snow load: Refer to the Structural Engineer's Information
   2.3. Permanent and wind loads: Refer to the Structural Engineer's Information
3. Wall cladding: Maximum permitted deflection under distributed loads as a multiple of span and due to:
   3.1. Wind load: Refer to the Structural Engineer's Information

## 192 Sound transmittance of cladding/ roofing system – weighted

1. Requirement:
    1.1. Minimum weighted sound reduction index (Rw): Refer to the Acoustic Strategy Report
    1.2. Minimum standardized level difference (DnTw): Refer to the Acoustic Strategy Report

## 193 Sound transmittance of cladding/ roofing system – frequency-specific

1. Requirement:
    1.1. Minimum sound reduction index (R): Refer to the Acoustic Strategy Report
        1.1.1. For one-third-octave band centre frequency (Hz) of: Refer to the Acoustic Strategy Report

## 194 Internal sound absorption of cladding/ roofing system

1. Requirement: Measure to BS EN ISO 354.
    1.1. Minimum sound absorption coefficient (alpha S): Refer to the Acoustic Strategy Report
        1.1.1. For one third octave band centre frequency (Hz) of: Refer to the Acoustic Strategy Report

## 198 Water penetration

1. Requirement: Under site exposure conditions, moisture must not penetrate onto internal surfaces, or into cavities not designed to be wetted.

## 200 Avoidance of interstitial condensation

1. Requirement: Determine interstitial condensation risk of cladding system using the method described in BS 5250. If necessary, provide a vapour control layer and/ or revise thermal insulation to ensure that damage and nuisance from interstitial condensation does not occur.
2. Outdoor psychrometric conditions (notional):
    2.1. Temperature: To be confirmed
    2.2. Relative humidity: To be confirmed
    2.3. Vapour pressure: To be confirmed
3. Indoor psychrometric conditions (notional):
    3.1. Temperature: To be confirmed
    3.2. Relative humidity: To be confirmed
    3.3. Vapour pressure: To be confirmed

## 202 Avoidance of surface condensation

1. Requirement: Determine surface condensation risk of cladding system using the method described in BS EN ISO 13788. If necessary, revise thermal insulation to provide satisfactory temperature factor (fmin). Ensure that damage and nuisance from surface condensation does not occur.

# Fixing cladding/ roofing

## 215 Painting structure

1. Sequence: Paint outer surface of supporting structure before fixing cladding/ roofing.

## 219 Fasteners

1. Unspecified fasteners: Where not specified individually, all fasteners should be as recommended for the purpose by the cladding/ roofing manufacturer or installer

## 221 Fittings and accessories

1. Unspecified fittings and accessories: Where not specified individually, all fittings and accessories should be as recommended for the purpose by the cladding/ roofing manufacturer or installer.

## 223 Prevention of electrolytic action

1. Isolating tape: As per the manufacturers recommendations

## 234 Gutters

1. Manufacturer: Submit proposals.
    1.1. Product reference: Submit proposals.
2. Sizes: Refer to Design Drawings.
3. Material: Double-skin GRP
    3.1. Gauge/ Thickness: Refer to Design Drawings.
    3.2. External finish: Bituminous coating
4. Insulation: As recommended by manufacturer
5. Internal liner sheet: As recommended by manufacturer
6. Jointing method: As recommended by manufacturer
7. Fixing method: As recommended by manufacturer
8. Accessories: As recommended by manufacturer

## 300 Profile fillers generally

1. Material: To be confirmed
2. Manufacturer: Submit proposals
    2.1. Product references: Submit proposals
3. Colour: White
4. Thickness: As recommended by manufacturer.
5. Fixing method: As recommended by manufacturer
    5.1. Requirement: To close cavities within the external envelope. Tight fit with no unintended gaps.

## 410 Fixing panels and sheets generally

1. Cut edges: Not permitted
2. Penetrations: Openings to minimum size necessary.
    2.1. Edge reinforcement: Angles
3. Orientation: Exposed joints of side laps away from prevailing wind unless shown otherwise on drawings.
4. Panel and sheet ends, laps and raking cut edges: Fully supported and with fixings at top of lap.
5. Fasteners: Drill holes. Position at regular intervals in straight lines, centred on support bearings.
    5.1. Position of fasteners in oversized drilled holes: Central.

5.2. Fasteners' torque: Sufficient to correctly compress washers.

6. Debris: Remove dust and other foreign matter before finally fixing panel and sheets.
7. Completion: Check fixings to ensure watertightness and that panels and sheets are secure.
8. Cut edges: Paint to match face finish.

## 419 Fixing plastics cladding/ roofing

1. Crown fixing: For panels/ sheets with a profile depth cavity greater than 20 mm, support crowns at primary fasteners with profile fillers.
2. Fastener holes: Sized in accordance with sheet manufacturer's recommendations.
3. End laps between plastics sheets: Use two strips of sealant tape, one along each edge of lap.

## 470 Structural movement joints

1. Type: Cover flashing fixed on one side over gap between panels.
2. Width of gap: To match structural movement joint requirements.
3. Requirement: Weathertight.

## 480 Flashings/ trims generally

1. Lap joint treatment
   1.1. Vertical and sloping flashings/ trims: End laps to be same as for adjacent panels.
   1.2. Horizontal flashings/ trims: End laps to be 150 mm, sealed and where possible arranged with laps away from prevailing wind.
2. Method of fixing: To structure in conjunction with adjacent panels. Otherwise to panels.
   2.1. Fasteners: Sealed aluminium rivets with adequately sized diameter head and plastics cap at centres recommended by the manufacturer.

## 481 Flashings/ trims

1. Fasteners: Sealed aluminium rivets with 16 mm-diameter head and blue plastics cap at 300 mm centres.

## 482 Butt-jointed flashings/ trims

1. Butt straps: 300 mm wide and made from sheet of same material and finish.
2. Butt joints: Seal

## 540 Abutments

1. Junctions with flashings: Weathertight and neatly dressed down.

## 550 Sealing external laps

1. Sealant tape: Types recommended by panel/ sheet manufacturer.
2. Position of tape: Below fixing positions in straight unbroken lines, parallel to and slightly back from edge of panel/ sheet.
3. Seal quality: Effective, continuous and not overcompressed.
4. End laps: Sealant tape positions:
   4.1. Single line tape: 10–15 mm below line of fasteners.
   4.2. Second line tape: 10–15 mm above line of fasteners.

4.3. Third line tape (silicone, where specified): 15 mm set back from the edge of external sheet.

5. Side laps: Sealant tape positions:

5.1. Single line tape: Outside line of fasteners.

5.2. Second line tape (where specified): On other side of fasteners.

## 670 Documentation

1. Certificates, records, guarantees and other documents: During initial design development, provide evidence that the insulating sandwich panel is manufactured to BS EN 14509.

Ω End of Section



make

# H51
# Natural Stone Slab Cladding/ Lining/ Features

## To be read with preliminaries/ general conditions.

### 5 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 10 Information to be provided with tender

1. General structural requirements in accordance with clause: Refer to the Structural Engineer's Information
2. Submit the following cladding particulars
   - 2.1. Typical plan, section and elevation drawings at suitable scales.
   - 2.2. Typical detailed drawings at large scales.
   - 2.3. Technical information and certification demonstrating compliance with specification of proposed incorporated products and finishes.
   - 2.4. Certification, reports and calculations demonstrating compliance with specification of proposed cladding.
   - 2.5. Proposals for connections to and support from the support structure/ background.
   - 2.6. Proposals for additional support structure/ background to that shown on preliminary design drawings.
   - 2.7. Schedule of builders' work, special provisions and special attendance by others.
   - 2.8. Examples of standard documentation from which project quality plan will be prepared.
   - 2.9. Prepare and submit a 'sampling plan' in accordance with BS 8298-2.
   - 2.10. Preliminary fabrication and installation method statements and programme.
   - 2.11. Proposals for replacing damaged or failed products.
   - 2.12. Areas of non-compliance with specification.

## Types of cladding

### 110 EWS-05 Rebuilt Historic Façade (Visual Intent Only)

1. Description: Rebuilt historic façade, as the Design Drawings.
2. Note:
   - 2.1. This description covers items specified by the Façade Consultant.
   - 2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Performance requirements: Refer to the Façade Performance Specification
5. Support structure: Refer to the Façade Performance Specification
6. Rebuilt façade:
   - 6.1. Type/ material: Refer to the Façade Performance Specification
   - 6.2. Finish/ colour: Refer to the Façade Performance Specification

7. Stone quality: Refer to the Façade Performance Specification

## 450 ESS-101 Portland Stone Soffit Panel System (Visual Intent Only)

1. Description: Portland stone soffit panel system, as the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings
5. Specific performance: Refer to the Façade Performance Specification
6. Colour/ finish/ sampling: Refer to the Façade Performance Specification
7. Fire, smoke and acoustic stopping: Refer to the Façade Performance Specification

# General requirements/ preparatory work

## 205 Contractor's design

1. Design responsibility: Determine sizes and thickness of units and type size, number and spacing of fixings to suit the secondary support system shown on the Design Drawings.
2. Structural and fire requirements
   2.1. Generally: Refer to the Fire Strategy Report and Refer to the Structural Engineer's Information
3. Performance requirements: In accordance with BS 8298-2.
4. Design requirements: As clause 110 and in accordance with clause 325.
5. Functional requirements: As specified in this section and Section A and Prelims, with fire stopping to the requirements of the Building Regulations.
6. Additional requirements: Responsibility and coordination at design interfaces and agreement on location of service openings.

## 207 Completion of design

1. Design responsibility: Prepare detail design, fabrication and installation details in accordance with the performance and design requirements. Coordinate with the detailed design of related and adjacent work
2. Performance requirement: In accordance with BS 8298-2
3. Structural requirements: Refer to the Structural Engineer's Information

## 245 Information to be provided during detailed design

1. Submit the following cladding particulars:
   - A schedule of detailed drawings and dates for submission for comment
   - A schedule of loads that will be transmitted from cladding to the support structure/ background
   - Proposed fixing details and systems relevant to structural design and construction with methods of adjustment and tolerances
   - Number, type and location of fixings for each slab
   - A schedule of fabrication tolerances/ size tolerances
   - A detailed testing programme in compliance with Main Contract Master programme

- A detailed fabrication and installation programme in compliance with Main Contract Master programme
- Proposals to support outstanding applications for Building Regulations consents or relaxations

2. Timing of submissions: Refer to Section A33.

## 247 Quality plan

1. Requirement: Submit during detailed design.
2. Content: In accordance with BS EN ISO 9001 and including the following: : To be confirmed

## 261 Indicative stone samples

1. Standard: In accordance with BS 8298-2
2. General: Before commencing detailed design, submit samples labelled in accordance with BS EN 12440 and BS EN 12670
3. Validation: Carry out supplementary initial testing under the supervision of a stone specialist to establish the suitability of the stone in the specific environmental conditions of the project site

## 262 Reference stone samples

1. Standard: In accordance with BS 8298-2
2. General: When the preferred stone is selected, record samples (representing the agreed stone type, source, colour, appearance and characteristics) should be retained by the manufacturer with a copy on site
3. Validation: Stone to be tested and carry a valid Declaration of Performance, in accordance with the requirements of BS EN 1469

## 263 Production control stone samples

1. Standard: In accordance with BS 8298-2
2. General: During the extraction, manufacture and installation of the stone production control samples, at defined intervals, are compared with the reference stone to ensure compliance. Notify CA of any variation from the reference sample.
3. Validation: Stone to be tested and carry a valid Declaration of Performance, in accordance with the requirements of BS EN 1469.

## 271 Fixings samples

1. Standard: In accordance with BS 8298-2
2. General: During detailed design, submit samples of every type. Clearly mark fixing to identify type and typical location. Include manufacturer's recommended torque figures.
3. Shims: Ensure that shim does not impact the embedment of the anchor and is of sufficient length to avoid introducing a turning moment on the connection.
4. Validation: Submit manufacturer's test data for the performance and durability of the article.

## 281 Mock-up sample panel/ preliminary panel installation

1. General: Complete areas of finished work to include all stone types, mortar joints, cornice and an external corner, and obtain approval of appearance before proceeding
2. Size: To be confirmed
3. Location: To be confirmed

4. Completion: Preliminary installation to be incorporated into the works

## Design/ performance requirements

### 325 Design requirements

1. Design standard: In accordance with BS 8298-2
2. Stone type:
   2.1. Denomination to BS EN 12440: Refer to the Design Drawings
   2.2. Name (traditional): Refer to the Design Drawings
3. Performance characteristics
   3.1. Geometric: Measured in accordance with BS EN 13373
   3.2. Tolerance on nominal thickness: To BS EN 1469, Table 1
   3.3. Flatness deviation: To BS EN 1469, clause 4.1.3
   3.4. Length, width and squareness tolerance: To BS EN 1469, Table 8
   3.5. Special shapes deviation: To BS EN 1469, clause 4.1.3
   3.6. Location of dowel holes: As Indicated on the Design Drawings and tolerances to BS EN 1469, clause 4.1.6
   3.7. Surface finish: As Indicated on the Design Drawings and to BS EN 1469, clause 4.1.8
4. Physical and mechanical characteristics
   4.1. Visual appearance: To BS EN 1469, clause 4.2.3
   4.2. Colour: Refer to the Design Drawings
   4.3. Veining: Refer to the Design Drawings
   4.4. Texture: Refer to the Design Drawings
   4.5. Flexural strength: To BS EN 12372
   4.6. Resistance to fixings: To BS EN 13364
   4.7. Water absorption at atmospheric pressure: To BS EN 13755
   4.8. Reaction to fire: To BS EN 13501-1, Class A1
   4.9. Water absorption by capillarity: To BS EN 1925
   4.10. Durability of flexural strength against freeze thaw: To BS EN 12371
   4.11. Resistance to thermal shock: To BS EN 14066
   4.12. Resistance to thermal and moisture cycling: To BS EN 16306
   4.13. Water vapour permeability: To BS EN ISO 12572 and/ or BS EN ISO 10456
   4.14. Direct airborne sound insulation: To BS EN 1936
   4.15. Thermal conductivity: To BS EN 1925
5. Impact resistance: In accordance with CWCT TN 75 and TN 76

### 326 Additional preliminary test information

1. Stone type:
   1.1. Denomination to BS EN 12440: Refer to the Design Drawings
   1.2. Name (traditional): Refer to the Design Drawings
   1.3. Petrographic name to BS EN 12407: Refer to the Design Drawings
2. Test and result: To BS EN 1469

### 330 Accuracy of erection

1. Elevation joints
   1.1. Widths: Within joint lengths, including in-line continuations across transverse joints, as follows.
   1.2. Tolerance: Greatest width not to exceed least width by more than 10%
   1.3. Variations: Evenly distribute, with no sudden changes.
2. Offset in elevation: Between nominally in-line edges across transverse joints not to exceed to be confirmed % width of joint.
3. Offset in plan or section: Between flat faces or adjacent panels across joints not to exceed to be confirmed % width of joint.
4. Sealant joints width limitations: To recommendations of sealant manufacturer.
5. Finished work: Square, regular, true to line and plane with satisfactory fit at junctions.

## Testing

### 415 Testing

1. Carried out by: A United Kingdom Accreditation Service (UKAS)-approved independent laboratory
2. Witnessed/ certified by: The contract administrator

### 425 Stone production tests

1. Frequency of tests: In accordance with BS 8298-2
2. Procedures: To BS EN 1925, BS EN 1936 and BS EN 12372.
3. Results: Submit in accordance with the approved sample plan

### 435 Site-testing of fixings to determine ultimate load

1. Number and location of test fixings: To be confirmed
2. Test method: To BS 8539 and Construction Fixings Association guidance note 'Procedure for site testing construction fixings'

### 445 Site-testing of fixings during installation

1. Number and location of test fixings: To be confirmed
2. Test method: To BS 8539 and Construction Fixings Association guidance note 'Procedure for site testing construction fixings'

## Fabrication and installation

### 510 Generally

1. Location of joints: Refer to the Design Drawings
2. Electrolytic corrosion: Isolate dissimilar metals.
3. Prefabrication: Machine cut and drill products in workshop wherever possible.
4. Identification: Mark or tag products. Do not mark surfaces visible in the complete installation.
5. Natural bed: Indicate on a non-exposed surface of each stone.
6. Cleanliness: Keep facework clean

7. Protection: Isolate and protect each stone with packers or protective coverings for transportation and in construction. Provide temporary protection over whole installation. Ensure that all arises are protective for as long as possible, with material which will not mark or stain the stone.

## 520 Cutting of stone – standard deviations

1. Standard: To BS 8298-2 and BS EN 1469 for production generally, including permissible deviations
2. Bedding: As per the manufacturers recommendations.

## 525 Cutting of stone – non-standard deviations

1. Standard: To BS 8298-2 for production generally
2. Bedding: As per the manufacturers recommendations.
3. Permissible deviations
    3.1. Height (+/- mm): As per the manufacturers recommendations.
    3.2. Width (+/- mm): As per the manufacturers recommendations.
    3.3. Squareness (+/- mm): As per the manufacturers recommendations.
    3.4. Dowel location along length or width (+/- mm): As per the manufacturers recommendations.
    3.5. Dowel location along thickness (+/- mm): As per the manufacturers recommendations.
    3.6. Dowel diameter (+/- mm): As per the manufacturers recommendations.
    3.7. Dowel depth (+/- mm): As per the manufacturers recommendations.
    3.8. Slot width (+/- mm): As per the manufacturers recommendations.
    3.9. Slot position (+/- mm): As per the manufacturers recommendations.
    3.10. Hole position on face (+/- mm): As per the manufacturers recommendations.
    3.11. Undercut anchor on rear (+/- mm): As per the manufacturers recommendations.
    3.12. Slot deviation: As per the manufacturers recommendations.
4. Leave selected stone units oversize, to accommodate deviations within building structure. Cut to precise dimensions taken on site.

## 530 Inspection of stone units

1. Give notice: At appropriate stages of production

## 540 Suitability of structure: contractor's survey

1. Programme: Not less than two weeks before commencement of cladding installation.
2. Scope: Geometric survey of supporting structure, checking line, level and fixing points.
3. Coordinate: With surveys for adjacent cladding.
4. Give notice: If the structure will not allow the required accuracy or security of erection.
5. Setting out: Establish erection datum points, lines and levels for a complete elevation at a time unless otherwise agreed.

## 550 Installation of interfaces

1. General: Locate flashings, closers, etc. correctly with neat overlaps to cladding to form weatherproof junctions.

## 560 Metalwork

1. Material standards and fabrication: As section Z11

## 570 Welding

1. In situ welding: Not permitted

## 580 Fixing

1. Torque figures and shim dimensions: Do not exceed fixing manufacturer's recommendations.
2. Grouting: Secure fixings in place in cladding and support structure/ background with cement:sand, epoxy or modified polymer mix, as recommended by the stone supplier.
3. External cladding: Do not use mortar spacer dabs. Keep cavity clear of debris.
4. Give notice: Before proceeding with next course on completion of every third stone course

## 585 Cement:lime:sand mortar jointing

1. General: As section Z21
2. Mix: Cement:lime:sand 1:1:5-6.
3. Preparation: Wet stones thoroughly.
4. Laying: Full mortar bed with joints and voids filled.
5. Cavities: Clear of mortar
6. Appearance: Neat and consistent.
7. Temporary distance spacers: Remove

## 590 Cement:lime:crushed stone mortar jointing

1. General: As section Z21
2. Mix: Cement:lime:crushed stone 1:2:8–9.
3. Crushed stone: Colour-matched and passing a 1.4 mm sieve.
   3.1. Colour matching: Submit samples
4. Preparation: Wet stones thoroughly.
5. Laying: Full mortar bed with joints and voids filled.
   5.1. Cavities: Clear of mortar
6. Appearance: Neat and consistent
7. Temporary distance spacers: Remove

## 595 Cement:sand/ cement:lime:sand/ cement:lime:crushed stone mortar jointing

1. General: As section Z21
2. Mix: 1:4
3. Preparation: Wet stones thoroughly.
4. Laying: Full mortar bed with joints and voids filled.
   4.1. Cavities: Clear of mortar
5. Appearance: Neat and consistent.
6. Temporary distance spacers: Remove

## 610 Sealant joints

1. Sealant: Silicone
   1.1. Standard: BS EN 15651-1
   1.2. Manufacturer: Submit proposals
   1.3. Product reference: Submit proposals
   1.4. Colour: To be confirmed
2. Application: As section Z22

## 630 Movement joints

1. Sealant: Silicone
   1.1. Standard: BS EN 15651-1
   1.2. Manufacturer: Submit proposals
   1.3. Product reference: Submit proposals
   1.4. Colour: To be confirmed
2. Application: As section Z22

Ω End of Section



make

# H62
# Natural Slate Roofing

## Types of slating

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 105 EWS-06 Slate Mansard With Dormers (Visual Intent Only)

1. Description: Slate tile mansard system with dormers configured as indicated on the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Substrate: Refer to the Façade Performance Specification
5. Pitch: Refer to the Façade Performance Specification
6. Underlay: Refer to the Façade Performance Specification
7. Counter battens/ battens/ clips: Refer to the Façade Performance Specification
8. Slates
    8.1. Type/ size: Refer to the Façade Performance Specification
    8.2. Colour: Refer to the Façade Performance Specification

## Performance - Not Used

## Slating generally

### 210 Basic workmanship

1. General: Fix slating and accessories to make the whole sound and weathertight at earliest opportunity.
2. Setting out: To true lines and regular appearance, with neat fit at edges, junctions and features.
3. Fixings for slating accessories: As recommended by manufacturer.
4. Gutters and pipes: Keep free of debris. Clean out at completion.

### 235 Vapour-permeable underlay

1. Standard: To BS EN 13859
    1.1. Reaction to fire: Manufacturer's standard
    1.2. Water vapour transmission (minimum): Manufacturer's standard
    1.3. Resistance to water penetration: Manufacturer's standard
    1.4. Tensile strength (minimum): Manufacturer's standard
    1.5. Tear resistance (minimum): Manufacturer's standard

1.6. Other BS EN 13859 characteristics: None

2. Weight (mass per unit area): 110 g/m²
3. Resistance to wind uplift: Greater than 1900 N/m²

## 240 Underlay

1. Handling: Do not tear or puncture.
2. Laying: Maintain consistent tautness.
3. Vertical laps (minimum): 100 mm wide, coinciding with supports and securely fixed.
4. Fixing: Galvanized steel, copper or aluminium 20 x 3 mm extra-large clout head nails.
5. Eaves: Where exposed, use an external grade (UV resistant) underlay or a proprietary eaves support product.
6. Penetrations: Use proprietary underlay seals or cut underlay to give a watertight fit around pipes and components.
7. Ventilation paths: Do not obstruct.

## 245 Battens/ counter battens

1. Timber: Sawn softwood.
   1.1. Species: In accordance with BS 5534, clause 4.11.1.
   1.2. Permissible characteristics and defects: Not to exceed limits in BS 5534, Annex D.
   1.3. Grading: Contractor's choice
   1.4. Moisture content at time of fixing and covering (maximum): 22%.
2. Preservative treatment: Contractor's choice

## 255 Counter battens on rigid sarking

1. Fixing: Through rigid sarking into rafters at not more than 300 mm centres.

## 257 Counter battens on structural metal liner trays

1. Fixing: To each tray upstand at not more than 600 mm centres.

## 259 Counter battens on rafters

1. Fixing: Into rafters at not more than 300 mm centres.

## 265 Batten fixing

1. Setting out: Align parallel to ridge in straight horizontal lines to gauge of slates. Align on adjacent areas.
2. Batten length (minimum): Sufficient to span over three supports.
3. Joints in length: Square cut. Butt centrally on supports. Joints must not occur more than once in any group of four battens on one support.
4. Additional battens: Provide where unsupported laps in underlay occur between battens.
5. Fixing: Each batten to each support. Splay fix at joints in length.

## 270 Battens fixed to masonry

1. Setting out: In straight horizontal lines. Align on adjacent areas.
2. Batten length (minimum): 3 m.

3. Fixing centres (maximum): 400 mm.

### 272 Timber for slating substrate work

1. Timber: Sawn softwood, free from wane, pitch pockets, decay and insect attack (ambrosia beetle excepted).
   1.1. Moisture content at time of fixing and covering (maximum): 22%.
2. Preservative treatment: As section Z12 and Wood Protection Association Commodity Specification C8.
   2.1. Type: Contractor's choice

### 275 Slate fixing

1. Setting out: Lay slates with an even overall appearance with slightly open (maximum 5 mm) butt joints. Align tails.
2. Slate thickness: Consistent in any one course. Lay with thicker end as tail.
3. Ends of courses: Use extra wide slates to maintain bond and to ensure that cut slates are as large as possible. Do not use slates less than 150 mm wide.
4. Top course: Head-nail short course to maintain gauge.
5. Fixing: Centre nail each slate twice through countersunk holes 20-25 mm from side edges.
   5.1. Nails: Copper clout to BS 1202-2 or aluminium clout to BS 1202-3.
   5.2. Nail dimensions: Determine in accordance with BS 5534 to suit site exposure, withdrawal resistance and slate supplier's recommendations.

### 290 Mortar bedding/ Pointing

1. Mortar: As section Z21, 1:3 cement:sand, with plasticizing admixtures permitted.
   1.1. Bond strength providing resistance to uplift: In accordance with BS 5534.
2. Weather: Do not use in wet or frosty conditions or when imminent.
3. Preparation of concrete and clay tile accessories to be bedded: Wet and drain surface water before fixing.
4. Appearance: Finish neatly as work proceeds and remove residue.

## Roof slating edges/ junctions/ features

### 305 Generally

1. Fittings and accessories: As recommended by slate supplier, do not improvise.
   1.1. Exposed fittings and accessories: To match slate colour and finish.
2. Cut slates: Cut only where necessary, to give straight, clean edges.
3. Flashings: Fix with or immediately after slating. Form neatly.

### 325 Fire-separating walls

1. Separating walls: Completely fill space between top of wall and underside of slates with mineral wool quilt to provide fire-stopping.
2. Boxed eaves: Completely seal air paths in plane of separating wall with wire-reinforced mineral wool, not less than 50 mm thick, fixed to rafters and carefully cut to shape fire-resisting board and quilt to provide fire-stopping.

## 345 Ventilated eaves with integrated grilles tray

1. Fascia grilles and ventilator trays
    1.1. Fix to carry underlay, form drip into gutter and provide free passage of air over insulation.
2. Undercourse and first course slates: Fix with tails projecting 50 mm over gutter or to centre of gutter, whichever dimension is the lesser.

## 355 Ventilated eaves with separated grilles/ trays

1. Fascia grilles
    1.1. Manufacturer: Contractors choice.
2. Product reference: Contractors choice.
3. Ventilator trays
    3.1. Manufacturer: Contractors choice.
        3.1.1. Product reference: Contractors choice.
    3.2. Fix to provide free passage of air over insulation.
4. Underlay support: Contractor's choice
    4.1. Continuous to prevent water retaining troughs.
5. Gutter: Dress underlay or underlay support tray to form drip into gutter.
6. Undercourse and first course slates: Fix with tails projecting 50 mm over gutter or to centre of gutter, whichever dimension is the lesser.

## 365 Unventilated eaves

1. Underlay support: Contractor's choice
    1.1. Continuous to prevent water retaining troughs.
2. Gutter: Dress underlay or underlay support tray to form drip into gutter.
3. Undercourse and first course slates: Fix with tails projecting 50 mm over gutter or to centre of gutter, whichever dimension is the lesser.

## 445 Mortar-bedded verges with bedded undercloak

1. Underlay: Carry 50 mm onto outer leaf of gable wall and bed on mortar.
2. Undercloak: Slates.
    2.1. Position: Over underlay, level with underside of slating battens, sloping towards verge.
    2.2. Projection beyond face of wall: 38-50 mm.
    2.3. Bedding: On mortar identical to that used in gable walling.
3. Slating battens: Carry onto undercloak and finish 100 mm from verge edge.
4. Verge slates
    4.1. Bedding: Flush with undercloak on 75 mm wide bed of mortar.
    4.2. Pointing: Flush profile.
    4.3. Fixing: Do not displace or crack mortar.

## 455 Mortar-bedded verges with nailed undercloak

1. Underlay: Carry over full width of verge.
2. Undercloak: Slates.
    2.1. Position: Over underlay, level with underside of slating battens, sloping towards verge.

2.2. Projection: 38-50 mm beyond face of Barge board or Gable slating.

2.3. Fixing: Nails.

3. Slating battens: Carry onto undercloak and finish 100 mm from verge edge.

4. Verge slates

4.1. Bedding: Flush with undercloak on 75 mm wide bed of mortar.

4.2. Pointing: Flush profile.

4.3. Fixing: Do not displace or crack mortar.

## 525 Mitred hips

1. Underlay: Lay courses over hip.

1.1. Overlaps (minimum): 150 mm.

2. Hip slate fixing battens: Contractor's choice

3. Mitred slates: Cut extra wide slates and fix to form a straight, close mitred junction.

4. Soakers: Interleave with mitred slates. Fix by turning down over head of mitred slates.

## 535 Lead roll hips

1. Underlay: Lay courses over hip.

1.1. Overlaps (minimum): 150 mm.

2. Hip slates: Cut close to timber roll and fix to form a straight junction.

## 555 Mortar-bedded and mechanically fixed tile hips

1. Underlay: Lay courses over hip.

1.1. Overlaps (minimum): 150 mm.

2. Hip tile fixing battens: Contractor's choice

3. Roof slates: Cut and fix closely at hip.

4. Hip irons: Galvanized steel in accordance with BS 5534, clause 4.15.4.

4.1. Fixing: To hip rafter or hip batten with not less than two zinc-coated steel screws.

5. Hip tiles

5.1. Manufacturer: Contractor's choice

5.1.1. Product reference: Contractor's choice

5.2. Bedding: On mortar, continuous to edges and solid to joints.

5.3. Fixing: Secure all hip tiles to hip rafters or hip tile fixing battens with self-sealing non-ferrous through fixings.

5.4. Bottom hip tiles: Shape neatly to align with corner of eaves and fill ends with mortar and slips of tile finished flush.

## 615 Metal valleys

1. Underlay: Cut to rake. Dress over tilting fillets to lap onto metal valley. Do not lay under metal.

2. Roof slates: Cut extra wide slates adjacent to valley to fit neatly.

2.1. Valley width between slates: Refer to the Design Drawings

## 635 Mitred valleys

1. Underlay: Lay strips not less than 600 mm wide centred on valleys. Overlap with general roof underlay.
2. Mitred slates: Cut extra wide slates and fix to form a straight, close mitred junction.
3. Soakers: Interleave with mitred slates. Fix by turning down over head of mitred slates.

## 645 Laced valleys

1. Preparatory work: Lay general roof underlay and slating battens into valleys.
2. Valley boarding: Timber 25 mm thick.
    2.1. Centre board: Minimum 250 mm wide.
    2.2. Side boards: 100 mm wide and feather-edged.
3. Underlay: Lay strips not less than 600 mm wide centred on valleys.
4. Valley slates: Form intersections of slate courses by laying square slates diagonally on centre of valley boarding. Sweep roof slate courses neatly upwards to meet the square slates at right angles, maintaining specified lap. Fix to battens or direct to valley boarding.

## 650 Swept valleys

1. Preparatory work: Lay general roof underlay and slating battens into valleys.
2. Valley boarding: Timber 25 mm thick.
    2.1. Centre board: Minimum 250 mm wide.
    2.2. Side boards: 100 mm wide and feather-edged.
3. Underlay: Lay strips not less than 600 mm wide centred on valleys.
4. Valley slates: Cut extra wide and/ or extra long slates to maintain bond and laps, and to give a smooth sweep through valleys. Fix to battens or direct to valley boarding.

## 660 Side abutments

1. Underlay: Turn up not less than 100 mm at abutments.
2. Abutment slates: Cut as necessary. Fix close to abutments.
3. Soakers: Interleave with abutment slates. Fix by turning down over head of abutment slates.

## 670 Top edge abutments

1. Underlay: Turn up not less than 100 mm at abutments.
2. Top slate courses: Fix close to abutments.

## 675 Top edge ventilated abutments

1. Underlay: Provide air gap at abutment as recommended by ventilator manufacturer.
2. Abutment ventilator
    2.1. Manufacturer: Contractor's choice
        2.1.1. Product reference: Contractor's choice
    2.2. Fixing: As recommended by manufacturer.

## 690 Roof windows

1. Underlay: Turn up not less than 100 mm at window surrounds under integral flashings/ soakers.
2. Roof slates: Cut as necessary and fix closely all round.

## 700 Dry ventilated tile ridges

1. Underlay: Provide air gap at apex as recommended by ridge tile manufacturer.
2. Dry ridge fixing battens: Contractor's choice
3. Dry ridge tiles
    3.1. Manufacturer: Contractor's choice
        3.1.1. Product reference: Contractor's choice
4. Ridge terminals
    4.1. Manufacturer: Contractor's choice
        4.1.1. Product reference: Contractor's choice

## 730 Lead roll ridges

1. Underlay: Lay courses over ridge.
    1.1. Overlap (minimum): 150 mm.
2. Top slate courses: Fit close to timber roll.

## 740 Mortar-bedded and mechanically fixed tile ridges

1. Underlay: Lay courses over ridge.
    1.1. Overlap (minimum): 150 mm.
2. Ridge tile fixing battens: Contractor's choice
3. Ridge tiles
    3.1. Manufacturer: Contractor's choice
        3.1.1. Product reference: Contractor's choice
    3.2. Bedding: On mortar, continuous to edges and solid to joints.
    3.3. Fixing: Secure all ridge tiles to ridge boards or supplementary ridge battens with self-sealing non-ferrous fixings.
    3.4. Gable end ridge tiles: Fill ends with mortar and slips of tiles finished flush.
4. Ridge terminals
    4.1. Manufacturer: Contractor's choice
        4.1.1. Product reference: Contractor's choice

## 770 Dry ventilated tile mono-ridges

1. Underlay: Provide air gap at apex.
2. Dry mono-ridge tiles
    2.1. Manufacturer: Contractor's choice
        2.1.1. Product reference: Contractor's choice

## 780 Mortar-bedded and mechanically fixed tile mono-ridges

1. Underlay: Lay not less than 100 mm over mono-ridge.
2. Mono-ridge tiles
    2.1. Manufacturer: Contractor's choice
        2.1.1. Product reference: Contractor's choice
    2.2. Bedding: On mortar, continuous to sloping edges and solid to joints.

2.3. Fixing: Secure vertical faces to ridge fixing battens with self-sealing non-ferrous fixings.

2.4. Gable end mono-ridge tiles: Fill ends with mortar finished flush.

## 800 Change of roof pitch

1. Slating substrate work: Fix timber tilting fillet to support flashing and courses of slates above change of pitch.
2. Roof slates: Make weathertight with flashing.
   2.1. Courses above change of pitch: Fix undercourse and first course slates.
   2.2. Courses below change of pitch: Cut top two courses as necessary to maintain gauge. Head-nail top course.

## 840 Ventilator slates

1. Ventilator slates
   1.1. Manufacturer: Contractor's choice
      1.1.1. Product reference: Contractor's choice
   1.2. Positions: Refer to the Design Drawings

## 850 Roof slope terminals

1. Ventilator slates:
   1.1. Manufacturer: Contractor's choice
      1.1.1. Product reference: Contractor's choice

# Vertical slating edges/ junctions

## 910 Bottom edges

1. Slating substrate work: Fix timber tilting fillet to support bottom course of slates in correct vertical plane. Fix flashing to tilting fillet.
2. Underlay: Dress over flashing.
3. Undercourse and bottom course slates: Fix with tails neatly aligned.

## 920 Top edges

1. Top slate courses: Fix under abutment and make weathertight with flashings dressed down not less than 150 mm.

## 930 Side abutments

1. Slating substrate work: Chase abutment wall and insert stepped flashing.
   1.1. Flashing: Return not less than 75 mm behind slating, overlapping underlay and battens. Turn back to form a vertical welt.
2. Abutment slates: Cut and fix neatly.

## 950 Angles with soakers

1. Angle slates: Cut extra wide slates and fix to form a straight, close mitred junction.
2. Soakers: Interleave with angle slates. Fix by nailing to battens at top edge.

## 960 Junction with roof verges

1. Slating substrate work: Fix additional slating batten parallel to and below verges.
2. Course end slates: Splay cut slate and a half width slates to angle of verge rake. Fix to additional slating batten with cut edge parallel to and below verge.

Ω End of Section

# H92
# Rainscreen Cladding

## Tendering

### 1 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 10 Information to be provided with tender

1. Submit the following cladding particulars
    1.1. Typical plan, section and elevation drawings at suitable scales.
    1.2. Typical detailed drawings at large scales.
    1.3. Technical information and certification demonstrating compliance with specification of proposed incorporated products and finishes.
    1.4. Certification, reports and calculations demonstrating compliance with specification of proposed cladding.
    1.5. Proposals for connections to and support from the primary support structure.
    1.6. Proposals for primary support structure additional to that shown on preliminary design drawings.
    1.7. Schedule of builders' work, special provisions and special attendance by others.
    1.8. Examples of standard documentation from which project quality plan will be prepared.
    1.9. Preliminary fabrication and installation method statements and programme.
    1.10. Proposals for replacing damaged or failed products.
    1.11. Areas of non-compliance with specification.

## Type(s) of rainscreen cladding

### 461 EWS-01A Woven Facade (Visual Intent Only)

1. Description: Woven façade cladding, as the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Support: Refer to the Façade Performance Specification
5. Façade/ cladding:
    5.1. Type/ material/ insulation/ joints/ fixings: Refer to the Façade Performance Specification
    5.2. Profile/ finish/ colour: Refer to the Façade Performance Specification

### 462 EWS-01B Woven Facade Corner Setback (Visual Intent Only)

1. Description: Woven façade corner setback cladding, as the Design Drawings.
2. Note:

2.1. This description covers items specified by the Façade Consultant.

2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.

3. Size and configuration: Refer to the Design Drawings
4. Support: Refer to the Façade Performance Specification
5. Façade/ cladding:
   5.1. Type/ material/ insulation/ joints/ fixings: Refer to the Façade Performance Specification
   5.2. Profile/ finish/ colour: Refer to the Façade Performance Specification

## 463 EWS-02A Barrel Curve Facade (Visual Intent Only)

1. Description: Barrel curve façade cladding, as the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Support: Refer to the Façade Performance Specification
5. Façade/ cladding:
   5.1. Type/ material/ insulation/ joints/ fixings: Refer to the Façade Performance Specification
   5.2. Profile/ finish/ colour: Refer to the Façade Performance Specification

## 464 EWS-02B Modern Mansard Facade (Visual Intent Only)

1. Description: Modern mansard façade cladding, as the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Support: Refer to the Façade Performance Specification
5. Façade/ cladding:
   5.1. Type/ material/ insulation/ joints/ fixings: Refer to the Façade Performance Specification
   5.2. Profile/ finish/ colour: Refer to the Façade Performance Specification

## 465 EWS-12 Core Cladding (Visual Intent Only)

1. Description: Core cladding, as the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Support: Refer to the Façade Performance Specification
5. Façade/ cladding:
   5.1. Type/ material/ joints/ fixings: Refer to the Façade Performance Specification

5.2. Profile/ finish/ colour: Refer to the Façade Performance Specification

# General requirements/ preparatory work

## 210 Design

1. Rainscreen cladding system and associated features: Complete detailed design in accordance with this specification and the preliminary design drawings and submit before commencement of fabrication.
2. Related works: Coordinate in detailed design.

## 215 Design proposals

1. Submission of alternative proposals: Preliminary design drawings indicate intent. Other reasonable proposals will be considered.

## 220 Specification

1. Compliance standards: In accordance with the Centre for Window and Cladding Technology (CWCT) Standard for systemised building envelopes.
2. Reference information: For the duration of the contract, keep available at the design office, workshop and on site copies of:
   2.1. The Centre for Window and Cladding Technology (CWCT) 'Standard for systemised building envelopes'.
   2.2. Publications invoked by the CWCT 'Standard for systemised building envelopes'.

## 230 Information to be provided during detailed design

1. Submit the following cladding particulars
   1.1. A schedule of detailed drawings and dates for submission for comment.
   1.2. A schedule of loads that will be transmitted from the rainscreen cladding to the structure.
   1.3. Proposed fixing details and systems relevant to the structural design and construction with methods of adjustment and tolerances.
   1.4. A schedule of fabrication tolerances/ size tolerances.
   1.5. A detailed testing programme in compliance with the main contract master programme.
   1.6. A detailed fabrication and installation programme in compliance with the main contract master programme.
   1.7. Proposals to support outstanding applications for Building Regulations consents or relaxations.

## 232 Quality management

1. Required documentation: Project specific method statement in accordance with CWCT Standard for systemised building envelopes and guidance in CWCT Technical Note 53.
2. Content: In accordance with BS EN ISO 9001 and including the following:
   2.1. Name of the quality manager.
   2.2. Quality assessment procedures.
   2.3. Inspection procedures to be adopted in checking the work.
   2.4. Stages at which check lists will be used and samples of the lists.
   2.5. List of work procedures on the correct use of materials or components, both off site and on site.

2.6. List of product information with latest revisions.

2.7. Subcontractors involved in the work.

2.8. Subcontractors quality plans.

2.9. Storage, handling, transport and protection procedures.

2.10. Procedure for registering and reporting non-compliances.

2.11. Maintenance procedures and calibration records.

2.12. Certification that completed work complies with specification.

2.13. Check list register to ensure all items have been inspected and non-compliances discharged.

3. Timing: Submit during detailed design

## 235 Information to be provided before commencement of testing or manufacture of rainscreen cladding system

1. Submit the following cladding particulars
   1.1. Detailed drawings to fully describe fabrication and installation.
   1.2. Detailed calculations to prove compliance with design/ performance requirements.
   1.3. Project-specific fabrication, handling and installation method statements.
   1.4. Certification for incorporated components manufactured by others confirming their suitability for proposed locations in the rainscreen cladding.
   1.5. Recommendations for spare parts for future repairs or replacements.
   1.6. Recommendations for safe dismantling and recycling or disposal of products.

## 240 Product samples

1. General: Before commencing detailed design, submit labelled samples of: all types specified.

## 250 Samples of fixings

1. General: During detailed design, submit labelled samples of each type of fixing, together with manufacturers' recommended torque figures.

## 260 Fabrication samples

1. General: During detailed design, submit samples of: To be confimred. Obtain approval of appearance before proceeding.

## 270 Mock-up

1. General: Construct during detailed design work in an agreed location. Satisfy purpose and obtain approval of appearance before proceeding. Retain undisturbed until completion of cladding installation.

# Design/performance requirements - Not Used

# Testing

## 510 Comparison (type) testing

1. Verification of performance
   1.1. Submit: Certification and reports satisfying CWCT Standard for systemised building envelopes, clause 1.5.5 items a and b

2. Commencement of fabrication and installation of rainscreen cladding: Not until test results and reports showing compliance with this specification have been submitted.

## 515 Project testing (laboratory)

1. Timing of testing: At an agreed stage in detailed design work arrange for laboratory testing of specimens of rainscreen cladding and components in accordance with relevant clauses of this specification.
2. Commencement of fabrication and installation of rainscreen cladding: Not until test results and reports showing compliance with this specification have been submitted.

## 520 Project testing (site)

1. Timing of testing: At an agreed stage during preliminary installation on site arrange for testing of a section of rainscreen cladding in accordance with relevant clauses of this specification.
2. Continuation of installation of general areas of rainscreen cladding: Not until site test results and reports showing compliance with this specification have been submitted.

## 530 Testing authority – UKAS-approved laboratory

1. Requirement: Project testing must be carried out by a United Kingdom Accreditation Service (UKAS)-approved independent laboratory.

## 535 Testing authority – cladding manufacturer/ contractor

1. Requirement: Project testing must be carried out by the rainscreen cladding manufacturer/ contractor and is to be witnessed and certified by: the Contract Administrator.

## 540 Large test specimen

1. Description: To be agreed with the Architect.
2. Arrangement and overall dimensions: Refer to the Design Drawings.

## 570 Test sequence – standard laboratory and discretionary tests

1. Requirement: To CWCT Standard for systemised building envelopes, sequence for discretionary tests, clause 8.13.5 and as follows:
2. Air permeability (infiltration): To CWCT Standard for systemised building envelopes, clause 8.6
   2.1. Test pressure: As clause 410.
   2.2. Method for identifying areas of concentrated air leakage: Submit proposals.
3. Air permeability (exfiltration): To CWCT Standard for systemised building envelopes, clause 8.6
   3.1. Allowable exfiltration rates: As clause 390.
   3.2. Method of identifying areas of concentrated air leakage: Submit proposals.
4. Water penetration resistance – static: To CWCT Standard for systemised building envelopes, clause 8.7.1
   4.1. Peak test pressure: As clause 420.
5. Wind resistance – serviceability: To CWCT Standard for systemised building envelopes, clause 8.8.
   5.1. Peak test pressure: Equal to design wind pressure.
   5.2. Position of displacement measuring devices: Submit proposals.
   5.3. Allowable elastic and residual deformation: Submit proposals.
6. Air permeability (infiltration): To CWCT Standard for systemised building envelopes, clause 8.6

6.1. Test pressure: As clause 410.

6.2. Method for identifying areas of concentrated air leakage: Submit proposals.

7. Air permeability (exfiltration): To CWCT Standard for systemised building envelopes, clause 8.6

7.1. Allowable exfiltration rates: As clause 390.

7.2. Method of identifying areas of concentrated air leakage: Submit proposals.

8. Water penetration resistance – static: To CWCT Standard for systemised building envelopes, clause 8.7.1

8.1. Peak test pressure: As clause 420.

9. Water penetration resistance – dynamic: To CWCT Standard for systemised building envelopes, clause 8.7.2

9.1. Test method: Dynamic aero engine.

9.2. Test pressure: As clause 420.

10. Impact serviceability: To CWCT Standard for systemised building envelopes, clause 8.10

10.1. Location, sequence and number of impacts to be applied: Submit proposals.

10.2. Type of soft body impactor: Loaded canvas spherical/conical bag as described in CWCT Standard test methods for building envelopes, clause 15.4.1

10.3. Energy of soft body impactor (E): Submit proposals.

10.4. Required hard body impact load category: Submit proposals.

10.5. Required soft body impact classification: Submit proposals.

11. Air permeability (infiltration): To CWCT Standard for systemised building envelopes, clause 8.6

11.1. Test pressure: As clause 410.

11.2. Method for identifying areas of concentrated air leakage: Submit proposals.

12. Air permeability (exfiltration): To CWCT Standard for systemised building envelopes, clause 8.6

12.1. Allowable exfiltration rates: As clause 390.

12.2. Method of identifying areas of concentrated air leakage: Submit proposals.

13. Water penetration resistance – static: To CWCT Standard for systemised building envelopes, clause 8.7.1

13.1. Peak test pressure: As clause 420.

14. Building movement regime: To CWCT 'Standard for systemised building envelopes', clause 8.13.3.

14.1. Displacements to test specimen supports: Submit proposals.

14.2. Condition of specimen for follow on air permeability and water penetration resistance tests: Submit proposals.

15. Air permeability (infiltration): To CWCT Standard for systemised building envelopes, clause 8.6

15.1. Test pressure: As clause 410.

15.2. Method for identifying areas of concentrated air leakage: Submit proposals.

16. Air permeability (exfiltration): To CWCT Standard for systemised building envelopes, clause 8.6

16.1. Allowable exfiltration rates: As clause 390.

16.2. Method of identifying areas of concentrated air leakage: Submit proposals.

17. Water penetration resistance – static: To CWCT Standard for systemised building envelopes, clause 8.7.1

17.1. Peak test pressure: As clause 420.

18. Thermal cycling regime: To CWCT Standard for systemised building envelopes, clause 8.13.4

18.1. External air temperature limits: Submit proposals.

18.2. Maximum solar radiation intensity on a vertical surface: Submit proposals.

18.3. Internal in use psychrometric conditions: Submit proposals.

18.4. Post-test requirement: Retain specimen in its final condition for subsequent air permeability and water penetration resistance tests.

19. Air permeability (infiltration): To CWCT Standard for systemised building envelopes, clause 8.6

19.1. Test pressure: As clause 410.

19.2. Method for identifying areas of concentrated air leakage: Submit proposals.

20. Air permeability (exfiltration): To CWCT Standard for systemised building envelopes, clause 8.6

20.1. Allowable exfiltration rates: As clause 390.

20.2. Method of identifying areas of concentrated air leakage: Submit proposals.

21. Water penetration resistance – static: To CWCT Standard for systemised building envelopes, clause 8.7.2

21.1. Peak test pressure: As clause 420.

22. Hose test: To CWCT Standard for systemised building envelopes, clause 8.16.2

22.1. Joints to be tested: Submit proposals.

23. Wind resistance – safety: To CWCT Standard for systemised building envelopes, clause 8.8.

23.1. Peak test pressure: Equal to 1.5 times design wind pressure.

23.2. Position of displacement measuring devices: Submit proposals.

23.3. Allowable residual deformation: Submit proposals.

24. Impact – safety: To CWCT Standard for systemised building envelopes, clause 8.10

24.1. Location, sequence and number of impacts to be applied: Submit proposals.

24.2. Type of soft body impactor: Loaded canvas spherical/ conical bag as described in CWCT Standard test methods for building envelopes, clause 15.4.1

24.3. Energy of soft body impactor (E): Submit proposals.

24.4. Required hard body impact load category: Submit proposals.

24.5. Required soft body impact classification: Submit proposals.

24.6. Load tests (other than wind load): Submit proposals.

25. Testing of wind loading on rainscreen panels: To CWCT Standard test methods for building envelopes, section 13

25.1. Peak test pressure: Equal to design wind pressure.

25.2. Position of displacement measuring devices: Submit proposals.

25.3. Allowable residual deformation: Submit proposals.

26. Dismantle inspect and record.

## 665 Wind load fatigue test, small specimen

1. Test standard: To CWCT Standard for systemised building envelopes, Standard test methods for building envelopes, section 14.

1.1. Test sequence: As CWCT Standard for systemised building envelopes, Table 8.1

1.2. Effective wind pressure: As clause 360.

## 670 Testing of fixings

1. Test standard: To CWCT Standard for systemised building envelopes, Standard test methods for building envelopes, section 19.
2. Type of test: Serviceability and Ultimate load

## 672 Site-testing of fixings

1. Test standard: To CWCT Standard for systemised building envelopes, Standard test methods for building envelopes, section 19.
2. Type of test: Serviceability and Ultimate load
3. Number and location of test fixings: To be agreed with the Architect.

## 680 Site spray bar test

1. Requirement: To CWCT Standard for systemised building envelopes, Standard test methods for building envelopes, section 10.
    1.1. Area to be tested: To be agreed with the Architect.
    1.2. Pressure difference across rainscreen wall: To be agreed with the Architect.

## 685 Site hose test

1. Requirement: To CWCT Standard for systemised building envelopes, Standard test methods for building envelopes, section 9.
    1.1. Joints to be tested: To be agreed with the Architect.

# Products

## 710 Aluminium alloy framing sections

1. Standards: To BS EN 755 alloy EN AW-6063 and suitable for the specified finish.
2. Structural members: To comply with BS EN 1999-1-1, BS EN 1999-1-3 and BS EN 1999-1-4.

## 712 Aluminium alloy sheet

1. Standards: To BS EN 485, BS EN 515 and BS EN 573.
2. Alloy, temper and thickness: Suitable for the application and specified finish.

## 715 Carbon steel framing sections/ reinforcement

1. Standards: To the relevant parts of BS 7668, BS EN 10029, BS EN 10025 or BS EN 10210.
2. Thickness: Suitable for the application, and for galvanizing or other protective coating.

## 717 Carbon steel sheet

1. Standards: To the relevant parts of BS 1449-1, BS EN 10048, BS EN 10051, BS EN 10111, BS EN 10131, BS EN 10132, BS EN 10139, BS EN 10140, BS EN 10149-1, BS EN 10149-2, BS EN 10149-3, BS EN 10209 or BS EN 10268.
2. Grade and thickness: Suitable for the application, and for galvanizing or other protective coating.

## 720 Stainless steel sheet

1. Standards: To the relevant parts of BS EN 10029, BS EN 10048, BS EN 10051, BS EN 10095 or BS EN ISO 9445-1 and BS EN ISO 9445-2.

2. Grade: To BS EN 10088-2, BS EN 10088-4, BS EN 10088-5, austenitic 1.4301 (304) generally, 1.4301 (316) when used externally or in severely corrosive environments.
3. Thickness: Suitable for the application.

## 725 Timber battens

1. General: Regularized softwood free from decay, insect attack and with no knots wider than half the width of the section.
2. Preservative treatment: As section Z12.
   2.1. Type: To be agreed with the Architect.
3. Moisture content at time of fixing (maximum): 19%.

## 730 Mechanical fixings – material requirements

1. Stainless steel: To BS EN ISO 3506 Grade A2 generally, Grade A4 when used in severely corrosive environments.
2. Carbon steel: To BS 4190 and suitable for galvanizing or other protective coating.
3. Aluminium: To BS EN 755.

## 732 Adhesives

1. General: Not degradable by moisture or water vapour.

## 735 Fixings and fasteners

1. Type and use: Reviewed and approved by manufacturers. Submit confirmatory information on request.
2. Dimensions: Not less than recommended by their manufacturers.
3. Adjustment capability: Sufficient in three dimensions to accommodate primary support structure and rainscreen cladding fabrication/ installation tolerances.

## 740 Glass generally

1. Standards: To BS 952 and the relevant parts of:
   1.1. BS EN 572 for basic soda lime silicate glass.
   1.2. BS EN 1096 for coated glass.
   1.3. BS EN 1863 for heat strengthening soda lime silicate glass.
   1.4. BS EN 12150-1 and BS EN 12150-2 for thermally toughened soda lime silicate glass.
   1.5. BS EN 13024 for thermally toughened borosilicate glass.
   1.6. BS EN ISO 12543 for laminated glass.
2. Glass quality: Clean and free from obvious scratches, bubbles, cracks, rippling, dimples and other defects.
3. Glass edges: Generally undamaged. Shells and chips not more than 2 mm deep and extending not more than 5 mm across surface are acceptable if ground out.

## 742 Heat-soaked thermally toughened glass

1. Standard: To BS EN 14179.
   1.1. Holding period: Two hours
2. Locations of heat-soaked glass: Refer to the Design Drawings.

## 745 Insulating glass units

1. Standard and labels for hermetically sealed units: To BS EN 1279.
2. Label: Each pane.
3. Colour of aluminium perimeter spacers: Natural
4. Perimeter taping: Not to be used.
5. Perimeter seals
   5.1. Resistant to UV light degradation on exposed edges.
   5.2. Compatible with structural, assembly and weather sealants.

## 760 Gaskets

1. Material: Cellular rubber to ASTM-C509-06
2. Durability: Resistant to oxidation, ozone and UV degradation.

## 765 Weatherstripping of opening units

1. Material: Cellular rubber to ASTM-C509-06
2. Installation: Fixed in undercut grooves in framing sections using preformed corners with any joints in the length.

## 770 General sealants

1. Selection: In accordance with BS 6213.
2. Classification and requirements: To BS EN ISO 11600.
3. Reaction to contact products and finishes: Stable and compatible.

## 775 Thermal insulation

1. Material: Mineral wool to BS EN 13162
   1.1. Properties: Durable, rot and vermin proof and not degradable by moisture or water vapour.
2. Fixing: Attached to the outer face or supported within the backing wall so as not to bulge, sag, delaminate or detach during installation or in situ during the life of the rainscreen cladding.

# Finishes

## 810 Protective coating of carbon steel framing sections/ reinforcement

1. Treatment:
   1.1. All surfaces to one of the following: Hot dip galvanized to BS EN ISO 1461

## 820 Protective coating of carbon steel mechanical fixings

1. Treatment:
   1.1. All surfaces to one of the following: Hot dip galvanized to BS EN ISO 1461

## 830 Powder coating

1. Requirement: As section Z31.

### 840 Anodizing

1. Requirement: As section Z33.

### 850 Polyvinylidene fluoride (PVDF) coating

1. Standard: To BS 4842 or AAMA 2604-22, subject to minimum coating thicknesses recommended by the manufacturer on all significant surfaces.
    1.1. Applicator: Submit proposals
        1.1.1. Product reference: Submit proposals
2. Process: Prepare base metals, prime, PVDF coat, test samples, protect components and repair damage in accordance with manufacturer's recommendations.
3. Sequence: Wherever possible, apply coatings after fabrication is complete.
    3.1. Fabrication of prefinished lengths: Submit proposals beforehand.
    3.2. Uncoated edges: Invisible in completed assemblies.

## Fabrication and installation

### 910 Generally

1. Electrolytic corrosion: Take necessary measures to prevent.
2. Identification of products: Mark or tag to facilitate identification during assembly, handling, storage and installation. Do not mark surfaces visible in the complete installation.

### 912 Metalwork

1. Requirement: As section Z11, unless specified otherwise in this section.

### 915 Timber treatment

1. Requirement: As section Z12, unless specified otherwise in this section.
2. Exposed cut and drilled surfaces: Treat with two flood coats of a solution recommended for the purpose by main treatment solution manufacturer.

### 920 Glazing

1. Requirement: As section L40, unless specified otherwise in this section.
2. Directional patterned/ wired glass: Fix generally parallel to surround and align adjacent panes where seen together at close quarters.

### 922 Fixings/ adhesives application

1. Requirement: As section Z20, unless specified otherwise in this section.

### 925 Sealant application

1. Requirement: As section Z22, unless specified otherwise in this section.

### 930 Assembly

1. Location: Carry out as much assembly as possible in the workshop.
2. Joints: Other than movement joints and designed open joints, must be rigidly secured, reinforced where necessary and fixed with hairline abutments.
3. Displacement of components in assembled units: Submit proposals for reassembly on site.



make

## 960 Preliminary rainscreen cladding installation

1. Requirement: Complete an area of cladding as set out below for inspection and approval of appearance as agreed with the Architect .

## 970 Rainscreen cladding installation

1. Tightening mechanical fasteners: To manufacturer's recommended torque figures. Do not overtighten fasteners intended to permit differential movement.
2. Protective coverings: Remove only where necessary to facilitate installation and from surfaces which will be inaccessible on completion.

## 975 Welding

1. In situ welding: Not permitted

## 980 Interfaces

1. Installation: Locate flashings, closers etc. correctly and neatly overlap cladding to form a weathertight junction.

## 985 Damage

1. Repairs: Do not repair cladding without approval.
   1.1. Approval: Will not be given where the proposed repair will impair performance or appearance.
2. Record of repairs: Prepare schedule or record on drawings for inclusion in the maintenance manual.

## 995 Maintenance

1. Maintenance manual: Incorporate details within the Building Manual in accordance with CWCT Standard for systemised building envelopes, clause 7.6.1.

Ω End of Section



make

# J30
# Waterproofing

## Types of tanking/ damp proofing

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 WPS-101 Waterproofing Systems (Generally)

1. Description: Waterproofing system providing a fully waterproof, damp-proof and water vapour impermeable barrier resistant to gases and contamination from the ground, to locations indicated on the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Waterproofing consultant.
    2.2. Refer to the Waterproofing Consultant Specification for detailed information and/ or performance requirements.
3. Substrate: Refer to the Design Drawings
    3.1. Preparation: The substrate should be free from irregularities such as voids or protrusions.
4. Membrane system:
    4.1. Type: Liquid or sheet-applied proofing where required and to be selected by the Contractor.
    4.2. Characteristics: Selected type to include all necessary ancillary items, to suit the conditions of use, collectively forming fully detailed, integrated and compatible Works to interface with other above-ground damp-proofing Works.
5. Application: In accordance with the manufacturer's written recommendations
6. Warranty: Provide a warranty/ guarantee for the complete tanking installation.
7. System accessories: Membranes shall be supplied and installed as complete integrated systems, including all other accessories/ components recommended/ supplied by the manufacturer.

## Execution

### 205 Suitability of substrate

1. Substrates generally: Smooth, even textured, clean, dry and frost-free
2. Moisture content and stability of substrate: Must not impair integrity of finished tanking/ damp-proofing
3. Vertical and horizontal surfaces: Correctly prepared and free from irregularities
4. Curing period for concrete substrates (minimum): Seven days

### 206 Adhesion testing

1. Tensile bond strength: Test in accordance with BS EN ISO 4624.
2. Test results: Submit.



make

## 207 Application of primers

1. Application: Uniform, continuous coverage.

## 210 Coating application

1. Adjacent surfaces exposed to view in finished work: Protect.
2. Coatings
   2.1. Coverage: As coating manufacturer's recommendations.
   2.2. Apply in dry atmospheric conditions when recommended by the manufacturer.
   2.3. Application generally: Uniform, continuous coverage. Do not allow to pool in hollows
3. Penetrations: Fully sealed and impervious.
4. Final covering: Apply as soon as possible after coating has hardened.

## 220 Application of cold-applied coatings

1. Thinning: Not permitted unless recommended by manufacturer.
2. Successive coats: Allow to dry before applying next.

## 230 Application of bituminous coatings

1. Air and surface temperatures (minimum): 5°C.
2. Weather conditions: Do not apply if there is a risk of rain during application and drying unless effective temporary cover is provided over working area.
   2.1. Unavoidable wetting of construction or coatings: Minimize and make good damage.

## 235 Application of modified coatings

1. Air and surface temperatures: Do not apply if below minimum recommended by coating manufacturer.
2. Curing: Keep dry until fully cured.

## 240 Reinforcement

1. Laying: Laid into second coat with third coating applied locally over fabric
2. Edge overlap (minimum): As per manufacturer's recommendations.
3. Finish: Smooth and free from imperfections.

## 250 Hot-applied coatings

1. Application: Heat to minimum temperature required for coating to pour easily.

## 260 Junctions with dpcs

1. Dpcs: Clean, all edges fully exposed.
2. Application: Fully coat dpc and overlap adjacent surfaces by (minimum): 75 mm.

## 270 Blinding

1. Coatings: As per manufacturers recommendations.
2. Surplus material: Remove when coatings are completely dry.

# Completion

## 310 Inspection

1. Interim and final inspections: Submit reports.

## 320 Electronic integrity test

1. Testing authority: UKAS-approved laboratory
2. Timing of test: Give notice
3. Condition of coating prior to testing
   3.1. Complete to a stage where integrity can be tested.
   3.2. Surface: Clean.
4. Test results and waterproof integrity certificate: Submit on completion of testing.

## 340 Backfilling to external coatings

1. Timing: Carry out as soon as possible after tanking and protection are complete.

Ω End of Section



make

# J31
# Liquid-Applied Waterproof Roof Coatings

## Types of coating

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Refer to and comply with A10.120 and A21.140.

### 130 RFS-101 Hot Melt Warm Roof System

1. Description: Hot melt warm roof system.
2. Manufacturer: Radmat, or acceptable equivalent
    2.1. Product reference: PermaQuik PQ6100
3. Substrate: Structural concrete deck, Refer to the Structural Engineer's documentation
4. Preparation:
    4.1. The surface shall be dry, clean, and free from all contaminants including oils, grease, laitance, dirt and debris.
    4.2. Advance bond tests must be carried out (and recorded for future reference) on all areas.
5. Primer: Surface conditioner, to the manufacturer's recommendations.
6. Waterproof coating:
    6.1. Product reference: PermaQuik PQ6100
    6.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        6.2.1. No. of coats: As per the manufacturer's recommendations
    6.3. Reinforcement: As per the manufacturer's recommendations
7. Protection sheet:
    7.1. Product reference: PQ1800
8. Insulation:
    8.1. Product reference: FOAMGLAS T3+
    8.2. Thickness: To be determined by the Contractor to achieve the performance requirements.
    8.3. Fixing: Loose laid.
    8.4. Insulation laid to falls.
9. Waterproof coating:
    9.1. Product reference: PermaQuik PQ6100
    9.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        9.2.1. No. of coats: As per the manufacturer's recommendations
    9.3. Reinforcement: As per the manufacturer's recommendations
10. Surface finish: Refer to the Landscape Architects documentation
11. Rainwater outlets:

11.1. Type: Insulated drainage outlet assemblies with compatible leaf/ gravel guards to be to the manufacturer's recommendations.

11.2. Manufacturer: Harmer, or acceptable equivalent

11.3. Spigots: Type and direction to suit pipelines with suitable adaptors and connections.

11.4. Integral accessories: As required to complete the installation.

12. Pressed metal components/ accessories:

12.1. Material: Aluminium

12.2. Finish: Polyester powder coated

12.3. Colour: Standard RAL colour.

13. Accessories: As required by the manufacturer to complete the installation, in accordance with their recommendations.

## 131 RFS-102 Hot Melt Warm Roof System - Concrete Pavers

1. Description: Hot melt warm roof system.
2. Manufacturer: Radmat, or acceptable equivalent

2.1. Product reference: PermaQuik PQ6100

3. Substrate: Structural concrete deck, Refer to the Structural Engineer's documentation
4. Preparation:

4.1. The surface shall be dry, clean, and free from all contaminants including oils, grease, laitance, dirt and debris.

4.2. Advance bond tests must be carried out (and recorded for future reference) on all areas.

5. Primer: Surface conditioner, to the manufacturer's recommendations.
6. Waterproof coating:

6.1. Product reference: PermaQuik PQ6100

6.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins

6.2.1. No. of coats: As per the manufacturer's recommendations

6.3. Reinforcement: As per the manufacturer's recommendations

7. Protection sheet:

7.1. Product reference: PQ1800

8. Insulation:

8.1. Product reference: FOAMGLAS T3+

8.2. Thickness: To be determined by the Contractor to achieve the performance requirements.

8.3. Fixing: Loose laid.

8.4. Insulation laid to falls.

9. Waterproof coating:

9.1. Product reference: PermaQuik PQ6100

9.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins

9.2.1. No. of coats: As per the manufacturer's recommendations

9.3. Reinforcement: As per the manufacturer's recommendations

10. Surface finish:

10.1. Support pedestals: Height adjustable support pedestals adhesive bonded and/ or mechanically fixed to substrates.

10.2. Slabs:

10.2.1. Size: 600 mm x 600 mm x 50 mm.

10.2.2. Colour/ finish: To be confirmed

10.2.3. Slip resistance: PTV >36.

11. Rainwater outlets:

11.1. Type: Insulated drainage outlet assemblies with compatible leaf/ gravel guards to be to the manufacturer's recommendations.

11.2. Manufacturer: Harmer, or acceptable equivalent

11.3. Spigots: Type and direction to suit pipelines with suitable adaptors and connections.

11.4. Integral accessories: As required to complete the installation.

12. Pressed metal components/ accessories:

12.1. Material: Aluminium

12.2. Finish: Polyester powder coated

12.3. Colour: Standard RAL colour.

13. Accessories: As required by the manufacturer to complete the installation, in accordance with their recommendations.

## 132 RFS-103 Overslab Roof System

1. Description: Overslab roof system
2. Substrate: Structural concrete deck, Refer to the Structural Engineer's documentation
3. Preparation:
   3.1. The surface shall be dry, clean, and free from all contaminants including oils, grease, laitance, dirt and debris.
   3.2. Advance bond tests must be carried out (and recorded for future reference) on all areas.
4. Primer: Surface conditioner, to the manufacturer's recommendations.
5. Waterproof coating: Refer to WPS-101 in J30
6. Insulation: Refer to INS-126 in P10
7. Overslab: Refer to the Structural Engineer's documentation
8. Accessories: As required by the manufacturer to complete the installation, in accordance with their recommendations.

## 133 RFS-104 Hot Melt Warm Roof System - Bio Solar

1. Description: Warm roof system, hybrid system that has waterproofing below and above.
2. Manufacturer: Radmat, or acceptable equivalent
   2.1. Product reference: PermaQuik PQ6100
3. Substrate: Structural concrete deck, Refer to the Structural Engineer's documentation
4. Preparation:
   4.1. The surface shall be dry, clean, and free from all contaminants including oils, grease, laitance, dirt and debris.
   4.2. Advance bond tests must be carried out (and recorded for future reference) on all areas.
5. Primer: Surface conditioner, to the manufacturer's recommendations.

6. Waterproof coating:
    6.1. Product reference: PermaQuik PQ6100
    6.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        6.2.1. No. of coats: As per the manufacturer's recommendations
    6.3. Reinforcement: As per the manufacturer's recommendations
7. Protection sheet:
    7.1. Product reference: PQ1800
8. Insulation:
    8.1. Product reference: FOAMGLAS T3+
    8.2. Thickness: To be determined by the Contractor to achieve the performance requirements.
    8.3. Fixing: Loose laid.
    8.4. Insulation laid to falls.
9. Waterproof coating:
    9.1. Product reference: PermaQuik PQ6100
    9.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        9.2.1. No. of coats: As per the manufacturer's recommendations
    9.3. Reinforcement: As per the manufacturer's recommendations
10. Surface finish: Green roof, Refer to the Landscape Architects documentation
11. Rainwater outlets:
    11.1. Type: Insulated drainage outlet assemblies with compatible leaf/ gravel guards to be to the manufacturer's recommendations.
    11.2. Manufacturer: Harmer, or acceptable equivalent
    11.3. Spigots: Type and direction to suit pipelines with suitable adaptors and connections.
    11.4. Integral accessories: As required to complete the installation.
12. Pressed metal components/ accessories:
    12.1. Material: Aluminium
    12.2. Finish: Polyester powder coated
    12.3. Colour: Standard RAL colour.
13. Accessories: As required by the manufacturer to complete the installation, in accordance with their recommendations.

## 134 RFS-105 Overrun Roof System

1. Description: Overrun warm roof system
2. Manufacturer: Radmat, or acceptable equivalent
    2.1. Product reference: PermaQuik PQ6100
3. Substrate: Structural concrete deck, Refer to the Structural Engineer's documentation
4. Preparation:
    4.1. The surface shall be dry, clean, and free from all contaminants including oils, grease, laitance, dirt and debris.
    4.2. Advance bond tests must be carried out (and recorded for future reference) on all areas.

5. Primer: Surface conditioner, to the manufacturer's recommendations.
6. Waterproof coating:
    6.1. Product reference: PermaQuik PQ6100
    6.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        6.2.1. No. of coats: As per the manufacturer's recommendations
    6.3. Reinforcement: As per the manufacturer's recommendations
7. Protection sheet:
    7.1. Product reference: PQ1800
8. Insulation:
    8.1. Product reference: FOAMGLAS T3+
    8.2. Thickness: To be determined by the Contractor to achieve the performance requirements.
    8.3. Fixing: Loose laid.
    8.4. Insulation laid to falls.
9. Waterproof coating:
    9.1. Product reference: PermaQuik PQ6100
    9.2. Type/ material: Hot applied roof waterproofing membrane manufactured from a rubberised bitumen modified with styrene resins
        9.2.1. No. of coats: As per the manufacturer's recommendations
    9.3. Reinforcement: As per the manufacturer's recommendations
10. Surface finish: N/A
11. Rainwater outlets:
    11.1. Type: Insulated drainage outlet assemblies with compatible leaf/ gravel guards to be to the manufacturer's recommendations.
    11.2. Manufacturer: Harmer, or acceptable equivalent
    11.3. Spigots: Type and direction to suit pipelines with suitable adaptors and connections.
    11.4. Integral accessories: As required to complete the installation.
12. Pressed metal components/ accessories:
    12.1. Material: Aluminium
    12.2. Finish: Polyester powder coated
    12.3. Colour: Standard RAL colour.
13. Accessories: As required by the manufacturer to complete the installation, in accordance with their recommendations.

## Performance

### 202 Contractor's design of roofing

1. Design responsibility: Determine methods in attaching roofing
2. Structural and fire requirements
    2.1. Generally: Refer to the Fire Strategy Report and Refer to the Structural Engineer's Information
    2.2. Modifications: None

2.3. Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.

3. Functional requirements
   3.1. Performance: As specified in this section
4. Additional requirements: None

### 205 Completion of roofing design

1. Requirement: Complete the detailed design to satisfy specified performance criteria and coordinate with the detailed design of related and adjacent work.
2. Structural requirements: Refer to the Structural Engineer's Information
3. Additional requirements: None

### 210 Roof performance

1. General: Firmly adhered, free-draining and weathertight.

### 225 Avoidance of interstitial condensation

1. Risk of interstitial condensation in roof construction: Assess in accordance with BS 5250.
2. Vapour control layer: If necessary, provide a suitable membrane so that damage and nuisance from interstitial condensation do not occur.

### 230 Thermal performance

1. Requirement: Determine type and thickness of insulation and integral or separate overlay to satisfy the following criteria:
   1.1. Thermal transmittance of roof (maximum): To be confirmed
   1.2. Compressive strength of insulation (minimum) at 10% compression: To be confirmed
   1.3. Substrate surface: Suitably even, stable and robust to receive roof coatings.
   1.4. Insulation compliance: To a relevant European Standard, or Agrément-certified.

### 240 Fire performance

1. Classification: Broof(t4) in accordance with BS EN 13501-5

## Products

### 310 Primer

1. Type: As per the system manufacturer's recommendations to suit the conditions of service.

### 315 Timber trims

1. Quality: Planed, free from wane, pitch pockets, decay and insect attack (except ambrosia beetle damage).
2. Moisture content at time of covering (maximum): 22%.
3. Preservative treatment: Not required

### 320 Plywood overlay to metal deck

1. Plywood: To BS EN 636, section 8 (plywood for use in humid conditions).
   1.1. Quality: Naturally durable timber, free from preservative.

2. Thickness: As per the system manufacturer's recommendations to suit the conditions of service

### 325 Oriented strand board (OSB) overlay to metal deck

1. Oriented strand board: To BS EN 300, Type: OSB/3.
2. Thickness: As per the system manufacturer's recommendations to suit the conditions of service

### 365 Stone ballast

1. Type: Washed, round aggregate
2. Size: Graded 20-40 mm, free from fines and sharps
3. Colour: To be confirmed

## Execution generally

### 410 Adverse weather

1. Do not apply coatings: In wet conditions or at temperatures below 5°C, unless otherwise permitted by coating manufacturer.
2. Unfinished areas of roof: Keep dry.

### 420 Suitability of substrates

1. Substrates generally: Secure, clean, dry, smooth; free from frost, contaminants, loose material, voids, protrusions and organic growths.
2. Complete preliminary work, including: Formation of upstands, kerbs, box gutters, sumps, grooves, chases and expansion joints.
3. Moisture content and stability: Must not impair integrity of roof.

## Existing substrates

### 510 Removing existing coverings

1. Mechanical stripping: Not permitted
2. Exposed substrate: Do not damage.

### 515 Existing flashings

1. General: Raise to facilitate cleaning of surfaces to receive coatings.
2. Timing: Leave raised during coating application and lower only after full curing.
3. Damaged lengths: Replace with new.

### 520 Preliminary power wash to existing coverings

1. Timing: Before renewing existing coverings, water jet-clean all areas. Allow to dry.

### 525 Renewing existing substrates/ coverings

1. Areas to be renewed: Submit proposals
2. Timing: Remove only sufficient substrates/ coverings as will be renewed and made weathertight on same day.

## 530 Making good existing liquid-applied waterproof roof coatings

1. General: Inspect for adherence and repair defective areas in accordance with proposed coating manufacturer's recommendations.

## 535 Making good existing reinforced bitumen membrane covering

1. Blisters: Star-cut, dry out and rebond.
2. Cracked and defective areas: Cut back to substrate.
3. Substrate: Dry out.
4. Bitumen membrane: Patch level with existing surface with layers of matching bitumen membrane, lapped minimum 100 mm onto existing membrane.

## 540 Making good existing mastic asphalt covering

1. Defective areas: Soften and carefully cut out.
    1.1. Hammers, chisels, etc: Do not use to cut cold mastic asphalt.
    1.2. Substrate: Dry out.
    1.3. Separating membrane: Make good.
    1.4. Mastic asphalt: Patch level with existing surface in two coats, the topcoat lapped minimum 75 mm onto existing mastic asphalt and to half its depth.

## 545 Making good existing fibre cement sheet covering

1. Loose and soft surfaces: Wet and remove. Allow to dry.
2. Cracked and damaged sheets: Replace.
3. Bolt heads: Tighten and crop where necessary.
    3.1. Corrosion and oxidation: Abrade back to bright metal.
4. Asbestos-based materials: Making good or removal must be carried out by a contractor licensed by the Health and Safety Executive prior to commencement of other works in their location.

## 550 Making good existing metal sheet covering

1. Loose coatings: Remove.
2. Corrosion and oxidation: Abrade back to bright metal.
3. Structurally unsound sheets: Replace.

## 555 Making good existing cementitious slabs/ screeds

1. Loose surfaces, sharp edges and projections: Remove.
2. Hollow surfaces, voids and cracks: Fill with cement-based repair mortar.

## 560 Existing edge trims

1. Fasteners: Check security. Replace as necessary.
2. Existing coverings: Cut out from edge trim recess sufficient to accommodate coatings.

## 565 Existing gutters/ outlets

1. Dirt, debris and build-up of previous coverings/ coatings: Remove to restore free flow of water.

### 570 Existing cracks/ gaps

1. General: Rake out, clean and make good with sealants or repair systems recommended by coating manufacturer.

### 575 Final power wash to existing coverings

1. General: Water jet-clean all areas. Allow to dry.

### 580 Sterilization treatment to existing coverings

1. Preliminary work: Complete, including making good and cleaning down.
2. Biocidal solution: Apply to all areas previously subject to organic growth. Allow to dry.

## New substrates/ vapour control layers/ warm deck roof insulation

### 610 Fixing timber trims

1. Fasteners: Sherardized steel screws.
2. Fixing centres (maximum): As per the system manufacturer's recommendations

### 620 Laying overlay to metal deck

1. Setting out: Lay boards with staggered joints and long edges at right angles to troughs in deck.
    1.1. Joints: 1 mm per metre panel size
    1.2. End joints: Centred over crowns of deck.
2. Fasteners
    2.1. Type: As recommended for purpose by fastener manufacturer
    2.2. Fixing centres: As per the system manufacturer's recommendations
    2.3. Fastener heads: Flush with, or below board surface.

### 630 Laying air and vapour control layer

1. Membrane: Loose-laid, flat and without wrinkles
2. Laps: Sealed using materials and method recommended by membrane manufacturer.
3. Upstands, kerbs and other penetrations: Enclose edges of insulation. Lap with coatings to form a complete seal.

### 640 Laying warm deck roof insulation

1. Setting out
    1.1. Long edges: Fully supported and run at right angles to Direction of span.
        1.1.1. Joints: Butted together.
    1.2. Ends: Adequately supported.
        1.2.1. Joints: Staggered.
2. Bedding: Not required
3. Mechanical fixing: Not required
4. Completion: Boards must be in good condition, well-fitting and stable.

### 650 Laying overlay to warm deck roof insulation

1. Setting out

1.1. Joints: Butted together.

1.2. End joints: Staggered to break joint with insulation.

2. Bedding: Not required

3. Mechanical fixing: Not required

### 660 Fixing perimeter trims

1. Setting out: 3 mm clear from wall or fascia.

2. Fasteners: 50 mm stainless steel countersunk wood screws

2.1. Fixing: 30 mm from ends and at 300 mm (maximum) centres.

3. Jointing

3.1. Sleeves: Fixed one side only.

3.2. GRP trims: Butt ends.

3.3. Aluminium trims: 3 mm gaps between ends.

4. Corner pieces: Purpose made.

### 670 Installing roof ventilators

1. Setting out: Position evenly over roof area.

1.1. Centres (maximum): 5 m square grid

1.2. Distance in from roof edge: 1 m

2. Holes for ventilators: Cut neatly to suit size of vents through insulation and air and vapour control layer.

3. Priming/ bonding

3.1. Substrate below vents: Do not prime or apply bonding compound.

3.2. Vent skirts: Primed before application of roof coating if recommended by manufacturer.

## Roof coating system

### 710 Adhesion tests

1. Requirement: Carry out a trial coating to determine priming requirements and/ or system suitability.

2. Nature of test: To be confirmed

3. Test results: Submit and arrange for inspection.

### 720 Applying primers/ conditioners

1. Coverage per coat (minimum): 0.1 L/m²

2. Surface coverage: Brushed well in to ensure local or full area coverage according to type.

3. Coats: Allow to dry before overcoating.

### 730 Laying carrier membrane

1. Bond: To the system manufacturers recommendations

2. Mechanical fixing: To the system manufacturers recommendations

### 740 Movement joints in substrate

1. Debonding tape: Apply over movement joints.



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2. Reinforcement strip: Apply over debonding tape.
    2.1. Bedding: Preliminary coating application.
    2.2. Joints: Lap in length.
    2.3. Bond: Continuous over whole surface, with no air pockets.
    2.4. Condition at completion: Smooth.

### 750 Preliminary local reinforcement

1. Reinforcement strip: Apply to junctions at upstands, penetrations and outlets, joints and fixings in discontinuous unit substrates.
    1.1. Bedding: Preliminary coating application.
    1.2. Joints: Lap in length.
    1.3. Bond: Continuous over whole surface, with no air pockets.
    1.4. Condition at completion: Smooth.

### 760 Application of roof coatings

1. Thickness: Monitor by taking wet/ dry film thickness readings.
2. Continuity: Maintain full thickness of coatings around angles, junctions and features.
3. Rainwater outlets: Form with watertight joints.
4. Drainage systems: Do not allow liquid coatings to enter piped rainwater or foul systems.
5. Edge trims: Apply coatings over horizontal leg of trim and into recess.

### 770 Skirtings and upstands

1. Top edges of coatings: Where not protected by flashings, apply into chases cut to a minimum depth of 10 mm.
2. Completion of chases: When coatings are fully cured, prepare chase and apply sealant as section Z22.
    2.1. Sealant: To BS EN ISO 11600
        2.1.1. Colour: As coating

## Surfacing

### 810 Blinding

1. Applying dusting powder: To coating surfaces at end of curing period to neutralize tackiness.

### 830 Laying inverted roof insulation

1. Condition of substrate: Clean.
2. Separating layer: Lay polyethylene sheet under insulation where required by coating manufacturer.
3. Setting out: Loose-lay with staggered joints. Minimize cutting and avoid small pieces at perimeters and penetrations.
    3.1. Joints: Butt together.
4. Projections, upstands, rainwater outlets, etc: Cut insulation cleanly and fit closely around.
5. Completion: Boards in good condition, well-fitting and stable.

### 840 Laying stone ballast

1. Condition of substrate: Clean.
2. Gravel guards: Fit to outlets.
3. Laying: Spread evenly. Do not pile to excessive heights.
    3.1. Depth (minimum): 50 mm

### 850 Laying precast concrete paving slabs

1. Condition of substrate: Clean.
2. Setting out: Minimize cutting.
3. Joints: Open.
    3.1. Width: 3 mm
4. Perimeter upstands: 300 mm wide, filled with ballast
5. Completion: Slabs must be level and stable.

### 860 Laying paving tiles

1. Condition of substrate: Clean.
2. Setting out: Minimize cutting.
3. Joints: Open.
    3.1. Width: 3 mm
4. Perimeter upstands: 300 mm wide, filled with ballast
5. Completion: Tiles must be level and stable.

### 880 Applying UV-resistant sealer

1. Coverage per coat (minimum): 1.2 L/m²
2. Surface coverage: Even and full.

### 890 Applying walkway protective coating

1. Coverage per coat (minimum): 1.5 L/m²
2. Surface coverage: Even and full.

## Completion

### 910 Inspection

1. Coating surfaces: Check when cured for discontinuities.
    1.1. Defective areas: Apply another coating.

### 920 Electronic roof integrity test

1. Testing authority: The roofing contractor
2. Timing of test: Prior to, and on completion of, access by other trades
3. Condition of roof prior to testing
    3.1. Coating: Complete to a stage where integrity can be tested.
    3.2. Surface: Clean.
4. Test results: Submit.



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5. Waterproof integrity certificate: Submit, on completion of testing.

## 930 Flood test

1. Condition of roof prior to testing
   1.1. Coating: Complete to a stage where integrity can be tested.
   1.2. Surface: Clean.
2. Outlets: Externally cover and seal. Protect against damage from water pressure using temporary kerbs. Do not use plugs to seal outlets.
3. Flood levels: Submit proposals. In no case higher than existing kerbs.
4. Flood duration: Two days
5. Inspection: Regular to detect leaks.
6. Completion of test: Slowly drain roof. Do not overload or flood outlets.
7. Test results: Submit.

## 940 Completion

1. Roof areas: Clean.
   1.1. Outlets: Clear.
   1.2. Flashings: Dressed into place.
2. Work necessary to provide a weathertight finish: Complete.
3. Storage of materials on finished surface: Not permitted.
4. Completed coatings: Protect against damage.

Ω End of Section



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# K10
# Gypsum Board Dry Linings/ Partitions/ Ceilings

## Types of dry lining

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 IWS-101/ LIN-101/ CLG-101 Gypsum Board Partition/ Linings/ Ceiling Systems (Generally)

1. Description: Metal stud plasterboard partitions/ wall linings/ ceiling types with acoustic slab insulation.
2. Manufacturer: British Gypsum, or acceptable equivalent.
3. Specific performance:
    3.1. Fire rating: Refer to the Fire Strategy Report
    3.2. Acoustic rating: Refer to the Acoustic Report
    3.3. Duty rating: Severe.
    3.4. Moisture resistance: To be confirmed.
4. Types: Refer to the Design Drawings for partitions/ wall linings/ ceiling types and locations.
5. Build-ups, construction, finishes, sizes and zone: Refer to the Design Drawings
6. Support framing: Varies, as per the manufacturer's recommendations to achieve the stipulated performance requirements.
7. Insulation: Refer to the Design Drawings
8. Linings: Refer to the Design Drawings
9. Patressing: Plywood pattress, timber noggings and fixing plates to be provided where necessary to support surface-mounted fixtures and fittings.
10. Finishing:
    10.1. Unless specified, or indicated otherwise, outer lining plasterboard shall be square edged, scrim taped and finished with plaster skim suitable to receive applied finishes.
    10.2. Generally, skim plaster shall not be applied to moisture resistant plasterboard in areas subject to high humidity.
    10.3. Flush jointing (tape and fill) needs to be undertaken in line with the manufacturer's recommendations and shall only be used if samples of the joints have been accepted by the Employer's Agent.
11. Other requirements: System shall include all studs/ framing, insulation, linings, fixings and accessories necessary to complete the installation.

### 119 IWS-181 Metal Stud Partition System - Shaftwall

1. Description: Metal stud plasterboard partition.
2. Manufacturer: British Gypsum, or acceptable equivalent.
3. Specific performance:
    3.1. Fire rating: Refer to the Fire Strategy Report

3.2. Acoustic rating: Refer to the Acoustic Strategy Report

3.3. Duty rating: Severe.

3.4. Moisture resistance: To be agreed with the Architect.

4. Partition height: Refer to the Design Drawings
5. Support framing: 1 No. row of Gypframe 70 S 50 metal 'C' studs at max 600mm centres with suitable retaining channel and framework.
6. Head condition:

6.1. Deflection allowance: +/- 25mm to be confirmed by the Structural Engineer.

7. Insulation (if required):

7.1. Manufacturer: Saint Gobain - Isover, or acceptable equivalent.

7.2. Product reference: Acoustic partition roll (APR 1200).

7.3. Thickness: Refer to the Design Drawings

8. Linings:

8.1. Shaft side: 1 No. layer of 19mm thick Gyproc 'CoreBoard'.

8.2. Room side:

8.2.1. Lining 1: 3 No. layers of 15mm thick Gyproc 'FireLine' plasterboard.

8.2.2. Lining 2: 1 No. layer of 12.5mm thick Gyproc 'WallBoard' plasterboard (sacrificial).

9. Finishing: Skim coat plaster.
10. Other requirements: System shall include all studs/ framing, insulation, linings, fixings and accessories necessary to complete the installation.

## 220 CLG-151 Mass Barrier Ceiling

1. Description: Suspended monolithic plasterboard ceiling lining fixed to suspended metal support system.
2. Standard: To BS EN 13964.
3. Evidence of compliance:: Submit Declaration of Performance (DoP).
4. Specific performance

4.1. Fire rating: Refer to the Fire Strategy Report

4.2. Acoustic rating: Refer to the Acoustic Strategy Report

5. Support framing: Suspended metal support framing.
6. Insulation: As required to meet acoustic and fire performance.

6.1. Thickness: As required to meet acoustic and fire performance.

7. Linings: 2 no layers of 12.5mm standard density plasterboard on isolated hangers.

7.1. Hanger manufacturer: British Gypsum, or acceptable equivalent.

7.1.1. Hanger product reference: GAH1.

7.2. Mass per unit: Circa 30kg/m²

8. Finishing: Skim coat plaster.
9. Perimeter trim/ shadow gap: Refer to the Design Drawings
10. Accessories/ Other requirements: System to include all supports, fixings, clips and all other associated accessories required to complete the installation.

## 265 IDS-521 Internal Ceiling Access Panel

1. Description: Plasterboard ceiling access panel.

2. Manufacturer: Access Panels Direct, or acceptable equivalent
3. Specific performance
   3.1. To maintain the surrounding wall system's stipulated performance criteria.
4. Frame: Flush finished beaded frame, to receive final finish to match surrounding ceiling system.
5. Hardware/ Ironmongery: Budget/ high security lock and hinges.

## General/ preparation

### 305 Compliance with performance requirements

1. Testing/ Assessment: Submit UKAS accredited laboratory reports for the following: Fire resistance: Partitions (including deflection heads and doorsets) and suspended ceilings (including access units)..
2. Materials, components and details: As used in testing/ assessment reports. If discrepancies arise, give notice.
3. Linings and Partitions to consist of complete integrated systems and maintain performance from floor slab to soffit.

### 325 Preparation of masonry to receive wall linings

1. General: Suitable to receive lining system. Redundant fixtures and services removed. Cutting, chasing and making good completed.
2. Holes, gaps, service penetrations, perimeter junctions and around openings: Seal.
3. Adhesive fixings: Prepare substrate to achieve effective bonding.
   3.1. Contaminants: Remove loose material, dirt, grease, oil, paper, etc.
   3.2. Absorption: Control by dampening, priming or applying bonding agents as necessary.

### 335 Additional supports

1. Framing: Accurately position and securely fix to give full support to:
   1.1. Partition heads running parallel with, but offset from main structural supports.
   1.2. Fixtures, fittings and service outlets. Mark framing positions clearly and accurately on linings.
   1.3. Board edges and lining perimeters, as recommended by board manufacturer to suit type and performance of lining.
2. Backing board:
   2.1. Use proprietary tile backing board as outer board where the system will receive a weight >= 32kg/m².
   2.2. Performance of the overall system to be maintained.

### 375 New wet laid bases

1. Dpcs: Install under full width of partitions/ freestanding wall linings.
   1.1. Material: Bituminous sheet or plastics.

### 390 Samples

1. General: Before placing orders submit representative samples of the following:
   1.1. Plasterboard: 500mm x 500mm of each.
   1.2. Framing: 500mm length.

1.3. Access panels/ grilles.

2. Delivered materials/ products: To match samples.

## 395 Control samples

1. General: Complete areas of finished work and obtain approval of appearance before proceeding.
   1.1. First structural bay of each type of partition system.
   1.2. First structural bay of each type of dry lining system.
   1.3. First structural bay of each type of ceiling system.

## 398 Performance and Testing

1. General:
   1.1. BS 5234 - Partitions.
2. Airtight Shafts and Stairs:
   2.1. Provide penetration seals as necessary to achieve the required performance.
   2.2. Leakage Rates:
      2.2.1. Builders work shafts: 0.23 litres/m2/second at 160 Pascals of differential pressure.
      2.2.2. Pressurisation shafts: 0.34 litres/m2/second at 50 Pascals of differential pressure.
3. Testing: To be carried out periodically during construction and upon completion of each shaft.

# Components

## 400 Gypsum boards generally

1. Standard
   1.1. Gypsum plasterboard to BS EN 520.
   1.2. Gypsum fibre board to BS EN 15283-2.
   1.3. Evidence of compliance: All sheets to be CE marked. Submit Declaration of Performance (DoP).
   1.4. BS 8212 - Code of practice for dry lining and partitioning using gypsum plasterboard.

## 401 Gypsum plasterboard

1. Type: To BS EN 520, type A
2. Core density (minimum): 650 kg/m³.
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 520 characteristics: None
7. Recycled content: Submit proposals
8. Exposed surface and edge profiles: Suitable to receive specified finish

## 402 Gypsum plasterboard (vapour control)

1. Type: To BS EN 520, type A
2. Core density (minimum): 650 kg/m³.
3. Reaction to fire: Class A2-s1, d0 or better

4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 520 characteristics: None
7. Recycled content: Submit proposals
8. Moisture vapour resistance of backing layer (minimum): 60 MNs/g.
9. Exposed surface and edge profiles: Suitable to receive specified finish

## 403 Gypsum plasterboard (moisture-resistant)

1. Type: To BS EN 520, type H1
2. Core: Moisture-resistant.
   2.1. Density (minimum): 710 kg/m³.
3. Paper facings: Moisture-resistant.
4. Reaction to fire: Class A2-s1, d0 or better
5. Water vapour resistance factor: Manufacturer's standard
6. Thermal conductivity: Manufacturer's standard
7. Other BS EN 520 characteristics: None
8. Recycled content: Submit proposals
9. Exposed surface and edge profiles: Suitable to receive specified finish

## 404 Gypsum plasterboard (improved fire protection)

1. Type: To BS EN 520, type F
2. Core: Including fibres and/ or other additives for improved cohesion.
   2.1. Density (minimum): 800 kg/m³.
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 520 characteristics: None
7. Recycled content: Submit proposals
8. Exposed surface and edge profiles: Suitable to receive specified finish

## 406 Gypsum plasterboard (improved fire protection and moisture-resistant)

1. Type: To BS EN 520, type H1 and F
2. Core: Moisture-resistant and including fibres and/ or other additives for improved cohesion.
   2.1. Density (minimum): 800 kg/m³.
3. Paper facings: Moisture-resistant.
4. Reaction to fire: Class A2-s1, d0 or better
5. Water vapour resistance factor: Manufacturer's standard
6. Thermal conductivity: Manufacturer's standard
7. Other BS EN 520 characteristics: None
8. Recycled content: Submit proposals
9. Exposed surface and edge profiles: Suitable to receive specified finish

## 407 Gypsum plasterboard (improved fire protection and vapour control)

1. Type: To BS EN 520, type F
2. Core density (minimum): 800 kg/m³.
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 520 characteristics: None
7. Recycled content: Submit proposals
8. Moisture vapour resistance of backing layer (minimum): 60 MNs/g.
9. Exposed surface and edge profiles: Suitable to receive specified finish

## 408 Gypsum plasterboard (impact-resistant)

1. Type: To BS EN 520, type I
2. Core density (minimum): 900 kg/m³.
3. Paper facings: Heavy duty.
4. Reaction to fire: Class A2-s1, d0 or better
5. Water vapour resistance factor: Manufacturer's standard
6. Thermal conductivity: Manufacturer's standard
7. Other BS EN 520 characteristics: None
8. Recycled content: Submit proposals
9. Exposed surface and edge profiles: Suitable to receive specified finish

## 409 Gypsum plasterboard (improved sound insulation)

1. Type: To BS EN 520, type D
2. Core density (minimum): 820 kg/m³.
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 520 characteristics: None
7. Recycled content: Submit proposals
8. Exposed surface and edge profiles: Suitable to receive specified finish

## 410 Gypsum plasterboard (improved sound insulation and moisture-resistant)

1. Type: To BS EN 520, type H1
2. Core: Moisture-resistant.
   2.1. Density (minimum): 820 kg/m³.
3. Paper facings: Moisture-resistant.
4. Reaction to fire: Class A2-s1, d0 or better
5. Water vapour resistance factor: Manufacturer's standard
6. Thermal conductivity: Manufacturer's standard
7. Other BS EN 520 characteristics: None
8. Recycled content: Submit proposals

9. Exposed surface and edge profiles: Suitable to receive specified finish

## 415 Gypsum plasterboard

1. Type: To BS EN 520, type R
2. Reaction to fire: Class A2-s1, d0 or better
3. Water vapour resistance factor: Manufacturer's standard
4. Thermal conductivity: Manufacturer's standard
5. Other BS EN 520 characteristics: None
6. Recycled content: Submit proposals
7. Exposed surface and edge profiles: Suitable to receive specified finish

## 420 Gypsum fibre board

1. Type: To BS EN 15283-2, type GF
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

## 421 Gypsum fibre board (enhanced density)

1. Type: To BS EN 15283-2, type GF-D
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

## 422 Gypsum fibre board (reduced water absorption)

1. Type: To BS EN 15283-2, type GF-H
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

## 423 Gypsum fibre board (enhanced surface hardness)

1. Type: To BS EN 15283-2, type GF-I
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard



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5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

### 424 Gypsum fibre board (enhanced strength)

1. Type: To BS EN 15283-2, type GF-R1
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

### 425 Gypsum fibre board (reduced surface water absorption)

1. Type: To BS EN 15283-2, type GF-W1
2. Thickness: 12.5 mm
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

### 426 Gypsum fibre board

1. Thickness: 12.5 mm
2. Paper facings: Moisture-resistant.
3. Reaction to fire: Class A2-s1, d0 or better
4. Water vapour resistance factor: Manufacturer's standard
5. Thermal conductivity: Manufacturer's standard
6. Other BS EN 15283-2 characteristics: None
7. Recycled content: Submit proposals

### 432 Metal studs

1. Compliance: To BS7364, BS EN 10162, BS EN 10143.
2. Requirements: Minimum 0.55mm hot-dip zinc/ iron zinc alloy coated steel.

## Installation

### 435 Dry linings generally

1. General: Use fixing, jointing, sealing and finishing materials, components and installation methods recommended by board manufacturer.
2. Cutting gypsum boards: Neatly and accurately without damaging core or tearing paper facing.
    2.1. Cut edges: Minimize and position at internal angles wherever possible. Mask with bound edges of adjacent boards at external corners.
3. Fixings boards: Securely and firmly to suitably prepared and accurately levelled backgrounds.

4. Finishing: Neatly to give flush, smooth, flat surfaces free from bowing and abrupt changes of level.
5. Wet or Humid conditions: : Outer layer of gypsum board to be moisture resistant with appropriate performance characteristics to suit the system.

## 445 Ceilings

1. Sequence: Fix boards to ceilings before installing dry lined walls and partitions.
2. Orientation of boards: Fix with bound edges at right angles to supports and with ends staggered in adjacent rows.
3. Two layer boarding: Stagger joints between layers.

## 455 Metal framing for partitions/ wall linings

1. Setting out: Accurately aligned and plumb.
    1.1. Frame/ Stud positions: Equal centres to suit specified linings, maintaining sequence across openings.
    1.2. Additional studs: To support vertical edges of boards.
2. Fixing centres at perimeters (maximum): 600 mm.
3. Fixings: Securely back to suitable structure accommodating all deflections.
4. Openings: Form and seal accurately with a water based acrylic mastic fire seal without detriment to the appearance or performance of the system.
    4.1. Doorsets: Use sleeved or boxed metal studs and/ or suitable timber framing to achieve strength grade requirements for framing assembly and adequately support weight of door.
    4.2. Services penetrations: Allow for associated fire-stopping.

## 465 Staggered stud partitions

1. Horizontal frame members (noggins, bearers, etc.) and boards: Fix between alternate studs and not touching adjacent offset studs.

## 475 Metal furrings for wall linings

1. Setting out: Accurately aligned and plumb.
    1.1. Vertical furring positions: Equal vertical centres to suit specified linings, maintaining sequence across openings. Position adjacent to angles and openings.
    1.2. Additional vertical furrings: To support vertical edges of boards and at junctions with partitions.
    1.3. Horizontal furring positions: To provide continuous support to edges of boards.
2. Adhesive bedding to furrings
    2.1. Dabs: Length 200 mm (minimum). Located at ends of furrings and thereafter at 450 mm (maximum) centres.
    2.2. Junctions with partitions: Continuous bed with no gaps across cavity.

## 485 Suspended ceiling grids

1. Setting out: Accurately aligned and level.
    1.1. Grid members and hangers: Centres to suit specified linings and imposed loads.
        1.1.1. Centres to suit specified linings and imposed loads.

1.1.2. Final locations to be agreed with Services Engineer and appointed sub-contractor.

1.2. Additional grid members: Provide bracing and stiffening at upstands, partition heads, access hatches, etc.

2. Fixing: Securely at perimeters, grid joints, top and bottom hanger fixings.
3. Compliance:: BS EN 13964 - Suspended ceilings.

## 505 Installing mineral wool insulation

1. Fitting insulation: Closely butted joints and no gaps. Use fasteners to prevent slumping or displacement.
2. Services
   2.1. Electrical cables overlaid by insulation: Sized accordingly.
   2.2. Ceilings: Cut insulation around electrical fittings, etc.

## 510 Sealing gaps and air paths

1. Location of sealant: To perimeter abutments and around openings.
   1.1. Pressurized shafts and ducts: At board-to-board and board-to-metal frame junctions.
2. Application: To clean, dry and dust free surfaces as a continuous bead with no gaps.
   2.1. Gaps greater than 6 mm between floor and underside of gypsum board: After sealing, fill with jointing compound.

## 530 Cavity fire barriers within partitions/ wall linings

1. Metal framed systems
   1.1. Material: Plasterboard 12.5 mm thick
   1.2. Installation: Form accurately and fix securely with no gaps to provide a complete barrier to smoke and flame.
2. Adhesive fixed wall lining systems
   2.1. Material: Adhesive compound.
   2.2. Installation: Form in a continuous line with no gaps to provide a complete barrier to smoke and flame.

## 545 Cavity fire barriers within suspended ceilings

1. Type: As recommended by board manufacturer to meet specified performance
2. Fire resistance: To BS EN 13501-2, EI 60 or better
3. Ceiling void subdivision: Fix barriers as drawings.
4. Fixing at perimeters and joints: Secure, stable and continuous with no gaps, to provide a complete barrier to smoke and flame.
5. Service penetrations: Cut and pack to maintain barrier integrity. Sleeve flexible materials. Adequately support services passing through barrier.
6. Ceiling systems for fire protection: Do not impair fire-resisting performance of ceiling system.

## 555 Fire-stopping at perimeters of dry lining systems

1. Material: Tightly packed mineral wool or intumescent mastic/ sealant.
2. Application: To perimeter abutments to provide a complete barrier to smoke and flame.



make

## 556 Fire-stopping and barriers generally

1. Comply with BS EN 1364 and BS EN 1366.
2. Fire/acoustic barriers, seals, strips: To be non-visible and appropriate at all conditions whilst maintaining performance.

## 560 Joints between boards

1. Tapered edged gypsum boards
    1.1. Bound edges: Lightly butted.
    1.2. Cut/ unbound edges: 3 mm gap.
2. Square edged plasterboards: 3 mm gap.
3. Square edged gypsum fibre boards: 5 mm gap.
4. Movement Joints: Invisible.

## 565 Vertical joints

1. Joints: Centre on studs.
    1.1. Partitions: Stagger joints on opposite sides of studs.
    1.2. Two layer boarding: Stagger joints between layers.

## 570 Horizontal joints

1. Surfaces exposed to view: Horizontal joints not permitted. Seek instructions where height of partition/ lining exceeds maximum available length of board.
2. Two layer boarding: Stagger joints between layers by at least 600 mm.
3. Edges of boards: Support using additional framing.
    3.1. Two layer boarding: Support edges of outer layer.

## 575 Plank plasterboard

1. First layer in two layer boarding: Square edged with long edges at right angles to studs.

## 580 Insulation backed plasterboard

1. General: Do not damage or cut away insulation to accommodate services.
2. Installation at corners: Carefully cut back insulation or plasterboard as appropriate along edges of boards to give a continuous plasterboard face, with no gaps in insulation.

## 590 Fixing gypsum board to metal framing/ Furrings

1. Partitions/ Wall linings: Fix securely and firmly at the following centres (maximum):
    1.1. Single layer boarding: To all framing at 300 mm centres. Reduce to 200 mm centres at external angles.
    1.2. Multi-layer boarding: Face layer at 300 mm centres, and previous layers around perimeters at 300 mm centres.
2. Ceilings: 230 mm. Reduce to 150 mm at board ends and at lining perimeters.
3. Position of screws from edges of boards (minimum): 10 mm.
    3.1. Screw heads: Set in a depression. Do not break paper or gypsum core.

## 592 Fixing insulation backed plasterboard to metal furrings

1. Fixing to furrings: In addition to screw fixings apply continuous beads of adhesive sealant to furrings.

## 595 Deflection heads

1. Fixing boards: Do not fix to head channels.
2. Provide:: Deflection details of all heads and abutments to suit deflection calculations and service conditions.
3. Visibility:: All works to be concealed.
4. Documentation:: Refer to the Structural Engineer's Information

## 610 Fixing gypsum board to timber

1. Fixing to timber: Securely at the following centres (maximum):
    1.1. Nails: 150 mm.
    1.2. Screws to partitions/ wall linings: 300 mm. Reduce to 200 mm at external angles.
    1.3. Screws to ceilings: 230 mm.
2. Position of nails/ screws from edges of boards (minimum)
    2.1. Bound edges: 10 mm.
    2.2. Cut/ unbound edges: 13 mm.
3. Position of nails/ screws from edges of timber supports (minimum): 6 mm.

## 620 Fixing gypsum board with adhesive dabs

1. Setting out boards: Accurately aligned and plumb.
2. Fixing to substrates: Securely using adhesive dabs.
3. Adhesive dab spacings for each board
    3.1. Horizontally: One row along top edge and one continuous dab along bottom edge.
    3.2. Vertically: One row along each edge and thereafter at intermediate spacings to suit size of board:
4. Thickness (mm) Width (mm) Dab centres (mm)
5. 9.5 1200 400
6. 9.5/12.5 900 450
7. 12.5 1200 600
8. Adhesive dab dimensions (width x length): At least 50-75 mm x 250 mm.
    8.1. Position of dabs from edges/ ends of boards (minimum): 25 mm.

## 625 Fixing insulation backed plasterboard with adhesive dabs

1. Fixing to substrates: In addition to adhesive dab fixings, secure boards with nailable plugs in locations recommended by board manufacturer.

## 630 Fixing insulation backed plasterboard with adhesive spots

1. Setting out boards: Accurately aligned and plumb.
2. Fixing to substrates: Securely using adhesive spots and mechanical fastenings.
3. Adhesive spot spacings to each board: Four vertical rows, at 400 mm centres in each row.

4. Adhesive spot diameters (minimum): 25 mm.
5. Mechanical fasteners: Nailable plugs in locations recommended by board manufacturer.

## 640 Miscellaneous

1. Sockets/ Outlets/ Data boxes/ Switches:
   1.1. Do not locate back to back.
   1.2. Detailing as per manufacturer's recommendations.
   1.3. Include necessary acoustic and putty pads where required.
2. Plates/ Patreses/ Noggings: Integrate within the system as required to support fixtures and fittings.
3. Loadings: Where partition / loading will receive wall tiling with a weight greater than 31kg/m2, tile backer boards to be used.
4. Access Panels: Included within systems whilst maintaining performance in locations agreed.

# Finishing

## 650 Level of dry lining across joints

1. Sudden irregularities: Not permitted.
2. Joint deviations: Measure from faces of adjacent boards using methods and straightedges (450 mm long with feet/ pads) to BS 8212, clause 3.3.5.
   2.1. Tapered edge joints
      2.1.1. Permissible deviation (maximum) across joints when measured with feet resting on boards: 3 mm.
   2.2. External angles
      2.2.1. Permissible deviation (maximum) for both faces: 4 mm.
   2.3. Internal angles
      2.3.1. Permissible deviation (maximum) for both faces: 5 mm.

## 670 Seamless jointing to gypsum boards

1. Cut edges of boards: Lightly sand to remove paper burrs.
2. Filling and taping: Fill joints, gaps and internal angles with jointing compound and cover with continuous lengths of paper tape, fully bedded.
3. Protection of edges/ corners: Reinforce external angles, stop ends, etc. with specified edge/ angle bead.
4. Finishing: Apply jointing compound. Feather out each application beyond previous application to give a flush, smooth, seamless surface.
5. Nail/ screw depressions: Fill with jointing compound to give a flush surface.
6. Minor imperfections: Remove by light sanding.

## 680 Skim coat plaster finish

1. Plaster type As recommended by board manufacturer.
   1.1. Thickness: 2-3 mm.
2. Joints: Fill and tape except where coincident with metal beads.
3. Finish: Tight, matt, smooth surface with no hollows, abrupt changes of level or trowel marks.
   3.1. Tight, matt, smooth surface with no hollows, abrupt changes of level or trowel marks.

3.2. Moisture resistant plasterboard, or tile backer boards are not to be plastered unless stated otherwise.

## 692 Rigid beads/stops

1. Internal: To BS EN 13658-1.
2. External: To BS EN 13658-2.

## 695 Installing beads/ Stops

1. Cutting: Neatly using mitres at return angles.
2. Fixing: Securely using longest possible lengths, plumb, square and true to line and level, ensuring full contact of wings with substrate.
3. Finishing: After joint compounds/ plasters have been applied, remove surplus material while still wet from surfaces of beads exposed to view.

## 725 Repairs to existing gypsum board

1. Performance of repairs must match original specified performances.
2. Filling small areas with broken cores: Cut away paper facing, remove loose core material and fill with jointing compound.
   2.1. Finish: Flush, smooth surface suitable for redecoration.
3. Large patch repairs: Cut out damaged area and form neat hole with rectangular sides. Replace with matching gypsum board.
   3.1. Fixing: Use methods to suit type of dry lining, ensuring full support to all edges of existing and new gypsum board.
   3.2. Finishing: Fill joints, tape and apply jointing compound to give a flush, smooth surface suitable for redecoration.

Ω End of Section



make

# K13
# Rigid Sheet Fine Linings and Panelling

## Types of lining and panelling

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 105 Wood-based sheets generally

1. Standard: To BS EN 13986.
    1.1. Evidence of compliance: All sheets to be UKCA/ UKNI/ CE marked. Submit Declaration of Performance (DoP).

### 141 LIN-106 Bronze Cladding - Lift Doors

1. Description: Internal bronze cladding
2. Sizes and configurations: Refer to the Design Drawings
3. Substrate: Refer to the Design Drawings
4. Support framing (if required): Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
5. Panels
    5.1. Type/ material: Bronze
    5.2. Thickness: To be confirmed
    5.3. Edge detail: 'V' cut, bent to a tight radius forming returns.
    5.4. Finish: Brushed
    5.5. Fabrication: As section Z12.
6. Joints: Butt jointed, Refer to the Design Drawings
7. Fixing: Concealed
8. Feature lighting (if required): Refer to the Services Engineer's documentation
9. System accessories: As required to complete the Works, to the manufacturer's recommendations.

### 142 LIN-107 Bronze Cladding - Lift Opening

1. Description: Internal bronze cladding
2. Sizes and configurations: Refer to the Design Drawings
3. Substrate: Refer to the Design Drawings
4. Support framing (if required): Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
5. Panels
    5.1. Type/ material: Bronze
    5.2. Thickness: To be confirmed
    5.3. Edge detail: 'V' cut, bent to a tight radius forming returns.

5.4. Finish: Brushed

5.5. Fabrication: As section Z12.

6. Joints: Butt jointed, Refer to the Design Drawings
7. Fixing: Concealed
8. Feature lighting (if required): Integrated strip lighting. Refer to the Services Engineer's documentation
9. Signage: Include engraved signage.
10. System accessories: As required to complete the Works, to the manufacturer's recommendations.

## 146 LIN-171 Wall Protection - Metal

1. Description: Internal metal wall protection
2. Sizes and configurations: Refer to the Design Drawings
3. Substrate: Refer to the Design Drawings
4. Support framing (if required): Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
5. Panels
   5.1. Manufacturer: Rimex, or acceptable equivalent.
      5.1.1. Product reference: ColourTex Bronze PinStripe.
   5.2. Type/ material: Stainless steel
   5.3. Thickness: 20-30 mm deep
   5.4. Edge detail: 'V' cut, bent to a tight radius forming returns.
   5.5. Finish: Polished
   5.6. Fabrication: As section Z12.
6. Joints: Butt jointed, Refer to the Design Drawings
7. Fixing: Concealed
8. System accessories: As required to complete the Works, to the manufacturer's recommendations.

## 151 LIN-236 Timber Veneered Feature Walls

1. Description: Timber veneered feature wall panels featuring curved blocks/ vertical timber battens and integrated precision joinery
2. Sizes and configuration: Refer to the Design Drawings
3. Substrate: Refer to the Design Drawings
4. Support framing: Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
5. Panels
   5.1. Core material: Medium-density fibreboard to BS EN 622-5, type MDF
      5.1.1. Thickness: 100 mm.
      5.1.2. Flame-retardant impregnation treatment: To BS EN 13501-1, Class B-s1, d0 or better
   5.2. Face veneer: Free from decay and holes.
      5.2.1. Wood species: English Oak.
      5.2.2. Cutting: Crown-cut.

5.2.3. Arrangement: Book-matched.

5.3. Backing veneer: As recommended by fabricator.

5.4. Edge treatment: Veneered strip.

5.5. Fabrication: As section Z10.

5.5.1. Adhesive: To BS EN 204; Type: To match durability class of core material.

5.6. Finish (to match approved sample): Two coats matt finish lacquer.

5.7. Moisture content at time of fixing: As recommended by fabricator to suit environmental conditions.

6. Joints: Refer to the Design Drawings
7. Fixing: Concealed
8. Fire and smoke stopping: As necessary, and concealed within the finished works.
9. System accessories: As required to complete the Works, to the manufacturer's recommendations.

## 152 LIN-237 Timber Veneered Wall Lining

1. Description: Solid hardwood timber panel wall lining, as shown on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Manufacturer: Deeside, or acceptable equivalent.
4. Substrate: Refer to the Design Drawings
5. Support framing: Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
6. Panels
   6.1. Core material: Medium-density fibreboard to BS EN 622-5, type MDF
      6.1.1. Thickness: 15 mm
      6.1.2. Flame-retardant impregnation treatment: To BS EN 13501-1, Class B-s1, d0 or better
   6.2. Face veneer: Free from decay and holes.
      6.2.1. Wood species: European Oak
      6.2.2. Appearance: Dark timber, with fine grain and no repeating grain.
      6.2.3. Quality: J2 to BS EN 942 or CSH to BS 1186.
      6.2.4. Cutting: Crown-cut.
      6.2.5. Arrangement: Book-matched.
   6.3. Backing veneer: As recommended by fabricator.
   6.4. Edge treatment: Veneered strip.
   6.5. Fabrication: As section Z10.
      6.5.1. Adhesive: To BS EN 204; Type: To match durability class of core material.
   6.6. Finish (to match approved sample): Oiled and with clear protective matt lacquer with fire resisting attributes.
   6.7. Moisture content at time of fixing: As recommended by fabricator to suit environmental conditions.
7. Joints: Refer to the Design Drawings
8. Samples: To be accepted by the Employer's Agent through range samples.
9. Skirting: Flush veneer skirting at base of lining with shadow gap.

10. Fixing: Concealed
11. Fire and smoke stopping: As necessary, and concealed within the finished works.
12. System accessories: As required to complete the Works, to the manufacturer's recommendations.

## 191 IWS-901 Internal Green Wall

1. Description: Internal preserved moss feature wall panels
2. Manufacturer: Vantage Spaces, or acceptable equivalent
3. Sizes and configuration: Refer to the Design Drawings
4. Substrate: Refer to the Design Drawings
5. Support framing: Corrosion resistant metal or timber support framing, mechanically secured to substrate and concealed in the finished works.
6. Panels
   - 6.1. Core material: Medium-density fibreboard to BS EN 622-5, type MDF
     - 6.1.1. Thickness: 15 mm
     - 6.1.2. Flame-retardant impregnation treatment: To BS EN 13501-1, Class B-s1, d0 or better
   - 6.2. Preserved moss:
     - 6.2.1. Type/ material: Soft, sculptural bun moss
     - 6.2.2. Arrangment: Refer to the Design Drawings
     - 6.2.3. Colour: Green
   - 6.3. Finish (to match approved sample): As supplied
7. Joints: Refer to the Design Drawings
8. Fixing: Concealed
9. System accessories: As required to complete the Works, to the manufacturer's recommendations.

## 236 LIN-646 GRG Panel Wall Lining

1. Description: GRG wall lining, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Fire performance: Refer to the Fire Strategy Report
4. Panels:
   - 4.1. Backing board: Suitable non-combustible core/ substrate board to suit the service conditions.
     - 4.1.1. Thickness: To suit the service conditions.
   - 4.2. Type/ material: GRG (Glass Reinforced Gypsum)
     - 4.2.1. Profile: Refer to the Design Drawings
     - 4.2.2. Texture: Refer to the Design Drawings
     - 4.2.3. Colour: Refer to the Design Drawings
     - 4.2.4. Fixing: Concealed fixed to the backing board as the Specialist Contractors recommendations to suit the service conditions.
5. Joints: Refer to the Design Drawings

## 241 LIN-651 Leather Panel Wall Lining

1. Description: Leather panel wall lining, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Fire performance: Refer to the Fire Strategy Report
4. Panels:
    4.1. Backing board: Suitable non-combustible core/ substrate board to suit the service conditions.
        4.1.1. Thickness: To suit the service conditions.
    4.2. Type/ material: Single piece Leather over suitable padded backing.
        4.2.1. Texture: Refer to the Design Drawings
        4.2.2. Colour: Refer to the Design Drawings
        4.2.3. Stitching: Double stitching with thread to match base colour of leather.
        4.2.4. Fixing: Padded leather to be suitably bonded/ concealed fixed to the backing board as the Specialist Contractors recommendations to suit the service conditions.
        4.2.5. To be constructed to allow for leather panel to be removed if removed but to be secure so that do not move after adjustment and alignment.
5. Details: Refer to the Design Drawings.
6. Joints: None. All panels to be full width as shown on the Design Drawings.
7. Fixing to joinery items: Suitable concealed fixings and support determined by the Specialist Contractor to suit the performance requirements and service conditions.
    7.1. Panels to be removable.
    7.2. All panels to be secure in place to avoid future movement or misalignment.

## 251 IWL-901 Durasteel Wall

1. Description: Proprietary fire rated wall system.
2. Manufacturer: Promat UK, or acceptable equivalent.
3. Standard: To BS EN 13964
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Report
    4.2. Acoustic rating: Refer to the Acoustic Report
5. Support structure:
    5.1. Type: Concealed metal grid support framing.
    5.2. Zone height: Refer to the Design Drawings
6. Sheet
    6.1. Product reference: DURASTEEL
    6.2. Type/ material: Composite panel of fibre-reinforced cement with mechanically bonded punched steel sheets on both surfaces.
    6.3. Size: 1200 mm (W) x 2500 mm (L).
    6.4. Shape: Rectangular
    6.5. Edge detail: Punched steel sheets
    6.6. Weight: 16.8 kg/m²
    6.7. Finish: Galvanised steel with two coats of paint.

7. System accessories: As required to complete the installation

## General requirements

### 210 Advance registration

1. Materials registered in advance by the employer: Obtain from supplier
    1.1. Ordering: Supersede employer's registration and take over responsibility by an order to the supplier covering price, supply and delivery to suit progress of the work.

### 220 Material samples

1. Representative samples of designated materials: Submit before placing orders.
2. Lining/ panelling: 300mm x 300mm of each type.
3. Demountable support system
4. Fixings and demountable supports.

### 230 Control samples

1. Sample area: Complete first finished full height or wall length of each type as part of the finished work.
2. Approval of appearance: Obtain before proceeding.

### 260 Environmental conditions

1. General requirements prior to starting work specified in this section: Building weathertight; wet trades completed and affected areas dried out.
2. Temperature and humidity before, during and after fixing lining/ panelling: Maintained at levels approximating to those which will prevail after building is occupied.

### 270 Heating system

1. Operating mode: Continuous
2. Design output: Refer to the Service Engineer's documentation
3. Operation up to Completion of the Works: Submit proposals.

### 280 Air conditioning system

1. Design output: Refer to the Service Engineer's documentation
2. Operation up to Completion of the Works: Submit proposals.

### 290 Testing

1. Prototypes: Verify the surface spread of flame performance of the works, to the satisfaction of the Building Inspector.

## Fabrication/ fixing/ finishing

### 310 Accuracy of fabrication

1. Site dimensions: Take as necessary before starting fabrication.
    1.1. Discrepancies with drawings: Report without delay and obtain instructions before proceeding.
2. Permissible deviations for panels

2.1. Length: ± 1.5 mm.

2.2. Width: ± 1.5 mm.

2.3. Squareness (taking the longer of 2 sides at a corner as a baseline and measuring the deviation of the shorter side from the baseline perpendicular): ± 1.5 mm in 1 m.

2.4. Flatness (of panels with a core thickness of 12 mm and over, as delivered to site): ± 1 mm under a 600 mm straightedge.

2.5. Maximum creep:: 1mm in 2.5m.

## 320 Laminated timber cores for wood-veneered panels

1. Face grain direction: Perpendicular to specified direction of veneer grain.
   1.1. Alternative arrangement: Panels cross-veneered before applying face veneer.

## 340 Holes/ cut-outs in laminated plastics veneers

1. Internal corners: Formed to a radius, minimum 5 mm when not specified otherwise.
2. Holes for fasteners: Formed slightly oversize.

## 350 Fixing linings and panelling

1. Setting out: Accurate, true to line and level, free from undulations and lipping, with lines and joints aligned, straight and parallel unless specified otherwise.
2. Movement allowance: Adequate for future moisture and temperature movement of boards.
3. Fixing of panels: Secure, to prevent pulling away, bowing, or other movement during use.
4. Methods of fixing and fasteners: As section Z20 unless specified otherwise.
5. Trims: Wherever possible, to be in unjointed lengths between angles or ends of runs.
   5.1. Running joints: Where unavoidable, submit proposals for location and method of jointing.
   5.2. Angle joints: Mitred, unless specified otherwise.
6. Linings to be from the same batch and match colour.

## 360 Open joints (joints without cover strips or similar)

1. General: Within a joint (including in-line continuations across transverse joints) greatest width must not exceed the least width.
2. Variations in width: Evenly distributed with no sudden changes. Joints with bevelled edges to be measured to the face arrises.

## 370 Sealant pointing

1. Sealant:
   1.1. Colour: To be confirmed
2. Application: As section Z22.

## 400 Door frames

1. Hardwood packing between frames and reveals: To give even joints of specified width.
   1.1. Position of packs: Where fixings tighten frame against structure.
2. Positions of frames: Accurate, plumb, level, and aligned.
   2.1. Fixing of frames: Secure to prevent pulling away, deflection, or other movement during use.

2.2. Fastener locations: 600 mm maximum centres with at least one fixing 150 mm from each end of jambs and one adjacent to each hanging point.

## 410 Fire-resisting frames

1. Gaps between frames and supporting construction: Filled as necessary in accordance with requirements for certification and/ or door/ doorset manufacturer's instructions.

## 430 Hanging of prefinished doors

1. Planing/ Sanding: Not permitted.

## 440 Ironmongery

1. Installation:
   1.1. Fasteners: Supplied by ironmongery manufacturer.
   1.2. Finish/ Corrosion resistance: To match ironmongery.
   1.3. Adjacent surfaces: Undamaged.
2. Moving parts: Adjusted, lubricated and functioning correctly at completion.

## 480 Clear finished wood

1. Nail holes: Filled with stopping coloured to match wood.
2. Prepared surface: Smooth, closed and free from sanding marks.
3. Finish: Smooth, free from brush marks, nibs, sags, runs and other defects.

Ω End of Section

# K20
# Timber Board Flooring

## To be read with preliminaries/ general conditions.

## 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

## 110 FLS-131 Battened Timber Sprung Flooring

1. Description: Hardwood sprung dance floor system.
2. Strips/ Boards
    2.1. Manufacturer/ Supplier: Junckers Ltd, or acceptable equivalent
        2.1.1. Product reference: Sprung solid hardwood dance floor systems (DuoBat 120+)
    2.2. System performance: To EN 14904, class A4.
    2.3. Vapour control layer: Junckers SylvaThene.
    2.4. Secondary supports or spacers: DuoBat 120+.
    2.5. Flooring
        2.5.1. Board type: SylvaBallet.
    2.6. System accessories: As required to complete the installation
    2.7. Appearance: Premium.
    2.8. Finish as delivered: UV light cured lacquer plus Junckers polyurethane Isolacquer - Silkmatt.
    2.9. Third-party certification: FSC, PEFC and Danish Indoor Climate certified, EN 15804 EPD.

## Workmanship

## 310 Workmanship generally

1. Protection during and after installation: Keep boards dry. Protect from dirt, stain and damage until completion.
2. Boards to be used internally: Do not install until building is watertight.
3. Methods of fixing, and fasteners: As section Z20 'Fixings and adhesives'.
4. Moisture content of timber supports at time of fixing boards: Not more than 18%.

## 320 Moisture content of new concrete/ screed substrates for floating floors

1. Test for moisture content
    1.1. Standard: To BS 8201, annex A, using an accurately calibrated hygrometer.
    1.2. Readings: Take in corners, along edges, and at random points over the area being tested.
2. Acceptability: Do not lay flooring until all readings show 75% relative humidity or less.



make

## 330 Moisture content of timber

1. Conditions during and after installation: Control ambient temperature and humidity conditions to maintain moisture content at average level specified in BS EN 942, Table B.1 for the relevant service condition until completion.
2. Test for moisture content: When instructed, using an approved moisture meter.

## 340 Installing vapour check membrane to floating floors

1. Location: Immediately below the floating layer.
2. Joints: Overlap by at least 150 mm and seal with vapour-resistant tape.
3. Perimeter/ upstands: Turn membrane up around perimeter of flooring and around any upstands and seal to top face of boards using: Polyethylene double-sided adhesive tape
   3.1. Excess material: Trim off neatly after fixing skirting boards/ cover beads.
4. Membrane condition: Intact, clean and dry prior to laying flooring.

## 345 Battens for floating floors

1. Quality of timber: Free from decay, insect attack (except ambrosia beetle damage) and with no knots wider than half the width of the section.
2. Surface finish: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
3. Treatment: Preservative impregnation.
   3.1. Standard: To section Z12 'Preservative/ flame-retardant treatment' and the Wood Protection Association (WPA) publication Code of Practice: Industrial Wood Preservation.
   3.2. Type: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
4. Moisture content at time of fixing boards: Not more than 16%.

## 347 Battens for floating floors

1. Quality of timber: Free from decay, insect attack (except ambrosia beetle damage) and with no knots wider than half the width of the section.
2. Surface finish: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
3. Moisture content at time of fixing boards: Not more than 16%.

## 350 Treated timber

1. Surfaces exposed by minor cutting and/ or drilling: Treat with two flood coats of a solution recommended by main treatment solution manufacturer.

## 360 Access panels

1. Size and position: Agree before fixing boards.
2. Additional noggings/ dwangs, battens, etc.: Provide as necessary.

## 370 Fixing boards

1. Environmental conditions: Do not fix boards when ambient temperature is at or below 0°C, or above 30°C.

2. Generally: Fix boards securely to each support to give flat, true surfaces free from undulations, lipping, splits and protruding fasteners.
3. Timber movement: Position boards and fixings to prevent cupping, springing, excessive opening of joints and other defects.
4. Heading joints: Tightly butted, central over supports and at least two board widths apart on any one support.
5. Edges: Plane off proud edges.
6. Exposed nail heads: Neatly punch below surface.

Ω End of Section



make

# K30
# Panel Partitions/ Glazed Screen Systems

## Timber requirements

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

## Types of partition

### 110 IWS-201 Aluminium Framed Glazed Screen System

1. Description: Internal metal-framed glazed screen with integrated doorset.
2. Manufacturer: Radii, or acceptable equivalent.
    2.1. Product reference: Fire Pro, Low Iron.
3. Sizes and configuration: Refer to the Design Drawings
4. Standard: To BS 5234-1
5. System performance:
    5.1. Fire rating: Refer to the Fire Strategy Report
    5.2. Acoustic rating: Refer to the Acoustic Strategy Report
6. Framing/ perimeters/ trims:
    6.1. Type/ material: Extruded aluminium profiles.
    6.2. Finish: Polyester powder coated.
    6.3. Colour: Standard RAL colour
    6.4. Texture: Matt
7. Glazed panels:
    7.1. Glass types: Low iron clear safety glass.
        7.1.1. Indicative manufacturer/ reference: AGC Pyrobel.
    7.2. Glazing build-up: To be confirmed by the Contractor to suit nominal zones, conditions of service and performance requirements.
    7.3. Manifestation: Refer to the Design Drawings
    7.4. Visible gaskets: Black
    7.5. Integral blinds: Manual blind tilt/ turn Venetian blinds with one blind control knob fixed to mullions per blind.
        7.5.1. Colour: To be agreed with the Architect.
8. Doorsets:
    8.1. Type: Internal framed glazed hinged doorset, integrated within the partition system.
    8.2. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
    8.3. Specific performance:
        8.3.1. Fire rating: To match surrounding partition system.
        8.3.2. Acoustic rating: To match surrounding partition system.
    8.4. Operation:

8.4.1. Type: Manual.

8.5. Door frame: To match surrounding partition system.

8.6. Door leaf:

8.6.1. Leaf primary and secondary framing, edging: To match the door frame.

8.6.2. Glazing: To match surrounding partition system.

8.6.3. Manifestation: Refer to the Design Drawings

8.6.4. Visible gaskets: To match surrounding partition system.

8.7. Meeting stiles (to door pairs): Ostensibly square with a subtle radiused profile, to suit the service conditions.

8.8. Ironmongery: To the acceptance of the Employer's Agent, by sampling and selected from the manufacturer's standard range.

8.9. Seals: Intumescent, smoke and acoustic seals to be of a type on which the manufacturer's certification is based.

8.10. Threshold: Low level

9. Fixing:

9.1. Concealed fixings: Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

10. Accessories: As required to complete the installation.

11. System to be designed and installed as complete integrated systems, including all necessary support structure, bracketry, fixing rails and plates, angles, cleats, fixings and fastenings, rivets, clips, firestops and cavity barriers, acoustic breaks, pressed metal components, closures, seals and sealants, gaskets, fillers, tapes, spacers, packers, shims, isolators and all other accessories and components necessary to complete the installation.

## 111 WIN-111 Steel Framed Glazed Window System

1. Description: Internal metal-framed glazed window
2. Manufacturer: Radii, or acceptable equivalent.

2.1. Product reference: FirePro, Low Iron.

3. Sizes and configuration: Refer to the Design Drawings
4. Standard: To BS 5234-1
5. System performance:

5.1. Fire rating: Refer to the Fire Strategy Report

5.2. Acoustic rating: Refer to the Acoustic Strategy Report

6. Framing/ perimeters/ trims:

6.1. Type/ material: Extruded steel profiles.

6.2. Finish: Polyester powder coated (Qualicoat class 2).

6.3. Colour: Standard RAL colour

6.4. Texture: Matt

7. Glazed panels:

7.1. Glass types: Low iron clear safety glass.

7.2. Glazing build-up: To be confirmed by the Contractor to suit nominal zones, conditions of service and performance requirements.

7.3. Indicative manufacturer/ reference: AGC Pyrobel.

7.4. Manifestation: Refer to the Design Drawings

7.5. Visible gaskets: Black

7.6. Integral blinds: Manual blind tilt/ turn Venetian blinds with one blind control knob fixed to mullions per blind.

7.6.1. Colour: To be agreed with the Architect.

8. Fixing:

8.1. Concealed fixings: Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

9. Accessories: As required to complete the installation.

10. System to be designed and installed as complete integrated systems, including all necessary support structure, bracketry, fixing rails and plates, angles, cleats, fixings and fastenings, rivets, clips, firestops and cavity barriers, acoustic breaks, pressed metal components, closures, seals and sealants, gaskets, fillers, tapes, spacers, packers, shims, isolators and all other accessories and components necessary to complete the installation.

## 121 IDS-701 Internal Sliding/ Folding Partition System

1. Description: Top hung, sliding/ folding partition system.
2. Manufacturer: Movawall, or acceptable equivalent.

2.1. Product reference: To be agreed with the Architect.

3. Sizes and configuration: Refer to the Design Drawings
4. System performance:

4.1. Fire rating: Refer to the Fire Strategy Report

4.2. Acoustic rating: Refer to the Acoustic Report

5. Height: To be agreed with the Architect.
6. Operation:

6.1. Type: To be agreed with the Architect.

7. Support framing:

7.1. Type/ mateiral: Concealed, heavy duty maintenance-free, mild steel trolley/ head track suspended from structural steel support framing fixed back to structural soffit.

7.2. Finish: Polyester powder coated, to match IDS-702.

7.3. Colour: Standard RAL colour to be selected, to match IDS-702.

7.4. Cutting void depth: Infilled with metal framed, insulated and boarded acoustic baffle.

7.5. Perimeter trim/ shadow gap: Refer to the Design Drawings

8. Panels:

8.1. Inner framing: Welded cross-braced metal framing.

8.2. Core: Fire-resistant sound absorbing insulation.

8.3. External facing: High density boards, isolated from the inner framing.

8.4. Thickness: To be agreed with the Architect.

8.5. Finish: Timber with patterened fabric inserts.

8.6. Colour: Warm brown and cream.

9. Panel-to-panel connection:

9.1. Type/ material: Interlocking concave/ convex extruded aluminium edge profiles with integral magnetic strip ensuring a fail-safe acoustic seal.

9.2. Finish: Satin anodised.

10. Other requirements:
    10.1. Appearance: Solid flush wall appearance when closed.
    10.2. Arrangement: Refer to the Design Drawings
11. Seals: Maintenance-free spring-loaded aluminium retractable acoustic seals to head and base, fitted with flexible sealing extrusions, manually operated with removable crank handle, to suit service conditions.
12. Ironmongery: All necessary ironmongery and hardware in satin finished stainless steel from the manufacturer's standard range to be included, as accepted by the Architect through sampling.
13. System to be designed and installed as complete integrated systems, including all fixings, framing, bracketry, jambs, trims, profiles, gaskets, extending pressure seals, counterweights, spacers and all other accessories and components necessary to complete the installation.

## 122 IDS-702 Internal Folding Partition Door System

1. Description: Bi-fold partition door system, as shown on the Design Drawings.
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Sizes and configuration: Refer to the Design Drawings
4. System performance:
    4.1. Fire rating: Refer to the Fire Strategy Report
    4.2. Acoustic rating: Refer to the Acoustic Report
5. Height: Varies. 6m and 2.7m.
6. Operation:
    6.1. Type: Powered.
7. Support framing:
    7.1. Type/ mateiral: Concealed, heavy duty maintenance-free, mild steel trolley/ head track suspended from structural steel support framing fixed back to structural soffit.
    7.2. Finish: Polyester powder coated
    7.3. Colour: Standard RAL colour to be selected
    7.4. Perimeter trim/ shadow gap: Refer to the Design Drawings
8. Panels:
    8.1. Inner framing: Welded cross-braced metal framing.
    8.2. Core: Fire-resistant sound absorbing insulation.
    8.3. External facing: High density boards, isolated from the inner framing.
    8.4. Thickness: To be agreed with the Architect.
    8.5. Finish: Timber with patterened fabric inserts.
    8.6. Colour: Warm brown and cream.
    8.7. Insets:
        8.7.1. Material: Fabric.
        8.7.2. Finish/ colour: To be agreed with the Architect.
9. Panel-to-panel connection:
    9.1. Type/ material: Interlocking concave/ convex extruded aluminium edge profiles with integral magnetic strip ensuring a fail-safe acoustic seal.
    9.2. Finish: Satin anodised.

10. Other requirements:
    10.1. Appearance: Solid flush wall appearance when closed.
    10.2. Arrangement: Leaves are to stacked to one side, Refer to the Design Drawings
11. Seals: Maintenance-free spring-loaded aluminium retractable acoustic seals to head and base, fitted with flexible sealing extrusions, manually operated with removable crank handle, to suit service conditions.
12. Ironmongery: All necessary ironmongery and hardware in satin finished stainless steel from the manufacturer's standard range to be included, as accepted by the Architect through sampling.
13. System to be designed and installed as complete integrated systems, including all fixings, framing, bracketry, jambs, trims, profiles, gaskets, extending pressure seals, counterweights, spacers and all other accessories and components necessary to complete the installation.

## Generally

### 310 Compliance with performance requirements

1. UKAS or equivalent-accredited laboratory test/ assessment reports: Submit for the following: Fire resistance including deflection heads, doorsets and over panels.
2. Materials, components and details: In accordance with test/ assessment reports. Give notice if discrepancies arise.

### 315 Acoustic field tests

1. Requirements: Refer to the Acoustic Strategy Report

### 320 Samples

1. Representative samples: Before placing orders, submit the following: ……..
   1.1.
   - 600mm x 600mm glazing, with coatings.
   - 600mm x 600mm framed glass panels.
   - 1000mm long fully finished framing member.
   - 1000mm long support framing.
2. Delivered materials/ products: To match samples.

### 330 Control samples

1. General: First complete bay of areas of finished work and obtain approval of appearance before proceeding.

### 340 Environmental conditions

1. General: Before, during and after installation, maintain temperature and humidity levels similar to those that will prevail after building is occupied.

### 375 New wet laid bases

1. Dpcs: Install under full width of partitions.
   1.1. Material: Bituminous sheet or plastics.

### 380 Coordination with others

1. Fire/ Sound barriers within floor and ceiling voids: Align accurately with partitions.

2. Service runs and outlets: Check designated routes. Do not obstruct outlets and access panels.

# Installation

## 410 Workmanship generally

1. Setting out: Plumb, true to line and level, and free from bowing, undulations and other planar distortions.
2. Joints: Align accurately with no lipping.
3. Stability: Fix securely, with additional supports where necessary at perimeters.
4. Moisture and thermal movement of framing/ panels: Make adequate allowance.

## 425 Installing relocatable partitions

1. Site dimensions: Check before installation. Give notice of discrepancies.
2. Perimeter abutments: Accommodate deviations whilst maintaining partition system performance. Form make up/ closer pieces accurately around projections and features.
3. Customization: Do not cut or otherwise alter framing/ panels on site, except where shown on drawings.
4. Intermediate joints in exposed frame members and trims: Keep to a minimum.
5. Electrical continuity and earth bonding: To BS 7671 (The IEE Wiring Regulations).

## 435 Installing plasterboard panel partitions

1. Timber or metal framing inserts: Size to give a firm press fit into the panel cavity. Use for:
    1.1. Panel joints.
    1.2. Floor, wall and ceiling perimeters.
    1.3. Support for fixtures and fittings: Mark positions of inserts on panel surface.
    1.4. Lining of openings.
2. Perimeter fixing
    2.1. Preparation: Fix continuous sole plate to floor at 600 mm centres, and 150 mm maximum from ends.
    2.2. Floor, wall and ceiling: Fix framing inserts to substrates at 600 mm centres. Locate panels into framing inserts and fix at 300mm centres with screws or 230 mm centres with nails.

## 436 Additional linings to plasterboard panel partitions

1. General: Bond additional linings to partitions with 60 mm-wide x 5 mm-thick continuous bands of adhesive at 300 mm centres, and centred on joints. Stagger joints between layers by minimum 150 mm.

## 445 Installing plasterboard laminated partitions

1. Timber battens: Size to suit plasterboard core layer. Use for:
    1.1. Floor, wall and ceiling perimeters.
    1.2. Lining of openings.
    1.3. Corners and angle junctions.
2. Perimeter fixing
    2.1. Floor, wall and ceiling: Fix timber battens to substrates at 600 mm centres.
3. Plasterboard layers

3.1. Bonding together: Apply adhesive in 60 mm-wide x 5 mm-thick continuous bands at 300 mm centres, and centred on joints. Stagger joints between layers by minimum 150 mm.

3.2. Nailing: Fix first and third layers to timber battens at 300 mm centres.

3.3. Additional mechanical fixings: Provide in accordance with partition manufacturer's recommendations to achieve specified fire resistance.

4. Complete partitions: Protect against movement for minimum four hours.

## 455 Taped seamless finish to plasterboard partitions

1. Cut edges of boards: Lightly sand to remove paper burrs.
2. Filling and taping: Fill joints, gaps and internal angles with jointing compound and cover with continuous lengths of paper tape, fully bedded.
3. Protection of edges/ corners: Reinforce stop ends, external angles, etc. with edge/ angle bead or tape.
4. Finishing: Apply jointing compound. Feather out each application beyond previous application to give a flush, seamless surface.
5. Nail/ screw depressions: Fill with jointing compound to give a flush surface.
6. Minor imperfections: Remove by lightly sanding.

## 470 Perimeter seals

1. Sealant material: A type recommended by partition/ panel manufacturer.
2. Application: Continuously, to clean, dry, dust-free surfaces, leaving no gaps. In accordance with the sealant manufacturer's instructions.

## 475 Fire-stopping

1. General: Seal gaps at junctions of partitions with perimeter abutments, cavity barriers, service penetrations, etc. to prevent penetration of smoke and flame.
2. Materials: Non-combustible or intumescent.
3. Fire-stopping: Cross-reference NBS section P12 for requirements of fire-stopping systems.

## 480 Glazing for relocatable partitions

1. Glass standards: To BS 952 and relevant parts of:
   1.1. BS EN 572 for basic soda lime silicate glass.
   1.2. BS EN 1096 for coated glass.
   1.3. BS EN 1748 for borosilicate glass.
   1.4. BS EN 1863 for heat-strengthened soda lime silicate glass.
   1.5. BS EN 12150 for thermally toughened glass.
   1.6. BS EN 12337 for chemically strengthened soda lime silicate glass.
   1.7. BS EN ISO 12543 for laminated glass.
   1.8. BS EN 13024 for thermally toughened borosilicate safety glass.
   1.9. BS EN 14178 for basic alkaline earth silicate glass.
   1.10. BS EN 14321 for thermally toughened alkaline earth silicate safety glass.
   1.11. BS EN 14449 for laminated glass and laminated safety glass.
2. Insulated glass unit standard: To relevant parts of BS EN 1279.
3. Pane size: Within ±2 mm of nominal size.

4. Pane condition: Clean and free from obvious scratches, bubbles, rippling, dimples and other defects.
5. Glazing standard: To relevant parts of BS 6262.
6. Preglazed panes: To be capable of being reglazed in situ.

## 490 Ironmongery for relocatable partitions

1. Frames and doors: Prepare accurately to accept ironmongery and associated fixings.
2. Ironmongery installation: Accurate and without damage to components or surrounding surfaces.
    2.1. Fasteners: Supplied by partition/ ironmongery manufacturer. Finish to match component.
3. Completion: Check, adjust and lubricate as necessary for correct operation.

## 495 Post-installation requirements for relocatable partitions

1. General: Inspect and adjust for correct fit and operation.
2. Documentation: Submit in accordance with the requirements of Regulation 38. Submit for each partition type, including:
    2.1. Partition type and construction.
    2.2. Manufacturer's identity.
    2.3. Sound and fire rating.
    2.4. Surface finishes and Light Reflectance Values (as manufactured).
    2.5. Requirements for fixtures and fittings.
    2.6. Test data and fire certificates.
    2.7. Special requirements.
    2.8. Dismantling and reassembly procedures.
    2.9. Maintenance: Replacement of panels, access to services, cleaning, etc.
3. Spares: Provide and hand over to the employer.

Ω End of Section



make

# K32
# Panel Cubicles/ Duct and Wall Linings/ Screens

## To be read with preliminaries/ general conditions.

## 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

## 151 LIN-501 Accessible Duct Panel Lining Systems (IPS)

1. Description: Accessible duct panel lining systems (IPS) with metal support frame.
2. Manufacturer: To be agreed with the Architect.
   2.1. Product reference: To be agreed with the Architect.
3. Sizes and configuration: Refer to the Design Drawings
4. Support framing:
   4.1. Site framing: Extruded aluminium base and headrail sections and rolled galvanised ultra steel wall channels.
   4.2. Baserail: 100mm high powder coated 2 part aluminium base rail to secure subframe units.
   4.3. Subframe: Rolled galvanised ultra steel subframe with riveted aluminium bracing bars, fitted to site frame with nylon click-fix locator brackets.
5. Panels: High performance Melamine Faced 660kg/m³ grade moisture resistant Chipboard (MFC) with Solid grade laminate finish.
   5.1. Thickness: 12.5 mm nominal.
   5.2. Colour/ finish: To be agreed with the Architect.
   5.3. Edge treatment: 1mm matching or contrasting ABS.
6. Flashgaps: 13 mm High density solid grade (SGL) laminate with decorative face both sides.
   6.1. Gap: Nominal 25 mm between each panel section.
   6.2. Colour/ finish: To be agreed with the Architect.
7. Fixing method:
   7.1. Panel sections fitted to subframe with concealed nylon lift-off fixing clips, using zinc plated steel rivets with butt joints fitted with zinc plated steel dowel clips and mating nylon dowel-docks.
   7.2. Riveted push-pull clips where applicable to allow bottom panels to be removed.
8. Pre-plumbing: Sanitary appliances pre-fixed and sealed to integrated plumbing system, refer to the Design Drawings and N13.
9. Ironmongery:
   9.1. Linear-bearing hinge: Zinc plated steel with chrome plated rivets.
   9.2. Lock: Micro panel lock in polished stainless steel to provide a tamper proof security fixing to the duct panels

## 210 Samples

1. General: Before placing orders submit representative samples of the following: Panel and door material and colours.

1.1. Panel: 300mm x 200mm.

1.2. Exposed ironmongery: 1 of each.

1.3. Exposed fixings: 1 of each.

1.4. Exposed cubicle accessories: 1 of each.

2. Delivered materials/ products: To match samples.

## 220 Control samples

1. General: Complete samples as part of finished work and obtain approval of appearance before proceeding.
2. Types: To be agreed with the Architect.
   2.1. Locations: To be agreed with the Architect.

## 250 Installation

1. Programming: Do not install cubicles or duct/ wall panels before building is weathertight, wet trades have finished their work, wall and floor finishes are complete, and the building is well dried out.
2. Accuracy: Set out to ensure frames and/ or panels and doors are plumb, level and accurately aligned.
3. Modifications: Do not cut, plane or sand prefinished components except where shown on drawings.
4. Fixing: Secure components using methods and fasteners recommended by the cubicle/ panel manufacturer. Prevent pulling away, bowing or other distortions to frames, panels and doors.
5. Moisture and thermal movement: Make adequate allowance for future movement.
6. Tolerances:
   6.1. Horizontal, vertical and in plane: ± 1 mm in 2 m.
   6.2. Interfaces: ± 1 mm.

Ω End of Section



make

# K40
# Demountable Suspended Ceilings

## Types of ceiling system

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 CLG-501 Acoustic Raft Ceiling System

1. Description: Acoustic raft ceiling system, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Manufacturer (fabric): Kvadrat, or acceptable equivalent.
4. Specific performance:
    4.1. Fire rating
        4.1.1. Surface spread of flame: Class B-s3, d0.
            4.1.1.1. Applicable to the complete system, including any backing boards and ancillary components as appropriate.
    4.2. Acoustic rating: Class A.
5. Support framing: Corrosion resistant rigid adjustable hangers, secondary and main runners, with appropriate fixings and other necessary ancillary components. To be coordinated with FCU's and where required incorporate additional secondary support/ uni-struts as needed to span beyond the FCU.
6. Raft:
    6.1. Type: Acoustic absorber.
    6.2. Material: High-density glass wool.
    6.3. Size: Refer to the Design Drawings
    6.4. Thickness: Refer to the Design Drawings
    6.5. Edges: Straight/ square and fully finished.
    6.6. Finish/ colour: Manufacturers standard
7. Lighting: Refer to the Services Engineer's Information
   All associated cabling / servicing to be fully concealed and inconspicuous in the finished works.
8. Fixing: Suitable concealed supports and fixings, to suit the service conditions and as the Manufacturer's recommendations.
9. Other requirements : System to be fully co-ordinated with lights and other services which are to be suspended independently of/ integrated within ceiling system.
10. System to include all supports, fixings, clips and all other associated accessories required to complete the installation.

### 113 CLG-503 Acoustic Raft Ceiling System

1. Description: Acoustic raft ceiling system, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Manufacturer: To be agreed with the Architect.

4. Specific performance:
    4.1. Fire rating
        4.1.1. Surface spread of flame: Class B-s3, d0.
            4.1.1.1. Applicable to the complete system, including any backing boards and ancillary components as appropriate.
    4.2. Acoustic rating: Class A.
5. Support framing: Corrosion resistant rigid adjustable hangers, secondary and main runners, with appropriate fixings and other necessary ancillary components. To be coordinated with FCU's and where required incorporate additional secondary support/ uni-struts as needed to span beyond the FCU.
6. Raft:
    6.1. Type: Acoustic absorber.
    6.2. Material: Cork with inset acoustic perforated timber panelling from Hunter Douglas or acceptable equivalent.
    6.3. Size: Refer to the Design Drawings
    6.4. Thickness: Refer to the Design Drawings
    6.5. Edges: Straight/ square and fully finished.
    6.6. Finish/ colour: Cork.
7. Lighting: Refer to the Services Engineer's Information
All associated cabling / servicing to be fully concealed and inconspicuous in the finished works.
8. Fixing: Suitable concealed supports and fixings, to suit the service conditions and as the Manufacturer's recommendations.
9. Other requirements : System to be fully co-ordinated with lights and other services which are to be suspended independently of/ integrated within ceiling system.
10. System to include all supports, fixings, clips and all other associated accessories required to complete the installation.

## 115 CLG-521 Metal Mesh Raft Ceiling System

1. Description: Metal mesh raft ceiling system.
2. Manufacturer: SAS International , or acceptable equivalent
    2.1. Product reference: Horizon.
3. Size and configuration: Refer to the Design Drawings
4. Form: Raft.
5. Rigidity: Rigid.
6. Application: Soffit mounting.
7. Framework material: Galvanized steel.
8. Acoustic performance
    8.1. Purpose: Absorber.
    8.2. Sound absorption: Class A.
9. Fire performance
    9.1. Reaction to fire: To BS EN 13501-1, A2-s1,d0.
10. Profile: Geometric profile.
11. Edges: Square.
12. Infill panels: Mesh.

13. Finish: Powder-coated.
14. Colour: RAL colour.
15. Accessories: Class A acoustic pads. Fittings for independent suspension.

## 151 CLG-601 Suspended Metal Ceiling Tile System

1. Description: Suspended metal ceiling tile system, as shown on the Design Drawings.
2. Manufacture: SAS International, or acceptable equivalent.
   2.1. Product reference: SAS System 330.
3. Size and configuration: Refer to the Design Drawings
4. Specific performance:
   4.1. Fire rating:
      4.1.1. Surface spread of flame: Refer to the Design Drawings
         4.1.1.1. Applicable to the complete system, including any backing boards and ancillary components as appropriate.
         4.1.1.2. Methodology: Method of achieving required surface spread of flame classification must not affect the visual appearance. - Submit proposals.
         4.1.1.3. Maintenance of system to be presented for review.
   4.2. Acoustic rating: Refer to the Acoustic Strategy Report
5. Support framing:
   5.1. Corrosion resistant mild steel rigid adjustable hangers, secondary and main runners, with appropriate fixings and other necessary ancillary components.
   5.2. Type: Suspended exposed metal grid support framing.
   5.3. Junctions: Butt-cut junctions.
6. Insulation: N/A.
7. Ceiling tiles:
   7.1. Type: Lay-in.
   7.2. Size(s): 600 x 600 mm,
   300 x 1200 mm.
   Refer to the Design Drawings
   7.3. Edge detail: Square edge.
   7.4. Micro perforations: To achieve acoustic rating.
   7.5. Finish: White, to be confirmed.
8. Exposed metalwork finish and colour: To match tile.
9. Acoustic insulation backing: To meet performance requirements.
10. Perimeter trim/ Shadow gap: 15mm shadow gap trim to plasterboard margin.
11. Access: Lift and tilt access.
12. System to include all supports, fixings, clips and all other associated accessories required to complete the installation.

## 175 CLG-701 Feature Acoustic Ceiling

1. Description: Feature acoustic ceiling, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Manufacturer: Kvadrat, or acceptable equivalent

3.1. Product reference: To be agreed with the Architect.

4. Specific performance:
   4.1. Fire rating:
      4.1.1. Surface spread of flame: Refer to the Design Drawings
         4.1.1.1. Applicable to the complete system, including any backing boards and ancillary components as appropriate.
         4.1.1.2. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
         4.1.1.3. Maintenance of system to be presented for review.
   4.2. Acoustic rating: Refer to the Acoustic Strategy Report
5. Support framing:
   5.1. Corrosion resistant metal support framing, mechanically secured to substrate and concealed in the finished works.
6. Insulation: Suitable non combustible insulation to achieve the performance requirements and suit the service conditions, where visible to be finished black using suitable proprietary facing.
7. Timber slats:
   7.1. Type: Timber effect throughs.
   7.2. Profile/ size: Refer to the Design Drawings
   7.3. Joints/ spacings: Refer to the Design Drawings
   7.4. Module size: Bespoke.
   7.5. Finish/ colour: White.
8. Fixing: Suitable concealed supports and fixings, to suit the service conditions and as the manufacturer's recommendations.
9. Perimeter trim/ Shadow gap: Refer to the Design Drawings
10. Access: Refer to the Design Drawings
11. Other requirements: System to be fully co-ordinated with lights and other services which are to be integrated within the ceiling throughs.
12. Make allowance for support and fixing of fixtures and fittings as shown on the Design Drawings, incorporating suitable concealed pattressing to suit the service conditions.
13. System to include all supports, fixings, clips and all other associated accessories required to complete the installation.

## 201 CLG-901 Durasteel Ceiling

1. Description: Proprietary fire rated suspended metal ceiling tile system.
2. Manufacturer: Promat UK, or acceptable equivalent.
3. Standard: To BS EN 13964
4. Specific performance:
   4.1. Fire rating: Refer to the Fire Strategy Report
   4.2. Acoustic rating: Refer to the Acoustic Report
5. Support structure:
   5.1. Type: Suspended concealed metal grid support framing.
   5.2. Zone height: Refer to the Design Drawings
6. Tile/ sheet

6.1. Product reference: DURASTEEL

6.2. Type/ material: Composite panel of fibre-reinforced cement with mechanically bonded punched steel sheets on both surfaces.

6.3. Size: 1200 mm (W) x 2500 mm (L).

6.4. Shape: Rectangular

6.5. Edge detail: Punched steel sheets

6.6. Weight: 16.8 kg/m²

6.7. Finish: Galvanised steel with two coats of paint.

7. Perimeter trim: Refer to the Design Drawings
8. Integrated services fittings: Refer to the Services Engineer's documentation
9. System accessories: System to include all fixings, framing, bracketry, support framing, grids, seals, insulation and all other components/ accessories necessary to complete the works, to the manufacturer's recommendations.

## General/ performance

### 205 Compliance with performance requirements

1. Testing/ assessment: Submit UKAS accredited laboratory reports for the following: Electrical continuity, earth bonding and fire performance.
2. Materials, components and details: Use those used in the test and identified in the assessment reports. If discrepancies arise, give notice.

### 210 Environment

1. Environmental classification to BS EN 13964: Class A
2. Release of Formaldehyde to BS EN 13964: Class E1
3. Susceptibility to growth of harmful micro-organisms to BS EN 13964: Level A

### 220 Structural performance

1. Loads: The ceiling system must safely support all anticipated loads, including services fittings:
   1.1. Class: 1
   1.2. Additional loads/ pressures to be sustained by ceiling system: None
      1.2.1. Ceiling system subject to wind pressure: No
2. Deflection (maximum) between points of support
   2.1. Span under 1200 mm: Span/400.
   2.2. Span 1200-1800 mm: Span/500.
   2.3. Span over 1800 mm: Span/600.
3. Test standard: To BS EN 13964.

### 225 Fire performance

1. Completed ceiling system: Not intended for fire protection
2. Reaction to fire ratings
   2.1. Ceiling soffit surfaces: To BS EN 13501-1, Class B-s3, d2 or better
   2.2. Ceiling void surfaces: To BS EN 13501-1, Class B-s3, d2 or better
3. Fire resistance



make

3.1. Overall fire resistance: To BS EN 13501-2, REI 60 or better

3.2. Ceiling resistance: To BS EN 13501-2, EI 30 or better

3.3. Other protection: Structural steel beams - to BS EN 13501-2, R 60 or better

4. Test reports or assessments: Include details of performance related to the particular elements of construction.

4.1. Ceilings with integrated luminaires: Test/ assess with luminaires in place.

## 230 Acoustic performance

1. Sound absorption to BS EN ISO 11654: Class A
2. Sound attenuation: Dnf, w (minimum) to BS EN ISO 717-1: Refer to the Acoustic Strategy Report
3. Other acoustic requirements: None

# Components

## 240 Samples

1. General: Submit representative samples of the following: …….. .

1.1. Infill units: 300mm x 200mm (include corners and adjacent edges).

1.2. Finishes.

1.3. Ceiling grid, support systems, access panels/ grilles.

1.4. Trims and other fixings that are visible.

1.5. Insulation and cavity barriers.

## 245 Standards

1. Steel panels: To BS EN 10346.
2. Aluminium sheet, strip and plate: To BS EN 485-1 and -2.
3. Aluminium bars, tubes and sections: To relevant parts of BS EN 515, BS EN 573, BS EN 755 and BS EN 12020.

## 255 Edge battens

1. Material: Planed softwood to BS EN 942, Class J10; moisture content at time of fixing: 15% ± 2%

1.1. Finished size: 21 x 44 mm

1.2. Finish (apply before ceiling grid is installed): Prime, undercoat and final coat of matt black to visible surfaces

## 260 Perimeter trims

1. Type: 32 x 19 mm steel angle trim, colour and finish to match exposed grid
2. Fixings

2.1. Fasteners: BZP raised head screws

2.2. Fixing centres (maximum): 450 mm

## 290 Air plenum barriers

1. Material: Rigid or semirigid non porous sheets with smooth nondusting surfaces.
2. Fire spread rating: Not less than ceiling materials exposed within the ceiling void.

# Execution

## 302 Control samples

1. General: Complete areas as part of the finished work in the following locations: To be confirmed.
2. Approval: Obtain before completing areas of similar work.

## 303 Installation Tolerances

1. Grid dimensions: ±1mm of nominal positions.
2. Final surface levels: ±2mm of nominal horizontal plane.
3. Visible gasket widths: ±5% nominal dimension.
4. Gridline creep: 1mm in 10,000mm.

## 305 Setting out

1. General: Completed ceiling should present, over the whole of its surface exposed to the room below, a continuous and even surface, jointed (where applicable) at regular intervals.
2. Infill and access units, integrated services: Fitted correctly and aligned.
3. Edge/ perimeter infill units size (minimum): Half standard width or length.
4. Corner infill units size (minimum): Half standard width and length.
5. Grid: Position to suit infill unit sizes. Allow for permitted deviations from nominal sizes of infill unit.
6. Infill joints and exposed suspension members: Straight, aligned and parallel to walls, unless specified otherwise.
7. Suitability of construction: Give notice where building elements and features to which the ceiling systems relate are not square, straight or level.

## 310 Bracing

1. General: Secure, with additional bracing and stiffening to give a stable ceiling system resistant to design loads and pressures.

## 315 Protection

1. Loading: Do not apply loads for which the suspension system is not designed.
2. Ceiling materials: When necessary, remove and replace correctly using special tools and clean gloves, etc. as appropriate.

## 320 Top fixing

1. Building structure: Verify suitability.
2. Structural soffit:
    2.1. Suitability to receive specified fixings: Evaluate and confirm.
3. Fixing generally: In accordance with BS EN 13964.
4. Fixing to
    4.1. Concrete: Drill and insert suitable expanding anchors.
    4.2. Aerated concrete: Fix through from the top of concrete units and provide a system of primary support channels.
    4.3. Structural steel: Drill, or use suitable proprietary clips/ adaptors.

4.4. Metal roof decking: Fix to sides of liner tray corrugations.

4.5. Timber: Fix to side of joists at least 50 mm from bottom edge. If ceiling system is intended for fire protection, fix into top third of joists.

4.6. Hollow structural members: Submit fixing proposals.

## 325 Installing hangers

1. Wire hangers: Straighten and tension before use.
2. Installation: Install vertical or near vertical, without bends or kinks. Do not allow hangers to press against fittings, services, or insulation covering ducts/ pipes.
3. Obstructions: Where obstructions prevent vertical installation, either brace diagonal hangers against lateral movement, or hang ceiling system on an appropriate rigid sub-grid bridging across obstructions and supported to prevent lateral movement.
4. Extra hangers: Provide as necessary to carry additional loads.
5. Fixing
   5.1. Wire hangers: Tie securely at top with tight bends to loops to prevent vertical movement.
   5.2. Angle/ strap hangers: Do not use rivets for top fixing.
6. Spacings: As per the system manufacturer's recommendations

## 330 Installing timber edge battens

1. Fixing: Firmly to perimeter wall or other building structure.
   1.1. Fasteners: As per the system manufacturer's recommendations
   1.2. Fixing centres (maximum): As per the system manufacturer's recommendations

## 335 Perimeter trims

1. Jointing: Neat and accurate, without lipping or twisting.
   1.1. External and internal corners: Mitre joints generally. Overlap joints at internal corners are not acceptable.
   1.2. Intermediate butt joints: Minimize. Use longest available lengths of trim. Align adjacent lengths.
2. Fixing: Fix firmly to perimeter wall, edge battens or other building structure.
   2.1. Fasteners: As per the system manufacturer's recommendations
   2.2. Fixing centres: As per the system manufacturer's recommendations

## 340 Exposed grids

1. Grid fixings: As per the system manufacturer's recommendations
2. Main runners: Install level. Do not kink or bend hangers.
   2.1. Spliced joints: Stagger.
   2.2. Wire hangers passing through main runners: Use sharp bends and tightly wrapped loops.
   2.3. Angle/ strap hangers: Do not use rivets for bottom fixing.
   2.4. Angular displacement of long axis of one runner in relation to next runner in line with it: Not visually apparent.
3. Cross members supported by main runners or other cross members: Install perpendicular to intersecting runners.
4. Cross tees: Flat and coplanar with flanges of main runners after panel insertion.

4.1. Cross tees over 600 mm long, cut and resting on perimeter trim: Provide an additional hanger.

5. Holding down clips: Locate to manufacturer's recommendations.
   5.1. Fire-protecting/ resisting ceiling systems: Use clip type featured in the fire test/ assessment.

## 345 Concealed grids

1. Grid fixings: As per the system manufacturer's recommendations
2. Primary support channels: Install level. Do not kink or bend hangers.
   2.1. Wire hangers wrapped around primary channels: Twice wrapped. Loops tightly formed.
   2.2. Angle/ Strap hangers: Do not use rivets for bottom fixing.
3. Splines: Locate between infill units to assist levelling of adjacent units and to resist air movement at joints.
4. Spring-tee grids: Do not omit primary channel.

## 355 Installing infill units

1. General
   1.1. Perimeter infill units: Trimmed, as necessary, to fully fill space between last grid member and perimeter trim. Prevent subsequent movement.
   1.2. Deeply textured infill units: Minimize variations in apparent texture and colour. In particular, avoid patchiness.
2. Concealed grids: Install infill units uniformly, straight and aligned. Avoid dimension creep.
   2.1. Infill units around recessed luminaires and similar openings: Prevent movement and displacement.

## 365 Installing metal infill units

1. Sound absorbing pads: Fit to prevent upward air movement through infill units. Cut or fold pads in cut perimeter infill units to full unit size. Reseal cut pads.
2. Perimeter infill units: Firmly wedge cut units into perimeter trim, or clip down.

## 375 Board ceiling systems

1. Cut boards: Neat and accurate.
2. Fixing to grid
   2.1. Board edges: Fully support. Screw to grid members. Set heads of screws below surface of boards and fill flush with surface.
   2.2. Boards applied in two or more layers: Stagger joints.
3. Movement joints: Provide as appropriate for the area of ceiling system and/ or to coincide with movement joints in surrounding structure.

## 385 Upstands and bulkheads

1. Vertical ceiling systems: Support and brace to provide alignment and stability.
2. High upstands: Provide support at base of upstand.

## 390 Openings in ceiling materials

1. General: Neat and accurate. To suit sizes and edge details of fittings. Do not distort ceiling system.

## 395 Integrated services

1. General: Position services accurately, support adequately. Align and level in relation to the ceiling and suspension system. Do not diminish performance of ceiling system.
2. Small fittings: Support with rigid backing boards or other suitable means. Do not damage or distort the ceiling.
    2.1. Surface spread of flame rating of additional supporting material: Not less than ceiling material.
3. Services outlets
    3.1. Supported by ceiling system: Provide additional hangers.
    3.2. Independently supported: Provide flanges to support ceiling system.

## 401 Ceiling-mounted luminaires

1. Support:
    1.1. Independently supported luminaires: Suspension adjusted to line and level of ceiling.
    1.2. Ceiling supported luminaires: Modifications and/ or extra support required: To each luminaire.
2. Surface mounted luminaires: Units installed so that in event of a fire the designed grid expansion provision is not affected.
3. Modular fluorescent recessed luminaires: Compatible with ceiling module. Extension boxes must not foul ceiling system.
4. Recessed rows of luminaires: Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in position with lateral restraint.
5. Fire-protecting/ resisting ceiling systems: Luminaires must not diminish protection integrity of ceiling system.
6. Access: Provide access for maintenance of luminaires.

## 406 Trunking

1. Recessed trunking: Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in position with lateral restraint.

## 411 Mechanical services

1. Fan coil units
    1.1. Inlet/ Outlet grilles: Trim ceiling grid and infill units to suit.
    1.2. Space beneath: Sufficient for ceiling system components.
    1.3. Suspension and connections: Permit accurate setting out and levelling of fan coil units.
2. Air grilles and diffusers
    2.1. Setting out: Accurate and level.
    2.2. Linear air diffusers: Retain in place with lateral restraint. Provide flanges for support of grid and infill units.
    2.3. Grille/ Diffuser ceiling joints: Provide smudge rings and edge seals.
3. Smoke detectors and PA speakers

3.1. Ceiling infill units: Scribe and trim to suit.

3.2. Independent suspension: Required

3.3. Flexible connections: Required.

4. Sprinkler heads: Carefully set out and level.

## 415 Installing insulation

1. Fitting: Fit accurately and firmly with butted joints and no gaps.
2. Insulation within individual infill units: Fit closely. Secure to prevent displacement when infill units are installed or subsequently lifted.

   2.1. Dustproof sleeving: Reseal, if cut.
3. Width: Lay insulation in the widest practical widths to suit grid member spacings.
4. Services: Do not cover electrical cables that have not been sized accordingly. Cut insulation carefully around electrical fittings, etc. Do not lay insulation over luminaires.
5. Sloping and vertical areas of ceiling system: Fasten insulation, to prevent displacement.

## 421 Ceiling systems intended for fire protection

1. Junctions of ceiling systems with perimeter abutments and service penetrations: Seal gaps with tightly packed mineral wool or intumescent sealant to prevent penetration of smoke and flame.
2. Ceiling system/ wall junctions: Maintain protective value of ceiling system.

   2.1. Fixings and grounds: Noncombustible.

   2.2. Metal trim: Provide for thermal expansion.
3. Access and access panels: Maintain continuity of fire protection.

## 425 Installing cavity fire barriers

1. Maximum ceiling void dimension in any direction: To comply with Approved Document B, volume 1, or volume 2.
2. Fixing:

   2.1. General: Fix barriers securely to channels or angles at abutments to building structure.

   2.2. At perimeters and joints: Provide permanent stability and continuity with no gaps to form a complete barrier to smoke and flame.
3. Joints: Form to preserve integrity in fire.
4. Service penetrations: Cut barriers neatly to accommodate services. Fit fire-resistant sleeves around flexible materials. Fill gaps around services to fire barrier manufacturer's recommendations to maintain barrier integrity. Adequately support services passing through the barrier.
5. Ceiling systems intended for fire protection: Do not impair fire-resisting performance of ceiling system.
6. Ceiling systems not intended for fire protection: Do not mechanically interlink barriers with ceiling system.

## 431 Installing sound barriers

1. Setting out: Align accurately with partition heads.
2. Fixing: Fix tightly at perimeters and joints using methods recommended by barrier manufacturer, including steel support sections as appropriate. Completed installation to be stable, secure and continuous, with no gaps.

3. Gaps at junctions with partition heads, ceiling system, structural soffit, walls, ducts, pipes, etc.: Seal with mineral wool or suitable sealant.

## 435 Installing air plenum barriers

1. Fixing: Fix tightly at perimeters and joints using methods recommended by the barrier manufacturer to ensure permanent stability.
2. Edges and joints: Seal effectively to prevent air leakage.

## 500 Electrical continuity and earth bonding

1. Inclusion in finished work: Not required

# Completion

## 505 Tools

1. Access tools: At Completion, supply one suction pad lifter for unit-suspended ceiling system.

## 520 User instructions

1. Contents: Include the following:
   1.1. Correct methods for removing and replacing infill units and other components.
   1.2. Cleaning methods and materials.
   1.3. Recommendations for redecoration.
   1.4. Ceiling systems intended for fire protection: Limitations placed on subsequent alterations and maintenance procedures, to ensure that their fire performance is not impaired.
   1.5. Maximum number, position and value of point loads that can be applied to ceiling system after installation.

## 530 Spares

1. General: At practical completion, supply the following: tiles for ceiling system.

Ω End of Section



make

# K41
# Raised Access Floors

## Types of raised access floor

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 FBS-201 Raised Access Flooring System - Medium Duty

1. Description: Fully accessible raised access flooring system.
2. Manufacturer: Tate global, or acceptable equivalent
3. Product reference: RMG 600+.
4. Standard: To MOB PF2 PS/SPU.
    4.1. Authority: References to authority in MOB PF2 PS/SPU are deemed to be to employer.
5. Specific performance:
    5.1. Fire rating: No target rating other than that provided inherently by the system.
    5.2. Acoustic rating: No target rating other than that provided inherently by the system.
    5.3. Structural grade: Medium duty, to comply with PSA MOB PF2 PS/SPU.
    5.4. Uniformly Distributed Load (UDL): Maximum of 3.0kN/m².
6. Total system depth:
7. Subfloor: Refer to the Design Drawings
    7.1. Preparation: To suit the service conditions.
8. Support:
    8.1. Type: Height adjustable support pedestals, adhesive bonded and/ or mechanically fixed to substrate.
    8.2. Fixing zone: Refer to the Design Drawings
    8.3. Material: Zinc plated steel, to the manufacturer's recommendations.
    8.4. Protective finish: Hot dip galvanised.
9. Floor panels:
    9.1. Type/ material: Galvanized steel sheets encapsulating high density particle board.
    9.2. Size: 600 mm x 600 mm.
    9.3. Thickness: 31 mm
    9.4. Edges: No sharp edges with ABS edge band
    9.5. Protective finish: Hot dip galvanised.
    9.6. Joints: To be constant with fixing details as the Design Drawings.
    9.7. Apertures/ cut-outs: To be precision factory-formed using rigid, manufacturer's templates. All exposed ends, cuts, etc. to be suitably finished and sealed.
10. Electrical, air supply, service outlets and earth bonding: Refer to the Services Engineer's documentation
11. Steps and ramps (if required): To include riser panels, vertical infill panels, flush nosing and general profiles, Refer to the Design Drawings



make

12. Seals and cavity barriers:
    12.1. Perimeter and abutments: Seal all perimeter and abutments with neoprene gaskets to meet air leakage requirements maintaining the integrity of the works.
    12.2. Cavity barriers: Permanently stable and continuous cavity barriers, to prevent penetration of smoke and flame.
    12.3. Gaps at junctions: Seal all gaps at the junction of the works with the walls with fireproof material to prevent penetration of smoke and flame.
13. Structural movement joints (where visible): Refer to the Design Drawings and Refer to the Structural Engineer's documentation

## General/ performance

### 210 General performance

1. Structural requirement: Refer to the Structural Engineer's Information

### 219 Installation

1. Completed installation: Clean and stable. Free from bounce and vibration. No lipping between floor panels.

### 229 Hygrothermal performance

1. Environmental conditions
    1.1. Consequential limits to use: Submit.
2. Deemed-to-satisfy test results: Submit, with testing authority's opinion.

### 234 Fire performance

1. Reaction to fire: To BS EN 13501-1, Class A1fl or better
2. Resistance to fire: To BS EN 13501-2, RE 30

### 239 Sound transmission

1. Laboratory system test: In accordance with BS EN ISO10848-2.
    1.1. Airborne sound insulation: Refer to the Acoustic Strategy Report
    1.2. Impact sound insulation: Refer to the Acoustic Strategy Report
    1.3. Test report: Submit.

### 249 Floor void (plenum) air leakage

1. Air leakage rate (maximum): Refer to the Services Engineer's Information
2. Design pressure (maximum): Refer to the Services Engineer's Information

## Components

### 310 Samples

1. General: Submit representative samples of the following: Floor panels with integral finishes and cable management accessories .

### 322 Pedestal adhesive

1. Adhesive: Compatible with subfloor surface finish

### 340 Cavity barriers

1. Type: Refer to the Fire Strategy Report
2. Performance
   2.1. General: Permanently stable, continuous, and an effective barrier to smoke and flame.
   2.2. Fire resistance: To BS EN 13501-2, EI 30
3. Spacing: Maximum dimension in any direction: 10 m.
4. Fixing: Fix securely so that their performance is unlikely to be affected by:
   4.1. Movement of the building due to subsidence, shrinkage or temperature change.
   4.2. Failure of their fixings in a fire.
   4.3. Failure in a fire of any abutting material or construction.
   4.4. Collapse in a fire of any services penetrating them.
5. Floor panels above cavity barriers: To be firmly secured.
6. Gaps between cavity barriers and other elements: Seal with mineral wool or other suitable material.
7. Notification: Give notice when cavity barriers have been installed.

## Installation

### 410 Control samples

1. General: Complete areas of finished work in the following locations: To be confirmed .

### 421 Preparation

1. Areas where flooring is to be installed: Clean before installation and keep clean during installation.
2. Setting out of flooring: Indelibly mark pedestal positions before installing services.
3. Fixtures: Before installing floor, complete fixtures that floor panels are to be cut around, or that are to be bridged by floor supports.

### 425 Environmental conditions

1. General: Dry, well ventilated, not subject to rapid variations or extremes of temperature or humidity.
2. RH of air (maximum): 75%.
3. Subfloors
   3.1. Moisture content: Test to BS 8201 using an accurately calibrated hygrometer
   3.2. RH (maximum): 75%.
   3.3. Temperature (minimum): 5°C.

### 431 Dust-proofing

1. Preparation: Surfaces to be sealed must be clean, dry and free from dust, grease and other contaminants.
2. Extent of sealing: Concrete and masonry surfaces within raised access floor void
3. Sealer: Recommended by raised access floor manufacturer. Compatible with materials used to pack and/ or fix pedestals.
   3.1. Colour: Tinted; different tint for each coat.

3.2. First coat: Apply before pedestals are erected.

3.3. Second coat: Apply after completion of services and other associated work.

## 433 Pedestal adhesive testing

1. Test method: To MOB PF2 PS/SPU
2. Rate of testing
   2.1. Pre-installation: 2% of total in each area, selected at random
   2.2. During installation: 1%, selected at random
3. Witnessing: Arrange for pre-installation testing to be witnessed by: Architect.
4. Results of testing during installation: Submit.

## 435 Cut floor panels

1. Size (minimum): Half standard width x half standard length.
2. Edges
   2.1. Burrs and rough edges: Make smooth.
   2.2. Sealing: Seal exposed cut edges of floor panels that have moisture sensitive or combustible cores.
      2.2.1. Sealer: Self-adhesive aluminium foil tape, to match fire rating of panel, or better.

## 441 Raised access floor levels

1. Permissible deviations in level
   1.1. Over set length: ±1.5 mm over 3m
   1.2. Overall: ±3 mm

## 445 Perimeters

1. Expansion gaps
   1.1. Size: 10 mm.
   1.2. Location: At abutments.
2. Expansion gap filling
   2.1. Filler type: Resilient closed cell.
   2.2. Filling: Before fixing skirtings and cover strips.

## 455 Level changes

1. Performance
   1.1. Ramps and steps: Comparable with performance of associated raised access floor.
   1.2. Balustrades: Structural and safety requirements in accordance with BS 6180.

## 460 Under-Floor plenum air barriers

1. Material: Rigid or semirigid nonporous sheets with smooth nondusting surfaces.
2. Performance: Permanently stable, continuous and airtight.
3. Fixing: Fix securely to subfloor, at joints and as necessary.
4. Edges and joints: Seal.

### 470 Integral finishes electrical resistance testing

1. Location: To be confirmed
2. Extent of testing: Test complete raised access floor.
3. Testing agent: Qualified electrician.

## Completion

### 510 Tools

1. Floor panel lifting devices: At completion, supply one set of suitable devices for each type of raised access floor finish installed. Train designated personnel in their use.
2. Pedestal locking: At completion, supply one set of tools for releasing pedestal locking.

### 515 User instructions

1. User manual contents: Include the following:
    1.1. Correct method for lifting and replacing floor panels and stringers.
    1.2. Limitations on sequence, number and positions of floor panels and stringers that can be removed safely at one time.
    1.3. Permissible loading, with guidance on use of spreader plates when shifting heavy equipment and subsequent maintenance.
    1.4. Methods for installing cabling and ducts to prevent damage to supporting structure.
    1.5. Methods for cleaning floor panels and integral finishes.
    1.6. Method for replacing integral floor panel coverings.
    1.7. Method for adjusting and locking pedestals.
    1.8. Recommended maintenance methods and frequency.
        1.8.1. Minimum maintenance-free life of raised access floor system.
        1.8.2. Minimum maintenance-free life of replaceable parts where this differs from that of the whole system.
        1.8.3. Minimum period during which replaceable components will be available.
    1.9. Installation instructions, including COSHH Assessment.

### 520 Spares

1. General: At completion, supply the following: floor panels for raised access floor type.

### 525 Cleaning

1. Subfloors: At completion, thoroughly clean accessible areas of subfloors and leave free of dust and debris.
2. Raised access floor: Before delivery of items carried by floor, clean thoroughly.

Ω End of Section

# L10
# Windows/ Rooflights/ Screens/ Louvres

## General

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions,
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 Evidence of performance

1. Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

### 115 Timber procurement

1. Timber (including timber for wood-based products): Obtained from well-managed forests and/ or plantations, in accordance with:
    1.1. The laws governing forest management in the producer country or countries.
    1.2. International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
2. Documentation: Provide either in accordance with chain of custody certification scheme requirements:
    2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied.
    2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. Chain of custody certification scheme: Forest Stewardship Council (FSC), Grown in Britain (GiB) or Programme for the Endorsement of Forest Certification (PEFC)
    3.1. Other evidence: UK Timber Procurement Policy (UKTPP), Category B

### 120 Pre-construction survey

1. Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
2. Primary support structure: Carry out survey sufficient to verify that required accuracy and security of erection can be achieved.
3. Timing: Before fabrication.

### 140 Control samples

1. Procedure
    1.1. Finalize component details.
    1.2. Fabricate one of each of the following designated items as part of the quantity required for the project.
    1.3. Obtain approval of appearance and quality before proceeding with manufacturer of the remaining quantity.

## 150 Daylight performance

1. Standard: In accordance with CIBSE Lighting Guide 10 and BS EN 17037.
2. Verification
   2.1. Submittals:
      - Design calculations in accordance with CIBSE 'Lighting Guide 10'.
      - Daylight factor.
      - BREEAM assessment.
      - LEED certification.
   2.2. Timing: Before completion of detailed design.

## 155 Views out

1. Windows/ opening sizes and position: Design in accordance with principles set out in BS EN 17037.
2. Verification
   2.1. Submittals:
      - Design plan and elevation drawings showing: Building location and proximity to external obstructions.
      - LEED 'Quality views' credits.
      - BREEAM 'View out' criteria for relevant building type.
      2.1.1. Other requirements, Design plans and elevations to show the following:
         - All BREEAM defined 'relevant areas' dependent on building type and room depths.
         - Actual or notional workstation/ desk layouts.
         - Window/ open areas.
   2.2. Timing: Before completion of detailed design.

## 160 Potential for natural ventilation

1. Design: In accordance with CIBSE AM10.
2. Verification: A copy of the results from a software modelling tool recommended in CIBSE AM10.
   2.1. Submittals:
      - Gross internal floor area of each occupied space.
      - Locations of openings.
      - Room depths.
      - Types of windows/ ventilators.
      - Total openable areas.
      - Types and degree of user-controls.
   2.2. Timing: Before completion of detailed design.

# Products

## 205 Materials selection

1. Third-party certification: Submit proposals.
2. Verification:

2.1. Submittals: BREEAM 'Green Guide to Specification Online'.
LEED materials and resources criteria.

2.2. Timing: Before completion of detailed design.

## 311 EWS-04A Side Hung Window (Visual Intent Only)

1. Description: Side hung window system.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
5. Performance requirments: Refer to the Façade Performance Specification
6. Frame: Refer to the Façade Performance Specification
   6.1. Profile: Refer to the Façade Performance Specification
   6.2. Finish as delivered: Refer to the Façade Performance Specification
7. Glazing details: Refer to the Façade Performance Specification

## 312 EWS-04B Curved Fixed Glazing - North West Setback Corner (Visual Intent Only)

1. Description: Curved fixed pane glazing to North West setback corner.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
5. Performance requirments: Refer to the Façade Performance Specification
6. Frame:
   6.1. Profile: Refer to the Façade Performance Specification
   6.2. Finish as delivered: Refer to the Façade Performance Specification
7. Glazing details: Refer to the Façade Performance Specification

## 313 EWS-04C Incline Fixed Glazing - Modern Mansard (Visual Intent Only)

1. Description: Incline fixed glazing to modern mansard.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification

4. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
5. Performance requirments: Refer to the Façade Performance Specification
6. Frame:
   6.1. Profile: Refer to the Façade Performance Specification
   6.2. Finish as delivered: Refer to the Façade Performance Specification
7. Glazing details: Refer to the Façade Performance Specification

## 314 EWS-04D Side Hung Window - Historic Façade (Visual Intent Only)

1. Description: Side hung window to historic façade.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
5. Performance requirments: Refer to the Façade Performance Specification
6. Frame:
   6.1. Profile: Refer to the Façade Performance Specification
   6.2. Finish as delivered: Refer to the Façade Performance Specification
7. Glazing details: Refer to the Façade Performance Specification

## 315 EWS-04E Side Hung Windows - Historic Mansard Dormers (Visual Intent Only)

1. Description: Side hung windows to historic mansard dormers.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Standard: Non-fire and/ or smoke-rated windows to BS EN 14351-1 and BS 6510
4. Manufacturer: Refer to the Façade Performance Specification
   4.1. Product reference: Refer to the Façade Performance Specification
5. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
6. Performance requirments: Refer to the Façade Performance Specification
7. Frame:
   7.1. Profile: Refer to the Façade Performance Specification
   7.2. Finish as delivered: Refer to the Façade Performance Specification
8. Glazing details: Refer to the Façade Performance Specification

## 316 EWS-04F Bi-fold Windows - Sky Lobby (Visual Intent Only)

1. Description: Bi-fold windows to sky lobby.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.

2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.

3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Window Schedule.
5. Performance requirments: Refer to the Façade Performance Specification
6. Frame:
   6.1. Profile: Refer to the Façade Performance Specification
   6.2. Finish as delivered: Refer to the Façade Performance Specification
7. Glazing details: Refer to the Façade Performance Specification

## 651 LVR-101 Shopfront Louvre - Ground Floor (Visual Intent Only)

1. Description: Shopfront louvre banding to ground floor.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
   5.1. Profile: Refer to the Façade Performance Specification
   5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 652 LVR-102 Louvre - Barrel Curve Façade (Visual Intent Only)

1. Description: Louvre to barrel curve façade.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
   3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
   4.1. Note: some louvres form part of removable façade panels for plant replacement. Refer to the Façade Performance Specification
5. Material: Refer to the Façade Performance Specification
   5.1. Profile: Refer to the Façade Performance Specification
   5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 653 LVR-103 Incline Architectural Louvres - Modern Mansard Façade (Visual Intent Only)

1. Description: Incline architectural louvres to modern mansard façade.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
    5.1. Profile: Refer to the Façade Performance Specification
    5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 654 LVR-104 Acoustic Louvres - Modern Mansard Façade (Visual Intent Only)

1. Description: Acoustic performance louvres behind incline architectural louvres to the modern mansard façade.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
    5.1. Profile: Refer to the Façade Performance Specification
    5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 655 LVR-105 Louvre Screening - Level 7 Plant (Visual Intent Only)

1. Description: Louvre screening to level 7 external plant.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
    5.1. Profile: Refer to the Façade Performance Specification
    5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 656 LVR-106 Louvre Screening - Adiabatic Cooler (Visual Intent Only)

1. Description: Louvre screening to adiabatic cooler.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
    5.1. Profile: Refer to the Façade Performance Specification
    5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 657 LVR-107 Louvre Screening - North Core Roof Plant (Visual Intent Only)

1. Description: Louvre screening to north core roof plant.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Material: Refer to the Façade Performance Specification
    5.1. Profile: Refer to the Façade Performance Specification
    5.2. Finish/ colour/ texture: Refer to the Façade Performance Specification
6. Air intake and exhaust performance: Refer to the Façade Performance Specification

## 658 LVR-108 UKPN Louvre Screening

1. Description: UKPN louvre screening.
2. Manufacturer: Sunray Doors.
    2.1. Product reference: UKPN-approved louvre system.
3. Size and configuration: Refer to the Design Drawings
4. Material: Extruded aluminium alloy frame and louvre blades, to BS EN 755-1.
    4.1. Profile: Refer to the Design Drawings
    4.2. Finish as delivered: Powder-coating to BS 6496.
    4.3. Finish/ colour/ texture: Standard RAL colour
    4.4. Texture: Matt.
5. Construction: Mitred corners, welded joints.
6. Air intake and exhaust performance: Refer to the Services Engineer's documentation
7. Calculated weighted sound reduction index (Rw): Refer to the Services Engineer's documentation

8. Weather performance: To BS EN 13030, class D. Refer to the Services Engineer's documentation
9. Number of louvre banks: Refer to the Design Drawings
10. Louvre blade pitch and angle: As required to suit the performance required and to match the accepted sample.
11. Birds and/ or insect mesh: Non-ferrous, coloured black, to the rear of the blades.
12. Blanking panels (if required): Aluminium sheet encapsulated stone wool insulation forming blanking panels when the louvre is inactive
13. Accessories/ other requirements: Incude all necessary pressed metal accessories including flashings, copings, cappings, cills, reveals and returns and other formed accessories.
14. Fixing: Concealed.

## 661 EWS-11 Roof Plant Screen (Visual Intent Only)

1. Description: Roof plant screening
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Screening:
    5.1. Type/ material: Refer to the Façade Performance Specification
    5.2. Profile: Refer to the Façade Performance Specification
    5.3. Finish as delivered/ colour: Refer to the Façade Performance Specification

## 666 EWS-09 Architectural Metal Patterned Screening (Visual Intent Only)

1. Description: Architectural metal patterned screening to north core roof plant.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Size and configuration: Refer to the Design Drawings
5. Decorative screen: Refer to the Façade Performance Specification
    5.1. Finish as delivered/ colour: Refer to the Façade Performance Specification

# Execution

## 710 Protection of components

1. General: Do not deliver components to site that cannot be installed immediately or placed in clean, dry-floored and covered storage.
2. Stored components: Stack them vertically or near-vertically on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

## 730 Priming/ sealing

1. Wood surfaces inaccessible after installation: Prime or seal as specified before fixing components.

## 740 Corrosion protection

1. Protective coating: Two coats of bitumen solution to BS 6949 or an approved mastic-impregnated tape.
    1.1. Timing of application: Before fixing components.

## 750 Building in

1. General: Not permitted unless indicated on drawings.
    1.1. Brace and protect components to prevent distortion and damage during construction of adjacent structure.

## 765 Window installation generally

1. Installation: Into prepared openings.
2. Gap between frame edge and surrounding construction
    2.1. Minimum: As per the system manufacturer's recommendations
    2.2. Maximum: As per the system manufacturer's recommendations
3. Distortion: Install windows without twist or diagonal racking.

## 766 Location of openable windows in naturally ventilated buildings

1. Location: Over 10 m from sources of external pollution.

## 770 Damp-proof courses in prepared openings

1. Location: Ensure correct positioning in relation to window frames. Do not displace during fixing operations.

## 782 Fixing of aluminium frames

1. Standard: As section Z20.
2. Fasteners: Suitable concealed fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
    2.1. Spacing: When not pre-drilled or specified otherwise, position fasteners not more than 250 mm from ends of each jamb, adjacent to each hanging point of opening lights, and at maximum 600 mm centres.

## 790 Fire-resisting frames

1. Gap between back of frame and reveal: Completely fill with intumescent mastic or tape/ plaster/ tightly packed mineral wool.

## 810 Sealant joints

1. Sealant
    1.1. Manufacturer: Submit proposals
        1.1.1. Product reference: Submit proposals.
    1.2. Colour: To be confirmed

1.3. Application: As section Z22 to prepared joints. Finish triangular fillets to a flat or slightly convex profile.

## 820 Ironmongery

1. Fixing: In accordance with any third-party certification conditions applicable. Assemble and fix carefully and accurately using fasteners with matching finish supplied by ironmongery manufacturer. Do not damage ironmongery and adjacent surfaces.
2. Checking/ adjusting/ lubricating: Carry out at completion and ensure correct functioning.

Ω End of Section



make

# L20
# Doors/ Shutters/ Hatches

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 Evidence of performance

1. Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

### 112 Timber procurement

1. Timber (including timber for wood-based products): Obtain from well-managed forests and/ or plantations in accordance with:
   1.1. The laws governing forest management in the producer country or countries.
   1.2. International agreements such as the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
2. Documentation: Provide either in accordance with chain of custody certification scheme requirements:
   2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied, or
   2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. Chain of custody certification scheme: In accordance with UK Government Timber procurement policy (UKTPP), i.e. FSC, GiB or PEFC

### 115 Fire-resisting and smoke control pedestrian doors/ door assemblies/ doorsets

1. UKCA/ UKNI/ CE marked fire-resisting and smoke control pedestrian doorsets: To BS EN 16034 and in conjunction with BS EN 13241, BS EN 14351-1 and BS EN 14351-2.
2. Door products: As defined in BS EN 12519.
3. Components, assemblies or sets will be marked to the relevant UKCA/ UKNI/ CE marking European product standard (hEN), national product standard and/ or third-party certification rating.
4. Verification: Refer to A sections.
   4.1. Submittals: Provide certified evidence, in the form of a product conformity certificate, directly relevant fire test report or engineering assessment, that each door/ door assembly/ doorset supplied will comply with the specified requirements for fire-resisting and/ or smoke control if tested to BS 476-22, BS EN 1634-1, BS EN 1634-3 or is UKCA/ UKNI/ CE marked to BS EN 16034. Specified values should not be a combination of both standards. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.
   4.2. Timing: Refer to A sections.

## 120 Non-fire-resisting pedestrian doors/ door assemblies/ doorsets

1. Evidence: Provide certified evidence, in the form of a product conformity certificate or engineering assessment, that each pedestrian door/ doorset/ assembly supplied will comply with the specified requirements to BS EN 14351-1. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.
2. Components and assemblies: Marked to the relevant UKCA/ UKNI/ CEI marking European product standard (hEN), national product standard and/ or third-party certification rating.

## 150 Site dimensions

1. Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.

## 160 Samples

1. General: Before placing orders submit representative samples of the following:
    1.1. One quarter finished section of each door type.
    1.2. Finished frame: 1000mm.
    1.3. Sealants and seals: 500mm.
    1.4. Finished glass: 500mm x 500mm.
2. Delivered materials/ products: To match samples.

## 170 Control samples

1. Procedure
    1.1. Finalize component details.
    1.2. Fabricate one of each of the following designated items as part of the quantity required for the project.
    1.3. Obtain approval of appearance and quality before proceeding with manufacture of the remaining quantity.
2. Designated items: First completed doorset of each type.

# Products

## 251 IDS-101 Internal Timber Hinged Doorset - Painted

1. Description: Internal timber doorset with paint finish.
2. Manufacturer: Shadbolt International, or acceptable equivalent
3. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
    4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.
5. Door frame:
    5.1. Type: Rebated profile, split frame to satisfy the wall/partition depth.
    5.2. Material: Hardwood fully compliant with the qualitative and sustainability criteria stipulated.
    5.3. Finish: Thoroughly sanded, factory primed and factory paint finished.
6. Door leaf:

6.1. Core: High density solid chipboard core

6.2. Lippings: Hardwood to all edges, applied before application of painting overlayment.

6.3. Finish: Factory priming applied to the overlayment and factory paint finish.

6.4. Thickness: As required to meet the performance requirements.

6.5. Vision panels (if required): Where shown on the Door Schedule and Design Drawings

6.5.1. Glass: Kitemarked safety glass to achieve the required performance, fixed with flush finished beading.

6.5.2. Apertures: Factory-formed.

6.5.3. Beading: Flush beading in hardwood to match the hardwood edging.

6.5.4. Bedding: Acoustic glazing and/ or intumescent compound or proprietary glazing channel, based on the manufacturer's performance certificate and recommendations

7. Access control: Refer to the Services Engineer's documentation
8. Overpanels: Where shown on the Design Drawings to be permanently fixed into the rebated frame and intumescently sealed.
9. Meeting stiles: Shall be ostensibly square with a subtle radiused profile in accordance with the acoustic seal manufacturers.
10. Perimeter seals: Intumescent, smoke and acoustic seals shall be of a type on which the manufacturer's certification is based on.
11. Ironmongery: Provide prototype lock cases, strikes, hinges and closers to the door manufacturer to enable factory machining/ fitting. Refer to the Ironmongery Schedule and P21.

## 252 IDS-102 Internal Timber Hinged Doorset - Veneered

1. Description: Internal timber doorset with veneer finish.
2. Manufacturer: Shadbolt International, or acceptable equivalent
3. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
4. Specific performance:

4.1. Fire rating: Refer to the Fire Strategy Reportand Door Schedule.

4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.

5. Door frame:

5.1. Type: Rebated profile, split frame to satisfy the wall/partition depth.

5.2. Material: Hardwood of minimum density, 650kg/m³ fully compliant with the qualitative and sustainability criteria stipulated herein.

5.3. Finish: Thoroughly sanded, factory primed and factory paint finished.

6. Door leaf:

6.1. Core: High density solid chipboard core

6.2. Lippings: Hardwood to all edges, applied before application of Veneer

6.3. Finish: Timber veneer

6.3.1. Species: European oak.

6.3.2. Cut: Crown cut.

6.3.3. Matching: Book matching.

6.3.4. Thicknesss: Min. 0.8 mm

6.4. Thickness: As required to meet the performance requirements.

6.5. Vision panels: Where shown on the Door Schedule and Design Drawings

6.5.1. Glass: Clear safety glass to achieve the required performance, fixed with flush finished beading.

6.5.2. Apertures: Factory-formed.

6.5.3. Beading: Flush beading in hardwood to match the hardwood edging.

6.5.4. Bedding: Acoustic glazing and/ or intumescent compound or proprietary glazing channel, based on the manufacturer's performance certificate and recommendations

7. Access control:
8. Overpanels: Where shown on the Design Drawings to be permanently fixed into the rebated frame and intumescently sealed.
9. Meeting stiles: Shall be ostensibly square with a subtle radiused profile in accordance with the acoustic seal manufacturers.
10. Perimeter seals: Intumescent, smoke and acoustic seals shall be of a type on which the manufacturer's certification is based on.
11. Ironmongery: Provide prototype lock cases, strikes, hinges and closers to the door manufacturer to enable factory machining/ fitting. Refer to the Ironmongery Schedule and P21.

## 253 IDS-103 Internal Timber Hinged Doorset - WC Cubicle

1. Description: Internal timber doorset with veneer finish to WC Cubicle.
2. Manufacturer: Shadbolt International, or acceptable equivalent
3. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
4. Specific performance:
   4.1. Fire rating: Refer to the Fire Strategy Reportand Door Schedule.
   4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.
5. Door frame:
   5.1. Type: Rebated profile, split frame to satisfy the wall/partition depth.
   5.2. Material: Hardwood of minimum density, 650kg/m³ fully compliant with the qualitative and sustainability criteria stipulated herein.
   5.3. Finish: Thoroughly sanded, factory primed and factory paint finished.
6. Door leaf:
   6.1. Core: High density solid chipboard core
   6.2. Lippings: Hardwood to all edges, applied before application of Veneer
   6.3. Finish: Timber veneer
      6.3.1. Species: European oak.
      6.3.2. Cut: Crown cut.
      6.3.3. Matching: Book matching.
      6.3.4. Thicknesss: Min. 0.8 mm
   6.4. Thickness: As required to meet the performance requirements.
7. Overpanels: Where shown on the Design Drawings to be permanently fixed into the rebated frame and intumescently sealed.
8. Perimeter seals: Intumescent, smoke and acoustic seals shall be of a type on which the manufacturer's certification is based on.

9. Ironmongery: Provide prototype lock cases, strikes, hinges and closers to the door manufacturer to enable factory machining/ fitting. Push plate, thumb lock, grab rail and handles. Refer to the Ironmongery Schedule and P21.

## 254 IDS-104 Internal Timber Hinged Doorset - Shower Cubicle

1. Description: Internal timber doorset with veneer finish to shower cubicle.
2. Manufacturer: Shadbolt International, or acceptable equivalent
3. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Reportand Door Schedule.
    4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.
5. Door frame:
    5.1. Type: Rebated profile, split frame to satisfy the wall/partition depth.
    5.2. Material: Hardwood of minimum density, 650kg/m³ fully compliant with the qualitative and sustainability criteria stipulated herein.
    5.3. Finish: Thoroughly sanded, factory primed and factory paint finished.
6. Door leaf:
    6.1. Core: High density solid chipboard core
    6.2. Lippings: Hardwood to all edges, applied before application of Veneer
    6.3. Finish: Timber veneer
        6.3.1. Species: European oak.
        6.3.2. Cut: Crown cut.
        6.3.3. Matching: Book matching.
        6.3.4. Thicknesss: Min. 0.8 mm
    6.4. Thickness: As required to meet the performance requirements.
7. Overpanels: Where shown on the Design Drawings to be permanently fixed into the rebated frame and intumescently sealed.
8. Perimeter seals: Intumescent, smoke and acoustic seals shall be of a type on which the manufacturer's certification is based on.
9. Ironmongery: Provide prototype lock cases, strikes, hinges and closers to the door manufacturer to enable factory machining/ fitting. Refer to the Ironmongery Schedule and P21.

## 255 IDS-406 Internal Timber Hinged Doorset - Feature Timber Lining

1. Description: Internal timber doorset with feature timber lining finish.
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
        4.1.1. Conforms to European class C -s3,d2.
    4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.
5. Door frame:
    5.1. Type: Rebated profile, split frame to satisfy the wall/partition depth.

5.2. Material: Hardwood fully compliant with the qualitative and sustainability criteria stipulated.

5.3. Finish: To match the surrounding timber panelling and appear inconspicuous in the finished works, providing a consistent overall elevation with the door leaf position visually concealed so far as is reasonably practicable.

6. Door leaf:
   6.1. Core: High density solid chipboard core
   6.2. Lippings: Hardwood to all edges, applied before application of painting overlayment.
   6.3. Finish: To match the surrounding timber panelling and appear inconspicuous in the finished works, providing a consistent overall elevation with the door leaf position visually concealed so far as is reasonably practicable.
   6.4. Thickness: As required to meet the performance requirements.
7. Access control: Refer to the Services Engineer's documentation
8. Overpanels: Where shown on the Design Drawings to be permanently fixed into the rebated frame and intumescently sealed.
9. Meeting stiles: Shall be ostensibly square with a subtle radiused profile in accordance with the acoustic seal manufacturers.
10. Perimeter seals: Intumescent, smoke and acoustic seals shall be of a type on which the manufacturer's certification is based on.
11. Ironmongery: Provide prototype lock cases, strikes, hinges and closers to the door manufacturer to enable factory machining/ fitting. Refer to the Ironmongery Schedule and P21.

## 302 IDS-302 Fire Rated Glazed Doors

1. Description: Internal fire rated glazed doors, as shown on the Design Drawings and Door Schedule.
   1.1. Size and configuration: Refer to the Design Drawings
   1.2. Visual reference: Radii, or acceptable equivalent.
   1.3. Specific performance:
      1.3.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
      1.3.2. Acoustic rating: Refer to the Acoustic Strategy Report and Door Schedule
   1.4. Door frame: Slim frame
      1.4.1. Type: Metal.
      1.4.2. Finish: Refer to the Design Drawings
      1.4.3. Colour: Brown.
   1.5. Door leaf:
      1.5.1. Leaf primary and secondary framing, edging: To match door frame.
      1.5.2. Glazing: To be confirmed by the Contractor to suit nominal zones, conditions of service and performance requirements.
      1.5.3. Manifestation: In accordance with Building Regulation requirements and to be agreed with the Architect.
      1.5.4. Two solid panels to be fitted on both sides of door to accommodate tablet.
   1.6. Overpanel/ sidelights (if required): Glazed to match door leaf.
      1.6.1. Performance: To match door leaf.

1.7. Seals: Intumescent, smoke and acoustic seals to be of a type on which the manufacturer's certification is based. Concealed when the door leaf is closed and inconspicuous when the door leaf is open.

1.8. Ironmongery/ hardware:

1.8.1. To be confirmed. Ironmongery/ hardware is generally not to be evident (except handles and locking devices).

1.8.2. Provide prototype lock cases, strikes, pivots and other ironmongery/ hardware to the door manufacturer to enable factory machining/ fitting.

1.8.3. Floor boxes for pivot mechanism to be inconspicuous in the finished works and concealed below the floor finishes. All apertures within floor finish for mechanism penetration(s) to be precision formed and finished neatly.

1.9. Other requirements: Silicone sealant.

## 303 IDS-303 Glazed Sliding Doors - Fire Rated

1. Description: Glazed sliding door system
2. Size and configuration: Refer to the Design Drawings and the Door Schedule.
3. Manufacturer: Radii, or acceptable equivalent.
4. Performance:
   4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
   4.2. Acoustic rating: Refer to the Acoustic Reportand Door Schedule.
5. Arrangement: Refer to the Design Drawings and Door Schedule
6. Door leaves
   6.1. Construction: Aluminium frame and doorset.
   6.2. Finish and colour: Polyester powder coated to a Standard RAL colour.
7. Operation: Manual/ Powered - to be agreed with the Architect.
8. Standard: In accordance to BS EN 16005, EN 1627, EN 356.
9. Configuration: Two sets of sliding doors.
10. Doors: Aluminium sections.
11. Glazing type: 8.76 mm laminated glass (vandal-resistant to BS EN 356, P2A)/ 10.28 mm laminated glass (vandal-resistant to BS EN 356, P5A) - to be agreed with the Architect.
12. Manifestation: Refer to the Design Drawings

## 401 IDS-201 Internal Metal Hinged Doorset

1. Description: Internal metal hinged doorset.
2. Manufacturer: HAG, or acceptable equivalent
   2.1. Product reference: Armourdoor AD70.
3. Sizes and configurations: Refer to the Design Drawings
4. Specific performance:
   4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
   4.2. Acoustic rating: Refer to the Acoustic Strategy Report and Door Schedule
   4.3. Security rating: Medium-security.
   4.4. Duty: Heavy.
5. Door frame:

5.1. Type: Rebated profile to facilitate opening, as the Design Drawings. Fully welded smooth finish with reinforced insulated (non-combustible) core.

5.2. Material: Magnelis steel.

5.3. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.

6. Door leaf:

6.1. Type/ material: Flat/ flush door leaf consisting of two skins with bevelled edges, connected with interlocking edge seams that are not evident in the finished works. Suitable to accommodate required finishes whilst maintaining the necessary performance(s).

6.2. Core: Stiffened; and including non-combustible material.

6.3. Facing material: Zinc coated mild steel to BS EN 10142.

6.4. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.

6.5. Thickness: To satisfy the performance criteria and accommodate the required ironmongery/ hardware dependent on location (refer to the designated ironmongery pack).Standard RAL colour.

7. Perimeter seals: To be of a type on which the manufacturer's certification is based, as applicable to the specific performance of the individual doorset.

8. Threshold: Slip resistant stainless steel/ aluminium threshold suitable for the service conditions.

9. Ironmongery:

- Anti-pick, high security cylinder lock, under the project Master Key plan.
- Heavy duty hinges with dog-bolts.
- Provide prototype ironmongery/ hardware to the door manufacturer to enable factory
- machining/ fitting.
- Refer to P21 and the Ironmongery Schedule.

## 402 IDS-202 Internal Insulated Metal Hinged Doorset

1. Description: Internal insulated metal hinged doorset.

2. Manufacturer: To be confirmed.

2.1. Product reference: To be confirmed.

3. Sizes and configurations: Refer to the Design Drawings

4. Specific performance:

4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.

4.2. Acoustic rating: Refer to the Acoustic Strategy Report and Door Schedule

4.3. Security rating: Medium-security.

4.4. Duty: Heavy.

5. Door frame:

5.1. Type: Rebated profile to facilitate opening, as the Design Drawings. Fully welded smooth finish with reinforced insulated (non-combustible) core.

5.2. Material: Magnelis steel.

5.3. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.

6. Door leaf:

6.1. Type/ material: Flat/ flush door leaf consisting of two skins with bevelled edges, connected with interlocking edge seams that are not evident in the finished works. Suitable to accommodate required finishes whilst maintaining the necessary performance(s).

6.2. Core: Stiffened; and including non-combustible material.

6.3. Facing material: Zinc coated mild steel to BS EN 10142.

6.4. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.

6.5. Thickness: To satisfy the performance criteria and accommodate the required ironmongery/ hardware dependent on location (refer to the designated ironmongery pack).Standard RAL colour.

7. Perimeter seals: To be of a type on which the manufacturer's certification is based, as applicable to the specific performance of the individual doorset.
8. Threshold: Slip resistant stainless steel/ aluminium threshold suitable for the service conditions.
9. Ironmongery:
    - Anti-pick, high security cylinder lock, under the project Master Key plan.
    - Heavy duty hinges with dog-bolts.
    - Provide prototype ironmongery/ hardware to the door manufacturer to enable factory
    - machining/ fitting.
    - Refer to P21 and the Ironmongery Schedule.

## 403 IDS-402 Internal Insulated Metal Hinged Riser Doorset

1. Description: Internal insulated metal hinged doorset.
2. Manufacturer: HAG, or acceptable equivalent
    2.1. Product reference: Armourdoor AD70.
3. Sizes and configurations: Refer to the Design Drawings
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Report and Door Schedule.
    4.2. Acoustic rating: Refer to the Acoustic Strategy Report and Door Schedule
    4.3. Security rating: Medium-security.
    4.4. Duty: Heavy.
5. Door frame:
    5.1. Type: Rebated profile to facilitate opening, as the Design Drawings. Fully welded smooth finish with reinforced insulated (non-combustible) core.
    5.2. Material: Magnelis steel.
    5.3. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.
6. Door leaf:
    6.1. Type/ material: Flat/ flush door leaf consisting of two skins with bevelled edges, connected with interlocking edge seams that are not evident in the finished works. Suitable to accommodate required finishes whilst maintaining the necessary performance(s).
    6.2. Core: Stiffened; and including non-combustible material.
    6.3. Facing material: Zinc coated mild steel to BS EN 10142.

6.4. Finish: Polyester powder coated, maximum 20% gloss. Colour to be confirmed from standard range.

6.5. Thickness: To satisfy the performance criteria and accommodate the required ironmongery/ hardware dependent on location (refer to the designated ironmongery pack).Standard RAL colour.

7. Perimeter seals: To be of a type on which the manufacturer's certification is based, as applicable to the specific performance of the individual doorset.
8. Threshold: Slip resistant stainless steel/ aluminium threshold suitable for the service conditions.
9. Ironmongery:
    - Anti-pick, high security cylinder lock, under the project Master Key plan.
    - Heavy duty hinges with dog-bolts.
    - Provide prototype ironmongery/ hardware to the door manufacturer to enable factory
    - machining/ fitting.
    - Refer to P21 and the Ironmongery Schedule.

## 480 IDS-512 Internal Wall Tiled Access Panel

1. Description: Proprietary non-rated visually discreet metal wall access panel for ceramic/ marble tiled walls.
2. Manufacturer: Profab Access, or acceptable equivalent
    2.1. Product reference: VISION 8000 - Non Fire Rated Riser Door (VISION 8000 NFR Riser Door - Single Door)
3. Size and configuration: Refer to the Design Drawings
4. Performance
    4.1. Fire resistance
        4.1.1. Integrity: Non-fire rated (NFR).
5. Door leaf
    5.1. Material: Electro-galvanized mild steel door tray, 1.5 mm thick.
    5.2. Finish
        5.2.1. Coating: Marine ply, 12.5 mm lining.
6. Finish: Powder-coated, matt, white (RAL 9010).
7. Acoustical Performance: To BS EN ISO 140-3, 30 dB (RW).
8. Sealing: Rubber compression gasket and moisture guard to all sides.
9. Collar: Stainless steel.
10. Door configuration: Single door (SD).
11. Position: Wall mounted.
12. Frame: Tiled frame.
13. Operation: Concealed pivot hinges outward opening (toward the user).
14. Latch or lock type: Three point locking (3PL).

## 481 IDS-401 Internal Riser Doorset

1. Description: Internal riser doorset, as shown on the Design Drawings.
2. Size(s) and configuration(s): Refer to the Design Drawings
3. Manufacturer: Selo, or acceptable equivalent.

3.1. Product reference: Quadra+.

4. Performance:
   4.1. Fire ratings (insulation and integrity): Refer to the Fire Strategy Report and Door Schedule
   4.2. Acoustic ratings: Refer to the Acoustic Report and Door Schedule
   4.3. Where indicated, provide Colt Defender F2 smoke control damper to smoke shaft openings, providing required fire integrity, insulation and smoke sealing. Coordinate with smoke control system controls and automatic operation.
5. Door leaf:
   5.1. Non-combustible insulated core.
   5.2. Mild steel encapsulated.
   5.3. Finish: Powder coated with a textured finish for on-Site painting to match the interfacing finished wall.
6. Door frame: Concealed.
7. Seals: To be of a type on which the manufacturer's certification is based, as applicable to the specific performance of the individual door set.
8. Sill: Low profile stainless steel, with slip resisting surface.
9. Ironmongery/ hardware:
   9.1. Operation: Square key drive.
   9.2. Operation/ Lock: 3 point locking system with suited Euro-cylinder, or square key drive as required, with spring loaded concealed lock operated using a touch latch with hinges to reveal the lock..
   9.3. Pivot hinges: Concealed in the finished works.
10. Signage:
    10.1. Statutory fire signage integrated within the leading edge of the door leaf.
    10.2. Other signage where shown on the Design Drawings.
11. Other requirements: Make allowance for support and fixing of fixtures and fittings as shown on the Design Drawings.

    Specialist to check and confirm in writing that accommodating such fixtures and fittings remains within the scope of the doorset testing, certification and warranty as applicable to each specific doorset. Specialist to provide adequate evidence to demonstrate compliance.

## 501 DRS-101 External Single Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification

6.2. Finish/ colour: Refer to the Façade Performance Specification

## 502 DRS-102 External Single Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 503 DRS-103 External Single Leaf Metal Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 504 DRS-104 External Double Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification

6.2. Finish/ colour: Refer to the Façade Performance Specification

## 505 DRS-105 External Double Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal framed opaque glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish: Refer to the Façade Performance Specification

## 506 DRS-106 External Double Leaf Metal Screen (Visual Intent Only)

1. Description: External double leaf metal screen, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 507 DRS-107 External Double Leaf Metal Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 508 DRS-108 External Bi-fold Doorset - Vehicle Lift (Visual Intent Only)

1. Description: External bi-fold doorset to vehicle lift, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 509 DRS-109 External Double Leaf Metal Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 510 DRS-110 Circle Slide Entrance Door (Visual Intent Only)

1. Description: Circle slide fully automated entrance doorset integrated within the curtain wall system where indicated on the Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
4. Sizes and configuration: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Doors
   6.1. Type: Refer to the Façade Performance Specification
   6.2. Glazing: Refer to the Façade Performance Specification

## 511 DRS-111 External Double Leaf Metal Framed Glazed Hinged Doorset - Curved (Visual Intent Only)

1. Description: External double leaf metal framed glazed hinged doorset - curved, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification

## 512 DRS-112 External Bi-fold Metal Framed Glazed Doorset (Visual Intent Only)

1. Description: External bi-fold metal framed doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification

## 513 DRS-113 External Single Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification

## 514 DRS-114 External Single Leaf Metal Framed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification

## 515 DRS-115 External Single Leaf Metal Framed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification

## 516 DRS-116 External Double Leaf Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification

6.2. Finish/ colour: Refer to the Façade Performance Specification

## 517 DRS-117 External Bi-fold Metal Framed Glazed Doorset (Visual Intent Only)

1. Description: External bi-fold metal framed doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 518 DRS-118 External Metal Framed Glazed Hinged Doorset (Visual Intent Only)

1. Description: External metal framed glazed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 519 DRS-119 External Integrated Green Wall Doorset (Visual Intent Only)

1. Description: External integrated green wall doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 520 DRS-120 Louvre Doorset (Visual Intent Only)

1. Description: Proprietary external metal louvred doorset to louvred screening.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish/ colour: Refer to the Façade Performance Specification
7. Door leaf:
    7.1. Material: Refer to the Façade Performance Specification
    7.2. Finish: Refer to the Façade Performance Specification

## 521 DRS-121 External Metal Framed Hinged Doorset (Visual Intent Only)

1. Description: External metal framed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Leaf:
        6.2.1. Type/ build-up: Refer to the Façade Performance Specification
        6.2.2. Finish/ colour: Refer to the Façade Performance Specification
    6.3. Finish/ colour: Refer to the Façade Performance Specification

## 522 DRS-122 External Single Leaf Metal Framed Hinged Doorset (Visual Intent Only)

1. Description: External single leaf metal framed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification

4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Leaf:
        6.2.1. Type/ build-up: Refer to the Façade Performance Specification
        6.2.2. Finish: Refer to the Façade Performance Specification
    6.3. Finish/ colour: Refer to the Façade Performance Specification

## 523 DRS-123 External Double Leaf Metal Framed Hinged Doorset (Visual Intent Only)

1. Description: External double leaf metal framed hinged doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Leaf:
        6.2.1. Type/ build-up: Refer to the Façade Performance Specification
        6.2.2. Finish: Refer to the Façade Performance Specification
    6.3. Finish/ colour: Refer to the Façade Performance Specification

## 524 DRS-124 External Single Riser Doorset (Visual Intent Only)

1. Description: External single riser doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Leaf:
        6.2.1. Type/ build-up: Refer to the Façade Performance Specification
        6.2.2. Finish: Refer to the Façade Performance Specification
    6.3. Finish/ colour: Refer to the Façade Performance Specification

## 525 DRS-125 External Double Riser Doorset (Visual Intent Only)

1. Description: External double riser doorset, as shown on the Design Drawings and Door Schedule.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame/ leaf/ seals:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Leaf:
        6.2.1. Type/ build-up: Refer to the Façade Performance Specification
        6.2.2. Finish: Refer to the Façade Performance Specification
    6.3. Finish/ colour: Refer to the Façade Performance Specification

## 526 DRS-126 Louvre Doorset (Visual Intent Only)

1. Description: Proprietary external metal louvred doorset to louvred screening.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Door frame:
    6.1. Type/ material: Refer to the Façade Performance Specification
    6.2. Finish: Refer to the Façade Performance Specification
7. Door leaf:
    7.1. Material: Refer to the Façade Performance Specification
    7.2. Finish: Refer to the Façade Performance Specification

## 527 DRS-127 Glazed Access Gate (Visual Intent Only)

1. Description: Glazed access gate to roof glazed balustrading.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer: Refer to the Façade Performance Specification
    3.1. Product reference: Refer to the Façade Performance Specification

4. Sizes and configurations: Refer to the Design Drawings and Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Gate leaf:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish: Refer to the Façade Performance Specification

## 528 DRS-128 External Top Hinged Delivery Hatch (Visual Intent Only)

1. Description: External top hinged delivery hatch, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
   4.1. Opening leaf can be fixed in open position to double as temporary canopy. Refer to the Design Drawings
5. Specific performance: Refer to the Façade Performance Specification
6. Hatch:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 529 DRS-129 External Dry Riser Cabinet (Visual Intent Only)

1. Description: External dry riser cabinet, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.
5. Specific performance: Refer to the Façade Performance Specification
6. Cabinet panels:
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 530 DRS-130 External Dry Riser/ Sprinkler Fill Cabinet (Visual Intent Only)

1. Description: External dry riser/ sprinkler fill cabinet, as shown on the Design Drawings and Door Schedule.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Manufacturer/ product reference: Refer to the Façade Performance Specification
4. Sizes and configurations: Refer to the Design Drawings and the Door Schedule.

5. Specific performance: Refer to the Façade Performance Specification
6. Cabinet panel :
   6.1. Type/ material: Refer to the Façade Performance Specification
   6.2. Finish/ colour: Refer to the Façade Performance Specification

## 611 FRP-301 Fire and Smoke Curtain Systems

1. Description: Fire curtain with smoke seals, as shown on the Design Drawings. Generally integrated within lift architraves or adjacent wall finishes.
2. Location: Refer to the Design Drawings
3. Manufacturer: Coopers, or acceptable equivalent.
4. Size and configuration: Refer to the Design Drawings
5. Automatic curtains to be designed and installed in accordance with BS-8524-1 (8) and BS 8524-2 (9) and are to operate in accordance with the requirements of the Fire Strategy.
6. Resistance to fire: Refer to the Fire Strategy Reportand Refer to the Design Drawings
7. Smoke control: Refer to the Fire Strategy Reportand Refer to the Design Drawings
8. Headbox : Fully concealed in the finished works, as show on the Design Drawings.
9. Arrangement: Refer to the Design Drawings
10. Shutter/ curtain material: To be proposed by Specialist to meet performance requirements.
11. Frame/ Guides: All framing/ guides and any other ancillary items are to be fully concealed within the finished works.
    11.1. Finish as delivered: To be colour matched to adjacent finishes.
12. Bottom bar:
    12.1. Finish: To be colour matched to adjacent finishes.
13. Control panel: Position to be concealed from general building users, position to be confirmed.
14. Operation, battery back up and accessories etc to be confirmed.
15. Other requirements: Refer to the Fire Strategy Report

## 636 RFL-101 Automatic Opening Vent

1. Description: Automatic opening vent.
2. Manufacturer: Colt International Ltd, or acceptable equivalent
3. Product reference: ECO / EN-Seefire (ECO) (Aluminium louvres)
4. Standard: To BS EN 12101-2.
5. Size/ opening: Refer to the Design Drawings.
6. Aerodynamic free area: Manufacturer's standard.
7. Reliability classification: Re 1000.
8. Snow load classification: Up to SL 2000.
9. Low ambient temperature classification: T (-25).
10. Wind load classification: Up to WL 4000.
11. Resistance to heat classification: B300.
12. Arrangement: Louvred.
13. Controls: Electric, 24 V d.c.
14. Materials
    14.1. Ventilator

14.1.1. Frame: Corrosion-resistant aluminium alloy grade 3005, insulated.

14.1.2. Louvres: Double-skin aluminium with insulation.

14.1.3. Finish: Mill finished.

14.1.4. Colour: Standard RAL colour

15. Seals: Low-loss brush seals.
16. Guards: Bird./ Burglar./ Insect.
17. Accessories: As required to complete the installation
18. Resistance to fire: E.
19. U-value: Up to 1.52 W/m²K (insulated); up to 1.52 W/m²K (uninsulated).
20. Sound insulation: Up to 11 dB Rw.
21. Actuator: Remote controlled linear actuator.

## 690 Performance

1. Standards: in accordance with Approved Document Part E of the Building Regulations, BS EN 1192, BS 6375, BS EN 12207, BS EN 12210, BS EN 12400, BS EN 13241, BS EN 14351, BS EN 16034, BS EN ISO 10077, BS 6262, BS EN 12600, BS 8214, WIS 1-47 - 'Timber external doors', published by TRADA Technology, BS 8300, LPS 1207 - Requirements for the LPCB Approval and Listing for fire performance of temporary protective covering materials for use in the interior of buildings, LPS 1215, BS EN 12217.
2. Environmental:
   2.1. External doorsets: minimum thermal performance 1.68W/m2K.
   2.2. External door air infiltration: maximum 2m3/hr/m.
   2.3. Meet requirements of BS 6375: Part 1 for external door weathertightness relating to exposure category.
3. Fire and Smoke:
   3.1. Doorsets and seals: fire resistance rating and smoke control in accordance with BS 476: Part 22 and Part 31.1 or BS EN 1634: Parts 1 and 3.
   3.2. Mark Fire doosets/doors sufficiently to show conformance to third party certifying schemes such as Exova BM TRADA Q-Mark Fire Door Manufacturer Scheme or BWF's (British Woodworking Federation) Certifire Fire Door and Doorset Scheme.
4. Acoustic:
   4.1. Doorsets and seals to achieve acoustic rating to BS EN ISO 717: Part 1.
   4.2. Door operation to be smooth and silent (maximum 68dB at 1000mm distance).
5. Mechanical strength requirements: Class 1/ 2/ 3/ 4 to BS EN 1192 - to be agreed with the Architect.
6. Security:
   6.1. External doorsets to be secure and lockable.
   6.2. In accordance with BS 6375: Part 3, Basic security. OR
   In accordance with PAS 24. OR
   In accordance with Loss Prevention Certification Board (LPCB): Security Rating 2/ 3/ 4 as LPS 1175. OR
   In accordance with burglar resistance Class 4/ 5/ 6 as BS EN 1627 - to be agreed with the Architect.

## 695 Materials

1. Standards: In accordance with BS EN 12519, BS EN 1313, BS 1245, BS 6206, BS 6262.
2. Solid core timber door panels of suitable/ specified thickness to have minimum moisture content of 12% ± 2% - internal use and 16% ± 3% - external use.
3. Hardwood to comply with sustainability and density (maximum 650kg/m3 for fire doors) requirements.
4. Metal frames manufactured in specialist facility to required profile. Door stops to be integrated. Metal frames manufactured from 1.6mm gauge zinc annealed steel with preformed fixing points and space for integral wiring where needed.
5. Connect any security doors to Building Management System.

# Execution

## 705 Workmanship Generally

1. Standards: In accordance with Door and Hardware Federation (DHF) 'Code of Practice for fire resisting metal doorsets', ASDMA documents - 'Guidance on Fire Door Installation' and 'Guidance and Recommendations for the Coordination of Bespoke Doorsets', BS 8213, BS 8214, BS 1186, BS 4787, BS 5277, EN 24, BS 5278, EN 25, BS EN 951, BS EN 1529, BS EN 1530.
2. Tolerances:
    2.1. Length and width - ±1mm.
    2.2. Thickness - ±0.5mm.
    2.3. Door to frame perimeter gaps and floor finish - 3mm (±1mm).
    2.4. Deviation from theoretical vertical plane - 1mm.
    2.5. Primary grid deviation - ±2mm.
    2.6. Plumb - ±1mm.
    2.7. Bow - ±1.5mm.
    2.8. Flatness/ straightness - +1.5mm in 2000mm in any direction.
    2.9. Perpendicular reference plane - ±2mm.
    2.10. Twist - 1° maximum.
    2.11. Cut-outs - ±1mm.
    2.12. Adjacent elements - ±2mm.

## 710 Protection of components

1. General: Do not deliver to site components that cannot be installed immediately or placed in clean, dry, floored and covered storage.
2. Stored components: Stacked on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

## 730 Priming and sealing

1. Wood surfaces inaccessible after installation: Primed or sealed as specified before fixing components.

## 740 Corrosion protection

1. Protective coating: Two coats of bitumen solution to BS 6949or an approved mastic impregnated tape.

1.1. Timing of application: Before fixing components.

## 750 Fixing doorsets

1. Timing: After associated rooms have been made weathertight and the work of wet trades is finished and dried out.

## 760 Building in

1. General: Not permitted unless indicated on drawings.

## 770 Damp proof courses associated with built-in wood frames

1. Method of fixing: To backs of frames using galvanized clout nails.

## 780 Damp-proof courses in prepared openings

1. Location: Correctly positioned in relation to door frames. Do not displace during fixing operations.

## 790 Fixing of wood frames

1. Spacing of fixings (frames not predrilled): Maximum 150 mm from ends of each jamb and at 600 mm maximum centres.

## 800 Fixing of loose thresholds

1. Spacing of fixings: Maximum 150 mm from each end and at 600 mm maximum centres.
2. Position and level: Ensure threshold is level and square to door frame and leaf.
3. Sealing: Ensure that voids are minimal and are sealed adequately.

## 809 Fire-resisting and smoke control doors/ door assemblies/ doorsets/ roller shutters and curtains – accredited installer

1. Installation: By a firm currently registered under a third-party-accredited fire door installer scheme in accordance with instructions supplied with the product conformity certificate, test report or engineering assessment.

## 810 Fire-resisting and smoke control doors/ door assemblies/ doorsets/ roller shutters and curtains – contractor-installed

1. Gaps between frames and supporting construction: Filled as necessary in accordance with requirements for certification and/ or door/ doorset manufacturer's instructions.

## 811 Fire-resisting and smoke control doorsets, industrial, commercial and garage doors

1. Installation: By manufacturer or their approved installers, in accordance with requirements of BS EN 16034 and in conjunction with BS EN 13241, including the Declaration of Performance (DoP) certification for the UKCA/ UKNI/ CE marked doorset.

## 820 Sealant joints

1. Sealant
   1.1. Colour: Refer to the Design Drawings
   1.2. Application: As section Z22 to prepared joints. Triangular fillets finished to a flat or slightly convex profile.

## 830 Fixing ironmongery generally

1. Fasteners: Supplied by ironmongery manufacturer.
   1.1. Finish/ Corrosion resistance: To match ironmongery.
2. Holes for components: No larger than required for satisfactory fit/ operation.
3. Adjacent surfaces: Undamaged.
4. Moving parts: Adjusted, lubricated and functioning correctly at completion.

## 840 Fixing ironmongery to fire-resisting door assemblies

1. General: All items fixed in accordance with door leaf manufacturer's recommendations ensuring that integrity of the assembly, as established by testing, is not compromised.
2. Holes for through fixings and components: Accurately cut.
   2.1. Clearances: Not more than 8 mm unless protected by intumescent paste or similar.
   2.2. Lock/ latch cases for fire doors requiring >60 minutes integrity performance: Coated with intumescent paint or paste before installation.

## 850 Location of hinges

1. Primary hinges: Where not specified otherwise, positioned with centre lines 250 mm from top and bottom of door leaf.
2. Third hinge: Where specified, positioned on centre line of door leaf .
3. Hinges for fire resisting doors: Positioned in accordance with door leaf manufacturer's recommendations.

## 860 Installation of emergency and panic exit devices

1. Standard: Emergency exit devices in accordance with BS EN 179.
   Panic exit devices in accordance with BS EN 1125.

Ω End of Section



make

# L30
# Stairs/ Ladders/ Walkways/ Handrails/ Balustrades

## Preliminary information/ requirements

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 105 Contractor's design

1. Design responsibility: Determine section sizes and strengths and type, sizes and numbers of fixings
2. Structural and fire requirements
   - 2.1. Generally: Refer to the Structural Engineer's Information
   - 2.2. Fire performance: Refer to the Fire Strategy Report
   - 2.3. Modifications: None.
   - 2.4. Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.
3. Functional requirements: To Building Regulations (Eng) Approved Documents K and M.
4. Additional requirements: None.
5. Design and production information: Refer to the A sections.
6. Timing of submissions: Refer to the A sections.

### 107 Completion of design

1. Requirement: Complete the detailed design to satisfy specified performance criteria and coordinate with the detailed design of related and adjacent work.
   - 1.1. Standard: To BS 8300.
2. Structural requirements: Refer to the Structural Engineer's Information
3. Fire performance: Refer to the Fire Strategy Report

### 115 Timber procurement

1. Timber (including timber for wood-based products): Obtained from well managed forests and/ or plantations in accordance with:
   - 1.1. The laws governing forest management in the producer country or countries.
   - 1.2. International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
2. Documentation: Provide either in accordance with chain of custody certification scheme requirements:
   - 2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied or
   - 2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. Chain of Custody Certification scheme: In accordance with UK Government Timber procurement policy (UKTPP), i.e. FSC, GiB or PEFC

## 130 Site dimensions

1. Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.

## 231 STR-301 Internal Metal Staircase

1. Description: Internal metal staircase and landings, as the Design Drawings
2. Performance requirements: Refer to the Structural Engineer's documentation
3. Size and configuration: Refer to the Design Drawings
4. Component materials/ grades
    4.1. Staircase structure:
        4.1.1. Type/ material: Metal support framing, Refer to the Structural Engineer's documentation
        4.1.2. Finish: Polyester powder coated.
        4.1.3. Colour: Standard RAL colour to be confirmed
    4.2. Treads/ landings:
        4.2.1. Type/ material: Refer to FLS-616 in M50.
        4.2.2. Nosing/ stair inlay:
            4.2.2.1. Type/ material/ finish: Contrasting slip proof metal nosing.
            4.2.2.2. Slip resistance value of integral nosing - water wet (minimum): PTV of 45 to BS 7976
    4.3. Risers: Closed.
    4.4. Strings: Refer to the Design Drawings
    4.5. Guarding
        4.5.1. Type/ material/ finish: Refer to BAL-141 in L30.
5. Feature lighting: Refer to the Services Engineer's Information
6. Workmanship: To section Z11.
7. Fixing: Concealed
    7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.
8. Other requirements: Include all fixings, bracketry, support framing and accessories necessary to complete the installation.

## 236 STR-302 Internal Feature Metal Stair

1. Description: Internal feature metal staircase and landings, as the Design Drawings
2. Performance requirements: Refer to the Structural Engineer's documentation
3. Size and configuration: Refer to the Design Drawings
4. Component materials/ grades
    4.1. Staircase structure:
        4.1.1. Type/ material: Metal support framing, Refer to the Structural Engineer's documentation
        4.1.2. Finish: Polyester powder coated.
        4.1.3. Colour: Standard RAL colour to be confirmed
    4.2. Treads/ landings:

4.2.1. Type/ material: Refer to FLS-201 in M40.

4.2.2. Nosing/ stair inlay:

4.2.2.1. Type/ material/ finish: Bronze inlay.

4.2.2.2. Slip resistance value of integral nosing - water wet (minimum): PTV of 45 to BS 7976

4.3. Risers: Closed.

4.4. Strings: Refer to the Design Drawings

4.5. Guarding

4.5.1. Type/ material/ finish: Refer to BAL-142 in L30.

5. Feature lighting: Integrated hidden LED lightsRefer to the Services Engineer's Information
6. Workmanship: To section Z11.
7. Fixing: Concealed

7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

8. Other requirements: Include all fixings, bracketry, support framing and accessories necessary to complete the installation.

## 240 STR-303 Internal Feature Metal Stair

1. Description: Internal feature metal staircase and landings, as the Design Drawings
2. Performance requirements: Refer to the Structural Engineer's documentation
3. Size and configuration: Refer to the Design Drawings
4. Component materials/ grades

4.1. Staircase structure:

4.1.1. Type/ material: Metal support framing, Refer to the Structural Engineer's documentation

4.1.2. Finish: Polyester powder coated.

4.1.3. Colour: Standard RAL colour to be confirmed

4.2. Treads/ landings:

4.2.1. Type/ material: Solid timber fixed to stair structure.

4.2.2. Finish: Charred

4.2.3. Nosing/ stair inlay:

4.2.3.1. Type/ material/ finish: Bronze inlay

4.2.3.2. Slip resistance value of integral nosing - water wet (minimum): PTV of 45 to BS 7976

4.3. Risers: Closed.

4.4. Strings: Refer to the Design Drawings

4.5. Guarding

4.5.1. Type/ material/ finish: Refer to BAL-143 in L30.

4.6. Cladding

4.6.1. Type/ material: Solid timber fixed to stair structure.

4.6.2. Profile/ arrangement: Refer to the Design Drawings

4.6.3. Finish: Charred

5. Feature lighting: Integrated hidden LED light at treads and balustrade,Refer to the Services Engineer's Information
6. Workmanship: To section Z11.
7. Fixing: Concealed
    7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.
8. Other requirements: Include all fixings, bracketry, support framing and accessories necessary to complete the installation.

## 251 STR-341 Internal In-Situ Concrete Stair

1. Description: Internal in-situ concrete staircase
2. Sizes and configurations: Refer to the Design Drawings
3. Stair support: Refer to the Design Drawings and Refer to the Structural Engineer's documentation
4. Components
    4.1. Treads, risers, landing:
        4.1.1. Refer to the Structural Engineer's documentation
        4.1.2. All visible concrete to have an acid etched finish. Treads/ landings to be heavily etched to provide the specified slip resistance.
    4.2. Balustrade/ handrail:
        4.2.1. Type: Refer to BAL-141.
5. Finish: Refer to CON-201 in E20
6. Colour: Natural
7. Flooring: Refer to FLS-616 in M50.
8. Nosing/ stair inlay:
    8.1. Type/ material/ finish: Contrasting slip proof metal nosing.
    8.2. Slip resistance value of integral nosing - water wet (minimum): PTV of 45 to BS 7976.
9. Soffit: Exposed steel staircase in same colour as handrail support.
    9.1. Colour: Dark bronze.
10. System accessories: Include all fixings, bracketry, support framing and accessories necessary to complete the Works in accordance with the Architectural Specification and the Design Drawings.

## 256 STR-501 Internal In-Situ Concrete Stair

1. Description: Internal in-situ concrete staircase
2. Sizes and configurations: Refer to the Design Drawings
3. Stair support: Refer to the Design Drawings and Refer to the Structural Engineer's documentation
4. Components
    4.1. Treads, risers, landing:
        4.1.1. Refer to the Structural Engineer's documentation
        4.1.2. All visible concrete to have an acid etched finish. Treads/ landings to be heavily etched to provide the specified slip resistance.
    4.2. Balustrade/ handrail:

4.2.1. Type: Refer to BAL-342 in L30.

5. Finish: Refer to CON-201 in E20
6. Colour: Natural
7. Flooring: Refer to FLS-201 in M40.
8. Nosing/ stair inlay:
   8.1. Type/ material/ finish: Bronze inlay.
9. Soffit: Plasterboard lined with painted steel returns at edges.
10. System accessories: Include all fixings, bracketry, support framing and accessories necessary to complete the Works in accordance with the Architectural Specification and the Design Drawings.

## 260 STR-506 Internal In-Situ Concrete Stair

1. Description: Internal in-situ concrete staircase
2. Sizes and configurations: Refer to the Design Drawings
3. Stair support: Refer to the Design Drawings and Refer to the Structural Engineer's documentation
4. Components
   4.1. Treads, risers, landing:
      4.1.1. Refer to the Structural Engineer's documentation
      4.1.2. All visible concrete to have an acid etched finish. Treads/ landings to be heavily etched to provide the specified slip resistance.
   4.2. Balustrade/ handrail:
      4.2.1. Type: Refer to BAL-141 in L30.
5. Finish: Refer to CON-201 in E20
6. Colour: Natural
7. Flooring: Refer to PNT-141 in M60.
8. Nosing/ stair inlay: Refer to TRM-311 in P20.
9. Soffit: Plasterboard lined with painted steel returns at edges.
10. System accessories: Include all fixings, bracketry, support framing and accessories necessary to complete the Works in accordance with the Architectural Specification and the Design Drawings.

## 501 FUS-111 External Maintenance Metal Staircase

1. Description: External maintenance metal staircase, as shown on the Design Drawings.
2. Location: Refer to the Design Drawings
3. Size and configuration: Refer to the Design Drawings
   3.1. Alignment: Stair landings to finish flush with adjoining surfaces as shown on the Design Drawings.
   3.2. Co-ordination: Make sufficient allowance within the design and fabrication programme to site measure/ co-ordinate with adjoining elements, such as where the stairs return and interface with door opening positions, as shown on the Design Drawings.
4. Support: To include secondary steelwork support where required.
   4.1. Permissible zones for structure: Refer to the Design Drawings
5. Stair structure/ component materials: Metal.

5.1. Type: To suit the service conditions.

5.2. Thickness/ gauge: To suit the service conditions.

5.3. Finish: Polyester powder coated.

6. Landings/ treads: Ball proof flat bar open mesh with solid perimeter edging as necessary, fixed to stair structure.

6.1. Slip resistance value of integral tread - water wet (minimum): PTV of 45 to BS 7976.

6.2. Nosings: Integrated proprietary hardwearing rigid nosings.

6.2.1. Fixing: Suitable mechanical fixings with colour matched inserts.

6.2.2. Slip resistance value - water wet (minimum): PTV of 45 to BS 7976.

6.2.3. Colour: LRV to BS 8493 contrast of 30 (minimum) with tread - Submit proposals.

7. Risers: Solid/ closed.

8. Balustrade/ handrail:

8.1. Configuration: Refer to the Design Drawings

8.2. Support: Side fixed to stair structure.

8.2.1. Where shown on the Design Drawings make allowance for a suitable kickplate, co-ordinated and aligned with the stair stringer/ support and fixed to the structural slab edge, to support the extension of the balustrade/ handrail beyond the stair configuration as necessary to complete the installation.

8.3. Installation height: Not less than the minimum heights stated in the Approved Documents of the Building Regulations.

8.4. Handrail/ mid-rail terminations: Include cover caps to all handrail/ mid-rail terminations.

8.4.1. Where shown on the Design Drawings the handrail is to return and extend to floor level, terminating at a suitable minimally sized circular fixing plate.

9. Fixings: To be determined by the Contractor to suit the service conditions and co-ordinated with supporting structures.

10. Other requirements: Include all fixings, bracketry, support framing and accessories necessary to complete the installation.

## 551 BAL-141 Metal Balustrade and Timber Handrail

1. Description: Metal balustrade and timber handrail as shown on the Design Drawings

2. Manufacturer: NT Handrail Design, or acceptable equivalent.

3. Performance requirements:

3.1. Loading requirements: To BS 6180.

4. Size and configuration: Refer to the Design Drawings

5. Component material, grade and finish as delivered

5.1. Guarding: Refer to the Design Drawings

5.1.1. Uprights:

5.1.1.1. Type/ material: Metal.

5.1.1.2. Profile: Refer to the Design Drawings

5.1.1.3. Size: Refer to the Design Drawings

5.1.1.4. Finish: Galvanised, to suit the service conditions.

5.1.2. Midrails:

5.1.2.1. Type/ material: Metal.

5.1.2.2. Profile: Refer to the Design Drawings

5.1.2.3. Size: Refer to the Design Drawings

5.1.2.4. Finish: Dark bronze.

5.1.3. Handrail:

5.1.3.1. Type/ material: Solid timber tubular handrail.

5.1.3.2. Profile: Refer to the Design Drawings

5.1.3.3. Size: Refer to the Design Drawings

5.1.3.4. Finish: Natural.

5.1.3.5. Installation height: Not less than the minimum heights stated in the Approved Documents of the Building Regulations.

5.1.3.6. Handrail corners: Curved and continuous as shown on the Design Drawings.

5.1.3.7. Fixing: Shall be determined by the Contractor.

5.1.4. Base:

5.1.4.1. Type/ material: Metal.

5.1.4.2. Thickness: Refer to the Design Drawings

5.1.4.3. Finish: Refer to the Design Drawings

6. Workmanship

6.1. Joinery: To section Z10.

6.2. Metalwork: To section Z11.

7. Fixing: Concealed.

7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

8. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 552 BAL-142 Metal Balustrade and Leather Finish Handrail

1. Description: Metal balustrade and leather finish handrail as shown on the Design Drawings
2. Manufacturer: NT Handrail Design, or acceptable equivalent.
3. Performance requirements:

3.1. Loading requirements: To BS 6180.

4. Size and configuration: Refer to the Design Drawings
5. Component material, grade and finish as delivered

5.1. Guarding: Refer to the Design Drawings

5.1.1. Uprights:

5.1.1.1. Type/ material: Metal.

5.1.1.2. Profile: Refer to the Design Drawings

5.1.1.3. Size: Refer to the Design Drawings

5.1.1.4. Finish: Galvanised, to suit the service conditions.

5.1.2. Midrails:

5.1.2.1. Type/ material: Metal.

5.1.2.2. Profile: Refer to the Design Drawings

5.1.2.3. Size: Refer to the Design Drawings

5.1.2.4. Finish: Dark bronze.

5.1.3. Handrail:

5.1.3.1. Type/ material: Mild steel

5.1.3.2. Profile: Refer to the Design Drawings

5.1.3.3. Size: Refer to the Design Drawings

5.1.3.4. Finish: Vegan leather. Sample to be submitted for review.

5.1.3.5. Stitching: Stitching details/ seam positioned to the underside of handrail.

5.1.3.5.1. Junctions/ terminations: End profiles or visible corner profiles of the leather to be joined with neat seam stitching detail in a thread colour to match the faux leather finish.

5.1.3.5.2. Pattern: In accordance with manufacturers recommendations and to be agreed via sampling.

5.1.3.5.3. Colour: Refer to the Design Drawings

5.1.3.6. Installation height: Not less than the minimum heights stated in the Approved Documents of the Building Regulations.

5.1.3.7. Handrail corners: Curved and continuous as shown on the Design Drawings.

5.1.3.8. Fixing: Shall be determined by the Contractor.

5.1.4. Base:

5.1.4.1. Type/ material: Metal.

5.1.4.2. Thickness: Refer to the Design Drawings

5.1.4.3. Finish: Refer to the Design Drawings

6. Workmanship

6.1. Joinery: To section Z10.

6.2. Metalwork: To section Z11.

7. Fixing: Concealed.

7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

8. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 552 BAL-171 External Glass Balustrade

1. Description: External glass balustrade as shown on the Design Drawings.
2. Performance requirements:

2.1. Loading requirements: To BS 6180.

3. Size and configuration: Refer to the Design Drawings.
4. Component material, grade and finish as delivered

4.1. Guarding:

4.1.1. Base channel:

4.1.1.1. Type/ material: Extruded profiles top mounted and bolted to the support structure.

4.1.1.2. Profile: Refer to the Design Drawings.

4.1.1.3. Finish: Brushed stainless steel.

4.1.1.4. Fixing: Concealed

4.1.2. Infills:

4.1.2.1. Type/ material: Clear low iron structural safety glass infill panels.

4.1.2.2. Panel joints: Open jointed.

4.1.2.3. Glass edge: Polished.

4.1.2.4. Thickness: To be determined by the Contractor. Glazing to be Kitemarked safety glass, fixed with flush matching beads.

4.1.3. Cap rail:

4.1.3.1. Type/ material: Extruded profile.

4.1.3.2. Profile: Refer to the Design Drawings.

4.1.3.3. Finish: Brushed stainless steel.

4.1.3.4. Fixing: As per the system manufacturers recommendations.

5. Workmanship

5.1. Metalwork: To section Z11

6. Fixing: Concealed

6.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

7. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 553 BAL-191 External Modular Guardrails

1. Description: External modular guardrails as shown on the Design Drawings
2. Performance requirements:

2.1. Loading requirements: To BS 6180.

3. Size and configuration: Refer to the Design Drawings
4. Component material, grade and finish as delivered

4.1. Guarding:

4.1.1. Uprights:

4.1.1.1. Type/ material: Metal.

4.1.1.2. Profile: Refer to the Design Drawings

4.1.1.3. Size: Refer to the Design Drawings

4.1.1.4. Finish: Galvanised, to suit the service conditions.

4.1.2. Handrail:

4.1.2.1. Type/ material: Metal.

4.1.2.2. Profile: Refer to the Design Drawings

4.1.2.3. Size: Refer to the Design Drawings

4.1.2.4. Finish: Galvanised, to suit the service conditions.

4.1.2.5. Installation height: Not less than the minimum heights stated in the Approved Documents of the Building Regulations.

4.1.2.6. Handrail corners: Mitred and welded, to suit the installation conditions.

4.1.2.7. Fixing: Concealed.

4.1.3. Base:

4.1.3.1. Type/ material: Metal.

4.1.3.2. Thickness: Refer to the Design Drawings

4.1.3.3. Finish: Refer to the Design Drawings

5. Workmanship
   5.1. Joinery: To section Z10.
   5.2. Metalwork: To section Z11.
6. Fixing: Concealed.
   6.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.
7. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 554 BAL-341 Leather Finish Handrail - Wall Mounted

1. Description: Leather finish wall mounted handrail as shown on the Design Drawings.
2. Manufacturer: NT Handrail Design, or acceptable equivalent.
3. Component material, grade and finish as delivered
   3.1. Handrails: Single piece of leather over concealed foam padding, wrapped around tubular handrail, with stitching to the underside.
      3.1.1. Tubular handrail: Mild steel
      3.1.2. Profile: Circular
      3.1.3. Diameter: Refer to the Design Drawings
      3.1.4. Finish: Vegan leather. Sample to be submitted for review.
      3.1.5. Stitching: Stitching details/ seam positioned to the underside of handrail.
         3.1.5.1. Junctions/ terminations: End profiles or visible corner profiles of the leather to be joined with neat seam stitching detail in a thread colour to match the faux leather finish.
         3.1.5.2. Pattern: In accordance with manufacturers recommendations and to be agreed via sampling.
      3.1.6. Colour: Refer to the Design Drawings
      3.1.7. Corners: Mitred and welded, to suit the installation conditions.
   3.2. Brackets:
      3.2.1. Type/ material: Metal elbow sections and cover plate, concealed and securely fixed to handrail and wall.
      3.2.2. Finish: Leather.
4. Installation height: Refer to the Design Drawings and not less than the minimum heights of the Approved Documents of the Building Regulations.
5. Workmanship
   5.1. Metalwork: To section Z11.
6. Fixing: Concealed.
   6.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.
7. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 555 BAL-342 Leather Finish Handrail - Floor Mounted

1. Description: Leather finish floor mounted handrail as shown on the Design Drawings.
2. Component material, grade and finish as delivered
    2.1. Handrails: Single piece of leather over concealed foam padding, wrapped around tubular handrail, with stitching to the underside.
        2.1.1. Tubular handrail: Mild steel
        2.1.2. Profile: Circular
        2.1.3. Diameter: Refer to the Design Drawings
        2.1.4. Finish: Vegan leather. Sample to be submitted for review.
        2.1.5. Stitching: Stitching details/ seam positioned to the underside of handrail.
            2.1.5.1. Junctions/ terminations: End profiles or visible corner profiles of the leather to be joined with neat seam stitching detail in a thread colour to match the leather finish.
            2.1.5.2. Pattern: In accordance with manufacturers recommendations and to be agreed via sampling.
        2.1.6. Colour: Refer to the Design Drawings
        2.1.7. Corners: Mitred and welded, to suit the installation conditions.
    2.2. Brackets:
        2.2.1. Type/ material: Metal elbow sections and cover plate, concealed and securely fixed to handrail and wall.
        2.2.2. Finish: Leather.
3. Installation height: Refer to the Design Drawings and not less than the minimum heights of the Approved Documents of the Building Regulations.
4. Workmanship
    4.1. Metalwork: To section Z11.
5. Fixing: Concealed.
    5.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.
6. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 556 BAL-401 Fluted Glass Balustrade

1. Description: Fluted glass balustrade as shown on the Design Drawings.
2. Manufacturer: Chelsea Artisan, or acceptable equivalent.
    2.1. Product reference : To be confirmed.
3. Performance requirements:
    3.1. Loading requirements: To BS 6180.
4. Size and configuration: Refer to the Design Drawings
5. Component material, grade and finish as delivered
    5.1. Guarding:
        5.1.1. Base channel:
            5.1.1.1. Type/ material: Extruded profiles top mounted and bolted to the support structure.

5.1.1.2. Profile: Refer to the Design Drawings

5.1.1.3. Finish: Brushed stainless steel.

5.1.1.4. Fixing: Concealed

5.1.2. Infills:

5.1.2.1. Type/ material: Clear low iron structural safety glass infill panels with a vertical fluted texture.

5.1.2.2. Panel joints: Open jointed.

5.1.2.3. Glass edge: Polished.

5.1.2.4. Thickness: Refer to the Design Drawings

5.1.3. Cap rail:

5.1.3.1. Type/ material: Extruded profile.

5.1.3.2. Profile: Refer to the Design Drawings

5.1.3.3. Finish: Brushed stainless steel.

5.1.3.4. Fixing: As per the system manufacturers recommendations.

5.1.4. Handrail:

5.1.4.1. Type/ material: Timber.

5.1.4.2. Profile: Refer to the Design Drawings

5.1.4.3. Finish: Suitable timber stain treatment, to be confirmed.

5.1.4.4. Fixing: As per the system manufacturers recommendations.

6. Workmanship

6.1. Metalwork: To section Z11

7. Fixing: Concealed

7.1. Type: To be determined by the Contractor. Where fixings cannot be concealed, finish shall be confirmed. Provide Working Drawings to show fixing locations for acceptance.

8. System accessories: As required to complete the installation, achieve the performance requirements and to suit the service conditions.

## 561 BAL-101 External Glazed Balustrade & Planting (Visual Intent Only)

1. Description: Purpose made external glass balustrade system with planting to weave parapet and listed cinema parapet interface, as shown on the Design Drawings.
2. Note:

2.1. This description covers items specified by the Façade Consultant.

2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.

3. Size and configuration: Refer to the Design Drawings
4. Component material, grade and finish as delivered

4.1. Guarding: Refer to the Façade Performance Specification

4.2. Integrated planting: Refer to the Façade Performance Specification

5. Metalwork finish: Refer to the Façade Performance Specification

## 562 BAL-102 Metal Balustrade System

1. Description: Metal balustrade system to historic mansard, modern mansard and roof, as shown on the Design Drawings.

2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Sizes and configurations: Refer to the Design Drawings
4. Manufacturer: Refer to the Façade Performance Specification
    4.1. Product reference: Refer to the Façade Performance Specification
5. Component material, grade and finish as delivered
    5.1. Guarding: Refer to the Façade Performance Specification
        5.1.1. Uprights: Refer to the Façade Performance Specification
        5.1.2. Infills: Refer to the Façade Performance Specification
    5.2. Handrails: Refer to the Façade Performance Specification
6. Metalwork finish: Polyester powder coated
7. Colour: Bronze.
8. Workmanship
    8.1. Metalwork: To section Z11
9. Fixing: Refer to the Façade Performance Specification
10. System accessories: As required to complete the installation.

## 563 BAL-103 External Glazed Inclined Balustrade & Planting (Visual Intent Only)

1. Description: Purpose made external glass inclined balustrade system with planting to modern mansard, as shown on the Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Size and configuration: Refer to the Design Drawings
4. Component material, grade and finish as delivered
    4.1. Guarding: Refer to the Façade Performance Specification
    4.2. Integrated planting: Refer to the Façade Performance Specification
5. Metalwork finish: Refer to the Façade Performance Specification

## 591 CAN-101 Prefabricated Metal Canopy System (Visual Intent Only)

1. Description: Bespoke-made cantilevered aluminium sheet canopy system, configured and to locations indicated as Design Drawings.
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements
3. Sizes and configurations: Refer to the Design Drawings
4. Component material, grade and finish as delivered
    4.1. Canopy structure:

4.1.1. Type/ material: Refer to the Façade Performance Specification

4.2. Aluminium cladding/ soffit:

4.2.1. Type/ material/ thickness: Refer to the Façade Performance Specification

5. Metalwork finish/ colour: Refer to the Façade Performance Specification

### 592 CAN-102 Prefabricated Metal Canopy System (Visual Intent Only)

1. Description: Bespoke-made cantilevered aluminium sheet canopy system, configured and to locations indicated as Design Drawings.
2. Note:
   2.1. This description covers items specified by the Façade Consultant.
   2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements
3. Sizes and configurations: Refer to the Design Drawings
4. Component material, grade and finish as delivered
   4.1. Canopy structure:
      4.1.1. Type/ material: Refer to the Façade Performance Specification
   4.2. Aluminium cladding/ soffit:
      4.2.1. Type/ material/ thickness: Refer to the Façade Performance Specification
5. Metalwork finish/ colour: Refer to the Façade Performance Specification

## Components - Not Used

## Installation

### 600 Samples

1. Metal elements: 500mm.
2. Panels: 500mm x 500mm.
3. All visible fixings and accessories.

### 601 Benchmarks

1. The first completed area/item of each system.

### 610 Moisture content

1. Temperature and humidity: Monitor and control internal conditions to achieve specified moisture content in wood components at time of installation.

### 620 Priming/ Sealing/ Painting

1. Surfaces inaccessible after assembly/installation: Before fixing components, apply full protective/decorative treatment/coating system.

### 630 Corrosion protection of dissimilar materials

1. Components/ substrates/ fasteners of dissimilar materials: Isolate using washers/ sleeves or other suitable means to separate materials to avoid corrosion and/ or staining.

## 635 Fabrication Tolerances

1. Squareness: ±1:1000 deviation to longest side.
2. Length: ±2mm.
3. Width: ±2mm.
4. Flatness: ±1mm under a 600mm straightedge.
5. Twist / Warp: < 2mm.

## 640 Installation generally

1. Fasteners and methods of fixing: To Section Z20.
2. Structural members: Do not modify, cut, notch or make holes in structural members, except as indicated on drawings.
3. Temporary support: Do not use stairs, walkways or balustrades as temporary support or strutting for other work.
4. Applied finishes: Substrates to be even, dry, sound and free from contaminants. Make good substrate surfaces and prepare/ prime as finish manufacturer's recommendation before application.
5. Visible fixings: : Stainless steel hex socket machine (Allen) screws, countersunk where possible.
6. Tolerances:
    6.1. Squareness: No diagonal panel deviation of ±3mm or ±0.075%., whichever is the lowest.
    6.2. Levelness: ±2mm from 2m straight edge
    6.3. Flatness: ±2mm from 2m straight edge
    6.4. Vertical joints: < 1.5mm from another joint.
    6.5. Vertical plane: <1 in 1000. Theoretical position ±1.5mm.
    6.6. Bow: 3mm or 0.075% whichever is the lowest.
    6.7. Twist: <1° from either end.
    6.8. Alignment: ±2mm of other elements.

## 670 Installation of tread inserts/ nosings

1. Treads: Fully cured, sound and level.

# Completion

## 910 Inspection

1. Timing: Two weeks after request by Contract Administrator
2. Period of notice (minimum): 5 working days

## 915 Slip resistance testing

1. Surfaces to be tested: All areas.
    1.1. Surface condition: Dry and wet.
2. Timing: As agreed with contract administrator
3. Period of notice (minimum): 5 working days
4. Test standard: To BS 7976
    4.1. Testing authority: A UKAS accredited laboratory

4.2. Witnessing/ Certification: Arrange for tests to be witnessed/ certified by: Contract administrator.

4.3. Report: Submit.

4.3.1. Format: As required under BS 7976

## 920 Documentation

1. Contents

1.1. Copies of structural design calculations/ test reports.

1.2. General product information.

1.3. Installation information.

1.4. Inspection and maintenance reports.

Ω End of Section



make

# L35
# Fixed Utilitarian Access Systems

## General

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 131 FUS-102 Fixed Vertical Ladder

1. Description: Fixed vertical ladder, as shown on the Design Drawings.
2. Substrate/ support: Refer to the Design Drawings
3. Size and configuration: Refer to the Design Drawings
4. To comply with BS 5395 Part 3, BS 4211, Class A with structural design to BS EN 1993.
5. Material: Painted galvanised steel.
6. Finish:
    6.1. Provide slightly textured slip resistant surface to rungs/treads and platform.
    6.2. Polyester powder coated to match cladding.
7. Fixing: System-compatible brackets mechanically anchored to support structure.
8. Access: To be removable as indicated on the Design Drawings, to provide access for plant replacement.
9. System to include fixings, framing, bracketry and other components/ accessories necessary to complete the installation.

### 141 FUS-141 Internal Metal Grate Platform

1. Description: Non-slip metal maintenance walkways.
2. Method of provision: Fabricated off site and assembled on site
3. Sizes and configurations: Refer to the Design Drawings
4. Performance: Minimum 25% free area
5. Basic component material: Galvanized carbon steel
6. Support:
    6.1. Metal channels and beams to form the structural frame.
    6.2. Mild steel flat plate and angle steel framing elements forming a site assembled decking assembly and flat mild steel base plates for bolting to structure.
7. Decking and flooring:
    7.1. Type: Welded mesh
    7.2. Manufacturer: Elefant Gratings, or acceptable equivalent
        7.2.1. Product reference: Type NP.
8. Stringers: Rectangular hollow sections.
9. Guarding system:
    9.1. Type: Balustrade and handrail/ guardrail on vertical posts fixed to cable trunking. Flat section kick rails fixed to inner face of vertical posts.

9.2. Material and protective coating: To match flooring and steps.

10. Assembly connectors:
11. Fixing to superstructure: Visible fixings shall be countersunk hex head Allen key galvanized mild steel bolts or machine screws.
12. Accessories: 150 mm high toe plate and gates at end of the walkway.
13. System to include integrated balustrades and handrails, steps and open grate platforms (where applicable) and all other accessories/ components to complete the works.

## 142 FUS-142 External Metal Grate Platform

1. Description: Non-slip metal maintenance walkways.
2. Method of provision: Fabricated off site and assembled on site
3. Sizes and configurations: Refer to the Design Drawings
4. Basic component material: Galvanized carbon steel
5. Support:
   5.1. Metal channels and beams to form the structural frame.
   5.2. Mild steel flat plate and angle steel framing elements forming a site assembled decking assembly and flat mild steel base plates for bolting to structure.
6. Decking and flooring:
   6.1. Type: Welded mesh
   6.2. Manufacturer: Elefant Gratings, or acceptable equivalent
      6.2.1. Product reference: Type NP.
7. Stringers: Rectangular hollow sections.
8. Guarding system:
   8.1. Type: Balustrade and handrail/ guardrail on vertical posts fixed to cable trunking. Flat section kick rails fixed to inner face of vertical posts.
   8.2. Material and protective coating: To match flooring and steps.
9. Assembly connectors:
10. Fixing to superstructure: Visible fixings shall be countersunk hex head Allen key galvanized mild steel bolts or machine screws.
11. Accessories: 150 mm high toe plate and gates at end of the walkway.
12. System to include integrated balustrades and handrails, steps and open grate platforms (where applicable) and all other accessories/ components to complete the works.

## 631 DRS-951 Gatic Hatch

1. Description: Waterproof gatic hatch for plant replacement
2. Manufacturer: Alumasc, or acceptable equivalent.
   2.1. Product reference: Gatic Composite.
3. Size and configuration: Refer to the Design Drawings
4. Hatch/ frame
   4.1. Orientation: Horizontal.
   4.2. Material: Metal
   4.3. Type: Gatic.
   4.4. Finish: To be confirmed

5. Gasket/ hinges/ latch/ lift assistance: As per the manufacturers recommendations to suit the conditions of service
6. Operation
   6.1. Mode: Manual.
7. Accessories: As required to complete the installation

## System performance

### 212 Contractor's design

1. Design responsibility: Determine section sizes and strengths and type, sizes and numbers of fixings
2. Structural requirements:
   2.1. Generally: Refer to the Structural Engineer's Information
   2.2. Design: Complete in accordance with the designated code of practice to satisfy specified performance criteria.
3. Functional requirements:
   3.1. Access to machinery: To BS EN ISO 14122-1, BS EN ISO 14122-2, and BS EN ISO 14122-3.
   3.2. Companionways: In accordance with BS 4211.
   3.3. External walkways: To BS 8300-1 and BS 8300-2.
   3.4. Helical and spiral stairs: In accordance with BS 5395-2.
   3.5. Industrial type flooring, walkways and stair treads: To BS 4592.
   3.6. Ladders for access to tall structures: To BS 4211.
   3.7. Straight stairs and winders: In accordance with BS 5395-1.
4. Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

### 214 Completion of design

1. Requirement: Complete the detailed design to satisfy specified performance criteria and coordinate with the detailed design of related and adjacent work.
   1.1. Standards:
      1.1.1. Access to machinery: To BS EN ISO 14122-1, BS EN ISO 14122-2, and BS EN ISO 14122-3.
      1.1.2. Companionways: In accordance with BS 4211.
      1.1.3. External walkways: To BS 8300-1 and BS 8300-2.
      1.1.4. Helical and spiral stairs: In accordance with BS 5395-2.
      1.1.5. Industrial type flooring, walkways and stair treads: To BS 4592.
      1.1.6. Ladders for access to tall structures: To BS 4211.
      1.1.7. Straight stairs and winders: In accordance with BS 5395-1.
      1.1.8. Requirements for fire safety and management: To BS 9999.
2. Structural requirements:
   2.1. Generally: Refer to the Structural Engineer's Information
   2.2. Design: Complete in accordance with the designated code of practice to satisfy specified performance criteria.
3. Timing of submissions: As Preliminaries section A31

## Products - Not Used

## Fabrication

### 510 Fabrication generally

1. Shop drawings: Submit.

## Execution

### 620 Execution generally

1. Structural members: Do not subject to nondesign loading. Do not modify, cut, notch or make unspecified holes.
2. Frameworks: Assemble and brace, including temporary members required for installation.
    2.1. Temporary support: Do not use access systems as temporary support or strutting for other work.
3. External durability of fastenings: Corrosion resistant material or with a corrosion resistant finish.
4. Bolted joints
    4.1. Contact between dissimilar metals: Avoid.
    4.2. Bolts and washers: Select types, sizes and quantities of fasteners or packings and spacings to retain supported components without distortion or loss of support.
5. Welded joints
    5.1. Standards
        5.1.1. Aluminium alloys: TIG or MIG welding to BS EN 1011-4.
        5.1.2. Carbon steel: Metal arc welding to BS EN 1011-1 and -2.
        5.1.3. Stainless steel: TIG welding to BS EN 1011-3.
    5.2. Surfaces to be jointed: Clean.
    5.3. Tack welds: Use only for temporary attachment.
    5.4. Traces of flux residue, slag and weld spatter: Remove.
    5.5. Surface of welds: Grind smooth.
    5.6. Joints: Fully bonded with no holes of cracks.
6. Finished components
    6.1. Free: From distortion, cracks, burrs and sharp arrises.
    6.2. Corner junctions of identical sections: Mitre.
    6.3. Handrails: Smooth and continuous, with no sharp edges.

### 660 Anchoring

1. Fixing positions: Coordinate location of holding down bolts and wall fixings with services fixing positions.
2. Edge distance and spacing (minimum): Refer to the Structural Engineer's Information Report any locations where the size or position of the member will prevent the correct positioning of anchorages

# Completion

## 910 Cleaning

1. General: Clean surfaces and wipe down finishes.

## 920 Inspection

1. Notice for inspection (minimum): 5 days

## 930 Documentation

1. Operation and maintenance instructions: Submit.
2. Record drawings: Submit.

Ω End of Section



make

# M10
# Cement Based Levelling/ Wearing Screeds

## Types of screed

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 106 SCR-101 Floating Screed

1. Description: Proprietary modified floating screed
2. Manufacturer: Flowcrete UK Ltd, or acceptable equivalent.
    2.1. Product reference: Isocrete K-Screed Heavy Duty.
3. Substrate: Refer to the Design Drawings
4. Screed construction: Fully bonded.
5. Preparation and priming: To the screed manufacturer's recommendations.
6. Drainage membrane: Refer to the Structural Engineer's documentation
7. Insulation:
    7.1. Manufacturer: Kingspan Insulation, or acceptable equivalent.
    7.2. Product reference: To be agreed with the Architect.
    7.3. Type: Fibre-free rigid thermoset phenolic core insulation.
    7.4. Thickness: To satisfy the stipulated performance criteria.
8. Screed:
    8.1. Type: Semi-dry cementitious screed.
    8.2. Reinforcement: Polypropylene fibres for crack control.
    8.3. Thickness: To satisfy the stipulated performance criteria.
9. Finish: Trowelled, to a uniform, smooth, but unpolished dense surface to suit final floor finish.

### 110 FBS-131 Self-Smoothing Levelling Compound

1. Description: Self smoothing levelling compound.
2. Manufacturer: Flowcrete UK Ltd, or acceptable equivalent.
    2.1. Product reference: Isocrete Self Level Plus.
3. Substrate: Refer to the Design Drawings
4. Preparation and priming: To the screed manufacturer's recommendations.
5. Screed:
    5.1. Type: High strength, fast drying, self-levelling, smoothing, cementitious screed.
    5.2. Thickness: To satisfy the stipulated performance criteria.
6. Finish: Trowelled, to a uniform, smooth, but unpolished dense surface to suit final floor finish.

### 111 SCR-201 Bonded Screed

1. Description: Modified bonded screed

2. Manufacturer: Flowcrete UK Ltd, or acceptable equivalent.
    2.1. Product reference: Isocrete K-Screed - to be confirmed.
3. Substrate: Refer to the Design Drawings
4. Screed construction: Fully bonded.
5. Preparation and priming: Solvent free epoxy resin bonding agent, to suit the service conditions.
6. Screed:
    6.1. Type: Semi-dry cementitious screed.
    6.2. Reinforcement: Polypropylene fibres for crack control.
    6.3. Thickness: To satisfy the stipulated performance criteria.
7. Finish: Trowelled, to a uniform, smooth, but unpolished dense surface to suit final floor finish.

## Generally/ preparation

### 205 Design life of screeds

1. Duration: 30 years
    1.1. Subject to reasonable wear and tear.
2. Location: Refer to the Design Drawings
3. Condition of use: Subject to correct loading and traffic usage throughout duration.

### 210 Suitability of substrates

1. General
    1.1. Suitable for specified levels and flatness/ regularity of finished surfaces. Consider permissible minimum and maximum thicknesses of screeds.
    1.2. Sound and free from significant cracks and gaps.
2. Concrete strength: In accordance with BS 8204-1, Table 2.
3. Cleanliness: Remove plaster, debris and dirt.
4. Moisture content: To suit screed type. New concrete slabs to receive fully or partially bonded construction must be dried out by exposure to the air for minimum six weeks.

### 215 Surface hardness of substrates to receive polymer modified wearing screeds

1. General: Substrates must restrain stresses that occur during setting and hardening of wearing screeds.
2. Test for surface hardness: To BS EN 12504-2 using a rebound hammer with compliance values as follows:
    2.1. Rebound hammer value (minimum)
        2.1.1. Screed thickness 15 mm or less: 25.
        2.1.2. Screed thickness greater than 15 mm: 30.
3. Report: Submit details of areas where substrate surface hardness does not comply with these values.

### 220 Proprietary levelling/wearing screeds

1. General: Materials, mix proportions, mixing methods, minimum/ maximum thicknesses and workmanship must be in accordance with recommendations of screed manufacturer.

2. Standard: In accordance with BS 8204-3 and BS EN 13813

## 225 Samples

1. General: Before placing orders submit representative samples of the following:
   1.1. Movement joints and edge restraints: 300mm long.
2. Delivered materials/ products: To match samples.

## 230 Control samples

1. General: Complete areas of finished work and obtain approval of appearance before proceeding.
2. Screed type: All types.

## 250 Conduits under floating screeds

1. Haunching: Before laying insulation for floating screeds, haunch up in 1:4 cement:sand on both sides of conduits.

## 251 Conduits cast into or under screeds

1. Reinforcement: Overlay with reinforcement selected from:
   1.1. 500 mm wide strip of steel fabric to BS 4483, reference D49, or
   1.2. Welded mesh manufactured in rolls from mild steel wire minimum 1.5 mm diameter to BS 1052, mesh size 50 x 50 mm.
2. Placing reinforcement: Mid depth between top of conduit and the screed surface.
   2.1. Width of reinforcement (minimum): 300 mm.
3. Screed cover over conduit (minimum): 50 mm

## 255 Pipe ducts/ trunking

1. Preformed access ducts: Before laying screed, fix securely to substrates and level accurately in relation to finished floor surface.

## 260 Fully bonded construction

1. Preparation: Generally in accordance with BS 8204-1.
2. Removing mortar matrix: Shortly before laying screed, expose coarse aggregate over entire area of hardened substrate.
3. Texture of surface: Suitable to accept screed and achieve a full bond over complete area.
4. Bonding coat: Manufacturer's standard

## 270 Partially bonded construction

1. Preparation: Generally in accordance with BS 8204-1.
2. Substrate surface: Brushed finish with no surface laitance.
   2.1. Texture of surface: Suitable to accept screed and achieve a bond over complete area.
3. Bonding coat: Manufacturer's standard

## 275 Cement slurry bonding

1. Slurry type: SBR polymer modified cement

2. Application: Shortly before laying screed, thoroughly wash clean the surface and keep well wetted for several hours. Remove free water then brush in cement slurry bonding coat of creamy consistency.
3. Screeding: While slurry is still wet.

## 280 Unbonded construction

1. Separation: Lay screed over a suitable sheet dpm or a separating layer.
   1.1. Type: Polyethylene sheet, minimum 125 micrometres thick (500 gauge)
2. Installation of separating layer: Lay on clean substrate. Turn up for full depth of screed at abutments with walls, columns, etc. Lap 100 mm at joints.

## 290 Floating construction

1. Insulation
   1.1. Type: 20 mm expanded polystyrene (EPS) boards to BS EN 13163, compressive strength 150 kPa
   1.2. Installation: Lay with tight butt joints. Continue up at perimeter abutments for full depth of screed.
2. Separating layer
   2.1. Type: Polyethylene sheet minimum 125 micrometres thick (500 gauge).
   2.2. Installation: Lay over insulation and turn up at perimeter abutments. Lap 100 mm at joints.

## 295 Floating construction (thin sheet impact sound insulation)

1. Substrate: Remove projections that may puncture the insulation.
2. Insulation
   2.1. Type: 5 mm extruded (closed cell) polyethylene foam, density 30-45 kg/m³
   2.2. Installation: Lay on substrate. Turn up for full depth of screed at perimeter abutments. Lap 100 mm at joints and seal with tape.
   2.3. Perimeter: Maintain isolation of screed.

# Batching/ mixing

## 302 Cements

1. Cement types: In accordance with BS 8204-1, clause 5.1.3.

## 305 Aggregates

1. Sand: To BS EN 13139.
   1.1. Grading limits: In accordance with BS 8204-1, Table B.1.
2. Coarse aggregates for fine concrete levelling screeds
   2.1. Standard: To BS EN 12620.
   2.2. Designation: 4/10.
3. Lightweight aggregates: In accordance with BS 8204-1, Annex A.

## 306 Proprietary polymer modified screeds

1. Cement types: In accordance with BS 8204-3.
2. Sand: To BS EN 13139:

2.1. Grading limits: 0/4 mm (MP) category 1

3. Aggregates: In accordance with BS 8204-3.

## 307 Admixtures

1. Standard: In accordance with BS 8204-1, Table 1.
2. Calcium chloride: Do not use in admixtures.

## 310 Batching with dense aggregates

1. Mix proportions: Specified by weight.
2. Batching: Select from:
   2.1. Batch by weight.
   2.2. Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.

## 311 Batching with lightweight aggregates

1. Standard: In accordance with BS 8204-1, Annex A.
2. Mix proportions: Specified by volume.
3. Batching: Use accurate gauge boxes.

## 330 Mixing

1. Water content: Minimum necessary to achieve full compaction, low enough to prevent excessive water being brought to surface during compaction.
2. Mixing: Mix materials thoroughly to uniform consistency. Mixes other than no-fines must be mixed in a suitable forced action mechanical mixer. Do not use a free fall drum type mixer.
3. Consistency: Use while sufficiently plastic for full compaction.
4. Ready-mixed retarded screed mortar: Use within working time and site temperatures recommended by manufacturer. Do not retemper.

## 335 In situ crushing resistance (ISCR)

1. Standards and category: In accordance with BS 8204-1, table 4.
   1.1. Testing of bonded and unbonded screeds: To Annex D.
   1.2. Testing of floating levelling screeds: To Annex E.

## 340 Adverse weather

1. Screeds surface temperature: Maintain above 5°C for a minimum of four days after laying.
2. Hot weather: Prevent premature setting or drying out.

# Laying

## 341 General

1. Maximise each pour size.
2. Bay and joints: As per working drawings or agreed with the Architect.
3. Edging: Straight, unblemished, clean and continuous. Dividing strips to be added.

## 345 Level of screed surfaces

1. Permissible deviation: (allowing for thickness of coverings) ±5 mm from datum.

## 350 Screeding to falls

1. Minimum screed cover: Maintain at the lowest point.
2. Falls: Gradual and consistent.

## 351 Screeding to ramps

1. Screed cover: Maintain consistent screed depth to graduated substrate
2. Falls: Gradual and consistent.

## 355 Flatness/ Surface regularity of floor screeds

1. Standard: In accordance with BS 8204-1, Table 5.
2. Test: In accordance with BS 8204-1, Annex C.
3. Sudden irregularities: Not permitted.

## 365 Flatness/ Surface regularity of roof screeds

1. Sudden irregularities: Not permitted.
2. Deviation of surface: Measure from underside of a 2 m straightedge (between points of contact), placed anywhere on surface.
    2.1. Permissible deviation (maximum): 6 mm.

## 375 Compaction of screeds

1. General: Compact thoroughly over entire area.
2. Screeds over 50 mm thick: Lay in two layers of approximately equal thickness. Roughen surface of compacted lower layer then immediately lay upper layer.

## 382 Stair screeds

1. Construction: Fully bonded to treads, risers and landings.
2. Risers: Form using fine finish formwork.
3. Wearing screed surfaces: Make good with compatible cement:sand mix. Wood float. When hardened remove laitance.

## 386 Coved in situ skirtings

1. Background: Refer to Design Drawings.
2. Mix: In accordance with BS 8204-3, Table 2. Aggregate size adjusted as necessary to suit skirting thickness.
3. Construction joint: Form at base.
4. Rendering: Apply bonding agent. Render skirting while bonding agent is still wet. Achieve a good bond. Allow each coat to set before applying subsequent coats. Render to true lines with a fine finish and an even consistent appearance.
5. Dimensions
    5.1. Thickness of each coat (maximum): 10 mm.
    5.2. Height: Refer to Design Drawings.

5.3. Cove radius: 30 mm.

5.4. Top edge: To be confirmed by the Architect.

## 392 General reinforcement

1. Steel fabric: To BS 4483.
2. Installation: In accordance with BS 8204-1.

## 395 Strip reinforcement

1. Location: Across day joints
2. Steel fabric: To BS 4483.
3. Installation: In accordance with BS 8204-1.

## 405 Joints in levelling screeds generally

1. Laying screeds: Lay continuously using 'wet screeds' between strips or bays. Minimize defined joints.
2. Daywork joints: Form with vertical edge.

## 406 Bay joints in levelling screeds

1. Screed type: Refer to Specification.
2. Bay sizes
   2.1. Area (maximum): 20 m²
   2.2. Length (maximum): 6 m
3. Location of bay joints: Coordinate with those required for substrate slab and floor covering.

## 415 Bay joints in bonded wearing screeds

1. Bay sizes
   1.1. Area (maximum): 15 m²
   1.2. Length: Breadth ratio (maximum): 3:2.
2. Location of bay joints: Submit proposals

## 428 Heated screeds

1. Substrate slab: Irregularities not permitted.
2. Screed bays, movement joints and other joints: Coordinate with heating circuits.
3. Heating elements: Secure properly. Prevent displacement.
4. Screed laying: Compact thoroughly around heating elements. Do not damage them.

## 435 Formed joints in wearing screeds

1. Temporary forms: Square edged with a steel top surface and in good condition.
2. Placing screed: Compact thoroughly at edges to give level, closely abutted joints with no lipping.

## 440 Crack inducing grooves in levelling screeds

1. Groove depth: At least half the depth of screed.
2. Cutting grooves: Straight, vertical and accurately positioned. Select from the following:

2.1. Trowel cut as screed is laid.

2.2. Saw cut sufficiently early after laying to prevent random cracking.

## 445 Crack inducing grooves in wearing screeds

1. Groove dimensions
   1.1. Depth: At least half the depth of wearing screed.
   1.2. Width: As narrow as possible
2. Cutting grooves: Straight, vertical and accurately positioned. Saw cut sufficiently early after laying to prevent random cracking.

## 450 Sealant for sawn joints in wearing screeds

1. Type: Two part polysulfide-based, colour to approval
2. Preparation and application: As section Z22.

## 452 Sealing strips for sawn joints in wearing screeds

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals
   1.2. Capping colour: To approval

## 460 Strip movement joints

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals
   1.2. Size: 5 mm
2. Installation: Set securely into screed to exact finished level of floor. Extend joints through to substrate.
   2.1. Secure fixing to substrate: To manufacturer's recommendation.

## 468 Sealant movement joints with metal edgings

1. Edging material: Stainless steel angle
   1.1. Size: 75 x 50 x 5 mm
   1.2. Bedding: 1:3 cement:sand.
2. Installation: Centre over joints in substrate. Set to exact finished level of floor.
   2.1. Secure fixing to substrate: Manufacturer's standard
3. Joint width: To match that of structural movement joint in substrate
4. Sealant
   4.1. Manufacturer: Submit proposals
   4.2. Product reference: Submit proposals
   4.3. Colour: Manufacturer's standard colour range
   4.4. Preparation and application: As section Z22.

## 476 Metal section movement joints

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals

1.2. Colour of insert: Manufacturer's standard colour range

2. Bedding: 1:3 cement:sand.
3. Installation: Centre over joint in substrate. Set to exact finished level of floor.
   3.1. Secure fixing to substrate: To manufacturer's standard

## Finishing/curing

### 510 Finishing generally

1. Timing: Carry out all finishing operations at optimum times in relation to setting and hardening of screed material.
2. Prohibited treatments to screed surfaces
   2.1. Wetting to assist surface working.
   2.2. Sprinkling cement.

### 520 Wood floated finish

1. Finish: Slightly coarse, even texture with no ridges or steps.

### 530 Smooth floated finish

1. Finish: Even texture with no ridges or steps.

### 540 Trowelled finish to levelling screeds

1. Floating: To an even texture with no ridges or steps.
2. Trowelling: To a uniform, smooth but not polished surface, free from trowel marks and other blemishes, and suitable to receive specified flooring material.

### 550 Trowelled finish to wearing screeds

1. Floating: To an even texture with no ridges or steps.
2. Trowelling: Successively trowel at intervals, applying sufficient pressure to close surface and give a uniform smooth finish free from trowel marks and other blemishes.

### 560 Dewatered trowelled finish to wearing screeds

1. Dewatering: Immediately after compaction of wearing screeds, remove water using a vacuum process.
2. Floating: Without delay, power float to an even texture with no ridges or steps.
3. Trowelling: Successively trowel at intervals, applying sufficient pressure to close surface and give a uniform smooth finish free from trowel marks and other blemishes.

### 600 Power ground finish to wearing screeds

1. Floating: To an even surface with no ridges or steps. Immediately commence curing.
2. Grinding: When concrete is sufficiently hard for sand particles not to be torn from surface, remove 1 - 2 mm from surface to give an even glass-paper texture, free from blemishes and trowel marks.
3. Cleaning: Remove dust and wash down. Resume curing without delay.

## 650 Curing

1. General: Prevent premature drying. Immediately after laying, protect surface from wind, draughts and strong sunlight. As soon as screed has set sufficiently, closely cover with polyethylene sheeting.
2. Curing period (minimum): Keep polyethylene sheeting in position for: period recommended by screed manufacturer.
3. Drying after curing: Allow screeds to dry gradually. Do not subject screeds to artificial drying conditions that will cause cracking or other shrinkage related problems.

## 670 Roof screeds

1. Protection: Cover screeds during wet weather. When weathertight coverings are laid, screeds must be as dry as practicable.

## 680 Surface sealer to wearing screeds

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals
2. Preparation: Clean cured screed surface to remove dirt, grease, oil and other surface contaminants.
3. Moisture content of screed: As recommended by sealer manufacturer. Test relative humidity in accordance with BS 8203, Annex A if required to verify suitability to receive sealer.
4. Application: Evenly to dry surfaces using sufficient coats to form an effective seal but without a glossy finish.

## 690 Slip resistance testing of wearing screeds

1. Test
   1.1. To BS 7976-1, -2 and -3 using a TRL Pendulum.
   1.2. When testing with a surface roughness meter, achieve minimum of 20 µm Rz.
   1.3. Make arrangements for test to be witnessed/ certified by: Consultant
2. Report: Submit. Include slip resistance values in the wet and dry states.
3. Pedestrian screed requirements:: 36PTV (wet and dry conditions).

## 700 Abrasion testing of wearing screeds

1. Test method: To BS EN 13892-4.

## 710 Impact testing

1. Use 4kg weight, to comply with the BRE Information Paper 11/84 - 'BRE screed tester: classification of screeds, sampling and acceptance limits'.

Ω End of Section

# M12
# Resin Flooring

## Types of flooring

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 FLS-401 Resin Floor Finish

1. Description: External grade, resin anti-slip floor finish, as shown on the Design Drawings.
2. Manufacturer: Flowcrete, or acceptable equivalent.
    2.1. Product reference: Flowcoat.
3. Substrate: Refer to the Design Drawings
4. Preparation/ primer: To manufacturers requirements and to suit the conditions of use.
5. Membrane coat: To manufacturers requirements and to suit the conditions of use.
6. Bodycoats: To manufacturers requirements and to suit the conditions of use.
7. Resin system/ topcoat:
    7.1. No. of coats: To manufacturers requirements and to suit the conditions of use.
    7.2. Thickness: Total system thickness and interface between vertical/ horizontal surfaces to be in accordance with manufacturers/ suppliers instructions.
    7.3. Surface finish/ treatment: Apply a graded aggregate between coats, in accordance with suppliers instructions, to achieve the slip resistance.
    7.4. Grade: External.
    7.5. Colour: Colours to be selected from manufacturer's standard colour range to suit the proposed color scheme, refer to the Design Drawings.
8. Line marking:
    8.1. Size and configuration: Refer to the Design Drawings
        8.1.1. Templates: Allow for templating for consistent applications of line marking.
    8.2. Preparation/ primer: To manufacturers requirements and to suit the conditions of use.
    8.3. Resin system/ line marker:
        8.3.1. Make allowance for the application of additional graded aggregate/ material in accordance with the manufacturers recommendations to maintain the required level of slip resistance following the application of line marking.
    8.4. Colour(s): Various bespoke.
    8.5. Application: In accordance with the manufacturers details and recommendations to achieve the performance requirements and suit the service conditions.
9. Joints and junctions: Not accepted, to represent a seamless appearance.
10. Slip resistance: Dry > 80/ Wet > 35 low slip potential according to HSE and UKSRG guidelines, having been tested in accordance with method described in BS 7976-2.
11. Application:
    11.1. Preparatory work and application in accordance with suppliers instructions.
12. Transition strips/ trims: : Refer to the Design Drawings

13. Skirting: Deckshield Rapide ED1-Ramp skirting applied minimum 150mm up wall face and finish to horizontal line.
14. System to include all fixings, sealants, adhesives, jointing/ transition strips, trims, and all other accessories/ components recommended by the manufacturer to suit the service conditions.

## Preparation of substrates

### 210 Testing moisture content of substrates

1. Drying aids: Remove minimum four days prior to test.
2. Test: In accordance with BS 8203, Annex B using an accurately calibrated hygrometer.
    2.1. Location of readings: Corners, along edges, and at various points over the test area.
3. Relative humidity before laying resin flooring (maximum): As per manufacturers recommendations.

### 220 Surface hardness of substrates

1. General: Substrates must restrain stresses that occur during setting and hardening of resin.
2. Test for surface hardness: To BS EN 12504-2 using a rebound hammer.
3. Test results: Submit.
4. Areas of non-compliance: Submit remedial proposals.

### 230 Preparation of substrates generally

1. Chases/ Saw cuts: Cut/ break out at skirtings, free edges, movement joints, etc. for termination of resin flooring.
2. Blow holes, cavities, cracks, etc.: Fill with repair product recommended by resin flooring manufacturer.
3. Cleanliness: Remove surface contaminants, debris, dirt and dust.
4. Surface texture: Suitable to accept resin flooring and achieve a full bond over the complete area.

### 240 Existing substrates

1. Preparation: Remove surface imperfections, ingrained contaminants, coatings and residues.
    1.1. Contaminated areas: Submit proposals for removal and repair.

### 250 Existing tile/ sheet floor coverings

1. Preparation: Remove coverings, residual adhesive, bedding, grouting and pointing.

### 260 Metal substrates

1. Cleaning: Shot blast to BS EN ISO 8501-1, Grade Sa2.5.
    1.1. Treated surfaces: Clean. Free from visible oil, grease and dirt, mill scale, rust, paint coatings and foreign matter.

## Laying flooring

### 310 Workmanship

1. Operatives
    1.1. Trained/ Experienced in the application of resin floorings.

1.2. Evidence of training/ experience: Submit on request.

2. Fillers and incorporated aggregates: Thoroughly mix in to ensure wetting. Avoid over-vigorous mixing resulting in excessive air entrainment.
3. Scattered aggregates: Broadcast onto wet surface of resin.
   3.1. Appearance: Consistent.
4. Curing: Allow appropriate periods between coats and before surface treatments and trafficking/ use.

## 315 Samples

1. General: Before placing orders submit representative samples of the following:
   1.1. 300 x 300mm of each complete type inclusive of skirtings.
2. Delivered materials/ products: To match samples.

## 320 Control samples

1. Complete areas of finished work in the following locations: All specified types.
2. Approval of appearance: Obtain before proceeding.

## 330 Primers

1. Application: Even to completely wet, penetrate and seal substrates.
2. Coverage per coat (minimum): As per manufacturers recommendations.
3. Curing period (maximum): 48 hours at 15-20°C before application of resin flooring or as per manufacturers recommendations.

## 340 Reinforcement

1. Fabric: Glass fibre mat
   1.1. Locations: Refer to Design Drawings.
   1.2. Laps (minimum): 50 mm.
   1.3. Bedding: Roll into preliminary thin layer of resin flooring.

## 350 Coated resin flooring

1. Application: Even, of uniform thickness, surface finish and colour.
2. Resin coatings will not deteriorate in appearance or performance when resisting water penetration from below.

## 355 Flow applied coated resin flooring

1. Application: Evenly, of uniform thickness, surface finish and colour.
2. Trapped air: Roll to release.

## 357 Trowel applied screeded resin flooring

1. Application: Even and thoroughly compacted, of uniform thickness, surface finish and colour.

## 360 Surface abrading/ polishing

1. Surface on completion: Uniform, fine and dust free.

## 370 Surface grinding/ grouting/ abrading/ polishing

1. Surface on completion
    1.1. Grinding: Regular appearance to exposed aggregate and dust free.
    1.2. Grouting: Surface defects filled.
    1.3. Abrading/ Polishing: Uniform, fine and dust free.

## 380 Surface sealer

1. Application: Even to completely wet resin surface.

## 390 Slip resistance testing of finished resin flooring

1. Standard: In accordance with BS 8204-6, Annex B and pendulum test to BS 7976-2.
2. Timing: Give notice.
3. Results: Submit pendulum test values, for both wet and dry states.
    3.1. When testing with a surface roughness meter, achieve minimum of 20 µm Rz.
    3.2. Pedestrian screed requirements: 36PTV (wet and dry conditions).

## 400 Bond strength of resin flooring

1. Contact surfaces: Substrate and fully cured resin flooring.
2. Bond: In accordance with manufacturer's performance data.
3. Test: To BS 8204-6, clause 11.4 and BS EN 1542.

## 410 Resin skirtings/ upstands

1. Quality: Render grade resin.
2. Colour: As flooring
3. Profile: Radius cove.
4. Top edge of skirting/ upstand: Retain with a stop bead
5. Transition with resin flooring: Strip movement joint

## 420 Free edges of resin flooring

1. Transition to abutting floor finishes: Straight and smooth.
2. Retention of resin edges: Chamfered edge chase in slab

## 430 Sealant movement joints

1. Location: Centre over movement joints in substrate.
2. Preparation and application: As section Z22.
3. Retention of resin edges: Chase in slab
4. Joint width: Refer to the Design Drawings
5. Sealant: Contractor's choice
    5.1. Colour: To be confirmed by Architect.

## 440 Strip movement joints

1. Location: Centre over movement joints in substrate.
2. Retention of resin edges: Chase in slab

3. Joint width: Refer to the Design Drawings
4. Strip
    4.1. Manufacturer: Submit proposals
        4.1.1. Product reference: Submit proposals
    4.2. Insert: Synthetic rubber
        4.2.1. Colour: To be agreed with the Architect.
    4.3. Fixing: As per manufacturers recommendations.

## 450 Applied slip resistant accessories

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals
2. Material: To be confirmed by Architect.
    2.1. Slip resistance: Pendulum test value to BS 7976-2: 40 (minimum) for wet and dry tests.
3. Size/ Coverage: In accordance with BS 8300-2
4. Colour: Submit proposals
5. Accessories/ Features: None

Ω End of Section

# M40
# Stone/ Concrete/ Quarry/ Ceramic/ Porcelain Tiling/ Mosaic

## Types of tiling/ mosaic

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 FLS-201 Stone Large Format Tile

1. Description: Stone large format tile as shown on the Design Drawings.
2. Substrate(s): Refer to the Design Drawings
    2.1. Use proprietary products to suit the substrate(s).
    2.2. Make allowance for suitable acoustic impact absorbent matting beneath the floor finish when applied to stairs.
3. Preparation: A levelling screed to be provided where necessary, in accordance with Section M10.
4. Adhesive/ primer: As recommended by the tile manufacturer, to suit the service conditions.
5. Size and configuration: Refer to the Design Drawings
6. Tiles:
    6.1. Type/ material: Ceramic.
    6.2. Face size(s): 1500 x 1500 mm.
    6.3. Thickness: To be agreed with the Architect.
    6.4. Colour: Light beige.
    6.5. Finish: Structured.
    6.6. Slip potential
        6.6.1. Slip resistance value (SRV) (minimum)/ Pendulum test value (PTV) (minimum) to BS 7976-1, -2, -3 or BS EN 14231 (natural stone only): To be agreed with the Architect.
        6.6.2. Surface roughness (Rz) (minimum) to BS 1134: To be agreed with the Architect.
7. Joint width: Nominal 2-3mm, to be agreed.
8. Movement joints: To visually match grout and to be agreed via sampling.
9. Grout:
    9.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    9.2. Colour: To match surrounding tile colour/ To be agreed.
10. Grout sealer: Proprietary penetrative sealer that does not impair the visual or performance requirements of the grout. Sealer to be resistant to water, oils, acid based contaminates, alkaline-based contaminants and staining, as recommended by the grout manufacturer.
11. Transition strips/ trims: Refer to the Design Drawings
12. Skirtings: Refer to the Design Drawings

13. The works to be laid in bays so that controlling dimensions can be checked and aligned with the planning grid.
14. Floor finish to be supplied and installed as complete integrated systems, including fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components recommended/ supplied by the flooring/ lining manufacturer to suit the service conditions.

## 112 FLS-202 Ceramic Tile Flooring - Large Format

1. Description: Ceramic tile floor finish as shown on the Design Drawings.
2. Substrate(s): Refer to the Design Drawings
    2.1. Use proprietary products to suit the substrate(s).
    2.2. Make allowance for suitable acoustic impact absorbent matting beneath the floor finish when applied to stairs.
3. Preparation: A levelling screed to be provided where necessary, in accordance with Section M10.
4. Adhesive/ primer: As recommended by the tile manufacturer, to suit the service conditions.
5. Size and configuration: Refer to the Design Drawings
6. Tiles:
    6.1. Type/ material: Ceramic.
    6.2. Face size(s): 1200 x 600 mm.
    6.3. Thickness: To be agreed with the Architect.
    6.4. Colour: To be agreed with the Architect.
    6.5. Finish: To be agreed with the Architect.
    6.6. Slip potential
        6.6.1. Slip resistance value (SRV) (minimum)/ Pendulum test value (PTV) (minimum) to BS 7976-1, -2, -3 or BS EN 14231 (natural stone only): To be agreed with the Architect.
        6.6.2. Surface roughness (Rz) (minimum) to BS 1134: To be agreed with the Architect.
7. Joint width: Nominal 2-3mm, to be agreed.
8. Movement joints: To visually match grout and to be agreed via sampling.
9. Grout:
    9.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    9.2. Colour: To match surrounding tile colour/ To be agreed.
10. Grout sealer: Proprietary penetrative sealer that does not impair the visual or performance requirements of the grout. Sealer to be resistant to water, oils, acid based contaminates, alkaline-based contaminants and staining, as recommended by the grout manufacturer.
11. Transition strips/ trims: Refer to the Design Drawings
12. Skirtings: Refer to the Design Drawings
13. The works to be laid in bays so that controlling dimensions can be checked and aligned with the planning grid.
14. Floor finish to be supplied and installed as complete integrated systems, including fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components recommended/ supplied by the flooring/ lining manufacturer to suit the service conditions.

## 114 FLS-204 Porcelain Tile Flooring

1. Description: Porcelain tile floor finish as shown on the Design Drawings.
2. Manufacturer: Iris Ceramica, or acceptable equivalent.
   2.1. Product reference: Victorian Stone Moon Cream.
3. Substrate(s): Refer to the Design Drawings
   3.1. Use proprietary products to suit the substrate(s).
   3.2. Make allowance for suitable acoustic impact absorbent matting beneath the floor finish when applied to stairs.
4. Preparation: A levelling screed to be provided where necessary, in accordance with Section M10.
5. Adhesive/ primer: As recommended by the tile manufacturer, to suit the service conditions.
6. Size and configuration: Refer to the Design Drawings
7. Tiles:
   7.1. Type/ material: Ceramic.
   7.2. Face size(s): 1200 x 600 mm,
   1200 x 1200 mm.
   Refer to the Design Drawings.
   7.3. Thickness: 20 mm.
   7.4. Colour: Mooncream.
   7.5. Finish: Antislip.
   7.6. Slip potential
      7.6.1. Slip resistance value (SRV) (minimum)/ Pendulum test value (PTV) (minimum) to BS 7976-1, -2, -3 or BS EN 14231 (natural stone only): To be agreed with the Architect.
      7.6.2. Surface roughness (Rz) (minimum) to BS 1134: To be agreed with the Architect.
8. Joint width: Nominal 2-3mm, to be agreed.
9. Movement joints: To visually match grout and to be agreed via sampling.
10. Grout:
    10.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    10.2. Colour: To match surrounding tile colour/ To be agreed.
11. Grout sealer: Proprietary penetrative sealer that does not impair the visual or performance requirements of the grout. Sealer to be resistant to water, oils, acid based contaminates, alkaline-based contaminants and staining, as recommended by the grout manufacturer.
12. Transition strips/ trims: Refer to the Design Drawings
13. Skirtings: Refer to the Design Drawings
14. The works to be laid in bays so that controlling dimensions can be checked and aligned with the planning grid.
15. Floor finish to be supplied and installed as complete integrated systems, including fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components recommended/ supplied by the flooring/ lining manufacturer to suit the service conditions.

## 141 LIN-301 Natural Stone Tile Wall Finish

1. Description: Natural stone tile wall finish, as the Design Drawings.

2. Substrate: Refer to the Design Drawings
3. Support framework (if required): To meet the structural requirements in accordance with the manufacturers recommendations.
4. Stone
    4.1. Name (traditional): Portland limestone - Jordans Whitbed Portland Stone.
    4.2. Manufacturer: Albion Stone, or acceptable equivalent.
        4.2.1. Product reference: Grove Whitbed Portland Stone
    4.3. Colour: To be agreed with the Architect.
    4.4. Finish: To be agreed with the Architect.
    4.5. Quality: Free from vents, cracks, fissures, discoloration, or other defects deleterious to strength/ colour.
    4.6. Size: To be agreed with the Architect.
    4.7. Thickness: To be agreed with the Architect.
5. Joint width: To be agreed with the Architect.
6. Grout:
    6.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    6.2. Colour: To match tiles.
7. Edge trims: Refer to the Design Drawingsand Section P20.

## 151 LIN-411 Porcelain Tile Wall Finish

1. Description: Large format porcelain tile wall finish, as the Design Drawings.
2. Substrate: Refer to the Design Drawings
3. Adhesive/ primer: To suit the service conditions.
4. Tiles:
    4.1. Manufacturer/ Supplier: Iris Ceramica/ Minoli, or acceptable equivalent.
        4.1.1. Product reference: Marvel Diva Taj Mahal.
    4.2. Colour: To be agreed with the Architect.
    4.3. Finish: To be agreed with the Architect.
    4.4. Size: To be agreed with the Architect.
    4.5. Thickness: To be agreed with the Architect.
    4.6. Recycled content: To be agreed with the Architect.
5. Joint width: To be agreed with the Architect.
6. Grout:
    6.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant as recommended by the tile manufacturer.
    6.2. Colour: To match tiles
7. Edge trims: Refer to the Design Drawings and Section P20.

## 152 LIN-412 Porcelain Tile Wall Finish

1. Description: Proprietary porcelain medium format wall tile finish, as the Design Drawings.
2. Substrate: Refer to the Design Drawings
3. Adhesive/ primer: To suit the service conditions.

4. Tiles:
    4.1. Manufacturer/ Supplier: Iris Ceramica, or acceptable equivalent.
        4.1.1. Product reference: Pietra de Bilbao.
    4.2. Colour: Ice/ or sand.
    4.3. Finish: To be agreed with the Architect.
    4.4. Size: 1200 mm x 1200 mm and/or 1200 mm x 600 mm.
    600 mm x 300 mm.
    4.5. Thickness: 9 mm.
    4.6. Recycled content: To be agreed with the Architect.
5. Joint width: To be agreed with the Architect.
6. Grout:
    6.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant as recommended by the tile manufacturer.
    6.2. Colour: To match tiles
7. Edge trims: Refer to the Design Drawings and Section P20.
8. Skirting: 9 mm recessed ceramic skirting at base of wall finish

## 153 LIN-413 Porcelain Tile Wall Finish

1. Description: Proprietary porcelain medium format wall tile finish, as the Design Drawings.
2. Substrate: Refer to the Design Drawings
3. Adhesive/ primer: To suit the service conditions.
4. Tiles:
    4.1. Manufacturer/ Supplier: Iris Ceramica, or acceptable equivalent.
        4.1.1. Product reference: Pietra de Bilbao.
    4.2. Colour: Ice/ or sand.
    4.3. Finish: To be agreed with the Architect.
    4.4. Size: 1200 mm x 1200 mm and/or 1200 mm x 600 mm.
    600 mm x 300 mm.
    4.5. Thickness: 9 mm.
    4.6. Recycled content: To be agreed with the Architect.
5. Joint width: To be agreed with the Architect.
6. Grout:
    6.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant as recommended by the tile manufacturer.
    6.2. Colour: To match tiles
7. Edge trims: Refer to the Design Drawings and Section P20.
8. Skirting: 9 mm recessed ceramic skirting at base of wall finish

## 190 Floor finish materials specification

1. Minimum BRE 'Green Guide to Specification Online' rating: Submit proposals

# General

## 210 Suitability of backgrounds/ bases

1. Background/ base tolerances: To permit specified flatness/ regularity of finished surfaces given the permissible minimum and maximum thickness of bedding.
2. New background drying times (minimum)
    2.1. Concrete walls: Six weeks.
    2.2. Brick/ block walls: Six weeks.
    2.3. Rendering: Two weeks.
    2.4. Gypsum plaster: Four weeks.
3. New base drying times (minimum)
    3.1. Concrete slabs: Six weeks.
    3.2. Cement:sand screeds: Three weeks.

## 215 Falls in the bases

1. General: Give notice if falls are inadequate.

## 220 Swimming pool tiling

1. Watertightness of pool base/ substrate structure: Test and prove to be watertight.
2. Time intervals between stages of the work (minimum)
    2.1. Completion (curing) of pool shell to start of rendering/ screeding/ tiling: Six weeks
    2.2. Completion of rendering/ screeding to start of tiling: Three weeks
    2.3. Completion of tiling to start of grouting: Three days
    2.4. Completion of grouting and sealing of movement joints to filling of pool: Three weeks.
3. Filling, emptying and heating pool: Minimize stresses.
    3.1. Maximum filling and emptying rate: 750 mm/24 hour.
    3.2. Maximum rate of heating water: 0.25°C/hour.

## 250 Samples

1. General: Submit representative samples of the following
    1.1. Minimum 10 No. full size tiles of each type.
    1.2. 1500 x 1000mm Sample board.

## 252 Natural stone samples

1. General: Submit samples in accordance with BS EN 12057, clause 4.2.3.2 for natural stone tiles and BS EN 12058, clause 4.2.3.2 for natural stone slabs.

## 260 Control samples

1. General: Complete sample areas, being part of finished work, in locations as follows: Refer to the Design Drawings.
    1.1. Approval of appearance: Obtain before proceeding.
2. Floor covering slip resistance testing:

## 270 Materials, Performance & Quality

1. Standards: In accordance with BS 5385, BS EN 1991, BS EN ISO 10545, BS ISO 13006. BS EN 14411, BS 8000, BS 8204.
2. Loadings: Floor tiles and components to resist the following loads:
   2.1. Dead loads: 3kN/m2
   2.2. Point loads: 10kN
   2.3. Imposed loads: 5kN/ m2
   2.4. Imposed loads when supporting partition systems: 6kN/m2
3. Waterproofing: In accordance with BS EN 14891.
4. Grouts / adhesives: In accordance with BS EN 12004, BS EN 13888, BS ISO 13007.

# Preparation

## 310 Existing backgrounds/ bases generally

1. Efflorescence, laitance, dirt and other loose material: Remove.
2. Deposits of oil, grease and other materials incompatible with the bedding: Remove.
3. Tile, paint and other nonporous surfaces: Clean.
4. Wet backgrounds: Dry before tiling.

## 320 Existing concrete/ screeds

1. Loose or hollow portions: Cut out.
2. Making good: Material recommended by tiling adhesive manufacturer.

## 330 Existing plaster

1. Defective areas: Remove plaster that is loose, soft, friable, badly cracked or affected by efflorescence. Cut back to straight horizontal and vertical edges.
2. Making good: Use plaster or nonshrinking filler.

## 340 Existing glazed brick

1. Defective areas: Cut out.
2. Making good: Material recommended by tiling adhesive manufacturer.

## 350 Existing tiles

1. Loose or hollow sounding tiles: Remove.
2. Making good: Material recommended by tiling adhesive manufacturer.

## 355 Old adhesive residues on concrete/ screed bases

1. Soft or unsound adhesive residues: Remove without damaging base.

## 360 Existing paint

1. Paint with unsatisfactory adhesion: Remove so as not to impair bedding adhesion.

## 370 New in situ concrete

1. Backgrounds/ bases to be tiled: Remove mould oil, surface retarders and other materials incompatible with bedding.

## 380 New plaster

1. Plaster: Dry, solidly bedded, free from dust and friable matter.
2. Plaster primer: Apply if recommended by adhesive manufacturer.

## 390 Plasterboard backgrounds

1. Boards: Dry, securely fixed and rigid with no protruding fixings and face to receive decorative finish exposed.

## 400 Backgrounds

1. Boards: Dry, securely fixed and rigid with no protruding fixings.
2. Surfaces to be tiled: Seal or prime if recommended by adhesive manufacturer.

## 410 Hacking for key

1. Keying: Roughen backgrounds thoroughly and evenly to a depth of 3 mm.

## 420 Raking out for key

1. Soft joints in existing masonry: Rake out to a depth of 13 mm (minimum).

## 430 Spatterdashing for key

1. Spatterdash mix
    1.1. Cement: As per the manufacturers recommendations
    1.2. Sand: Clean coarse.
    1.3. Proportions (cement:sand): As per the manufacturers recommendations
    1.4. Admixture: BBA-certified SBR bonding agent
    1.5. Consistency: Thick slurry with no segregation.
2. Application/finish: Thrown.
    2.1. Thickness: 3–5 mm.
3. Curing/ drying: Keep damp until hardened. Dry out to provide securely bonded finish.

## 431 Stippling for key

1. Stipple mix
    1.1. Cement: As per the manufacturers recommendations
    1.2. Sand: Clean coarse.
    1.3. Proportions (cement:sand): As per the manufacturers recommendations
    1.4. Admixture: BBA-certified SBR bonding agent
2. Application/ finish: Brush applied to a deep close texture.
3. Curing/ drying: Keep damp until hardened. Dry out to provide securely bonded finish.

## 438 Preparing concrete bases for fully bonded bedding

1. Surface cement:sand matrix: Remove to expose coarse aggregate.

2. Surface preparation: Suitable to achieve a full bond with bedding. Select from:
   2.1. Keep well wetted for several hours. Remove free water then brush in a slurry bonding coat.
      2.1.1. Slurry: SBR polymer-modified cement
   2.2. Prepare, prime as necessary and apply a bonding agent.
      2.2.1. Bonding agent: BBA-certified SBR bonding agent

## 444 Preparing concrete bases for partially bonded bedding

1. Surface finish: Brushed, with no laitance.
2. Surface preparation: Suitable to achieve a full bond with bedding. Select from:
   2.1. Keep well wetted for several hours. Remove free water then brush in a slurry bonding coat.
      2.1.1. Slurry: SBR polymer-modified cement
   2.2. Prepare, prime as necessary and apply a bonding agent.
      2.2.1. Bonding agent: BBA-certified SBR bonding agent

## 450 Preparing concrete bases for unbonded bedding – without separating layer

1. Surface finish: Smooth.
2. Surface preparation: Dampen lightly before laying mortar bed.

## 451 Preparing concrete bases for unbonded bedding

1. Separating layer: Polyethylene sheet.
   1.1. Thickness: 125 micrometres (500 gauge)
   1.2. Lap at joints: 100 mm.

## 460 Smoothing underlayment

1. Type: Recommended by adhesive manufacturer.
2. Condition: Allow to dry before tiling.

## 470 Intermediate substrate

1. Type: Refer to the Design Drawings
   1.1. Manufacturer: Submit proposals
      1.1.1. Product reference: Submit proposals.
   1.2. Thickness: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
2. Fixing system: Suitable fixing system in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
   2.1. Manufacturer: Submit proposals.
      2.1.1. Product reference: Submit proposals.
   2.2. Application: As recommended by manufacturer
   2.3. Substrate surface: Secure, true and even.
3. Joints: Close butt.
   3.1. Treatment: As recommended by manufacturer
4. Penetrations: Seal.

5. Accessories: None

# Fixing

## 510 Fixing generally

1. Colour/ shade: Unintended variations within tiles for use in each area/ room are not permitted.
   1.1. Variegated tiles: Mix thoroughly.
2. Adhesive: Compatible with background/ base. Prime if recommended by adhesive manufacturer.
3. Use of admixtures with cementitious adhesives: Only admixtures approved by adhesive manufacturer.
4. Cut tiles: Neat and accurate.
5. Fixing: Provide adhesion over entire background/ base and tile backs.
6. Final appearance: Before bedding material sets, make adjustments necessary to give true, regular appearance to tiles and joints when viewed under final lighting conditions.
7. Surplus bedding material: Clean from joints and face of tiles without disturbing tiles.

## 530 Setting out

1. Joints: True to line, continuous and without steps.
   1.1. Joints on walls: Horizontal, vertical and aligned round corners.
   1.2. Joints in floors: Parallel to the main axis of the space or specified features.
2. Cut tiles: Minimize number, maximize size and locate unobtrusively.
3. Joints in adjoining floors and walls: Align.
4. Joints in adjoining floors and skirtings: Align.
5. Movement joints: Where locations are not indicated, submit proposals.

## 540 Level of floor tiling

1. Permissible deviation in level from datum To be confirmed.

## 550 Flatness/ Regularity of tiling/ mosaics

1. Sudden irregularities: Not permitted.
2. Deviation of surface: Measure from underside of a 2 m straightedge with 3 mm-thick feet placed anywhere on surface. The straightedge should not be obstructed by the tiles and no gap should be greater than 6 mm, i.e. a tolerance of +/- 3 mm.

## 560 Level of tiling across joints

1. Deviation (maximum) between tile surfaces either side of any type of joint
   1.1. 1 mm for joints less than 6 mm wide.
   1.2. 2 mm for joints 6 mm or greater in width.

## 570 Mortar bedding

1. Bedding mix
   1.1. Cement: Portland to BS EN 197-1 Type CEM 1/42.5.
   1.2. Sand for walls: Fine aggregate to BS EN 13139.
      1.2.1. Grading designation: 0/2 (CP or MP) Category 2 fines.

1.3. Sand for floors: Fine aggregate to BS EN 13139.

1.3.1. Grading designation: 0/4 (MP) Category 1 fines and between 20%–66% passing a 0.5 sieve.

2. Batching: Select from:

2.1. Batch by weight.

2.2. Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.

3. Mixing: Mix materials thoroughly to uniform consistence. Use a suitable forced action mechanical mixer. Do not use a free fall type mixer.

4. Application: At normal temperatures use within two hours. Do not use after initial set. Do not retemper.

## 578 Crack control reinforcement

1. Type to BS 4483:
2. Installation: Place centrally in depth of bed. Lap not less than 100 mm and securely tie together with steel wire.
3. Corners: Avoid a four-layer build at corners.

## 580 Porous tiles

1. Tiles to be bedded in cement:sand: Soak in clean water for at least 30 minutes. Fix as soon as surface water has drained.

## 590 Coved tile skirtings

1. Sequence: Bed solid to wall before laying floor tiles.
2. Bedding: 1:4 cement:sand mortar

## 600 Sit-on tile skirtings

1. Sequence: Bed solid to wall after laying floor tiles.
2. Bedding: 1:4 cement:sand mortar

## 610 Multistorey external tiling

1. Sequence: Start fixing at the highest level and work downwards. Protect exposed top edges.
2. Copings, sills, etc.: Complete as soon as practicable.

## 650 Adhesive bed – notched trowel method (walls)

1. Application: By 3 mm floated coat of adhesive to dry background in areas of approximately 1 m². Comb surface.
2. Tiling: Press tiles firmly onto float coat.

## 651 Adhesive bed – notched trowel and buttering method (walls)

1. Application: By floated coat of adhesive to dry background in areas of about 1 m². Comb surface.
2. Tiling: Apply thin even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles. Press tiles firmly onto float coat.
3. Finished adhesive thickness: 3 mm or within the range allowed by the adhesive manufacturer.

## 652 Adhesive bed – buttering method (walls)

1. Tiling: Apply even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles.
2. Finished adhesive thickness: 3 mm or within the range allowed by the adhesive manufacturer.

## 660 Adhesive bed – notched trowel method for mesh-backed mosaic (walls)

1. Application: By 3 mm floated coat of adhesive to dry background. Comb surface.
2. Placing mosaic sheets: Hang in horizontal rows, working downwards. Stagger vertical joints. Prevent slippage of sheets. Lightly beat mosaics into adhesive.
3. Width, plane and alignment of joints between sheets: To match joints between mosaic tiles.

## 661 Adhesive bed – notched trowel method for paper-faced mosaic (walls)

1. Application: By 3 mm floated coat of adhesive to dry background. Comb surface.
2. Preparing mosaic sheets: Pre-grout. Remove surplus before fixing.
3. Placing mosaic sheets: Hang in horizontal rows, working downwards. Stagger vertical joints.
4. Width, plane and alignment of joints between sheets: To match joints between mosaic tiles.
5. Paper face: Before adhesive hardens completely, remove paper face. Complete grouting. Wash off glue from face of mosaic.

## 690 Cement:sand mortar bedding (walls)

1. Preparation: Dampen background.
2. Application: By floated coat to background: 1:3–4 cement:sand mortar bedding.
   2.1. Thickness (maximum): 10 mm.
   2.2. Finish: Equivalent to wood float. Before tiling allow to stiffen slightly.
3. Tiling: Without delay, apply 2 mm-thick coat of 1:1 cement:fine sand mortar to backs of tiles, filling keys. Press tiles firmly onto float coat. Tap firmly into position.

## 710 Adhesive bed – notched trowel and buttering method (floors)

1. Application: Floated coat of adhesive to dry base and comb surface.
2. Tiling: Apply coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles. Press tiles firmly onto float coat.
3. Finished adhesive thickness: Within range allowed by manufacturer.

## 711 Adhesive bed – buttering method (floors)

1. Tiling: Apply even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles.
2. Finished adhesive thickness: Within the range allowed by the adhesive manufacturer.

## 720 Cement:sand mortar bedding (floors)

1. Mortar bedding mix: 1:3–4 cement:sand.
   1.1. Consistency: Stiff plastic.
2. Laying: Lay suitably small working areas of screeded bed. Compact thoroughly to level.
   2.1. Finished bed thickness
      2.1.1. For tiles up to 10 mm thick: 10–15 mm.

2.1.2. For tiles greater than 10 mm thick: 15–20 mm.

3. Tiling: Within two hours and before bedding sets, evenly coat backs of tiles with adhesive. Press tiles firmly into position.
4. Finished adhesive thickness: Within range recommended by manufacturer.

## 730 Semi-dry cement:sand bedding (floors)

1. Mortar bedding mix: 1:3.5-4 cement:sand.
   1.1. Water content: A film of water must not form on surface of bed when fully compacted.
2. Preparation: Dampen base.
3. Laying: Lay suitably small working areas of screeded bed. Compact thoroughly to level.
   3.1. Finished bed thickness (minimum): As per the manufacturers recommendations to suit the conditions of service
   3.2. Finished bed thickness (maximum): As per the manufacturers recommendations to suit the conditions of service
4. Tiling: Within two hours and before bedding sets, evenly coat backs of tiles with neat cement slurry. Beat tiles firmly into position.

## 740 Cement:lime:sand mortar bedding (floors)

1. Mortar bedding mix:
   1.1. Cement: Portland to BS EN 197-1, Type CEM I/42.5.
   1.2. Lime:sand (coarse stuff): Ready-mixed to BS EN 998-2.
   1.3. Sand: Fine aggregate to BS EN 13139.
   1.4. Grading designation: 0/4 (MP) Category 1 fines and between 20–66% passing a 0.5 sieve.
   1.5. Consistency: Stiff plastic. A film of water must not form on surface of bed when fully compacted.
2. Preparation: Dampen base.
3. Laying: Lay and compact bed to level.
   3.1. Finished bed thickness (minimum): 15 mm
   3.2. Finished bed thickness (maximum): 50 mm
4. Tiling: Tamp tiles firmly into position.

## 750 Natural hydraulic lime mortar bedding (floors)

1. Mortar: As section Z21.
   1.1. Mix: 2:5 NHL5 hydraulic lime:sharp well-graded sand.
   1.2. Sand: Fine aggregate to BS EN 13139, annex B.
   1.3. Consistency: Stiff plastic. A film of water must not form on the surface of bed when fully compacted.
2. Laying: Lay and compact bed to level.
   2.1. Finished bed thickness (minimum): 15 mm
   2.2. Finished bed thickness (maximum): 50 mm
3. Tiling: Tamp tiles firmly into position.

# Movement joints/ grouting/ completion

## 805 Sealant movement joints with metal edgings

1. Edging material: Stainless steel angle
   1.1. Size: Refer to the Design Drawings
   1.2. Bedding: Bed in 1:3 cement:sand.
2. Installation: Centre over joints in base. Set to exact finished level of floor.
   2.1. Fixing to base: As per the manufacturers recommendations to suit the conditions of service
3. Joint width: To match that of structural movement joint in base
4. Sealant: Submit proposals.
   4.1. Colour: Refer to the Design Drawings
   4.2. Preparation and application: As section Z22

## 815 Sealant movement joints

1. Joints: Extend through tiles and bedding to base/ background. Centre over joints in base/ background.
   1.1. Width: As required to meet the performance requirements.
2. Sealant: Submit proposals.
   2.1. Colour: Refer to the Design Drawings
   2.2. Preparation and application: As section Z22.

## 825 Strip movement joints

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
   1.2. Colour: Refer to the Design Drawings
2. Joint width: As required to meet the performance requirements.
3. Fixing to base: Suitable concealed fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
4. Joints: Extend through tiles and bedding to base.

## 835 Metal section movement joints

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
   1.2. Colour: Refer to the Design Drawings
2. Bedding: Bed in 1:3 cement:sand.
3. Installation: Centre over joints in base. Set to exact finished level of floor.
4. Fixing to base: Suitable concealed fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.

## 855 Cement:sand grouting mix

1. Grout mix
   1.1. Cement: Portland cement to BS EN 197-1, Type CEM I/42.5.

1.2. Sand (Fine aggregate)

1.2.1. Joint widths of 6 mm or greater: To BS EN 13139, grading designation 0/2 (FP or MP), Category 2 fines.

1.2.2. Joint widths of 3-6 mm: To BS 5385-5, Table 1.

1.3. Proportions (cement:sand): As per manufacturers recommendations.

1.4. Admixture: None.

2. Mixing: Mix thoroughly. Use the minimum of clean water needed for workability.

## 865 Cement:lime sand grouting mix

1. Grout mix

1.1. Cement: Portland cement to BS EN 197-1, Type CEM I/42.5.

1.2. Hydrated non-hydraulic lime: To BS EN 459-1, BS EN 459-2 and BS EN 459-3.

1.3. Sand

1.3.1. Joint widths of 6 mm or greater: To BS EN 13139, grading designation 0/2 (FP or MP), Category 2 fines.

1.3.2. Joint widths of 3-6 mm: To BS 5385-5, Table 1.

1.4. Proportions (cement:lime:sand): As per manufacturers recommendations.

1.5. Admixture: None.

2. Mixing: Mix thoroughly. Use the minimum of clean water needed for workability.

## 875 Grouting

1. Sequence: Grout when bed/adhesive has set sufficient to prevent disturbance of tiles.
2. Joints: 6 mm deep (or depth of tile if less). Free from dust and debris.
3. Grouting: Fill joints completely, tool to profile, clean off surface. Leave free from blemishes.

3.1. Profile: Flush

4. Polishing: When grout is hard, polish tiling with a dry cloth.

## 885 Coloured grout

1. Staining of tiles: Not permitted
2. Evaluating risk of staining: Apply grout to a few tiles in a small trial area. If discoloration occurs apply a protective sealer to tiles and repeat trial.

# Performance

## 900 Pendulum floor covering slip resistance testing

1. Floor covering(s) to be tested: M40/ all types.
2. Testing authority: A UKAS-accredited laboratory
3. Test: To BS EN 16165

3.1. Floor covering condition: Dry and wet

3.2. Requirements: 40 PTV.

3.3. Witnessing/ certification: Arrange for tests to be witnessed/ certified by:Contract administrator.

3.4. Report: Submit.

## 910 Pendulum floor covering slip resistance testing

1. Natural stone floor covering(s) to be tested: M40/ all types.
2. Testing authority: A UKAS-accredited laboratory
3. Test: To BS EN 14231
   - 3.1. Floor covering condition: Dry and wet
   - 3.2. Requirements: 40 PTV.
   - 3.3. Witnessing/ certification: Arrange for tests to be witnessed/ certified by: Contract administrator.
   - 3.4. Report: Submit in addition to the requirements of the standard include:
     - 3.4.1. Location of the site and a drawing showing the positions of the tests.
     - 3.4.2. Description of the surface tested and its condition.
     - 3.4.3. Slider material used.
     - 3.4.4. PTV at each position tested.
     - 3.4.5. Temperature of the test surface.

Ω End of Section

# M41
# Terrazzo Tiling/ In-Situ Terrazzo

## Types of terrazzo

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 FLS-205 Precast Terrazzo Tile Floor Finish

1. Description: Slip resistant precast terrazzo tile floor finish
2. Substrate: Refer to the Design Drawings
3. Preparation: Levelling screed to be provided where necessary.
4. Adhesive/ bedding compound: To suit the service conditions.
5. Tiles:
    5.1. Standard: To BS EN 13748-1 for internal use/ To BS EN 13748-2 for external use.
    5.2. Manufacturer: Parkside architectural tiles, or acceptable equivalent
        5.2.1. Product reference: Ethical Stone Terrazzo or Arizona.
    5.3. Recycled content: Ethical Stone Terrazzo offers a unique, 100% British stone, crafted with sustainability at its core. Sourced from local aggregates across the British Isle or Arizon is approximately 50% recycled content including aggregates reclaimed from building site waste and other secondary materials.
    5.4. Size: 1200 x 600 mm.
    5.5. Thickness: To be agreed with the Architect.
    5.6. Colour/ style: Varies. To be confirmed.
6. Joint width: Nominal 3mm.
7. Grout:
    7.1. Type: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    7.2. Colour: To compliment surrounding tile colour.
8. Movement joints: To be agreed with the Architect.
9. Finishing
    9.1. Finish: Honed.
10. Skirting: Matching tile skirting (where required).
11. Trims: Satin stainless steel.
12. Other requirements: System to include all fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components as required to suit the service conditions.

### 111 FLS-206 Precast Terrazzo Tile Floor Finish

1. Description: Slip resistant precast terrazzo tile floor finish
2. Substrate: Refer to the Design Drawings
3. Preparation: Levelling screed to be provided where necessary.

4. Adhesive/ bedding compound: To suit the service conditions.
5. Tiles:
    5.1. Standard: To BS EN 13748-1 for internal use.
    5.2. Manufacturer: Cancian, Diespeker & Co, or acceptable equivalent
        5.2.1. Product reference: Precast Bespoke range.
    5.3. Recycled content: To be agreed with the Architect.
    5.4. Size: 1200 x 1200 mm.
    5.5. Thickness: 20 mm.
    5.6. Colour/ style: Varies. To be confirmed.
6. Joint width: Nominal 3mm.
7. Grout:
    7.1. Type: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    7.2. Colour: To compliment surrounding tile colour.
8. Movement joints: To be agreed with the Architect.
9. Finishing
    9.1. Finish: To be agreed with the Architect.
10. Skirting: Matching tile skirting (where required).
11. Trims: Satin stainless steel.
12. Other requirements: System to include all fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components as required to suit the service conditions.

## 116 LIN-421 Precast Terrazzo Tile Wall Finish

1. Description: Precast terrazzo tile wall finish
2. Substrate: Refer to the Design Drawings
3. Adhesive: To suit the service conditions.
4. Tiles:
    4.1. Standard: To BS EN 13748-1 for internal use
    4.2. Manufacturer: Duracryl, or acceptable equivalent
        4.2.1. Product reference: Durabella Seamless Terrazzo - A-114-M
    4.3. Size: To be agreed with the Architect.
    4.4. Thickness: To be agreed with the Architect.
    4.5. Colour/ style: A-114-M
5. Joint width: Nominal 3mm.
6. Grout:
    6.1. Type: Proprietary non-staining, flexible, water impervious, mould and chemical resistant.
    6.2. Colour: To compliment surrounding tile colour.
7. Finishing
    7.1. Finish: To be agreed with the Architect.
8. Trims: Satin stainless steel.
9. Other requirements: System to include all fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components as required to suit the service conditions.

## 170 Floor finish materials specification

1. Minimum BRE 'Green Guide to Specification Online' rating: Submit proposals.

# Generally

## 210 Suitability of substrates

1. General: Suitably dry and clean for specified bedding method.
2. Existing substrates: Free from loose areas and significant cracks and gaps.
3. New in situ concrete substrates: Allow for initial shrinkage.
   3.1. Air drying after curing: (minimum) Six weeks.
   3.2. Other methods: Submit proposals.

## 220 Existing concrete/ Screeds

1. Loose or hollow portions: Cut out and make good
2. Material for making good: 1:3 cement:sand mortar

## 230 Samples

1. Type of tile: Six of each type including grout.
2. Timing: Submit before placing orders.

## 235 Control samples

1. General: Complete sample areas, being part of the finished work, and obtain approval of appearance before proceeding.
2. Terrazzo type: All types including skirtings and dividing strips.
   2.1. Location/ Size: To be confirmed

## 242 Cement

1. Standard: To BS EN 197-1

## 245 Fine aggregate (Sand)

1. Standard: To BS EN 13139

## 248 Admixtures

1. Water reducing and set retarding admixtures: To BS EN 934-2 and compatible with other mortar constituents.

## 280 Separating layer for unbonded bedding

1. Type: Polyethylene sheet not less than 65 micrometres thick (250 gauge).
2. Laps at joints (minimum): As per the manufacturer's recommendations.

# Laying

## 305 Laying generally

1. Tiles: Fully adhered to bedding
2. Cut tiles: Neat and accurate



make

3. Joints: True to line, continuous and without steps
    3.1. Joints in floors: Parallel to main axis of the space or specified features.
4. Dividing strips/ Movement joints: If locations are not indicated, submit proposals.

## 330 Level of finished floor

1. Permissible deviations from datum: To BS 5385-3.

## 340 Flatness/ Surface regularity of finished floor

1. Sudden irregularities: Not permitted
2. Deviation (maximum) of surface: To BS 5385-3

## 345 Level of unground tiling across joints

1. Deviations (maximum) between surfaces either side of any type of joint
    1.1. Joints less than 6 mm wide: To BS 5385-3
    1.2. Joints 6 mm wide or greater: To BS 5385-3

## 390 Mixing bedding mortar

1. Batching
    1.1. Batch by volume: Permitted on the basis of previously established weight: volume relationships of the particular materials. Allow for bulking of damp sand
2. Water content: Minimum necessary to achieve full compaction
3. Mixes: Of uniform consistency, free from lumps. Do not retemper or reconstitute mixes.

## 395 Crack control reinforcement

1. Standard: To BS 4483
2. Location: To be confirmed

## 405 Tiling on mortar bedding

1. Compaction of bedding: Compact thorough.
2. Tiling: Lay tiles on layer of wet slurry and beat tiles firmly into position
    2.1. Timing: Before bedding sets and to achieve effective bond between bedding and tiles
    2.2. Slurry type: 1:1 cement: fine sand.
3. Grouting: Fill joints. Remove surplus and allow grout to cure before finishing
    3.1. Grout colour: To match tile

## 430 In situ terrazzo

1. Mixing terrazzo materials: Before adding water, mix in a dry state to a uniform colour and with even distribution of different size aggregates
    1.1. Water content: Minimum necessary to achieve full compaction.
2. Bedding: Compact thoroughly.
3. Terrazzo: Lay on layer of wet slurry and compact level with the dividing strips.
    3.1. Timing: To achieve effective bond to bedding.
    3.2. Slurry type: 1:1 Cement fine sand.
4. Surfaces: Free from excess laitance

4.1. Aggregate: Regularly distributed with minimum amount of cement matrix visible.

## 440 Dividing strips

1. Material: Aluminium
   1.1. Thickness: 3 mm
2. Installation: Set to required finished floor level.

## 445 Strip movement joints

1. Joint width: Refer to the Design Drawings
2. Installation: Extend through terrazzo and bedding to base. Set to required finished floor level.
   2.1. Fixing: Integral anchors at 300 mm centres.

## 455 Sealant movement joints with metal edgings

1. Edging angle material: To be confirmed
   1.1. Angle size: To be confirmed
2. Joint width: To match that of structural movement joint in base
3. Installation: Centre over joint in base. Set to exact finished level of floor.
   3.1. Mortar levelling bed: 1:3 cement:sand.
   3.2. Fixing: Brass screws to plugs at 600 mm centres.
4. Sealant: To BS EN ISO 11600.
   4.1. Colour: To be confirmed
   4.2. Preparation and application: As section Z22

## 465 Metal section movement joints

1. Joint width: To suit structural movement joints in base.
2. Installation: Centre over joint in base. Set to exact finished level of floor.
   2.1. Mortar levelling bed: 1:3 cement:sand.
   2.2. Fixing: Stainless steel screws, washers and plugs at 300 mm centres.

## 470 Applied slip resistant accessories

1. Material: Rubber composite inserts
   1.1. Format: Disc for insertion, 6.5 mm thick, prefinished
   1.2. Slip resistance value/ pendulum test value – water wet (minimum) to BS EN 16165: SRV/ PTV of 45
2. Size/ Coverage: Refer to the Design Drawings
3. Colour: To be confirmed.

## 520 Grinding

1. Mechanical grinding: Use wet process. Dry process permitted only where it is not practicable to use wet process.
2. Coarse grinding: Remove all surface grout, correct unevenness and remove lipping between tiles/ slabs
   2.1. Residue and dust: Remove. Wash and rinse surface
3. Holes: Fill with matching grout.

4. Polishing: Use wet process.
    4.1. Residue and dust: Remove. Wash and rinse surface and leave to dry naturally.
5. Protection: Cover flooring and keep free from traffic for at least two days.

## Completion

### 525 Slip resistance testing

1. Test: To BS EN 16165
2. Report
    2.1. Name and address of testing authority: Required.
    2.2. Date of tests: Required.
    2.3. Location of test: Required.
    2.4. Temperatures of test surfaces: Required.
    2.5. The slider material used on the pendulum tester: Required.
    2.6. Slip resistance values in wet and dry conditions: Required.

### 530 Impact testing

1. Test: BRE Information Paper 11/84 - Impact testing.

Ω End of Section



make

# M50
# Rubber/ Plastics/ Cork/ Lino/ Vinyl/ Carpet Tiling/ Sheeting

## Types of covering

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 121 FLS-601 Rubber Sheet Flooring

1. Description: Rubber sheet flooring, as shown on the Design Drawings.
2. Manufacturer: Interface, or acceptable equivalent.
    2.1. Project reference: Norament 926 arago.
3. Substrate: Refer to the Design Drawings
4. Base:
    4.1. Preparation: Where necessary and as recommended by the rubber sheet manufacturer, apply a smoothing compound.
5. Adhesive/ primer: To suit the service conditions.
6. Flooring roll: PVC to BS EN 14041 and BS EN 13553.
    6.1. Evidence of compliance: Submit.
    6.2. Reaction to fire classification: A1.
    6.3. BS EN ISO 10874 class: To be agreed with the Architect.
    6.4. Slip potential
        6.4.1. Slip resistance value (SRV) (minimum)/ Pendulum test value (PTV) (minimum) to BS 7976-1, -2 and -3: To be agreed with the Architect.
    6.5. Recycled content: To be agreed with the Architect.
    6.6. Width: To be agreed with the Architect.
    6.7. Thickness: To be agreed with the Architect.
    6.8. Colour/ pattern: To be agreed with the Architect.
7. Seam welding: Hot welding with complimentary coloured rod
8. Transition strips/ trims/ skirtings: Refer to the Design Drawings and Section P20.

### 126 FLS-616 Woven Vinyl Sheet Floor Finish

1. Description: Slip resisting vinyl sheet floor finish.
2. Manufacturer: Bolon or acceptable equivalent.
    2.1. Product reference: BKB.
3. Substrate(s):
    3.1. Refer to the Design Drawings
    3.2. Use preparatory products and processes to suit substrates.
    3.3. Where necessary and as recommended by the vinyl sheet manufacturer install a levelling screed/ compound or plywood sheet where on raised flooring system.
4. Primer(s)/ adhesive(s): As recommended by the vinyl sheet manufacturer.

5. Vinyl sheet:
   5.1. Roll width: 2000 mm.
   5.2. Thickness: 2,3 mm.
   5.3. Colour/ pattern: Sisal Dune.
   5.4. Slip resistance: Achieve a minimum slip resistance value of 36PTV in both dry and wet conditions when carrying out tests using a TRL Pendulum Test.
6. Joints/ junctions:
   6.1. Minimised. Where not shown on the Design Drawings agree locations with the Architect.
   6.2. To be cut in, grooved and hot welded with an appearance to complement the vinyl sheet.
7. Skirting: To be agreed with the Architect.
8. Joints/ edgings/ transition strips: To be agreed with the Architect.
9. System to include all fixings, sealants, adhesives, jointing/ transition strips, trims, skirtings and all other accessories/ components recommended by the manufacturer to suit the service conditions.

### 195 Floor finish materials specification

1. Minimum BRE 'Green Guide to Specification' (online) rating: Refer to the Sustainability Report

## General requirements

### 210 Workmanship generally

1. Base condition after preparation: Rigid, dry, sound, smooth and free from grease, dirt and other contaminants
2. Finished coverings: Accurately fitted, tightly jointed, securely bonded, smooth and free from air bubbles, rippling, adhesive marks and stains
3. Operatives: Appropriately skilled and experienced for the type of work

### 220 Samples

1. Covering samples: Submit samples representative of each floor type prior to placing orders
   1.1. Size (minimum): 1 no. carpet tile from each range specified.
2. Purpose: For use as reference sample, and to obtain approval of appearance from client.

### 230 Control samples

1. General: Complete areas of finished work in approved locations as follows, and obtain approval of appearance before proceeding.
   1.1. Location: Refer to the Design Drawings
   1.2. Accessories: Include coved skirtings and edging strips.

### 250 Layout – roll materials

1. General: Layout to be planned carefully to avoid wastage and unnecessary seams. All material laid in a location to come from the same batch.
2. Seams: Agree setting out for sheeting types and minimize occurrences of seams and cross seams, where possible.
   2.1. Position: With traffic flow, along the length of the corridor, and ensure that joints in hardwood base and seams do not coincide.

2.1.1. Cross seams: Away from heavy traffic areas, doorways, lift entrances, stair noses and areas of narrow access

3. Pattern: In accordance with BS 5325. Agree setting out for covering types.
4. Transition joints:

## 270 Extra material

1. Requirement: Provide extra material for each type of floor covering installed.
   1.1. Quantity: 10% extra carpet tile.
   1.2. Timing: Hand over on completion.

## 330 Commencement

1. Required condition of works prior to laying floor-covering materials
   1.1. Building is weathertight and well dried out.
   1.2. Wet trades have finished work.
   1.3. Paintwork is finished and dry.
   1.4. Conflicting overhead work is complete.
   1.5. Floor service outlets, duct covers and other fixtures around which materials are to be cut are fixed.
   1.6. Provisions have been made to allow for ventilation during and after laying of materials.
2. Notification:

## 340 Conditioning

1. Prior to laying: Condition materials in accordance with manufacturer's instructions.
2. Conditioning time and temperature (minimum): As recommended by manufacturer, with time extended by a factor of two for materials stored or transported at a temperature of less than 10°C immediately prior to laying.

## 350 Environmental conditions

1. Temperature and humidity: In accordance with BS 8203 and BS 5325 and as per manufacturer's instructions.
   1.1. Subfloor : Minimum temperature of 10°C.
2. Ventilation: Maintain adequate provision before, during and after laying the floor covering.

## 360 Floors with underfloor heating

1. Commencement of laying: Not before a period of 48 hours after heating has been turned off.
2. Post-laying start up of heating system: Slowly return heating to its operative temperature not less than 48 hours after completing laying.
   2.1. Maximum temperature: As per manufacturer's instructions.

## 370 Fire Performance

1. Carpet Flammability: Low radius of char ignition factor rating when tested using the hot metal nut method to BS 4790.
2. Reaction: Carpet to achieve minimum Class Cfl rating to BS EN 13501: Part 1.

## 375 Antistatic Performance

1. Static dissipative or static conductive floor coverings in accordance with BS 2050 and BS ISO 2878.

## 380 Testing

1. Where required, in accordance with BS EN ISO 10580, BS EN 13893, BS EN 1081, BS EN 1815, PSA MOB PF2 PS/ SPU, clause T43.00, BS 4790, BS EN ISO 9239: Part 1.

## 390 Materials

1. Where required, in accordance with BS EN 14041, BS EN ISO 24342, BS EN 650, BS EN 651, BS EN 652, BS EN 653, BS EN ISO 10581, BS EN ISO 10582, BS EN ISO 10595,BS ISO 11638, BS EN 13413, BS EN 13553, BS EN 13845, BS EN 14565, BS EN ISO 26986, BS EN 12199, BS EN 1816, BS EN 1817, BS EN 14521, BS EN 686, BS EN 687, BS EN ISO 24011, BS EN 655, BS EN 12104, BS EN 688, BS EN ISO 20326, BS ISO 2424, BS EN 14499.
2. Usage classifications:
    2.1. Resilient floor: BS EN ISO 10874.
    2.2. Carpet: BS EN 1307.

# Preparing bases

## 405 Workmanship Generally

1. In accordance with BS 5325, BS 8203, BS ISO 13746.

## 410 New bases

1. General: Designer and those responsible for the construction of the base to ensure suitability.
2. Suitability of bases and conditions within any area: Base to be sound, rigid, level, properly secured and dry ad in accordance with BS 5325 or BS 8203.

## 420 Existing bases

1. Notification: Before commencing work, the contractor is to confirm that existing bases will, after preparation, be suitable to receive coverings.
2. Suitability of bases and conditions within any area: Base to be sound, rigid, level, properly secured and dry ad in accordance with BS 5325 or BS 8203.

## 430 New wet-laid bases

1. Artificial base drying aids: Not used for at least four days prior to moisture content testing
2. Base moisture content test: In accordance with BS 5325, annex B or in accordance with BS 8203, annex B.
    2.1. Locations for readings: In all corners, along edges, and at various points over area being tested.
3. Commencement of laying coverings: Not until all readings show 75% relative humidity or less

## 440 Substrates to receive thin coverings

1. Trowelled finishes: Uniform, smooth surface free from trowel marks and other blemishes. Abrade suitably to receive specified floor-covering material.

## 460 Smoothing/ levelling underlayment compound

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Type: As recommended by covering manufacturer.
   2.1. Thickness: As per manufacturer's recommendations.

## 470 Bases from which existing floor coverings have been removed

1. Substrate: Clear of covering and as much adhesive as possible. Skim with smoothing underlayment compound to give smooth, even surface.

## 480 Existing floor coverings to be overlaid

1. Substrate: Make good by local re-sticking and patching or filling with smoothing underlayment compound to give smooth, even surface.

## 510 Wood block flooring

1. Substrate: Clean and free from wax, with all blocks sound and securely bonded. Adequately protected from moisture. Fill hollows with smoothing underlayment compound to give smooth, even surface.
2. Missing and loose blocks: Replace and reset in adhesive to match existing. Sand or plane to make level.

## 520 Timber boarding/ strip flooring

1. Substrate: Boards/ strips securely fixed and acceptably level with no protruding fasteners. Plane, sand or apply smoothing underlayment compound to give a smooth, even surface.

## 530 Particleboard flooring

1. Grade: To BS 8203 and BS 5325
2. Substrate: Boards securely fixed, level and free from surface sealers and contaminants.
   2.1. Gaps between boards: As per manufacturer's instructions.
   2.2. Level: Sand off areas raised up to 1 mm.
   2.3. Priming: As required by covering adhesive manufacturer.
3. Equilibrium moisture content at time of laying covering: As in-service conditions.

## 535 Particleboard floating flooring

1. Substrate: Boards securely fixed and free from surface sealers and contaminants.
2. Level: Sand to remove areas raised more than 1 mm.
   2.1. Gaps between boards: Not more than 1 mm.
   2.2. Priming: As required by covering adhesive manufacturer.
3. Equilibrium moisture content at time of laying covering: As in service conditions.

## 550 Hardboard underlay

1. Manufacturer: Submit proposals
   1.1. Product reference : Submit proposals.
2. Standard: To BS EN 622-2

3. Properties
   3.1. Type: Standard hardboard.
   3.2. Reaction-to-fire classification: To BS EN 13501-1.
   3.3. Thickness: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
   3.4. Sheet size: As per manufacturer's recommendations.
4. Substrate: Refer to the Design Drawings
5. Conditioning sheets: As clause 340.
   5.1. Requirement: Install sheets while slightly expanded
6. Laying sheets: As per manufacturer's recommendations.
   6.1. Joints: No gap or step between joints of sheets.
      6.1.1. Requirement: Staggered joints with none coincident with joints in base.
   6.2. Perimeter gap: Required
      6.2.1. Width: As per manufacturer's instructions.
7. Fasteners: Suitable concealed fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
   7.1. Spacing: In accordance with the manufacturers details and pecommendations to achieve the performance requirements and to suit the service conditions.
   7.2. Placement: Not to project above sheet surface or through underside of base. Not deformed.
8. Hardboard surface: Free from sealants, coatings and residues.
   8.1. Requirement: Do not lay coverings until underlay is fully dry.

## 560 Plywood underlay

1. Manufacturer: Submit proposals
   1.1. Product reference : submit proposals.
2. Standard: To BS EN 636
3. Properties
   3.1. Material: Refer to the Design Drawings
   3.2. Reaction-to-fire classification: To BS EN 13501-1.
   3.3. Surface appearance class: To BS EN 635-1
   3.4. Finish: Sanded.
   3.5. Thickness: Refer to the Design Drawings
4. Substrate: Existing floor boards securely fixed and acceptably level, with no gross irregularities or protruding fasteners.
5. Conditioning sheets: As clause 340 and refer to BS 8201.
6. Laying sheets
   6.1. Joints: No gap or step between joints of sheets.
      6.1.1. Requirement: Staggered joints with none coincident with joints in base.
   6.2. Perimeter gap: Required
      6.2.1. Width: As per manufacturer's instructions.
7. Fasteners: Suitable concealed fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.

7.1. Spacing: In accordance with the manufacturers details and pecommendations to achieve the performance requirements and to suit the service conditions.

7.2. Placement: Not to project above sheet surface or through underside of base. Not deformed.

8. Plywood surface: Free from sealants, coatings and residues.

## Laying coverings

### 610 Setting out tiles

1. Method: Refer to the Design Drawings
   1.1. Requirement: Setting out to be approved by client prior to installation.

### 620 Colour consistency

1. Finished work in any one area/ room: Colour of individual pieces of floor covering to match. Finished floor to be free from banding or patchiness. No colour inconsistencies between individual pieces of floor-covering material unless they are part of the design
   1.1. Requirement: Use material from the same batch by the same manufacturer

### 640 Adhesive fixing generally

1. Primer: Type as recommended by adhesive manufacturer.
   1.1. Application: As per manufacturer's instructions.
2. Adhesive
   2.1. Type: As recommended by floor-covering/ underlay manufacturer
   2.2. Application:
      2.2.1. Method: As per manufacturer's instructions.
      2.2.2. Temperature: Between 5°C and 35°C.
      2.2.3. Open time: Maximum 30 minutes.
3. Laying of floor covering: In accordance with manufacturer's instructions.
   3.1. Fixtures: Floor-covering material to be cut around all permanent fixtures
4. Finished surface: Free from trowel ridges, high spots caused by particles on the substrate, and other irregularities.

### 645 Laying interlocking tiles and planks

1. Preparation: Subfloor to be sound and free from surface irregularities or protruding fasteners. Clean the area to be covered with flooring, removing all debris and dust.
2. Setting out: Refer to the Design Drawings
3. Laying of floor covering: Lay tiles out from the starting point. Ensure that all tiles are at right angles to each other. Connect tiles and tap joints together with a rubber mallet.
   3.1. Fixing temperature: 15°C (minimum).
   3.2. Fixtures: Floor-covering material to be cut around all permanent fixtures
4. Perimeter gap : As per manufacturer's instructions.

### 650 Seams

1. Requirement: Closely butted with pile securely held in place. Direction of pile lay matching.
2. Patterns: Refer to the Design Drawings

3. Joints: Neat, smooth, strongly bonded, flush with finished surface

## 680 Seam welding – resilient floor coverings

1. Floor covering:
   1.1. Method: As per manufacturer's instructions.
2. Commencement: At least 24 hours after laying, or after adhesive has set.
3. Welding: As per manufacturer's instructions.
   3.1. Trimming: Required
4. Accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

## 690 Seam bonding carpet

1. Seaming method: As per manufacturer's instructions.
   1.1. Seaming adhesive: Submit proposals.
      1.1.1. Application: Continuous bead to edges.
      1.1.2. Joints: Securely bonded, free of air bubbles.
2. Accessories: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.

## 700 Loose laying of tiles

1. Preparation: Subfloor to be sound and free from surface irregularities or protruding fasteners. Clean the area to be covered with flooring, removing all debris and dust.
2. Setting out: Refer to the Design Drawings
3. Laying of floor-covering tiles: Lay tiles out from the starting point. Ensure that all tiles are at right angles to each other.
   3.1. Areas of adhered tiles: As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions.
   3.2. Joints: Butted
      3.2.1. Perimeter gap: As per manufacturer's instructions.
   3.3. Fixtures: Floor-covering material to be cut around all permanent fixtures

## 720 Doorways

1. Joint location: On centre line of door leaf.

## 745 Laying of floor coverings on stairs

1. Preparation: Subfloor to be sound and free from surface irregularities or protruding fasteners. Clean area to be covered with flooring, removing all debris and dust.
2. Fixing: As per manufacturer's instructions.
   2.1. Requirement: Floor-covering material accurately cut. Fit risers before treads.

## 780 Trafficking after laying

1. Traffic-free period: 7 hours

# Completion

## 820 Finishing

1. Cleaning:
   1.1. Wash floor with water containing neutral (pH 6–9) detergent. If necessary, lightly scrub heavily soiled areas.
   1.2. Rinse with clean water, removing surplus to prevent damage to adhesive. Allow to dry.
2. Treatment: Two coats of a type of emulsion polish recommended by covering manufacturer.
3. Temporary protection: Cover all trafficked areas with a suitable carpet protection film.

## 825 Finishing non-textile antistatic flooring

1. Cleaning operations
   1.1. Wash floor with water containing neutral (pH 6-9) detergent. If necessary, lightly scrub heavily soiled areas.
   1.2. Rinse with clean water, removing surplus to prevent damage to adhesive. Allow to dry for not less than 24 hours before testing.

## 830 Finishing rubber flooring

1. Cleaning: As per manufacturer's recommendations.
2. Final treatment: As per manufacturer's recommendations.

## 840 Finishing rubber flooring – polish finish

1. Cleaning: As per manufacturer's recommendations.
2. Polishing: As per manufacturer's recommendations.

## 850 Finishing cork tile flooring – polish finish

1. Preparation: Lightly sand flooring and joints to remove any lipping and imperfections. Match original finish.
2. Cleaning: As per manufacturer's recommendations.
3. Sealing: As per manufacturer's recommendations.
4. Polish: As per manufacturer's recommendations.

## 851 Finishing cork tile flooring – polyurethane finish

1. Preparation: Lightly sand flooring and joints to remove any lipping and imperfections. Match original finish.
2. Cleaning: As per manufacturer's recommendations.
3. Polyurethane finish: As per manufacturer's recommendations.

## 860 Testing antistatic flooring

1. Test methods: In accordance with BS 2050
2. Electrical resistivity test: Agree and submit date and time of test. Record and submit results.
3. Marking: Before fixing, agree location and submit details required for marking in accordance with BS 2050, a brass plate, size 50 x 75 mm, engraved with the following details:
   3.1. The words 'ANTISTATIC FLOOR'.
   3.2. Manufacturer and reference of flooring.

3.3. Name of flooring contractor.

3.4. Date when laid.

## 861 Slip resistance testing

1. Testing authority: A UKAS-accredited laboratory
2. Test: To BS EN 16165
   2.1. Method: To be confirmed
   2.2. Floor covering condition: Dry and wet
      2.2.1. When testing with a surface roughness meter, achieve minimum of 20 µm Rz.
      2.2.2. When testing with TRL Pendulum Test achieve : 36PTV (wet and dry conditions).
   2.3. Room temperature: 15–25°C.
   2.4. Witnessing/ Certification: Arrange for tests to be witnessed/ certified by Consultant.
3. Verification
   3.1. Submittals: Slip-resistance test report.
   3.2. Timing: Before placement of orders.

## 880 Leftover material

1. Spare covering material: Retain suitable material for patching/ repairs. On completion, submit pieces for selection. Hand over selected pieces to the employer.
2. Waste disposal: In accordance with relevant regulations.

Ω End of Section



make

# M60
# Painting/ Clear Finishing

## Coating systems

### 100 Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 116 PNT-141 Floor Paint

1. Description: Floor paint to floor surfaces.
2. Manufacturer: AkzoNobel, or acceptable equivalent.
    2.1. Product reference: Dulux Trade High Performance Floor Paint.
3. Surfaces:
    3.1. Preparation: Thoroughly clean down the surfaces to remove all dirt, grease and surface contaminants and allow to dry.
4. Composition: Acrylic emulsion with fungicide/ algicide.
5. Finish: Smooth
6. Colour: Refer to the Design Drawings
7. Initial coats: As per the manufacturer's recommendations.
8. Finishing coats: As per the manufacturer's recommendations.
9. Application: As per the manufacturer's recommendations.

### 117 PNT-156 Concrete Dust Sealer

1. Description: Quick drying, low VOC, water-based sealer system to be applied to visual concrete surfaces.
2. Manufacturer: Keim, or acceptable equivalent.
    2.1. Product reference: Keim Concretal Lasur.
3. Surfaces:
    3.1. Preparation: Thoroughly clean down the surfaces to remove all dirt, grease and surface contaminants and allow to dry.
4. Finish: Clear matt (low glare)
5. Initial coats:
    5.1. Primer: Where required, to the manufacturer's recommendations, to suit the concrete type/ substrate type.
6. Finishing coats: As per the manufacturer's recommendations.
7. Application: Brush and roller or spray.

### 126 PNT-101 Emulsion Paint

1. Description: Water based emulsion paint.
2. Manufacturer: AkzoNobel, or acceptable equivalent.
    2.1. Product reference: Dulux Trade Vinyl Matt Emulsion.



make

3. Surfaces:
    3.1. Preparation: Thoroughly clean down the surfaces to remove all dirt, grease and surface contaminants and allow to dry.
4. Composition: Acrylic copolymer
5. Finish: Matt
6. Colour: Refer to the Design Drawings
7. Initial coats:
    7.1. Spot prime any bare metal, metal fixings and nail heads with 1 No. coat of Dulux Trade Metalshield QD Metal Primer.
    7.2. Prime overall with 1 No. coat of Dulux Trade Dry Wall Primer Sealer.
8. Finishing coats: Minimum 2 No. coats to provide a consistent finish with full depth of colour.
9. Application: Brush and roller or spray.

## 127 PNT-157 Metalwork Paint

1. Description: Metalwork paint, to exposed ceiling services/ ductwork.
2. Manufacturer: AkzoNobel, or acceptable equivalent.
    2.1. Product reference: Dulux Trade Metalshield Satin
3. Surfaces:
    3.1. Preparation: Thoroughly clean down the surfaces to remove all dirt, grease and surface contaminants and allow to dry.
4. Composition: Acrylic copolymer.
5. Sheen: Satin
6. Colour: Refer to the Design Drawings
7. Initial coats: Provide 1 no. initial coat in accordance with the manufacturer's recommendations.
8. Finishing coats: Provide 2 no. full finishing coats to provide a consistent finish with full depth of colour.
9. Application: Spray.

## 131 PNT-201 Feature Stair Finish

1. Description: Water based emulsion paint.
2. Manufacturer: AkzoNobel, or acceptable equivalent.
    2.1. Product reference: Dulux Trade Vinyl Matt Emulsion.
3. Surfaces:
    3.1. Preparation: Thoroughly clean down the surfaces to remove all dirt, grease and surface contaminants and allow to dry.
4. Composition: Acrylic copolymer
5. Finish: Matt
6. Colour: Refer to the Design Drawings
7. Initial coats:
    7.1. Spot prime any bare metal, metal fixings and nail heads with 1 No. coat of Dulux Trade Metalshield QD Metal Primer.
    7.2. Prime overall with 1 No. coat of Dulux Trade Dry Wall Primer Sealer.
8. Finishing coats: Minimum 2 No. coats to provide a consistent finish with full depth of colour.

9. Application: Brush and roller or spray.

## Generally

### 210 Coating materials

1. Selected manufacturers: Submit names before commencement of coating work.

### 215 Handling and storage

1. Coating materials: Deliver in sealed containers, labelled clearly with brand name, type of material and manufacturer's batch number.
2. Materials from more than one batch: Store separately. Allocate to distinct parts or areas of the work.

### 220 Compatibility

1. Coating materials selected by contractor
   1.1. Recommended by their manufacturers for the particular surface and conditions of exposure.
   1.2. Compatible with each other.
   1.3. Compatible with and not inhibiting performance of preservative/fire-retardant pretreatments.

### 280 Protection

1. 'Wet paint' signs and barriers: Provide where necessary to protect other operatives and general public, and to prevent damage to freshly applied coatings.

### 290 Samples

1. General: Submit representative samples of the following:
   1.1. 600mm x 600mm of each paint finish to the relevant substrate.
2. Delivered materials/ products: To match samples.

### 300 Control samples

1. Sample areas of finished work: Carry out, including preparation, as follows:
   1.1. First 9m2 of each coating.
2. Approval of appearance: Obtain before commencement of general coating work.

### 310 Supervised control samples

1. Sample areas of finished work: Carry out, including preparation, as follows:
2. Inspection: Give notice when each stage is ready for inspection.
3. Approval of appearance: Obtain before commencement of general coating work.

### 320 Inspection by coating manufacturers

1. General: Permit manufacturers to inspect work in progress and take samples of their materials from site if requested.

### 321 Inspection of work stages

1. Programme for inspections: Submit as follows:

2. Inspection: Give prior notice when each stage is ready for inspection.

## 330 Slip Resistance

1. Testing of floor paints:
   1.1. Surface roughness meter: minimum of 20 µm Rz.
   1.2. Pendulum Test: 36PTV (wet and dry conditions).
2. Report: Submit.

## 340 Emissions

1. Comply with BS EN 13300 and Paints Directive 2004/42/CE.
2. Only fungi/ algae resistant products to be used. Products must demonstrate low VOC levels.

## 350 Testing

1. Standard: In accordance with BS EN 16402, BS EN ISO 11890: Part 2, BS EN ISO 17895, BS EN ISO 2808.
2. Tests and inspections to be carried out as BS EN ISO 2808.

## 360 Materials

1. Standard: In accordance with BS 2015, BS 7664, BS EN ISO 4618, BS EN ISO 2813.
2. Tests and inspections to be carried out as BS EN ISO 2808.

# Preparation

## 400 Preparation generally

1. Standard: In accordance with BS 6150, BS 8000: Part 12, BS EN ISO 2808
2. Refer to any pre-existing CDM Health and Safety File.
3. Refer to CDM Construction Phase Plan where applicable.
4. Suspected existing hazardous materials: Prepare risk assessments and method statements covering operations, disposal of waste, containment and reoccupation, and obtain approval before commencing work.
5. Preparation materials: Types recommended by their manufacturers and the coating manufacturer for the situation and surfaces being prepared.
6. Substrates: Sufficiently dry in depth to suit coating.
7. Efflorescence salts: Remove.
8. Dirt, grease and oil: Remove. Give notice if contamination of surfaces/ substrates has occurred.
9. Surface irregularities: Remove.
10. Joints, cracks, holes and other depressions: Fill flush with surface, to provide smooth finish.
11. Dust, particles and residues from preparation: Remove and dispose of safely.
12. Water based stoppers and fillers
    12.1. Apply before priming unless recommended otherwise by manufacturer.
    12.2. If applied after priming: Patch prime.
13. Oil based stoppers and fillers: Apply after priming.
14. Doors, opening windows and other moving parts
    14.1. Ease, if necessary, before coating.

14.2. Prime resulting bare areas.

## 420 Fixtures and fittings

1. Removal: Before commencing work remove: Coverplates, grilles, wall clocks, and other surface mounted fixtures.
2. Replacement: Refurbish as necessary, refit when coating is dry.

## 425 Ironmongery

1. Removal: Before commencing work: Remove ironmongery from surfaces to be coated.
2. Hinges: Remove
3. Replacement: Refurbishment as necessary; refit when coating is dry.

## 430 Existing ironmongery

1. Refurbishment: Remove old coating marks. Clean and polish.

## 440 Previously coated surfaces generally

1. Preparation: In accordance with BS 6150, clause 11.5.
2. Contaminated or hazardous surfaces: Give notice of:
    2.1. Coatings suspected of containing lead.
    2.2. Substrates suspected of containing asbestos or other hazardous materials.
    2.3. Significant rot, corrosion or other degradation of substrates.
3. Suspected existing hazardous materials: Prepare risk assessments and method statements covering operations, disposal of waste, containment and reoccupation, and obtain approval before commencing work.
4. Removing coatings: Do not damage substrate and adjacent surfaces or adversely affect subsequent coatings.
5. Loose, flaking or otherwise defective areas: Carefully remove to a firm edge.
6. Alkali affected coatings: Completely remove.
7. Retained coatings
    7.1. Thoroughly clean to remove dirt, grease and contaminants.
    7.2. Gloss-coated surfaces: Provide key.
8. Partly removed coatings
    8.1. Additional preparatory coats: Apply to restore original coating thicknesses.
    8.2. Junctions: Provide flush surface.
9. Completely stripped surfaces: Prepare as for uncoated surfaces.

## 451 Previously coated surfaces – blast cleaning

1. Operatives
    1.1. Trained/ experienced in blast cleaning.
    1.2. Submit evidence of training/ experience on request.
2. Dust and nuisance: Minimize.

## 456 Previously coated surfaces – burning off

1. Risk assessment and method statement: Prepare, and obtain approval before commencing work.
2. Adjacent areas: Protect from excessive heat and falling scrapings.
3. Exposed resinous areas and knots: Apply two coats of knotting.
4. Removed coatings: Dispose of safely.

## 461 Previously coated wood

1. Degraded or weathered surface wood: Take back to provide suitable substrate.
2. Degraded substrate wood: Repair with sound material of same species.
3. Exposed resinous areas and knots: Apply two coats of knotting.

## 471 Preprimed wood

1. Areas of defective primer: Take back to bare wood and reprime.

## 481 Uncoated wood

1. General: Provide smooth, even finish with arrises and moulding edges lightly rounded or eased.
2. Heads of fasteners: Countersink sufficient to hold stoppers/fillers.
3. Resinous areas and knots: Apply two coats of knotting.

## 490 Previously coated steel

1. Defective paintwork: Remove to leave a firm edge and clean bright metal.
2. Sound paintwork: Provide key for subsequent coats.
3. Corrosion and loose scale: Take back to bare metal.
4. Residual rust: Treat with a proprietary removal solution.
5. Bare metal: Apply primer as soon as possible.
6. Remaining areas: Degrease.

## 500 Preprimed steel

1. Areas of defective primer, corrosion and loose scale: Take back to bare metal. Reprime as soon as possible.

## 511 Galvanized, sherardized and electroplated steel

1. White rust: Remove.
2. Pretreatment: Apply one of the following:
    2.1. Mordant solution to blacken whole surface.
    2.2. Etching primer recommended by coating system manufacturer.

## 521 Uncoated steel – manual cleaning

1. Oil and grease: Remove.
2. Corrosion, loose scale, welding slag and spatter: Remove.
3. Residual rust: Treat with a proprietary removal solution.
4. Primer: Apply as soon as possible.

## 531 Uncoated steel – blast cleaning

1. Oil and grease: Remove.
2. Blast cleaning
    2.1. Atmospheric conditions: Dry.
    2.2. Abrasive: Suitable type and size, free from fines, moisture and oil.
    2.3. Surface finish: To BS EN ISO 8501-1
3. Primer: Apply as soon as possible and within four hours of blast cleaning.

## 541 Uncoated aluminium/ copper/ lead

1. Surface corrosion: Remove and lightly key surface.
2. Pretreatment: Etching primer if recommended by coating system manufacturer.

## 552 Uncoated PVC-U

1. Dirt and grease: Remove. Do not abrade surface.

## 560 Uncoated concrete

1. Release agents: Remove.

## 570 Uncoated masonry/ Rendering

1. Loose and flaking material: Remove.

## 580 Uncoated plaster

1. Nibs, trowel marks and plaster splashes: Scrape off.
2. Overtrowelled 'polished' areas: Key lightly.

## 590 Uncoated plasterboard

1. Depressions around fixings: Fill with stoppers/ fillers

## 601 Uncoated plasterboard – to receive textured coating

1. Joints: Fill, tape and feather out with materials recommended by textured coating manufacturer.

## 611 Wall coverings

1. Retained wall coverings: Check that they are in good condition and well adhered to substrate.
2. Previously covered walls: Wash down to remove paper residues, adhesive and size.

## 622 Organic growths

1. Dead and loose growths and infected coatings: Scrape off and remove from site.
2. Treatment biocide: Apply appropriate solution to growth areas and surrounding surfaces.
3. Residual effect biocide: Apply appropriate solution to inhibit re-establishment of growths.

## 631 Previously painted window frames

1. Paint encroaching beyond glass sight line: Remove.
2. Loose and defective putty: Remove.

3. Putty cavities and junctions between previously painted surfaces and glass: Clean thoroughly.
4. Finishing
    4.1. Patch prime, reputty as necessary, and allow to set.
    4.2. Seal and coat as soon as fully set.

### 640 External pointing to existing frames

1. Defective sealant pointing: Remove.
2. Joint depth: Approximately half joint width; adjust with backing strip if necessary.
3. Sealant
    3.1. Preparation and application: As section Z22.

### 645 Sealing of internal movement joints

1. General: To junctions of walls and ceilings with architraves, skirtings and other trims.
2. Sealant: Waterborne acrylic.
    2.1. Preparation and application: As section Z22.

### 651 Existing gutters

1. Dirt and debris: Remove from inside of gutters.
2. Defective joints: Clean and seal with suitable jointing material.

## Application

### 711 Coating generally

1. Application standard: In accordance with BS 6150, clause 9.
2. Conditions: Maintain suitable temperature, humidity and air quality during application and drying.
3. Surfaces: Clean and dry at time of application.
4. Thinning and intermixing of coatings: Not permitted unless recommended by manufacturer and agreed with the Architect.
5. Overpainting: Do not paint over intumescent strips or silicone mastics.
6. Priming coats
    6.1. Thickness: To suit surface porosity.
    6.2. Application: As soon as possible on same day as preparation is completed.
7. Finish
    7.1. Even, smooth and of uniform colour.
    7.2. Free from brush marks, sags, runs and other defects.
    7.3. Cut in neatly.
8. Doors, opening windows and other moving parts: Ease before coating and between coats.
9. Application:: Material from a single batch, providing an even finish, free from any defects.
10. Thickness:: Paint thickness must always be consistent and checked with a wet film thickness wheel or comb gauge.
11. Drying:: Must never be accelerated through artificial means such as heaters or dehumidifiers.
12. Final Coat:: Applied as a single coat and once dried it shall be viewed from 2 metres to determine colour consistency and quality.

13. Inspection: : Conducted on-site by appropriately qualified person to inspect the work/ quality control documentation/ test results.

## 720 Priming joinery

1. Preservative treated timber: Retreat cut surfaces with two flood coats of a suitable preservative before priming.
2. End grain: Coat liberally allow to soak in, and recoat.

## 730 Workshop coating of concealed joinery surfaces

1. General: Apply coatings to all surfaces of components.

## 731 Site-coating of concealed joinery surfaces

1. General: After priming, apply additional coatings to surfaces that will be concealed when fixed in place.

## 740 Concealed metal surfaces

1. General: Apply additional coatings to surfaces that will be concealed when component is fixed in place.

## 751 Staining wood

1. Primer: Apply if recommended by stain manufacturer.
2. Application: Apply in flowing coats and brush out excess stain to produce uniform appearance.

## 760 Varnishing wood

1. First coat: Thin with plant oil-based thinner
    1.1. Brush well in and lay off avoiding aeration.
2. Subsequent coats: Provide light key and smooth along the grain between coats.

## 770 External doors

1. Bottom edges: Prime and coat before hanging doors.

## 780 Bead glazing to coated wood

1. Before glazing: Apply first two coats to rebates and beads.

## 790 Linseed oil putty glazing

1. Setting: Allow putty to set for seven days.
2. Sealing
    2.1. Within a further 14 days, seal with a solvent-borne primer.
    2.2. Fully protect putty with coating system as soon as it is sufficiently hard.
    2.3. Extend finishing coats on to glass up to sight line.

## 800 Glazing

1. Etched, sand blasted and ground glass: Treat or mask edges before coating to protect from contamination by oily constituents of coating materials.

## 810 Water-repellent

1. Application: Liberally flood surface, giving complete and even coverage.

Ω End of Section



make

# M61
# Intumescent Coatings for Fire Protection of Steelwork

## Protective coating systems

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 160 FPS-201 Intumescent Coating Systems

1. Description: Factory-applied two pack fast curing epoxy zinc phosphate, intumescent, recoatable polyurethane compliant coating.
2. Manufacturer: AkzoNobel, or acceptable equivalent
3. Use/ location: Internal exposed steelwork plus internal non-exposed steelwork.
4. Surface: Blast cleaned structural steelwork.
5. Prmer:
    5.1. Product reference: Intercure 200.
    5.2. Dry film thickness: 75 microns nominal as required for section size, to be determined by the Sub-contractor to the acceptance of the Contractor and Structural Engineer.
    5.3. C1/C2 environment: For a C1/C2 environment to BS EN ISO 12944 the Intercure 200 can be replaced with Intergard 269 at a nominal dry film thickness of 40 microns. If the intumescent is to be Site applied, then shop apply Intercure 200 at 75 microns nominal dry film thickness, regardless of environment.
6. Intumescent coat:
    6.1. Product reference: Interchar 2060.
    6.2. Dry film thickness: As required for section size, to be determined by the Sub-contractor to the acceptance of the Contractor and Structural Engineer.
    6.3. Fire resistance: 60 minutes/ 90 minutes, to BS 476: Parts 20 and 22, Refer to the Fire Strategy Report
7. Topseal:
    7.1. Product reference: Interthane 990.
    7.2. Dry film thickness: 50 microns nominal - two coats required for external exposed steelwork, as required for section size, to be determined by the Sub- contractor to the acceptance of the Contractor and Structural Engineer
    7.3. Colour: As required by the Contractor
    7.4. Finish: Decorative
    7.5. Site touch up: Repair any intumescent paint damaged during transit or during erection, to the manufacturer's working procedures.
    7.6. Bolt caps: Interchar bolt caps to be snapped into place over all structural steel bolt connections.
    7.7. Refer to manufacturer's working procedures for handling, transportation and storage of steelwork treated with Interchar 2060 off Site.

# Performance and general requirements

## 202 Fire performance

1. Reaction to fire: To BS EN 13501-1, A1
2. Fire resistance to BS EN 13501-2: To BS EN 13501-2, REI 120

## 204 Exposure and durability

1. Environmental exposure: C1
2. Durability classification: W/Y

## 205 Validation of materials

1. Project-specific evaluation of intumescent coating materials
   1.1. Standard: In accordance with BS EN 16623, clause 4.
   1.2. Test results: Submit on request.

## 210 Working procedures

1. Standard: In accordance with BS EN 16623.
2. Give notice: Before commencing surface preparation and coating application.
3. Quality control: Record project specific procedures for surface preparation and coating application.

## 215 Working conditions

1. General: Maintain manufacturer's recommended temperature, humidity and air quality conditions during application and drying.
2. Surface condition: Clean and dry at time of application.

## 220 Applicator's personnel

1. Operatives: Trained/ experienced in anticorrosive and intumescent coatings.
2. Evidence of training/ experience: Submit on request.

## 250 Sprayed coating application on site

1. Standard: In accordance with BS EN 16623.
2. Spray drift: Minimize.
3. Masking: Protect designated adjacent surfaces.
   3.1. Designated surfaces: Fair-faced brickwork and Hardwood window frames

## 255 Sprayed coating application off site

1. Standard: In accordance with BS EN 16623.

## 260 Control samples

1. General: Carry out sample areas of finished work as follows: One column and two adjoining beams.
2. Approval of appearance: Obtain before commencement of general application.

## 270 Inspection

1. Permit intumescent manufacturer to
    1.1. Inspect work in progress.
    1.2. Inspect quality control records.
    1.3. Take dry film thickness and other measurements.
    1.4. Take samples of products.
2. Intumescent manufacturer's inspection reports: Submit without delay.

## 280 Off-site-coated steel

1. Handling and erection: Use methods and devices designed to minimize damage to intumescent coatings.
2. Damaged areas of coating: Reinstate in accordance with coating manufacturer's recommendations.

# Preparation of surfaces

## 315 New steel – blast-cleaning

1. Preparation: Remove oil, grease and contaminants.
2. Blast-cleaning
    2.1. Atmospheric condition: Dry.
    2.2. Abrasive: Suitable type and size, free from fines, moisture and oil.
    2.3. Finish: To BS EN ISO 8501-1, preparation grade SA2½, with an average profile of approximately 75 micrometres.
    2.4. Abrasive residues and moisture: Remove.
3. Primer: Apply as soon as possible after cleaning and before gingering or blackening appears.

## 320 Existing steel – blast-cleaning

1. Preparation: Remove oil, grease and contaminants.
2. Blast-cleaning: Remove existing coatings and rust.
    2.1. Atmospheric condition: Dry.
    2.2. Abrasive: Suitable type and size, free from fines, moisture and oil.
    2.3. Finish: To BS EN ISO 8501-1, preparation grade SA2½, with an average profile of approximately 75 micrometres.
    2.4. Abrasive residues and moisture: Remove.
3. Primer: Apply as soon as possible after cleaning and before gingering or blackening appears.

## 330 Existing steel – manual cleaning

1. Preparation: Remove oil, grease and contaminants.
2. Loose or unsound coatings: Remove to a firm edge.
3. Finish: To BS EN ISO 8501-1, preparation grade St2. Leave a clean but unpolished dry surface.
4. Primer: Apply as soon as possible after cleaning and before gingering or blackening appears. Remove coating edges that lift as a result of priming, and reprime.

### 340 Existing steel – preparation for overcoating

1. Preparation: Remove oil, grease and contaminants.
2. Loose or unsound coatings: Remove to a firm edge.
3. Exposed steel finish: Manually clean to BS EN ISO 8501-1, grade St2. Leave a clean but unpolished dry surface.
4. Existing coatings finish: Abrade to give a good key. Leave a clean, dry surface.
5. Primer: Apply one brush coat to bare steel areas. Remove coating edges that lift as a result of priming, and reprime.

## Application of castings

### 402 Intumescent casting thickness

1. Required thickness: Determine for every steel member to give specified period of fire resistance. Use intumescent casting manufacturer's current published loading tables.
    1.1. Special sections and partial fire exposure conditions: Obtain required thickness in writing from manufacturer.
2. Schedule and drawings: Submit at least two weeks before starting work.
    2.1. Schedule content: Member sizes, weights/ thicknesses, loading conditions, etc. showing, for each variant, the exposed perimeter/ sectional area (Hp/A) ratio and required casting thickness.
    2.2. Drawing content: Steelwork drawings marked in colour to show required thickness for each member.

### 404 Intumescent casting installation

1. Assembly: Adhesive fix casting sections tight to steel substrate.
2. Filling: Apply intumescent filler to all joints and deformations to produce a smooth and uniform finish.

## Application of coatings

### 410 Intumescent coating dry film thickness (dft)

1. Required dft: Determine for every steel member to give specified period of fire resistance. Use intumescent coating manufacturer's current published loading tables.
    1.1. Special sections and partial fire exposure conditions: Obtain required dft in writing from manufacturer.
2. Schedule and drawings: Submit at least two weeks before starting work.
    2.1. Schedule content: Member sizes, weights/ thicknesses, loading conditions, etc. showing, for each variant, the exposed perimeter/ sectional area (Hp/A) ratio and required dft.
    2.2. Drawing content: Steelwork drawings marked in colour to show required dft for each member.

### 420 Measurement of intumescent dft

1. Primer dft: Determine average dft (for deduction from total dft after application of intumescent).
2. Intumescent dft: Determine at:

2.1. 500 mm centres along each coated plane of universal sections (8 planes), and rectangular hollow sections (4 planes).

2.2. 125 mm centres along coated circular hollow sections, spread evenly around circumference.

3. Acceptance standard

3.1. Average intumescent dft: Not less than required dft (exclusive of primer and top sealer).

3.2. Local intumescent dft: Not less than 80% of required dft. Areas greater than 100 mm equivalent diameter with a dft of less than 80% of required dft must be brought up to thickness.

## 440 Basic finish

1. Definition: Reasonably smooth and even. Orange peel, other texture, minor runs and similar minor defects are acceptable.

## 450 Normal decorative finish

1. Definition: Good standard of cosmetic finish generally, when viewed from a distance of 5 m or more. Minor orange peel or other texture is acceptable.

## 460 High decorative finish

1. Definition: High standard of evenness, smoothness and gloss when viewed from a minimum distance of 2 m.

## 490 Top sealer coat

1. Application: To achieve dft recommended by manufacturer and to give an even, solid, opaque appearance, free from runs, sags and other visual defects.

## 520 Completion of off-site-coated steel

1. Uncovered areas, including fixings: Following erection of steelwork, apply intumescent coating locally.
2. Unscheduled additional connections to erected steelwork: Remove and reinstate intumescent coating locally.

# Completion

## 530 Records of intumescent application

1. On completion of intumescent work, submit

1.1. Accurate surface preparation, coating and intumescent application records.

1.2. Fire resistance certificates.

1.3. Intumescent manufacturer's recommendations for maintenance and overcoating.

Ω End of Section

# N10
# General Fixtures/ Furnishings/ Equipment/ Custom-made Joinery

## Products

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 FFE-306 Joinery Units

1. Description: Joinery units as shown on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Fire rating:
    3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
        3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
        Maintenance of system to be presented for review.
4. Support framing:
    4.1. Type and material: To be determined by the Contractor to provide a stable ground for the specified applied finishes.
    4.2. Include all necessary support framing, fixings and bracketry, which is to be concealed unless indicated otherwise on the Design Drawings.
    4.3. The frame is to form fixing grounds for the applied surface finishes, internal secure stowage and cable throughways and is to be completely concealed by the architectural finishes.
5. Carcass/ core:
    5.1. Type/ material: Medium-density fibreboard (MDF)/ Moisture resistant, medium-density fibreboard (MDF.H)/ Water and boil proof (WBP) plywood - To be determined by the Contractor to suit the performance requirements and service conditions.
    5.2. Thickness: Refer to the Design Drawings
    5.3. Fixing: Suitably concealed fixed.
6. External (visible) linings/ facings/ finishes:
    6.1. Type/ material: Refer to the Design Drawings
    6.2. Species: Refer to the Design Drawings
    6.3. Finish: Refer to the Design Drawings
7. Internal linings:
    7.1. Type/ material: Refer to the Design Drawings
    7.2. Finish/ colour: Refer to the Design Drawings
    7.3. Colour: Refer to the Design Drawings
8. Sliding doors/ shelving:
    8.1. Type/ material: Refer to the Design Drawings
    8.2. Finish/ colour: Refer to the Design Drawings

8.3. Colour: Refer to the Design Drawings

9. Worktop:
   9.1. Type/ material: Terrazzo.
   9.2. Edge: Metal trim.
   9.3. Finish/ colour: Refer to the Design Drawings
   9.4. Colour: Refer to the Design Drawings
10. Fixings: Concealed
11. Joinery workmanship: As section Z10.
12. Metalwork materials and workmanship: As section Z11.
13. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
14. Services and lighting: Refer to the Services Engineer's Information
15. Other requirements: To include all accessories, fixings and components to complete the installation.

## 112 FFE-311 Joinery Unit and Soft Seating

1. Description: Joinery unit and soft seating, as shown on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Fire rating:
   3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
4. Support framing:
   4.1. Type and material: To be determined by the Contractor to provide a stable ground for the specified applied finishes.
   4.2. Include all necessary support framing, fixings and bracketry, which is to be concealed unless indicated otherwise on the Design Drawings.
   4.3. The frame is to form fixing grounds for the applied surface finishes, internal secure stowage and cable throughways and is to be completely concealed by the architectural finishes.
5. Carcass/ core:
   5.1. Type/ material: Medium-density fibreboard (MDF)/ Moisture resistant, medium-density fibreboard (MDF.H)/ Water and boil proof (WBP) plywood - To be determined by the Contractor to suit the performance requirements and service conditions.
   5.2. Thickness: Refer to the Design Drawings
   5.3. Fixing: Suitably concealed fixed.
6. External (visible) linings/ facings/ finishes:
   6.1. Type/ material: Refer to the Design Drawings
   6.2. Species: Refer to the Design Drawings
   6.3. Finish: Refer to the Design Drawings
7. Internal linings:
   7.1. Type/ material: Refer to the Design Drawings
   7.2. Finish/ colour: Refer to the Design Drawings

7.3. Colour: Refer to the Design Drawings

8. Shelving:
   8.1. Type/ material: Refer to the Design Drawings
   8.2. Finish/ colour: Refer to the Design Drawings
   8.3. Colour: Refer to the Design Drawings
9. Seating:
   9.1. Material: Refer to the Design Drawings
   Stitching details, thread colour to match base body fabric colour. Stitching type to be agreed.
   9.2. Type: Refer to the Design Drawings
   9.3. Finish/ colour: Refer to the Design Drawings
   9.4. Backing: Upholstery foam, density to be agreed through sampling on suitable concealed rigid substrate board, secured back to support below using suitable concealed fixings.
10. Fixings: Concealed
11. Joinery workmanship: As section Z10.
12. Metalwork materials and workmanship: As section Z11.
13. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
14. Services and lighting: Refer to the Services Engineer's Information
15. Other requirements: To include all accessories, fixings and components to complete the installation.

## 113 FFE-316 Integrated Joinery Unit and Planter

1. Description: Bespoke fabricated moveable planter with integrated timber seating as indicated on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Fire rating:
   3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
4. Support framing:
   4.1. Type and material: To be determined by the Contractor to provide a stable ground for the specified applied finishes.
   4.2. Include all necessary support framing, fixings and bracketry, which is to be concealed unless indicated otherwise on the Design Drawings.
   4.3. The frame is to form fixing grounds for the applied surface finishes, internal secure stowage and cable throughways and is to be completely concealed by the architectural finishes.
5. Carcass/ core:
   5.1. Type/ material: Medium-density fibreboard (MDF)/ Moisture resistant, medium-density fibreboard (MDF.H)/ Water and boil proof (WBP) plywood - To be determined by the Contractor to suit the performance requirements and service conditions.
   5.2. Thickness: Refer to the Design Drawings
   5.3. Fixing: Suitably concealed fixed.

6. External (visible) seat linings/ facings/ finishes:
    6.1. Type/ material: Timber.
    6.2. Species: Refer to the Design Drawings
    6.3. Finish: Refer to the Design Drawings
7. Internal finish:
    7.1. Type/ material: Refer to the Design Drawings
    7.2. Finish/ colour: Refer to the Design Drawings
    7.3. Colour: Refer to the Design Drawings
8. Seating:
    8.1. Type/ material: Refer to the Design Drawings
    8.2. Finish/ colour: Refer to the Design Drawings
    8.3. Colour: Refer to the Design Drawings
9. Movability: Mounted on castors with brakes.
10. Fixings: Concealed
11. Joinery workmanship: As section Z10.
12. Metalwork materials and workmanship: As section Z11.
13. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
14. Services and lighting: Refer to the Services Engineer's Information
15. Other requirements: To include all accessories, fixings and components to complete the installation.

## 114 FFE-321 Loose seating

1. Description: Bespoke fabricated moveable stool with upholstered seat cushion as indicated on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Support framing: Specialist fabricator designed metal or timber support frame anchored to primary structure before the final floor finish is laid.
4. External (visible) linings/ facings/ finishes:
    4.1. Type/ material: Timber.
    4.2. Species: Oak.
    4.3. Colour: Refer to the Design Drawings
5. Seat:
    5.1. Type: Upholstered and leather covered seat.
    5.2. Foam infill:
        5.2.1. Type: CMHR, to CRIB 5 ignition standard to BS 6807.
    5.3. Fabric: Leather.
    5.4. Provide an interliner between the fabric and CMHR foam.
    5.5. Fabric to be wrapped tautly across the seat cushion and retained on the underside in a concealed manner.
    5.6. Sticthing/ specialist details: Refer to the Finishes Schedule and Refer to the Design Drawings
6. Fixings: Concealed

7. Joinery workmanship: As section Z10.
8. Other requirements: To include all accessories, fixings and components to complete the installation.

## 115 FFE-325 Seating With Integrated Planter Box

1. Description: Bespoke fabricated planter boxes with integrated timber seating as indicated on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Support framing: Specialist fabricator designed metal or timber support frame anchored to primary structure before the final floor finish is laid.
4. Carcass/ core:
    4.1. Type/ material: Refer to the Design Drawings
    4.2. Thickness: Refer to the Design Drawings
    4.3. Fixing: Suitably concealed fixed.
5. External (visible) seat linings/ facings/ finishes:
    5.1. Type/ material: Solid timber
    5.2. Species: English Oak
        5.2.1. Source: Crown forest.
    5.3. Finish: Lacquered
6. External (visible) seat base linings/ facings/ finishes:
    6.1. Type/ material: Metal
    6.2. Finish/ colour: Polyester powder coated
    6.3. Colour: Brown
7. External (visible) planter linings/ facings/ finishes:
    7.1. Type/ material: Metal
    7.2. Finish/ colour: Polyester powder coated
    7.3. Colour: Green
8. Fixings: Concealed
9. Joinery workmanship: As section Z10.
10. Metalwork materials and workmanship: As section Z11.
11. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
12. Services and lighting: Refer to the Services Engineer's Information
13. Other requirements: To include all accessories, fixings and components to complete the installation.

## 118 FFE-336 Seating With Integrated Feature Planter Box

1. Description: Bespoke fabricated feature planter boxes with integrated timber seating as indicated on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Support framing: Specialist fabricator designed metal or timber support frame anchored to primary structure before the final floor finish is laid.
4. Carcass/ core:
    4.1. Type/ material: Refer to the Design Drawings

4.2. Thickness: Refer to the Design Drawings

4.3. Fixing: Suitably concealed fixed.

5. External (visible) seat linings/ facings/ finishes:
   5.1. Type/ material: Solid timber
   5.2. Species: English Oak.
   5.3. Finish: Lacquered
6. External (visible) planter linings/ facings/ finishes:
   6.1. Type/ material: Patinated metal
   6.2. Finish/ colour: Brushed bronze
7. Fixings: Concealed
8. Joinery workmanship: As section Z10.
9. Metalwork materials and workmanship: As section Z11.
10. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
11. Services and lighting: Refer to the Services Engineer's Information
12. Other requirements: To include all accessories, fixings and components to complete the installation.

## 121 FFE-341 Seating With Integrated Planter Box

1. Description: Bespoke fabricated planter boxes with integrated timber seating as indicated on the Design Drawings.
2. Sizes and configuration: Refer to the Design Drawings
3. Support framing: Specialist fabricator designed metal or timber support frame anchored to primary structure before the final floor finish is laid.
4. Carcass/ core:
   4.1. Type/ material: Refer to the Design Drawings
   4.2. Thickness: Refer to the Design Drawings
   4.3. Fixing: Suitably concealed fixed.
5. External (visible) seat linings/ facings/ finishes:
   5.1. Type/ material: Solid timber dowel cladding
   5.2. Finish: Lacquered
6. External (visible) planter linings/ facings/ finishes:
   6.1. Type/ material: Patinated metal
   6.2. Finish/ colour: Bronze
7. Seat/ back rest:
   7.1. Type: Upholstered and fabric covered seat/ back rest.
   7.2. Foam infill:
      7.2.1. Type: CMHR, to CRIB 5 ignition standard to BS 6807.
   7.3. Fabric: To comply with BS EN 1021: Parts 1 and 2 and BS 7176 (Low Hazard).
      7.3.1. Type: Leather
      7.3.2. Colour: Black
   7.4. Provide an interliner between the fabric and CMHR foam.

7.5. Fabric to be wrapped tautly across the seat cushion and retained on the underside in a concealed manner.

7.6. Sticthing/ specialist details: Refer to the Design Drawings

8. Fixings: Concealed
9. Joinery workmanship: As section Z10.
10. Metalwork materials and workmanship: As section Z11.
11. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
12. Services and lighting: Refer to the Services Engineer's Information
13. Other requirements: To include all accessories, fixings and components to complete the installation.

## 132 FFE-217 Built-In Wet Vanity Unit

1. Description: Proprietary vanity unit as indicated on the Design Drawings.
2. Manufacturer: To be agreed with the Architect.

2.1. Product reference: To be agreed with the Architect.

3. Form: Wall-mounted.
4. Dimensions: Refer to the Design Drawings
5. Worktop: Refer to the Design Drawings

5.1. Finish/ Colour: Refer to the Design Drawings

6. Sink/ bowl/ tap: Refer to N13 and the Sanitaryware Schedule.
7. Mirror: Refer to the Design Drawings

7.1. Safety requirements: To BS 6262-4 and BS 6262-6.

7.2. Material: Float glass, silvered to give maximum reflection, free from tarnishing, discolouration, scratches and other defects visible in the designed viewing conditions.

7.3. Thickness: Nominal 6 mm.

7.4. Backing: Laminated with 3M safety film.

7.5. Edges: Polished.

8. Doors/ Drawers/ Shelves

8.1. Material: Refer to the Design Drawings

8.2. Finish/ Colour: Refer to the Design Drawings

9. Supports: Refer to the Design Drawings
10. Other requirements: As required to complete the installation

## 134 FFE-219 Purpose-Made Built-In Dry Vanity unit

1. Description: Purpose-made built-in dry vanity unit with integrated mirror, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Support:

3.1. Suitable support system, either corrosion resistant or treated to resist rot.

3.2. Carcassing/ support linings to be an appropriate structural grade of Class 3 plywood as a minimum.

3.3. Concealed in the finished works, and co-ordinated with services, refer to the Services Engineer's documentation.

4. Vanity counter with integral sinks, shelf and splashback:
    4.1. Material: Ultracompact product formed using a high tech process (Sinterized Particle Technology) which represents an accelerated version of the metamorphic change that natural stone undergoes when subjected to high temperatures and pressure over a long duration:
        4.1.1. Colour: Refer to the Design Drawings
        4.1.2. Finish: Refer to the Design Drawings
        4.1.3. Thickness: Refer to the Design Drawings
    4.2. Element fabrication:
        4.2.1. The material is cut in to required pieces, running joints to not be accepted on joints/ junctions at changes in direction.
        4.2.2. Permitted joints/ junctions are to be suitably bonded and abraded to form an inconspicuous interface matching the general material in appearance. Edges are to be lightly arrised.
5. Cabinetry:
    5.1. Carcassing/ cabinetry interior:
        5.1.1. Material: Suitable core material such as Class 3 plywood, minimum 22mm thick.
        5.1.2. Lining/ facing: High pressure laminate (HPL).
        5.1.3. Fixings, cramps to be stainless steel and concealed in the finished works.
    5.2. Access panels/ visible external panelling:
        5.2.1. Material: Suitable core material such as Class 3 plywood, minimum 18mm thick.
        5.2.2. Lining/ facing: High pressure laminate (HPL).
        5.2.3. Finish/ colour:
            5.2.3.1. Colour to be confirmed by the Architect through sampling.
        5.2.4. Fixings, cramps to be stainless steel and concealed in the finished works.
    5.3. Integrated seating
        5.3.1. Material: Refer to the Design Drawings
        Stitching details, thread colour to match base body fabric colour. Stitching type to be agreed.
        5.3.2. Type: Refer to the Design Drawings
        5.3.3. Finish/ colour: Leather.
    5.4. Integrated mirror:
        5.4.1. Material: Low iron back silver safety glass, impact resistant.
        5.4.2. Thickness: Minimum 6mm.
        5.4.3. Edges: Chamfered and polished.
    5.5. Adhesives: Type to match durability class of core material, compliant with BS EN 204.
    5.6. Edges, including rebated edges, are to be fully laminated, lipped and de-arrised along all edges. The appearance of laminate finishes to be uniform and continuous at junctions without visible join lines, when visible to the general building user.
    5.7. Hardware, to include:
        5.7.1. Suited stainless steel locking mechanism where the key remains in place to provide a 'handle', inconspicuous in the finished works. Used by maintenance staff to gain access to the service to the rear, to be confirmed.

5.7.2. Stainless steel hinge mechanisms suitable for required functionality, concealed when the access panels are in the closed position.

5.7.3. Magnetic door closers to access panel.

## 135 FFE-221 Built-In Shelf

1. Description: Built-in shelf as indicated on the Design Drawings.
2. Manufacturer: To be agreed with the Architect.
   2.1. Product reference: To be agreed with the Architect.
3. Form: Wall-mounted.
4. Dimensions: Refer to the Design Drawings
5. Material: Timber.
   5.1. Finish/ Colour: Refer to the Design Drawings
6. Supports: Refer to the Design Drawings
7. Other requirements: As required to complete the installation

## 141 FFE-231 Reception Desk

1. Description: 2 tier reception desk, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Fire rating:
   3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
4. Support framing:
   4.1. Type and material: To be determined by the Contractor to provide a stable ground for the specified applied finishes.
   4.2. Include all necessary support framing, fixings and bracketry, which is to be concealed unless indicated otherwise on the Design Drawings.
   4.3. The frame is to form fixing grounds for the applied surface finishes, internal secure stowage and cable throughways and is to be completely concealed by the architectural finishes.
5. Carcass/ core/ substrate:
   5.1. Material: Medium-density fibreboard (MDF)/ Moisture resistant, medium-density fibreboard (MDF.H)/ Water and boil proof (WBP) plywood - To be determined by the Contractor to suit the performance requirements and service conditions.
   5.2. Thickness: Refer to the Design Drawings
   5.3. Fixing: Suitably conceal fixed.
   5.4. Integrated power and cable grommet: Refer to the Design Drawings
6. External (visible) linings/ facings/ finishes:
   6.1. Material: Stone.
   6.2. Profile: Fluted.
   6.3. Type: Refer to the Design Drawings
   6.4. Finish/Colour: Refer to the Design Drawings
7. Internal linings/ facings/ finishes:

7.1. Material: Refer to the Design Drawings

7.2. Type: Refer to the Design Drawings

7.3. Finish/Colour: Refer to the Design Drawings

8. Benchtops:

8.1. Upper:

8.1.1. Material: Stone.

8.1.2. Thickness: Refer to the Design Drawings

8.1.3. Finish/Colour: Refer to the Design Drawings

8.1.4. Edge detail: Refer to the Design Drawings

8.1.5. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive.

8.2. Lower:

8.2.1. Material: Refer to the Design Drawings

8.2.2. Thickness: Refer to the Design Drawings

8.2.3. Finish/Colour: Refer to the Design Drawings

8.2.4. Edge detail: Refer to the Design Drawings Visible edges to be finished with soft corners.

8.2.5. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive. Timbers to be butt jointed with grain directions to match.

9. Skirting/ plinth:

9.1. Material: Refer to the Design Drawings

9.2. Finish: Refer to the Design Drawings

9.3. Fixing: Suitably concealed fixings.

10. Trims/shadow gaps: Where shown on the Design Drawings.

11. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.

12. Joinery workmanship: As section Z10.

13. Metalwork materials and workmanship: As section Z11.

14. Fixings: Suitable concealed fixings and support determined by the Specialist Contractor to suit the performance requirements and service conditions.

15. Services : All services, power/ plumbing/ lighting etc as applicable to each system, to be fully integrated as shown on the Design Drawings. Refer to the Design Drawings and Refer to the Services Engineer's Information

16. System to include all fixings, bracketry, support framing and all other components/ accessories necessary to complete the works to suit the conditions of service.

## 142 FFE-232 Reception Desk

1. Description: 2 tier reception desk, as shown on the Design Drawings.

2. Size and configuration: Refer to the Design Drawings

3. Fire rating:

3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.

3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
Maintenance of system to be presented for review.



make

4. Support framing:
    4.1. Type and material: To be determined by the Contractor to provide a stable ground for the specified applied finishes.
    4.2. Include all necessary support framing, fixings and bracketry, which is to be concealed unless indicated otherwise on the Design Drawings.
    4.3. The frame is to form fixing grounds for the applied surface finishes, internal secure stowage and cable throughways and is to be completely concealed by the architectural finishes.
5. Carcass/ core/ substrate:
    5.1. Material: Medium-density fibreboard (MDF)/ Moisture resistant, medium-density fibreboard (MDF.H)/ Water and boil proof (WBP) plywood - To be determined by the Contractor to suit the performance requirements and service conditions.
    5.2. Thickness: Refer to the Design Drawings
    5.3. Fixing: Suitably conceal fixed.
    5.4. Integrated power and cable grommet: Refer to the Design Drawings
6. External (visible) linings/ facings/ finishes:
    6.1. Material: Stone.
    6.2. Profile: Fluted.
    6.3. Type: Refer to the Design Drawings
    6.4. Finish/Colour: Refer to the Design Drawings
7. Internal linings/ facings/ finishes:
    7.1. Material: Refer to the Design Drawings
    7.2. Type: Refer to the Design Drawings
    7.3. Finish/Colour: Refer to the Design Drawings
8. Benchtop:
    8.1. Material: Stone.
    8.2. Thickness: Refer to the Design Drawings
    8.3. Finish/Colour: Refer to the Design Drawings
    8.4. Edge detail: Refer to the Design Drawings
    8.5. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive.
9. Podium:
    9.1. Material: Charred timber.
    9.2. Finish: Refer to the Design Drawings
    9.3. Fixing: Suitably concealed fixings.
10. Trims/shadow gaps: Where shown on the Design Drawings.
11. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
12. Joinery workmanship: As section Z10.
13. Metalwork materials and workmanship: As section Z11.
14. Fixings: Suitable concealed fixings and support determined by the Specialist Contractor to suit the performance requirements and service conditions.
15. Services : All services, power/ plumbing/ water supply/ lighting etc as applicable to each system, to be fully integrated as shown on the Design Drawings. Refer to the Design Drawings and Refer to the Services Engineer's Information

16. System to include all fixings, bracketry, support framing and all other components/ accessories necessary to complete the works to suit the conditions of service.

## 143 FFE-241 Purpose-Made Towel Unit

1. Description: Purpose-made towel unit with concealed towel collection bins, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Unit with integrated towel bin cabinet and fixed shelves:
    3.1. Material: Refer to the Design Drawings
    3.2. Type / species: Refer to the Design Drawings
        3.2.1. Cutting method: To be confirmed.
            3.2.1.1. To be suitably treated to provide protection in wet/ humid environment.
    3.3. Finish: Clear satin varnish to suit wet conditions.
    3.4. Doors: Part-height doors as shown on the Design Drawings, to give access to a towel bin concealed within the unit. Hung with concealed hinges and secured in place using a suitable stainless steel locking mechanism, inconspicuous in the finished works and suited for use by maintenance staff only.
    3.5. Integrated towel bin: Proprietary towel collection bin on wheels (with ability to lock in position), concealed within unit and accessible hinged door, size and configuration to be agreed via sampling and mock-up.
4. System to include all fixings, adhesives and all other accessories/ components recommended to suit the service conditions.

## 144 FFE-281 Joinery Seating Integrated Waterpoint

1. Description: Joinery seating integrated waterpoint within porcelain counter top, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Support:
    3.1. Suitable support system, either corrosion resistant or treated to resist rot.
    3.2. Carcassing/ support linings to be an appropriate structural grade English Oak.
    3.3. Concealed in the finished works, and co-ordinated with services, refer to the Services Engineer's documentation.
4. Vanity counter with integral sinks, waterpoint, shelf and splashback:
    4.1. Material: Vanity counter and associated top surfaces to be porcelain or stone like appearance.
        4.1.1. Colour: Refer to the Design Drawings
        4.1.2. Finish: Refer to the Design Drawings
        4.1.3. Thickness: Refer to the Design Drawings
    4.2. Element fabrication:
        4.2.1. The material is cut in to required pieces, running joints to not be accepted on joints/ junctions at changes in direction.
        4.2.2. Permitted joints/ junctions are to be suitably bonded and abraded to form an inconspicuous interface matching the general material in appearance. Edges are to be lightly arrised.

4.3. Sanitaryware: Apertures formed for integration of sanitary appliances/ access to accessories, refer to the Design Drawings and Section N13.

5. Cabinetry:
   5.1. Carcassing/ cabinetry interior:
      5.1.1. Material: Suitable core material such as English Oak.
      5.1.2. Lining/ facing: High pressure laminate (HPL).
      5.1.3. Fixings, cramps to be stainless steel and concealed in the finished works.
   5.2. Access panels/ visible external panelling:
      5.2.1. Material: Suitable core material such as English Oak.
      5.2.2. Lining/ facing: High pressure laminate (HPL).
      5.2.3. Finish/ colour:
         5.2.3.1. Colour to be confirmed by the Architect through sampling.
      5.2.4. Fixings, cramps to be stainless steel and concealed in the finished works.
   5.3. Seating:
      5.3.1. Material: Refer to the Design Drawings
      5.3.2. Thickness: Refer to the Design Drawings
      5.3.3. Fixing: Suitably concealed fixed.
6. Fixings: Concealed.
7. Joinery workmanship: As section Z10.
8. Apertures/ cut-outs (where required): To be precision factory-formed using rigid, manufacturer's templates.
9. Services and lighting: Refer to the Services Engineer's Information
10. Other requirements: To include all accessories, fixings and components to complete the installation.

## 221 FFE-111 Locker - Clothes

1. Description: Lockers, as shown on the Design Drawings.
2. Manufacturer: Helmsman, or acceptable equivalent.
   2.1. Product reference: Z Lockers.
3. Size(s) and configuration(s): Refer to the Design Drawings
4. Fire rating:
   4.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      4.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
5. Carcass: Refer to the Design Drawings
6. Doors/ end panels/ overpanel:
   6.1. Type: Z locker.
   6.2. Material: Timber veneer.
   6.3. Colour/ finish: Refer to the Design Drawings
   6.4. Fixings: Concealed.
   6.5. Fittings: Locking strategy to lockers to be confirmed.

6.6. Overpanel/ ceiling infill: Provide continuous timber-veneered over panel/ extension above locker.

7. Plinth:
   7.1. Type: Refer to the Design Drawings
   7.2. Colour/ finish: Refer to the Design Drawings
   7.3. Fixings: Concealed.
8. Ironmongery/ accessories : To be confirmed.
   8.1. Numbering/ signage: Numbering and ancillary signage/ graphics to be bespoke.
9. Skirting
   9.1. Material: Refer to the Design Drawings
   9.2. Type: Refer to the Design Drawings
   9.3. Finish/ colour: Refer to the Design Drawings
10. System to include components and fixings as required and all other accessories/ components recommended to suit the service conditions

## 222 FFE-112 Locker - Shoes

1. Description: Lockers, as shown on the Design Drawings.
2. Size(s) and configuration(s): Refer to the Design Drawings
3. Fire rating:
   3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
4. Carcass: Refer to the Design Drawings
5. Doors/ end panels/ overpanel:
   5.1. Type: Shoes locker.
   5.2. Material: Timber veneer.
   5.3. Colour/ finish: Refer to the Design Drawings
   5.4. Fixings: Concealed.
   5.5. Fittings: Locking strategy to lockers to be confirmed.
6. Plinth:
   6.1. Type: Refer to the Design Drawings
   6.2. Colour/ finish: Refer to the Design Drawings
   6.3. Fixings: Concealed.
7. Ironmongery/ accessories : To be confirmed.
   7.1. Numbering/ signage: Numbering and ancillary signage/ graphics to be bespoke.
8. Skirting
   8.1. Material: Refer to the Design Drawings
   8.2. Type: Refer to the Design Drawings
   8.3. Finish/ colour: Refer to the Design Drawings
9. System to include components and fixings as required and all other accessories/ components recommended to suit the service conditions

## 223 FFE-113 Locker - Bicycles

1. Description: Internal folding cycle locker with integrated key or pad lockable system.
2. Size and configuration: Refer to the Design Drawings
3. Carcass: Refer to the Design Drawings
4. Doors/ end panels/ overpanel:
    4.1. Material: Metal.
    4.2. Finish: Polyester powder coated.
    4.3. Colour: Standard RAL colour to be selected.
5. Plinth:
    5.1. Material: Refer to the Design Drawings
    5.2. Finish/ colour: Refer to the Design Drawings
    5.3. Fixing: Concealed.
6. Ironmongery/ accessories:
    6.1. Numbering/ signage: Numbering/ signage
7. Skirting:
    7.1. Type: Refer to the Design Drawings
    7.2. Material: Refer to the Design Drawings
    7.3. Finish/ colour: Refer to the Design Drawings
8. System to include components and fixings, as required, and all other accessories/ components recommended to suit the service conditions.

## 224 FFE-116 Drying Lockers - Clothes Steaming

1. Description: Drying lockers, as shown on the Design Drawings.
2. Manufacturer: Samsung, or acceptable equivalent
    2.1. Product reference: AirDresser Clothing Care System
3. Size(s) and configuration(s): Refer to the Design Drawings
4. Fire rating:
    4.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
        4.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
        Maintenance of system to be presented for review.
5. Carcass:
    5.1. Body: As supplied
    5.2. Finish/ colour: As supplied
6. Doors:
    6.1. Material: Crystal mirror
    6.2. Colour/ finish: As supplied
    6.3. Panel display: Touch + 2.25" LCD
    6.4. Fittings: Locking strategy to lockers to be confirmed.
7. System to include components and fixings as required and all other accessories/ components recommended to suit the service conditions

## 226 FFE-121 Drying Lockers

1. Description: Drying lockers, as shown on the Design Drawings.
2. Manufacturer: Five At Heart, or acceptable equivalent
   2.1. Product reference: Airing Station
3. Size(s) and configuration(s): Refer to the Design Drawings
4. Fire rating:
   4.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      4.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
5. Carcass:
   5.1. Type/ material: Particleboard
   5.2. Thickness: As supplied
   5.3. Finish/ colour: White
6. Doors:
   6.1. Type/ material: MDF
   6.2. Thickness: As supplied
   6.3. Finish: Laminated
   6.4. Colour: Standard RAL colour
   6.5. Ironmongery: Pull handle
7. Plinth:
   7.1. Type: Moisture resistant base
   7.2. Colour: Black
   7.3. Fixings: Concealed.
8. Accessories:
   8.1. Clothes hanging rail
   8.2. Storage baskets (Shoe store)
   8.3. Concealed air supply inlet
   8.4. "Airing station" decal on door
9. System to include components and fixings as required and all other accessories/ components recommended to suit the service conditions

## 232 FFE-142 Framed Mirrors

1. Description: Framed, wall mounted mirror with discreet back lighting
2. Standards: To BS EN 1036-2.
3. Sizes and configurations: Refer to the Design Drawings
4. Frame:
   4.1. Material: Metal
   4.2. Profile: Refer to the Design Drawings
   4.3. Finish/ colour: Standard RAL colour
5. Mirror:
   5.1. Safety requirements: To BS 6262-4 and BS 6262-6.

5.2. Material: Float glass, silvered to give maximum reflection, free from tarnishing, discolouration, scratches and other defects visible in the designed viewing conditions.

5.3. Thickness: Nominal 6 mm.

5.4. Backing: Laminated with 3M safety film.

5.5. Edges: Polished

6. Back lighting: Refer to the Services Engineer's documentation
7. Fixing: Refer to the Design Drawings

## 235 FFE-301 Locker

1. Description: Triple stack locker, as shown on the Design Drawings.
2. Size(s) and configuration(s): Refer to the Design Drawings
3. Fire rating:
   3.1. Surface spread of flame: Surface spread of flame: to meet Approved Document Part B.
      3.1.1. Methodology: Method of achieving required surface spread of flame classification must not affect visual appearance - Submit proposals.
      Maintenance of system to be presented for review.
4. Carcass: Refer to the Design Drawings
5. Doors/ end panels/ overpanel:
   5.1. Type: Triple stack locker.
   5.2. Material: Timber veneer.
   5.3. Colour/ finish: Refer to the Design Drawings
   5.4. Fixings: Concealed.
   5.5. Fittings: Locking strategy to lockers to be confirmed.
6. Plinth:
   6.1. Type: Refer to the Design Drawings
   6.2. Colour/ finish: Refer to the Design Drawings
   6.3. Fixings: Concealed.
7. Ironmongery/ accessories : To be confirmed.
   7.1. Numbering/ signage: Numbering and ancillary signage/ graphics to be bespoke.
8. Skirting
   8.1. Material: Refer to the Design Drawings
   8.2. Type: Refer to the Design Drawings
   8.3. Finish/ colour: Refer to the Design Drawings
9. System to include components and fixings as required and all other accessories/ components recommended to suit the service conditions

## 241 FFE-421 Blind Box

1. Description: Blind box, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Position: Refer to the Design Drawings
4. Dimensions: Refer to the Design Drawings
5. Blind boxes:
   5.1. Material: Aluminium.

5.2. Finish: Polyester Powder Coated.

5.3. Colour: To be confirmed by the Architect.

6. Accessories/ Other requirements: As recommended by the manufacturer.

## 242 FFE-422 Blind Box - Angled

1. Description: Blind box fitted at angle, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Position: Refer to the Design Drawings

3.1. Recessed blind box formed within ceiling lining to fully conceal the blind headrail, brackets and operating mechanism. Blind box to be fitted at the angle and coordinated with adjacent finishes as shown on the Design Drawings.

4. Dimensions: Refer to the Design Drawings
5. Blind boxes:

5.1. Material: Aluminium.

5.2. Finish: Polyester Powder Coated.

5.3. Colour: To be confirmed by the Architect.

6. Accessories/ Other requirements: As recommended by the manufacturer.

## 251 FFE-401 Curtain Track and Sheer Curtain

1. Description: Hand operated, curtain track and sheer curtain, as shown on the Design Drawings.

1.1. Size and configuration: Refer to the Design Drawings

1.2. Track:

1.2.1. Type: To be agreed with the Architect.

1.2.2. Profile: Refer to the Design Drawings

1.2.3. Colour: To be selected from the manufacturer's standard range.

1.3. Curtain/ fabric:

1.3.1. Curtain:

1.3.1.1. Material: To be agreed with the Architect.

1.3.1.2. Colour: To be agreed with the Architect.

1.4. Fixing: Suitably concealed fixed.

## 255 FFE-411 Privacy Screen

1. Description: Privacy screen, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Framing:

3.1. Type/ material: To be agreed with the Architect.

3.2. Finish: To be agreed with the Architect.

4. Screen:

4.1. Type/ material: To be agreed with the Architect.

4.2. Finish: To be agreed with the Architect.

5. Fixing: Suitably concealed fixed.
6. Accessories: System to include all accessories, fixings and components to complete the installation.



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## 271 FFE-141 Wall Hung Backlit Artwork Panel

1. Description: Wall mounted, framed, fabric artwork with integrated lighting/ backlighting, as shown on the Design Drawings.
2. Size and configuration: Refer to the Design Drawings
3. Frame:
    3.1. Material: Refer to the Design Drawings
    3.2. Finish: Refer to the Design Drawings
4. Panel:
    4.1. Material: Refer to the Design Drawings
    4.2. Finish: Backlit light box.
5. Lighting: Refer to the Services Engineer's Information
6. Fixing: Suitably concealed fixings.

## 301 FLS-581 Entrance Barrier Matting

1. Description: Entrance barrier matting as shown on the Design Drawings.
2. Manufacturer: Gradus, or acceptable equivalent
    2.1. Product reference: Esplanade 1000.
3. Size and configuration: Refer to the Design Drawings
4. Specific performance:
    4.1. Fire rating: Refer to the Fire Strategy Report
5. Arrangement: Inset internal.
6. Perimeter frame:
    6.1. Type/ material: Recessed, finished flush with adjacent floor finishes, stainless steel perimeter frame.
    6.2. Finish: Brushed.
7. Matting:
    7.1. Type: Aluminium base with ribbed carpet insert.
    7.2. Thickness: To be agreed with the Architect.
    7.3. Wiper design: To be agreed with the Architect.
    7.4. PVC hinge colour: Black.
    7.5. Wiper colour: Black.
8. Fixing: To the manufacturer's requirements.

## 401 FFE-151 Internal Cycle Stands

1. Description: Proprietary sheffield bicycle stands.
2. Size: Refer to the Design Drawings
3. Material: Stainless steel
4. Finish/ colour: Satin
5. Fixing: As per the manufacturer's recommendations.

## 402 FFE-152 Internal Cycle Stands

1. Description: Proprietary double Stacking (two level) bicycle parking system.

2. Manufacturer: Five at Heart, or acceptable equivalent
    2.1. Product reference: ARC
3. Size: Refer to the Design Drawings
4. Type: Two tier cycle rack, single sided.
5. Material: Steel
6. Finish/ colour: Powder-coated standard RAL colour.
7. Base: Galvanized steel.
8. Bike centres: 400 mm.

## 404 FFE-156 Cycle Repair Station

1. Description: Proprietary internal cycle maintenance station that features a bike pump and essential tool kit with stainless-steel cables.
2. Manufacturer: Falco, or acceptable equivalent.
    2.1. Product reference: FalcoFix 2.0 Cycle Repair Stand
3. Size: Refer to the Design Drawings
4. No of bikes: Different positions to support the bike
5. Material: Zintec steel
6. Finish/ colour: Powder-coated standard RAL colour.
7. Fixing: Surface fixed, to manufacturer's recommendations.
8. Accessories:
    8.1. Base module: Enables you to lift your cycle higher.
    8.2. Logo module: For branding and advertising.
    8.3. Pump: Universal pump head accommodates all valve stem types
    8.4. Tools: Most commonly needed tools for basic cycle maintenance secured with stainless steel braided aircraft cables.

## 416 AMS-201 Refuge Call Point

1. Description: Proprietary refuge call point
2. Size: Refer to the Design Drawings
3. Material: Stainless steel
4. Finish: Brushed
5. Fixing: Surface fixed, to manufacturer's recommendations.
6. Associated signage: Brushed stainless steel

## 490 FFE-331 Bronze Finish Speed Gates

1. Description: Proprietary internal sensor-controlled optical speed gates with automatic door elements.
2. Manufacturer: Boon Edam, or acceptable equivalent.
    2.1. Product reference: Signature Speedlane.
3. Size and configurations: Refer to the Design Drawings
4. Housing
    4.1. Type: Solid frame.
    4.2. Material: Stainless steel.

4.3. Finish: PVD coating

4.4. Colour: Bronze

5. Panel/ barrier

5.1. Type: Double leaf.

5.2. Rotation: Swing.

5.3. Material and finish: Glass.

6. Security: Tailgating, safety and object detection sensor sets included.
7. Credential Reader

7.1. Number of transactions (minimum): 20–30 people per minute.

8. Control unit: Universal control panel.
9. Power supply: 110-240 VAC, 50/60 Hz.
10. Fixing: To the manufacturer's recommendations.
11. Accessories: As required to complete the installation.

## 491 FFE-332 Bronze Finish Speed Gates - Bike Entrance

1. Description: Proprietary internal sensor-controlled optical speed gates with automatic door elements.
2. Manufacturer: Boon Edam, or acceptable equivalent.

2.1. Product reference: Lifeline Speedgate Swing.

3. Size and configurations: Refer to the Design Drawings
4. Housing

4.1. Type: Solid frame.

4.2. Material: Stainless steel.

4.3. Finish: PVD coating

4.4. Colour: Bronze

5. Panel/ barrier

5.1. Type: Double leaf.

5.2. Rotation: Swing.

5.3. Material and finish: Glass.

6. Security: Tailgating, safety and object detection sensor sets included.
7. Credential Reader

7.1. Number of transactions (minimum): 20–30 people per minute.

8. Control unit: Universal control panel.
9. Power supply: 110-240 VAC, 50/60 Hz.
10. Fixing: To the manufacturer's recommendations.
11. Accessories: As required to complete the installation.

## 600 Samples

1. Before placing orders submit representative samples of each item specified.

## 605 Control Samples

1. Sample area: Each item specified.



make

2. Approval of appearance: Obtain before proceeding.

## 610 Quality

1. Compliance:
   1.1. BS EN 13120 - Internal blinds.
   1.2. BS EN 13561 - External blinds and awnings.
   1.3. BS 5867 - Textiles and textile products.
   1.4. BS 4680 - Specification for clothes lockers.
   1.5. BS 7953 - Entrance flooring systems.

# Execution

## 710 Moisture content of wood and wood-based boards

1. Standard: To BS EN 942
2. Moisture content on delivery: 9–13%
3. Temperature and humidity: During delivery, storage, fixing and to handover maintain conditions to suit specified moisture contents of timber components.

## 720 Installation generally

1. General: As Preliminaries
2. Fixing and fasteners: As section Z20.
3. Services: Refer to the Services Engineer's Information
4. Tolerances:
   4.1. Vertical and horizontal positions within ±1mm up to 2000mm.
   4.2. Maximum offset in plane, level or section between adjacent elements ±1mm and cut-outs ±1mm.

## 760 Sealant bedding and pointing

1. Application: As Section Z22.
2. Bedding: Built-in hob to top of worktop, sinks to top of worktops and sinks to underside of worktops
3. Pointing: Joints between worktop and splash backs, joints between units and floor finish and joints between plinth and floor finish

## 770 Trims

1. Lengths: Wherever possible, unjointed between angles or ends of runs.
2. Running joints: Where unavoidable, obtain approval of location and method of jointing.
3. Angle joints: Mitred.

# Completion

## 910 General

1. Doors and drawers: Accurately aligned, not binding. Adjusted to ensure smooth operation.
2. Ironmongery: Checked, adjusted and lubricated to ensure correct functioning.

## 920 Appliances

1. Test: Ensure that all functions and features work correctly.
2. Documentation: Submit guarantees, instruction manuals, etc.

Ω End of Section



make

# N11
# Domestic Kitchen Fittings, Furnishings and Equipment

## Products

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 311 KTN-111 Kitchenettes

1. Description: Kitchenettes configured as indicated on the Design Drawings
2. Standard: To BS EN 14749.
3. Sizes and configurations: Refer to the Design Drawings
4. Carcass/ core:
    - 4.1. Material: Melamine faced chipboard (MFC)
    - 4.2. Thickness: 16 mm.
5. Doors and drawer fronts
    - 5.1. Material: Melamine faced chipboard (MFC)
    - 5.2. Thickness: 16 mm.
    - 5.3. Finish and colour: To be agreed with the Architect.
    - 5.4. Edges: To be agreed with the Architect.
6. Side panels, plinths and shelves
    - 6.1. Material: Melamine faced chipboard (MFC)
    - 6.2. Thickness: 16 mm.
    - 6.3. Finish and colour: To match doors/ drawer fronts.
    - 6.4. Edges: To be agreed with the Architect.
7. Worktop
    - 7.1. Material: To be agreed with the Architect.
    - 7.2. Thickness: To be agreed with the Architect.
    - 7.3. Colour: To be agreed with the Architect.
    - 7.4. Finish: To be agreed with the Architect.
    - 7.5. Edges: To be agreed with the Architect.
    - 7.6. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive. Refer to the Design Drawings.
8. Splashback
    - 8.1. Material: To be agreed with the Architect.
    - 8.2. Thickness: To be agreed with the Architect.
    - 8.3. Colour: To be agreed with the Architect.
    - 8.4. Finish: To be agreed with the Architect.
    - 8.5. Edges: To be agreed with the Architect.
9. Appliances/ equipment: Integrated appliances.

10. Fixing: All concealed fixed.
11. Apertures (where required): To be precision factory-formed using rigid, manufacturer's templates.
12. Services/ lighting: Refer to the Services Engineer's documentation
13. Accessories:
    13.1. Drawer system: Blum Tandem with soft closer. The drawer box to be fully extendable and to have soft close pneumatic action, to comply with BS 6222: Part 2 strength designation Level H.
    13.2. Hinges: Concealed self and soft closing BLUM hinges, to comply with BS 6222: Part 2 strength designation Level H.
    13.3. Metal fittings: All metal fittings and screws used in manufacture are to be plated.
    13.4. Adhesive: Low formaldehyde.

## 312 KTN-112 Teapoint/ Kitchenette

1. Description: Teapoint/ kitchenette configured as indicated on the Design Drawings
2. Standard: To BS EN 14749.
3. Sizes and configurations: Refer to the Design Drawings
4. Carcass/ core:
    4.1. Material: Melamine faced chipboard (MFC)
    4.2. Thickness: 16 mm.
5. Doors and drawer fronts
    5.1. Material: Melamine faced chipboard (MFC)
    5.2. Thickness: 16 mm.
    5.3. Finish and colour: To be agreed with the Architect.
    5.4. Edges: To be agreed with the Architect.
6. Side panels, plinths and shelves
    6.1. Material: Melamine faced chipboard (MFC)
    6.2. Thickness: 16 mm.
    6.3. Finish and colour: To match doors/ drawer fronts.
    6.4. Edges: To be agreed with the Architect.
7. Tables/ rear upstands/ undershelves/ wall shelving
    7.1. Material: Stainless steel.
    7.2. Thickness: To be agreed with the Architect.
    7.3. Finish and colour: To be agreed with the Architect.
    7.4. Edges: To be agreed with the Architect.
8. Worktop
    8.1. Material: To be agreed with the Architect.
    8.2. Thickness: To be agreed with the Architect.
    8.3. Colour: To be agreed with the Architect.
    8.4. Finish: To be agreed with the Architect.
    8.5. Edges: To be agreed with the Architect.
    8.6. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive. Refer to the Design Drawings.

9. Splashback
    9.1. Material: To be agreed with the Architect.
    9.2. Thickness: To be agreed with the Architect.
    9.3. Colour: To be agreed with the Architect.
    9.4. Finish: To be agreed with the Architect.
    9.5. Edges: To be agreed with the Architect.
10. Appliances/ equipment: Integrated appliances.
    Sink.
    Fridge.
    Soap and paper towel dispensers.
    60L waste bin.
11. Fixing: All concealed fixed.
12. Apertures (where required): To be precision factory-formed using rigid, manufacturer's templates.
13. Services/ lighting: Refer to the Services Engineer's documentation
14. Accessories:
    14.1. Drawer system: Blum Tandem with soft closer. The drawer box to be fully extendable and to have soft close pneumatic action, to comply with BS 6222: Part 2 strength designation Level H.
    14.2. Hinges: Concealed self and soft closing BLUM hinges, to comply with BS 6222: Part 2 strength designation Level H.
    14.3. Metal fittings: All metal fittings and screws used in manufacture are to be plated.
    14.4. Adhesive: Low formaldehyde.

## 313 KTN-113 Kitchenette

1. Description: Kitchenette configured as indicated on the Design Drawings
2. Standard: To BS EN 14749.
3. Sizes and configurations: Refer to the Design Drawings
4. Carcass/ core:
    4.1. Material: Melamine faced chipboard (MFC)
    4.2. Thickness: 16 mm.
5. Doors and drawer fronts
    5.1. Material: Melamine faced chipboard (MFC)
    5.2. Thickness: 16 mm.
    5.3. Finish and colour: To be agreed with the Architect.
    5.4. Edges: To be agreed with the Architect.
6. Side panels, plinths and shelves
    6.1. Material: Melamine faced chipboard (MFC)
    6.2. Thickness: 16 mm.
    6.3. Finish and colour: To match doors/ drawer fronts.
    6.4. Edges: To be agreed with the Architect.
7. Worktop
    7.1. Material: To be agreed with the Architect.

7.2. Thickness: To be agreed with the Architect.

7.3. Colour: To be agreed with the Architect.

7.4. Finish: To be agreed with the Architect.

7.5. Edges: To be agreed with the Architect.

7.6. Joints (where required): Seamless joints, sealed with a colour matched joint adhesive. Refer to the Design Drawings.

8. Splashback

8.1. Material: To be agreed with the Architect.

8.2. Thickness: To be agreed with the Architect.

8.3. Colour: To be agreed with the Architect.

8.4. Finish: To be agreed with the Architect.

8.5. Edges: To be agreed with the Architect.

9. Appliances/ equipment: Integrated appliances.
Sink.
Fridge.
Paper towel dispensers.
Stainless steel waste bin.

10. Fixing: All concealed fixed.

11. Apertures (where required): To be precision factory-formed using rigid, manufacturer's templates.

12. Services/ lighting: Refer to the Services Engineer's documentation

13. Accessories:

13.1. Drawer system: Blum Tandem with soft closer. The drawer box to be fully extendable and to have soft close pneumatic action, to comply with BS 6222: Part 2 strength designation Level H.

13.2. Hinges: Concealed self and soft closing BLUM hinges, to comply with BS 6222: Part 2 strength designation Level H.

13.3. Metal fittings: All metal fittings and screws used in manufacture are to be plated.

13.4. Adhesive: Low formaldehyde.

## Execution

### 610 Moisture content of wood and wood-based boards

1. Control and monitoring

1.1. Method statement: Submit.

### 620 Installation generally

1. Fixings and adhesives: As section Z22.
2. Services: Refer to the Services Engineer's Information

### 630 Installing units and worktops

1. General: Fitted to be stable, secure and level, with no unplanned gaps between adjacent units and between units and building fabric.
2. Assembly and fixing: Surfaces designed to falls and drain as intended.

## 640 Installing appliances

1. Unless otherwise stated, kitchen sinks shall be stainless steel inset sink to the approval of the Employer and to BS 1244: Parts 1 and 2.
    1.1. The edges of the worktop cut-out shall be sealed with a thick layer of PVA (visibly apparent) before the kitchen sink is fitted.
    1.2. The drainer of the sink should not project into a corner.
2. The minimum required distance between the cooker and cooker isolation switch is 300 mm.

## 650 Installing sinks, taps and wastes

1. Water supply: To BS EN 806-2 and BS EN 806-4.
2. Taps
    2.1. Fixing: Secure, watertight seal with the appliance.
    2.2. Positioning: Hot tap to left of cold tap as viewed by the user of the appliance.
3. Wastes
    3.1. Bedding: Waterproof jointing compound.
    3.2. Fixing: With resilient washer between appliance and backnut.

## 660 Sealant bedding and pointing

1. Application: As section Z22.
2. Bedding: As manufacturers recommendations.
3. Pointing: As manufacturers recommendations.

## 670 Installing trims and mouldings

1. Lengths: Unjointed between angles or ends of runs.
2. Angle joints: Mitred.

# Completion

## 910 General

1. Doors and drawers: Accurately aligned, not binding. Adjusted to ensure smooth operation.
2. Ironmongery: Checked and adjusted to ensure correct functioning.

## 920 Appliance commissioning

1. Appliance operation, functions and controls: Verify.
2. Documentation: Submit guarantees, instruction manuals, etc.

Ω End of Section

# N13
# Sanitary Appliances and Fittings

## Products

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 301 SAN-101 Standard WC Package

1. Description: Standard WC Suite including pan, seat, cistern, manual flush plate, toilet roll holder, coat hooks, sanitary bin, wall mounted mirror, paper towel dispenser, soap dispenser, toilet brush and holder and shelf as shown on the Design Drawings.
2. Manufacturer: Varies, see below
3. Specific items:
    3.1. Dolphin or acceptable equivalent for wall hung softclosing WC pan, concealed cisterns, basin mounted Dolphin sensor taps with temperature control.Splashlab double toilet roll holder in satin, Dolphin mini electric flushplate, UHPC cast vanity countertop with built in basin and bin chute with bespoke joinery cabinetry with hinge doors below housing the sanitary waste bin and 40L bin from Rodan or suitable equivalent, Dolphin wall mounted Alavo mirror with integrated paper towel and soap resevoir, Dolphin sensor soap dispensor, contrasting toilet seat ring, x2 coat hooks, Terrazzo splashback finish from the counter to the ceiling behind mirror

### 302 SAN-102 Accessible WC Package

1. Description: Doc M pack accessible WC room suite, refer to the Design Drawings.
2. Size and configuration: As shown on the Design Drawings.
3. Manufacturer: Dolphin, or acceptable equivalent
    3.1. Product reference: Doc M Accessible WC Pack
4. Specific items:
    4.1. The sanitaryware and washroom package includes wall-hung toilet pan, contrasting toilet seat ring , sensor-activated cistern with matching sensor flush plate, basin, sensor tap, TMV3 thermostatic mixing valve, waste trap, and waste grate. Accessibility fittings comprise four 600mm grab bars, 500mm grab bar, drop-down hinged support rail, back rest, and emergency alarm pull cord. Washroom accessories consist of sensor activated soap dispenser integrated within mirror supplied with anti-bacterial soap cartridge, toilet tissue dispenser, paper towel dispenser integrated within the mirror, waste bin, 200mm vanity shelf, 400mm vanity shelf, toilet brush, two coat hooks, 1000mm x 400mm mirror on terrazzo backing

### 303 SAN-103 Ambulant WC Package

1. Description: Ambulant WC room suite including pan, seat, cistern, manual flush plate, toilet roll holder, coat hooks, sanitary bin, wall mounted mirror, paper towel dispenser, soap dispenser, toilet brush and holder, shelf and horizontal grab rails as shown on the Design Drawings.
2. Manufacturer: Varies, see below
3. Specific items:

3.1. Dolphin or acceptable equivalent for wall hung soft closing WC pan, concealed cisterns, basin mounted Dolphin sensor taps with temperature control. Splashlab double toilet roll holder in satin, Dolphin mini electric flush plate, UHPC cast vanity countertop with built in basin and bin chute with bespoke joinery cabinetry with hinge doors below housing the sanitary waste bin and 40L bin from Rodan or suitable equivalent, Dolphin wall mounted Alavo mirror with integrated paper towel and soap reservoir, Dolphin sensor soap dispenser, contrasting toilet seat ring,, contrasting toilet seat ring, grab rails and hinged support rail, x2 coat hooks

## 304 SAN-104 Accessible WC Package

1. Description: Doc M pack accessible WC room suite, refer to the Design Drawings.
2. Size and configuration: As shown on the Design Drawings.
3. Manufacturer: Dolphin, or acceptable equivalent
   3.1. Product reference: Doc M Accessible WC Pack
4. Specific items:
   4.1. The sanitaryware and washroom package includes wall-hung toilet pan, contrasting toilet seat ring , sensor-activated cistern with matching sensor flush plate, basin, sensor tap, TMV3 thermostatic mixing valve, waste trap, and waste grate. Accessibility fittings comprise four 600mm grab bars, 500mm grab bar, drop-down hinged support rail, back rest, and emergency alarm pull cord. Washroom accessories consist of soap dispenser supplied with anti-bacterial soap cartridge, toilet tissue dispenser, paper towel dispenser, high-speed hand dryer, waste bin, 200mm vanity shelf, 400mm vanity shelf, toilet brush, two coat hooks, 1000mm x 400mm mirror on terrazzo backing. This room is larger than the standard accessible WC and also contains a fold down baby changing table.

## 351 SAN-141 Standard Shower Area

1. Description: Enclosure and shower system, as shown on the Design Drawings
2. Manufacturer: Dolphin or acceptable equivalent
3. Specific items:
   3.1. Package includes glazed door with double hinges and screen, timber bench, door stop, wall hook, slip-resistant floor, 900mm x 1100mm flush shower tray, shower control valve (WRAS No. 1911347), arm-mounted round shower head with ceramic finish (WRAS No. 2003342), 200mm ceiling-mounted shower arm, 1500mm flexible hose, hand-held shower head, and wall outlet with hand-held shower bracket, soap shelf to match tile wall lining.

## 352 SAN-142 Accessible Shower Area

1. Description: Part M compliant WC and shower room pack, as shown on the Design Drawings
2. Manufacturer: Dolphin Solutions, or acceptable equivalent.
3. Specific items:
   3.1. Sanitaryware includes wall-hung toilet pan, grey contrasting toilet seat ring (specified in grey to ensure visual contrast compliance; do not substitute with white), support frame for wall-hung pan, sensor-activated cistern with matching sensor flush plate, basin, sensor tap, TMV3 thermostatic mixing valve, waste trap, and waste grate. Accessibility fittings comprise seven 600mm grab bars, 500mm grab bar, three drop-down hinged support rails, folding shower seat, bench, toilet back rest, shower back rest, and two emergency alarm pull cords. Washroom accessories consist of sensor activated soap dispenser integrated within mirror supplied with anti-bacterial soap cartridge, toilet tissue dispenser, paper towel dispenser integrated within the mirror, hair dryer, shaving socket, waste bin, 200mm vanity shelf, 400mm vanity shelf, toilet brush, two coat hooks, and 1000mm x

400mm mirror. Shower system includes shower control valve, arm-mounted shower head, shower arm, shower outlet, flexible shower hose, foldable contrasting wooden changing bench, hand-held shower head, slider rail and holder, shower curtain rail, shower curtain rail ceiling stay, two shower curtains, and shower curtain hooks, soap shelf to match tile wall lining.

## 353 SAN-143 Ambulant Shower Area

1. Description: Enclosure and shower system, as shown on the Design Drawings
2. Manufacturer: Dolphin Solutions, or acceptable equivalent.
3. Specific items:
    3.1. Package includes glazed door with double hinges and screen, timber bench, door stop, wall hook, slip-resistant floor, 900mm x 1450mm flush shower tray, shower control valve (WRAS No. 1911347), arm-mounted round shower head with ceramic finish (WRAS No. 2003342), 200mm ceiling-mounted shower arm, 1500mm flexible hose, hand-held shower head, and wall outlet with hand-held shower bracket. 4 grab rails, soap shelf to match tile wall lining.

## 361 SAN-161 Cleaners Sink

1. Description: Cleaners sink, as shown on the Design Drawings
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect. To be confirmed
3. Size and configuration: Refer to the Design Drawings

## 366 SAN-186 Hairdryer

1. Description: Hairdryer, as shown on the Design Drawings
2. Manufacturer: Dyson, or acceptable equivalent
    2.1. Product reference: To be agreed with the Architect.
3. Size and configuration: Refer to the Design Drawings

## 367 SAN-187 Shaver Sockets

1. Description: Shaver sockets, as shown on the Design Drawings
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Size and configuration: Refer to the Design Drawings

## 369 SAN-189 Hand Dryer

1. Description: Hand dryer, as shown on the Design Drawings
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Size and configuration: Refer to the Design Drawings

## 371 SAN-191 Wall Hooks

1. Description: Wall hooks, as shown on the Design Drawings
2. Manufacturer: C.P Hart, or acceptable equivalent
    2.1. Product reference: To be agreed with the Architect.

3. Size and configuration: Refer to the Design Drawings

## 372 SAN-192 Door Stops

1. Description: Door stops, as shown on the Design Drawings
2. Manufacturer: To be agreed with the Architect.
   2.1. Product reference: To be agreed with the Architect.
3. Size and configuration: Refer to the Design Drawings

## 600 Samples

1. Before placing orders submit representative samples of each appliance or fitting specified.

## 605 Control Samples

1. Sample area: Each item specified.
2. Approval of appearance: Obtain before proceeding.

# Execution

## 610 Installation generally

1. Standards: In accordance with BS 6465-1, -2 and -3.
2. Assembly and fixing: Surfaces designed to falls to drain as intended.
3. Fasteners: Non-ferrous or stainless steel.
4. Fixing: Fix appliances securely to structure. Do not support on pipework.
5. Jointing and bedding compounds: Recommended by manufacturers of appliances, accessories and pipes being jointed or bedded.
6. Appliances: Do not use. Do not stand on appliances.
7. Supply and discharge pipework: Fix before appliances.
8. On completion: Components and accessories working correctly with no leaks.
9. Labels and stickers: Remove.
10. Compliance: BS 3402 - Specification for quality of vitreous china sanitary appliances.
    BS EN 33 - WC pans and WC suites.
    BS EN 997 - WC pans and WC suites with integral trap.
    BS EN 13407 - Wall-hung urinals.
    BS 1206 - Specification for fireclay sinks.
    BS EN 14688 - Sanitary appliances.
    BS EN 31 - Wash basins. BS EN 14055 - WC and urinal flushing cisterns.
    BS EN 200 - Sanitary tapware.
    BS EN ISO 1456 - Metallic and other inorganic coatings.
    BS EN 14516 - Baths for domestic purposes.
    BS EN 198 - Sanitary appliances.BS EN 14428 - Shower enclosures.
    BS EN 14527 - Shower trays for domestic purposes.
    BS 6340: Part 4 - Shower units.
11. Public mains water supply: Water fittings to comply with Water Supply (Water Fittings) Regulations and have Water Regulations Advisory Scheme (WRAS) approval when connecting to/from a public mains water supply.

## 613 Compatibility of components

1. General: Each sanitary assembly must consist of functionally compatible components, preferably obtained from a single manufacturer.
    1.1. Exceptions: Water supply fittings, wastes and traps

## 620 Noggings and bearers

1. Noggings, bearers, etc. to support sanitary appliances and fittings: Position accurately. Fix securely.

## 630 Tiled backgrounds other than splashbacks

1. Timing: Complete before fixing appliances.
2. Fixing appliances: Do not overstress tiles.

## 650 Installing WC pans

1. Floor-mounted pans: Screw fix and fit cover caps over screw heads. Do not use mortar or other beddings.
2. Seat and cover: Stable when raised.

## 660 Installing slab urinals

1. Waterproofing of walls and floor (specified elsewhere): Completed before fixing urinal components.
2. Gap between components: 3 mm.
3. Space behind channels and slabs: Grout with 1:5 cement:sand grout.
4. Pointing: Rake out joints to 10 mm depth. Point flush with waterproof jointing compound recommended by urinal manufacturer.

## 665 Installing wet room floor former trays

1. Preparation:
    1.1. Overboard lower area of new joists with 18 mm WBP plywood.
    1.2. Remove existing floorboards and nails from joists over area of wet room former and make good edges.
2. Waterproofing/ Tanking:
    2.1. Tape wall/ floor joint and other gaps.
    2.2. Apply primer to porous surfaces.

## 670 Installing cisterns

1. Cistern operating components: Obtain from cistern manufacturer.
2. Inlet and flushing valves: Match to pressure of water supply.
3. Internal overflows: Into pan, to give visible warning of discharge.
4. External overflows: Fix pipes to falls and locate to give visible warning of discharge. Agree location where not shown on drawings.

## 710 Installing taps

1. Fixing: Secure against twisting.
2. Seal with appliance: Watertight.

3. Positioning: Hot tap to left of cold tap as viewed by user of appliance.

## 720 Installing wastes and overflows

1. Bedding: Waterproof jointing compound.
2. Fixing: With resilient washer between appliance and backnut.

## 725 Installing hand dryers

1. Fused connection units
   1.1. Type: Switched
   1.2. Engraving: With 'HAND DRYER'.
   1.3. Location: Submit proposals
2. Final connection: Concealed.
   2.1. Containment: 25 mm HG galvanized conduit

## 755 Sealant bedding and pointing

1. Bedding: Bed and point basins to underside of vanity units and bed sinks to top of worktops
2. Pointing: Joints between appliances and splashbacks

Ω End of Section



make

# N14
# General Internal Signage Systems

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

## System performance

### 210 General requirements

1. Signage system: Complete to BS 559, including facing information, components, inserts, accessories and fixings necessary to complete the system.
    1.1. Comply with the requirements of: Refer to the Fire Strategy Report and wayfinding strategy
2. Geometric shapes, colours and layout: In accordance with BS ISO 7001.
3. Design standard for disabled people: In accordance with BS 8300-2.
4. Proposals: Submit drawings, schedules, technical information, calculations and manufacturer's literature.

### 240 Fire performance

1. Fire performance:
    1.1. Reaction to fire: Refer to the Fire Strategy Report

### 280 Design life

1. Description: Refer to A Section.
2. Condition of use: Subject to regular maintenance.

### 290 Signage samples

1. Sign type:
    1.1. Action: Submit labelled samples.
    1.2. Conformity: Retain samples on site for the duration of the contract, or until instructed to remove them.
    1.3. Delivered product: To conform with labelled samples.

### 295 Signage samples board

1. Samples board: Submit.
    1.1. Content: Selected labelled signs, showing methods of fixing.
    1.2. Conformity: Retain samples on site for the duration of the contract or until instructed to remove.
    1.3. Delivered product: To conform with labelled samples.

## Products - Not Used

## Execution

### 610 Fixing signs generally

1. Installation: Secure, plumb and level.
2. Strength of fasteners: Sufficient to support all live and dead loads.
3. Fasteners and or adhesives: As section Z20.
4. Fixings showing on surface of sign: Must not detract from the message being displayed.

### 620 Fixing signs for the visually impaired

1. Protection of users
    1.1. Fasteners for tactile/ Braille signs must not have sharp edges or protrusions that would cause confusion or injury to users.

### 670 Electrical and data services

1. Services connection required: Power for lighting
2. Standard: In accordance with BS 7671.
3. Coordinate with services trades.

## Completion

### 910 Documentation

1. Submit
    1.1. Manufacturer's maintenance instructions.
    1.2. Guarantees, warranties, test certificates, record schedules and logbooks.

### 920 Spares

1. Supply as follows
    1.1. Quantity: Five modular sets containing all alphabetical and numeral characters

### 930 Specialist tools

1. Supply as follows: Two sets of nameplate adjustment tools

Ω End of Section

# N15
# Internal Fire and Safety Signage Systems

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

## System performance

### 205 Design of internal signage systems

1. Design: Complete detailed design and submit before commencing work.
2. Content: Signs including facing information, components, inserts, accessories and fixings necessary to complete the system.
3. Proposals: Submit drawings, schedules, technical information, calculations and manufacturer's literature before commencing work.

### 210 General requirements

1. Signage and way guidance system design:
    1.1. For fire escape and evacuation signage: In accordance with: BS 5499-4 or BS ISO 16069.
    1.2. For way guidance systems: In accordance with BS ISO 16069.
    1.3. For safety signs other than escape route signage: In accordance with: BS 5499-10.
2. Comply with the requirements of: Refer to the Fire Strategy Report

### 220 Sign design and format

1. Format: In accordance with BS EN ISO 7010
2. Geometric shapes, colours and layout: In accordance with BS ISO 3864-1.
3. Design principles for graphical symbols: In accordance with BS ISO 3864-3.
4. Colorimetric and photometric properties of safety sign materials: In accordance with BS ISO 3864-4.
5. Water safety: In accordance with BS EN ISO 7010.

### 260 Photoluminescent properties

1. Luminance delay classification to BS ISO 17398: A

### 270 Fire performance

1. Description: Refer to the Fire Strategy Report
2. Reaction to fire
    2.1. Standard: To BS EN 13501-1 In accordance with Building Regulations guidance

### 280 Design life

1. Description: Refer to A Section.
2. Condition of use: Subject to regular maintenance.

## 290 Signage samples

1. Sign type:
    1.1. Action: Submit labelled samples.
    1.2. Conformity: Retain samples on site for the duration of the contract, or until instructed to remove them.
    1.3. Delivered products: To conform with labelled samples.

# Products

## 360 General Materials, Performance & Quality

1. Standards:
    - BS 559 - Specification for the design and construction of signs for publicity, decorative and general purposes.
    - HSE directives and regulations.
    - The European Sign Directive Format.
    - Health and Safety at Work Act 1974 - Lighting ballast.
    - BS 5266: Part 1 - Emergency lighting. Code of practice for the emergency lighting of premises.
    - BS EN 1838 - Emergency lighting.
    - BS EN 50172 - Emergency escape lighting systems.
    - BS 5499: Part 4 - Code of practice for escape route signing.
    - BS 5499: Part 10 - Guidance for the selection and use of safety signs and fire safety notices.
    - BS ISO 17398 - Safety colours and safety signs. Classification, performance and durability of safety signs.
    - BS ISO 3864 - Graphical symbols. Safety colours and safety signs.
    - BS EN ISO 7010 - Graphical symbols. Safety colours and safety signs. Registered safety signs.
    - BS ISO 20712: Part 1 - Specifications for water safety signs used in workplaces and public areas.
    - BS 4533 - Luminaires.
    - BS EN 60598 - Luminaires.

# Execution

## 610 Fixing signs generally

1. Installation: Secure, plumb and level.
2. Fasteners and adhesives: As section Z20.
3. Strength of fasteners: Sufficient to support live and dead loads.
4. Fixings showing on surface of sign: Must not detract from the message being displayed.
5. Tolerances:
    5.1. Glass and acrylic: ±1mm length/ width and 0.5mm in 1000mm flatness.
    5.2. Metal sheets: ±1mm length/ width and 1mm in 1000mm flatness.

# Completion

## 910 Documentation

1. Submit
    1.1. Manufacturer's maintenance instructions.
    1.2. Guarantees, warranties, test certificates, record schedules and logbooks.

Ω End of Section



make

# N25
# Permanent Access and Safety Equipment

## Types of system/ equipment

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 211 ASE-101 Mansafe System (Visual Intent Only)

1. Description: Lifeline/ Fall prevention system
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. System arrangement: Refer to the Design Drawings and Refer to the Façade Performance Specification
4. Structural anchors: Refer to the Façade Performance Specification
5. Anchorage device: Refer to the Façade Performance Specification
6. Turnbuckle, swage and clevis assemblies: Refer to the Façade Performance Specification
7. Transfastener: Refer to the Façade Performance Specification
8. System attachment: Refer to the Façade Performance Specification

### 221 ASE-201 BMU Restraint (Visual Intent Only)

1. Description: BMU restraint system
2. Note:
    2.1. This description covers items specified by the Façade Consultant.
    2.2. Refer to the Façade Performance Specification for detailed information and/ or performance requirements.
3. Anchor type: Refer to the Façade Performance Specification
4. Dimensions: Refer to the Design Drawings and Refer to the Façade Performance Specification
5. Material: Refer to the Façade Performance Specification
6. Finish: Refer to the Façade Performance Specification
7. Location and co-ordination: Refer to the Design Drawings and Refer to the Façade Performance Specification

## Design/ performance requirements - Not Used

## Fabrication, assembly and installation

### 510 Fabrication and assembly generally

1. Machine cutting, drilling and assembly: Carry out as much as possible in the workshop. Obtain approval for any reassembly on site.

2. Dissimilar metal surfaces of assembly components/ supports/ fixings: Isolate to prevent electrolytic corrosion.

## 520 Protection

1. General: Do not deliver to site any components or assemblies that cannot be installed immediately or unloaded into a suitable well protected storage area.

## 530 Suitability of structure/ fabric

1. Visual and geometric survey of supporting structure and fabric: Carry out before commencing installation of access/ safety system. Report immediately if structure/ fabric will not allow required accuracy/ security of erection/ fixing and if structural testing is required.

## 535 Execution generally

1. Structural members: Do not modify, cut notch or make holes in structural members without permission.
2. Frameworks: Assemble and brace, including temporary members required for installation.
    2.1. Temporary support: Do not use access systems as temporary support or strutting for other work.
3. Bolted joints
    3.1. Contact between dissimilar metals: Avoid.
    3.2. Bolts and washers: Select types, sizes and quantities of fasteners or packings and spacings to retain supported components without distortion or loss of support.
4. Welded joints: Comply with latest edition of National Structural Steelwork Specification (NSSS), Section 5.
5. Finished components: Smooth, free from distortion, cracks, burrs and sharp arrises.

## 540 Mechanical fixings

1. Materials: Unless otherwise recommended by equipment manufacturer:
    1.1. Connecting bolts and other fixings fully accessible for inspection: Carbon steel hot dip galvanized to BS 7371-6.
2. Nuts: Tapped after galvanizing.
    2.1. Cast-in anchors and other fixings not accessible for routine inspection: Austenitic stainless steel, grade 1.4401 (316) to BS EN 10088-1.

## 550 Fasteners, inserts and bolts for building in

1. Supplier: Equipment manufacturer/ supplier.

## 560 Fixings for securing equipment

1. Adjustment capability: Adequate three dimensional adjustment to accommodate building structure/ fabric irregularities.

## 570 Fixing anchor installation

1. Site drilling or cutting into structure/ fabric: Permitted only in approved locations.
2. Distance between all fixing devices and edges of supporting material: Not less than recommended by fixing manufacturer.

## 605 Identification and registration labels for suspended access system

1. Provision
    1.1. Suspension rig and any non-dedicated track systems: Provide and fix a permanent label giving:
        1.1.1. a) Manufacturer's name, address and telephone number.
        1.1.2. b) Name and/ or reference code of model.
        1.1.3. c) Serial number and year of manufacture.
        1.1.4. d) Rated load of suspended platform (where the platform is permanently attached to the system) or maximum total suspended load (where the platform can be derigged).
    1.2. Suspended platform: As a), b) and c) above, plus
        1.2.1. d) Rated load and self weight in kg of the platform.
        1.2.2. e) Maximum number of persons allowed on the platform.
        1.2.3. f) Identification mark or number of the suspension rig or rigs with which the platform is compatible (where the platform can be derigged).
    1.3. Suspension and safety ropes (detachable): Marked with length and unique identification number.
2. Location: In positions such that labels can be easily read.

## 610 Identification and registration labels for other equipment

1. Provision: Provide and fix to each piece of equipment a permanent label giving:
    1.1. Manufacturer's name, address and telephone number.
    1.2. Name and/ or reference code of model.
    1.3. Serial number and year of manufacture.
    1.4. Maximum load (in kilograms) that may be sustained by the equipment.
2. Location: In positions such that labels can be easily read.

## 615 UKCA/ UKNI/ CEmarking and declaration of conformity of permanently installed suspended access equipment

1. Marking: Equipment to be clearly labelled with a UKCA/ UKNI/ CE mark indicating conformity with the current edition of the Supply of Machinery (Safety) Regulations.
2. Declaration of conformity: Provide for the complete installation a declaration of conformity with Machinery Directive 2006/42/EC together with certificates of incorporation for parts not supplied or installed by the main equipment manufacturer or supplier.

## 630 Lifting equipment configuration

1. General: Where the rated load of machinery for lifting depends on its configuration, the identification and registration labels and rated load notices must indicate the rated load for each configuration.

## 640 Marking of anchor devices

1. Provision: Provide on or near each anchor device a label or other clear marking giving:
    1.1. Manufacturer's name and telephone number.
    1.2. Serial number and year of manufacture of device.
    1.3. Maximum number of personnel that may be attached to the device at any one time.

1.4. Requirements for energy absorbers, ground clearance, etc.

2. Anchor devices intended solely for use with personal protective equipment: Indicate restriction of use by pictogram or other suitable marking on or near the device.

## 710 Electrical installation

1. General: Install, test and commission electrical work forming part of the access equipment system up to and including wiring of isolating switch or switches provided by the electrical contractor, all in accordance with BS 7671.
2. Components/ Fasteners: Use waterproof components and corrosion resisting fasteners throughout. Avoid contact between dissimilar metals.
3. Control system: Designed so that an electrical supply failure, or the failure or malfunction of any electrical component, will cause the system to fail to safety.
4. Failsafe provision: In the event of power failure the system must be capable of delivering the working platform to a position of safety either by hand winding or by a controlled gravity descent device.
5. Isolating switches: Clearly labelled to identify purpose. Labels to display an emergency contact telephone number.

## 810 Service/ Maintenance

1. General: Following acceptance of the completed installation, service and maintain the equipment for the period stated below as and at intervals recommended by the manufacturer. Such maintenance to include a 'call-out' service during normal working hours to maintain the equipment in an acceptable and safe condition.
2. Service/ Maintenance period: 12 months

## 820 Operating instructions

1. Equipment and accessories: Where appropriate, mark in such a way that it is possible to identify the correct mode of operation for their safe use.

## 830 Operating and maintenance manual

1. General: Provide, for inclusion in the Building Manual, printed instructions and recommended procedures to be established by the Employer for operating and routinely maintaining the equipment. Provide diagrams where appropriate.
2. Content
   2.1. Instructions for assembling/ erecting equipment for use.
   2.2. Comprehensive operating instructions, including safety and emergency procedures, for all motions including upward, downward and lateral travel, and slew.
   2.3. Servicing and planned maintenance procedures, including assembly instructions where maintenance necessitates dismantling of machinery parts.
   2.4. List of replacement parts, with references.
   2.5. Recommended procedures for testing equipment.

## 840 As installed drawings

1. General: After commissioning/ testing of the equipment provide as installed drawings for inclusion in the Building Manual.
2. Drawing content
   2.1. Contractor's name and contract number.

2.2. Location and date of installation.

2.3. Manufacturer's name, model and type numbers.

2.4. General arrangement of the complete installation.

2.5. Electrical circuit wiring diagrams complete with details and ratings of all items of equipment.

Ω End of Section

# P10
# Sundry Insulation

## Types of insulation

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 111 INS-101 Column/ Wall Insulation

1. Description: Mineral wool insulation
2. Manufacturer: Rockwool Ltd, or acceptable equivalent
   2.1. Product reference: Rockwool RWA 45.
3. Standard: To BS EN 13162; ISO 14001.
4. Performance requirements:
   4.1. Fire performance: Euroclass A1 non-combustible.
5. Thickness: To be determined by the Sub-contractor, to suit the service conditions
6. Thermal conductivity (maximum): 0.035 W/mK.
7. Facing: Plain (unfaced).
8. Edges: Square.
9. Density: 45 kg/m³.

### 112 INS-111 Soffit Insulation

1. Description: Mineral wool insulation with fibre cement board facing
2. Manufacturer: Rockwool Ltd, or acceptable equivalent
   2.1. Product reference: Rockwool High Impact Soffit Slab
3. Standard: To BS EN 13162.
4. Performance requirements:
   4.1. Fire performance: To BS EN 13501-1, Euroclass A1 non-combustible.
5. Thickness: To be determined by the Sub-contractor, to suit the service conditions
6. Thermal conductivity (maximum): 0.034 W/mK.
7. Facing: 6 mm-thick, rigid fibre cement board, off-white.
8. Edges: Square

### 116 INS-116 Underfloor Insulation

1. Description: Dual-density, stone wool board
2. Manufacturer: ROCKWOOL Ltd, or acceptable equivalent
   2.1. Product reference: Thermal RockFloor®
3. General requirements: Insulation products generally.
4. Standard: To BS EN 13162; ISO 14001.
5. Thickness: To be determined by the Sub-contractor, to suit the service conditions



make

6. Facing: Tissue face finish on the top surface.
7. Edges: Square.
8. Thermal conductivity (maximum): 0.038 W/m·K.
9. Fire performance: To BS EN 13501-1, Euroclass A1.
10. Size:: 600 x 1000 mm.

## 121 INS-121 Expanded Polystyrene (EPS) Insulation - Pourable Void Former

1. Description: Pourable insulation
2. Manufacturer: Energystore Ltd, or acceptable equivalent
   2.1. Product reference: energystore TLA® + (TLA110 Grade)
3. Standard: To BS EN 13501-1, BS EN 13163, BS EN13165, BS EN 16025-1, BS EN 16809-1.
4. Third-party certification: KIWA Certificate BAF-18-042-S-A-UK / BRE GreenGuide A+ rating / LABC warranty / NHBC Approval / BM TRADA system certification.
5. Fire performance: A2-s1, d0 to BS EN 13501-1. Non combustible.
6. Compressive strength (minimum) at 10% compression: 200 kPa.
7. Thermal conductivity (maximum): 0.043 W/m·K.
8. Water vapour permeability: 5—20 μ.
9. Type/ material: Pourable insulation formed from EPS beads coated in a specially formulated additive with cement.
10. Thickness: Refer to the Design Drawings
11. Cement: Dosage: 110 kg/m³, minimum grade Portland 32.5 Cem I or Cem II.
12. Dry density: 130 kg/m³ (approximately).
13. Purpose: Suitable for ground floor insulation replacement.
14. Beads: Bonded with cement.
15. Shrinkage: <0.5%

## 126 INS-126 Cellular Glass Insulation

1. Description: Cellular glass insulation
2. Manufacturer: FOAMGLAS®, or acceptable equivalent
   2.1. Product reference: FOAMGLAS® T3+ (600 x 450 mm slabs)
3. Standard: To BS EN 13167.
4. Reaction to fire: Non-Combustible Euroclass A1 To EN 13501-1.
5. Third-party product certification: Environmental Product Declaration (ISO 14025 and EN 15804): EPD-PCE-20150042-IBA1-DE.
6. Thermal conductivity (maximum): λD ≤0.036 W/m·K.
7. Thickness: Refer to the Design Drawings
8. Compressive strength (minimum): ≥500 kPa to EN 826 annexe A.
9. Density: 95 kg/m³.
10. Facing: Unfaced.
11. Edges: 90 degrees.
12. Environmental information
    12.1. Recycled content: ≥50%.
13. Water vapour resistance: μ = ∞.

## 500 Samples

1. Insulation: 500mm x 500mm
2. 1 of each visible fixings, accessories and other visible elements.

## 510 Control Samples

1. First of each installed type.

## 590 Performance Generally

1. Deflections: The maximum allowable deflections of the works to be L/200, or 10mm (whichever is the lesser).

## 600 Workmanship

1. Refer to Z01.
2. Accurately install all elements as indicated on the working Drawings.
3. Tolerance: ±2mm or 0.1% of overall length, whichever is the lowest.
4. The performance of interfacing elements must not be adversely affected.

Ω End of Section

# P12
# Fire-Stopping Systems

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 FPS-101 Fire-stopping (Generally)

1. Description: General firestopping, Refer to the Design Drawings
2. All fire and smoke stopping systems to interfaces, services and other penetrations to maintain fire compartmentation.
3. Fire and Smoke stopping systems and products: Submit proposals
4. Comply with and Refer to the Fire Strategy Report, all necessary Building Regulations and the recommendations of the Passive Fire Protection Federation.
5. All systems to include everything necessary to provide a safe, fully rated, continuous barrier, that meets all of the relevant performance requirements.
6. All products proposed for the works to maintain the fire performance of the building.

## System performance

### 210 Design

1. Design: Complete the design of the fire-stopping system.
2. Verification
   - 2.1. Submittals: Submit drawings, technical information, calculations and manufacturers' literature.
   - 2.2. Timing: To comply with the programme.
   - 2.3. Format: Drawings and specification in BIM-compliant format.

### 240 Fire performance

1. Resistance to fire: To BS EN 13501-2, EI 120 or better
2. Reaction to fire: To BS EN 13501-1, Class A1
3. Smoke resistance
   - 3.1. Air leakage rate (maximum): 3 $m^3/m^2 \cdot h$

### 260 Design life

1. Effective design life: Refer to the A Sections.

## Products

### 305 Product certification

1. Certification: For products specified generically, submit evidence of compliance with the specification.



make

2. Acceptable evidence: Third-party certification
    2.1. Timing: To comply with the programme and before completion of detailed design.

## 310 Boards

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.

## 330 Flexible intumescent gap sealers

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.
2. Strip width: Wider than joint width.

## 335 Intumescent foam fillers

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.

## 338 Intumescent mastic

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.

## 340 Intumescent mortars

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.

## 342 Fire-resisting mortars

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.

## 345 Intumescent pillows

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.
2. Composition: Sealed polyethylene bags containing graphite and vermiculite granules treated with fire activated chemicals.
3. Integral reinforced eyelets: Required.
4. Size: In accordance with system manufacturers details/ recommendations and the requirements of the testing/ certification criteria applicable to each system to achieve the specific performance requirements and to suit the service conditions of each individual instance.
5. Linking cable: Non-corrosive cotton-coated wire.

## 350 Intumescent putties

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals.



make

## 360 Mineral wool rigid batts

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Standard: To BS EN 13162.
3. Recycled content: Submit proposals.

## 370 Pipe collar

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Type:
3. Size: In accordance with system manufacturers details/ recommendations and the requirements of the testing/ certification criteria applicable to each system to achieve the specific performance requirements and to suit the service conditions of each individual instance.

## 385 Sealant-backing materials

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.

## 390 Sealants

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals.
2. Type: Fire-resisting silicone.

## 500 Samples

1. Provide:
   1.1. 500mm x 500mm - Proposed material.
   1.2. 1 No. - Fixings and accessories.

## 510 Control Samples

1. The first area of each type.

# Execution

## 610 Third-party-certified installers

1. Certification: For the technical competency of the installer, evidence of compliance with a third-party installation certification scheme to be provided.
2. Acceptable evidence: FIRAS installer certification/ LPCB certification/ UKAS Certificate of Accreditation

## 620 Workmanship generally

1. Gaps: Seal between building elements and services, to provide effective resistance to fire and the passage of smoke. Allow for capping sealants where required. Finish flush with surrounds.
2. Adjacent surfaces: Prevent overrun of filler, sealant or mortar on to finished surfaces. Any overrun to be cleaned up in accordance with manufacturer's instructions.



make

## 640 Installing boarding

1. Position of boarding: Across face to full width of opening.
2. Framing: Provide framing to all edges of boarding and intermediate studs and rails at centres not exceeding 450 mm.
3. Bedding: Bed boarding on fire-resisting adhesive/ sealant.
4. Multiple board layers: Stagger joints between layers.
   4.1. Joints: Seal with board adhesive
5. Fixing: Suitable fixings in accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
6. Other requirements: Countersink screw heads to and fill and finish joint sealant flush with boards

## 650 Installing flexible intumescent gap sealer

1. Fitting of strips: Compress strips and fit into gap so that, as they decompress, the strips wedge themselves in the void.
2. Shrink-wrapping: Do not remove.
3. Joints
   3.1. Ends of strips: Fit intumescent 'end piece' at both ends of the run of the firestop laminate.
   3.2. Joints between strips: Fit two intumescent 'end pieces' at each butt joint.

## 660 Applying intumescent foam

1. New joints: Remove builders' debris, mortar droppings, grease and other contaminants.
2. Old joints: Clean and remove existing sealant from each joint.
3. Priming: Lightly moisten substrate with water.
4. Application: Fill joint to approximately half its depth, and allow foam to expand to face of joint.
5. Trimming: Trim excess foam to give a neat, flush appearance

## 670 Applying intumescent mortar

1. Sequence: Install mortar after services are permanently installed.
2. Loose dust and combustible materials: Remove from the opening, along with any other debris.
3. Shuttering: Install suitable shuttering panels to the faces of the opening.
4. Temperature: Do not apply mortar when it could be damaged by frost.
5. Mortar cure: Do not disturb mortar before final set has taken place.
6. Shuttering removal: Remove after mortar has cured.

## 680 Installing intumescent pillows

1. Number of pillows (per m² of opening): Number necessary to achieve fire resistance.
2. Orientation of bags: Parallel to the plane of the construction element containing the opening.

## 690 Applying intumescent putty

1. Sequence: Install putty after services are permanently installed.
2. Loose dust and combustible materials: Remove from the opening, along with any other debris.
3. Putty cure: Do not disturb putty before final set has taken place.

## 710 Installing mineral wool batts

1. Installing batts: Fit tight into void between the penetrating services and the surrounding construction to form a solid barrier.
    1.1. Brackets: Impale batts on proprietary pressed steel brackets at 500 mm maximum centres and not greater than 250 mm from ends of batts.
2. Face of batts: Flush with the surface of wall, floor or soffit.
3. Joints between batts: Butt joints; seal with fire-resisting sealant
4. Gaps between services and barrier: Seal with fire-resisting sealant.

## 730 Fixing pipe collars

1. Collar fixing: Screw fixed, to manufacturers recommendations to suit service conditions.
2. Gap around collar: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
3. Length of wraps: Project 50 mm from each side of the element.

## 740 Inserting sealant-backing material

1. Preparation: Remove debris from service penetration.
2. Installation: Insert joint filler to full depth of joint, leaving sufficient depth to apply sealant

## 745 Applying sealants generally

1. Application: As section Z22.

## 750 Applying capping sealant

1. Preparation: Degrease using cleaner recommended by sealant manufacturer
2. Priming: Primer recommended by sealant manufacturer.
3. Width of sealant: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
4. Depth of sealant: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
5. Temperature: Do not apply water-based sealants when they could be damaged by frost.

# Completion

## 910 Cleaning

1. Masking tapes: Remove.
2. Cleaning: Clean off splashes and droppings. Wipe down finishes.

## 920 Inspection

1. Notice for inspection (minimum): Five working days

Ω End of Section



make

# P20
# Unframed Isolated Trims/ Skirtings/ Sundry Items

## To be read with preliminaries/ general conditions

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 120 TRM-160 Medium-Density Fibreboard (MDF) Skirting

1. Description: Medium density fibreboard (MDF) skirting to receive a painted finish
2. Sizes and configurations: Refer to the Design Drawings
3. Standard: To BS EN 622-5.
   3.1. Type: Moisture resistant medium density fibreboard (MDF).
   3.2. Formaldehyde class: To BS EN 622-1, class E1.
4. Reaction to fire rating: To BS EN 13501-1, Class B
5. Thickness: 12 mm
6. Edges: Square.
7. Profile: Refer to the Design Drawings
8. Finish: Factory prepared, primed and painted on-site
9. Joints:
   9.1. Straight runs: To be formed in single lengths wherever possible.
   9.2. Running joints: Scarfe joints, location of running joints to be agreed.
   9.3. Joints at angles: Mitred, unless otherwise shown on the Design Drawings.
   9.4. Corner joints: Ease arrises.
10. Fixing: Countersunk screwed and filled.

### 121 TRM-161 Solid Timber Skirting - Feature Walls

1. Description: Solid hardwood timber skirting.
2. Sizes and configurations: Refer to the Design Drawings
3. Quality of wood and fixing:
   3.1. Species: Oak.
   3.2. Class: To be agreed with the Architect.
4. Moisture content at time of fixing: Moisture and crack resistant.
5. Profile: Curved to suit feature wall. 150mm thick at base with slight recess. Refer to the Design Drawings
6. Finish as delivered: Factory prepared, primed and painted on-site
7. Joints:
   7.1. Straight runs: To be formed in single lengths wherever possible.
   7.2. Running joints: Scarfe joints, location of running joints to be agreed.
   7.3. Joints at angles: Mitred, unless otherwise shown on the Design Drawings.

7.4. Corner joints: Ease arrises.

8. Fixing: Countersunk screwed, and filled.

## 122 TRM-162 Porcelain/ Stone Skirting

1. Description: Porcelain/ stone skirting, as the Design Drawings.
2. Sizes and configurations: Refer to the Design Drawings
3. Adhesive: To suit the service conditions.
4. Manufacturer: To match floor tiling
   4.1. Product reference: To match floor tiling
5. Colour: To match floor tiling
6. Finish: To match floor tiling
7. Joint width: 2-3 mm.
8. Grout:
   8.1. Type/ classification: Proprietary non-staining, flexible, water impervious, mould and chemical resistant as recommended by the tile manufacturer.
   8.2. Colour: To match floor tiling

## 161 TRM-311 PVC Stair Nosing

1. Description: PVC stair nosing.
2. Manufacturer: Gradus, or acceptable equivalent.
   2.1. Product reference: Flexible PVC stair endings.
3. Material: PVC.
4. Profile: Refer to the Design Drawings
5. Size: Refer to the Design Drawings
6. Finish: Black.
7. Slip resistance: To be agreed with the Architect.
8. Fixing: Suitably concealed fixed.

## 170 TRM-172 Corner Guard - Metal

1. Description: Corner wall protection guards.
2. Manufacturer: Construction specialities, or acceptable equivalent.
   2.1. Product reference: A Series Stainless Steel.
3. Material: Stainless steel to BS EN 10088-2, grade 1.4301 (304)
4. Size: Refer to the Design Drawings
5. Thickness: Refer to the Design Drawings
6. Finish: Brushed
7. Corner radius: 5 mm.
8. Fixing: Suitably concealed fixed.

## 171 TRM-173 Wall Protection - Metal

1. Description: Wall protection
2. Manufacturer: Construction Specialities, or acceptable equivalent.
   2.1. Product reference: UK Range - LLRSS/ CRS

3. Material: Stainless steel to BS EN 10088-2, grade 1.4301 (304)
4. Size: Refer to the Design Drawings
5. Thickness: Refer to the Design Drawings
6. Finish: Brushed
7. Fixing: Suitably concealed fixed.

## 454 Samples

1. General: Before placing orders submit representative samples of the following:
   1.1. 500mm of any visible element.
2. Delivered materials/ products: To match samples.

## 455 Control Samples

1. General: Complete areas of finished work and obtain approval of appearance before proceeding.

# Execution

## 510 Installation generally

1. Joinery workmanship: As section Z10.
2. Metal workmanship: As section Z11.
3. Methods of fixing and fasteners: As section Z20 where not specified.
4. Straight runs: To be in one piece, or in long lengths with as few joints as possible.
5. Running joints: Location and method of forming to be agreed where not detailed.
6. Joints at angles: Mitre, unless shown otherwise
7. Position and level: To be agreed where not detailed.
8. Fixing:: Maintain straight, vertical and horizontal lines within ±1mm from notional positions including plane and cut-outs.

Ω End of Section



make

# P21
# Door/ Window Ironmongery

## Pre-tender

### 10 Quantities and locations

1. Quantities and locations of ironmongery: Refer to the ironmongery schedule.
2. Fixing: As sections L10 and L20.

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 Ironmongery Scope (Generally)

1. Description: Refer to the Ironmongery Schedule for materials, finishes, product codes, manufacturer(s) and door locations for fixing.
2. Material/ finish: Brushed stainless-steel, Grade 1.4401 (316) ironmongery, to come from a single manufacturer and range.

### 141 Sample boards

1. General: Before placing orders with suppliers submit a sample board, containing labelled samples of ironmongery and showing methods of fixing
2. Range: include To be confirmed .
    2.1. Conformity: Retain board on site in an approved location for the duration of the contract. Ensure conformity of ironmongery as delivered with labelled samples

### 165 Control Samples

1. Submit first item of ironmongery installed of each type.

### 170 Ironmongery for fire doors

1. Relevant products: Ironmongery fixed to, or morticed into, the component parts of a fire-resisting door assembly
2. Compliance: Ironmongery included in successful tests to BS EN 1634-1 on door assemblies similar to those proposed
    2.1. Certification: Submit evidence of successful testing by UKAS-accredited laboratory
3. Melting point of components (except decorative non-functional parts): 800°C minimum.

### 180 Strength class or category of duty for door ironmongery

1. Requirement: As schedule.
2. General: Durability of ironmongery components to be compatible with stated category of duty of each door leaf.
    2.1. Exclusions: Ironmongery with specific duty or 'category of use' defined elsewhere.

2.2. Documentation: Before placing orders with suppliers submit documentation showing product compliance with stated category of duty.

## 190 General Performance

1. Standards: In accordance with:

1.1.

- Construction Products Regulations (CPR) 2013 - CE marking.
- Building Regulations Approved Document 7 2013 - CE marking.
- Approved Document Part M of the Building Regulations.
- Code of Practice: Hardware for Fire and Escape Doors' produced jointly by the Door and Hardware Federation (DHF) and the GAI.
- BS EN 179 - Emergency exit devices operated by a lever handle or push pad, for use on escape routes.
- BS 476: Part 22 - Method for determination of the fire resistance of non-loadbearing elements of construction.
- BS EN 1125 - Panic exit devices operated by a horizontal bar, for use on escape routes.
- BS EN 1154 - Controlled door closing devices.
- BS EN 1155 - Electrically powered hold-open devices for swing doors.
- BS EN 1158 - Door coordinator devices.
- BS EN 1192 - Doors. Classification of strength requirements.
- BS EN 1303 - Cylinders for locks.
- BS EN 1527 - Hardware for sliding doors and folding doors.
- BS EN 1634 - Fire resistance and smoke control tests for door and shutter assemblies, openable windows and elements of building hardware.
- Recommendations of the GAI.
- BS EN 1670 - Corrosion resistance.
- BS EN 1906 - Lever handles and knob furniture.
- BS EN 1935 - Single-axis hinges.
- 'PAS 3621 - Multipoint locking assemblies. Keyed egress.
- BS 3621 - Lock assemblies operated by key from both the inside and outside of the door.
- BS 6375: Part 2 - Performance of windows and doors. Classification for operation and strength characteristics and guidance on selection and specification.
- BS 8300 - Design of an accessible and inclusive built environment.
- BS 8424 - Pull handles.
- PAS 8621 - Multipoint locking assemblies. Keyless egress.
- BS 8621 - Lock assemblies operated by key from the outside of the door and by handle or thumb turn from the inside of the door.
- PAS 10621 - Multipoint locking assemblies. Dual mode egress.
- BS EN 12051 - Door and window bolts.
- BS EN 12209 - Mechanically operated locks and locking plates.
- BS EN 12320 - Padlocks and padlock fittings.
- BS EN 13126 - Hardware for windows and door height windows.
- prEN 13633 - Electrically controlled panic exit systems for use on escape routes.
- BS EN 13637 - Electrically controlled exit systems for use on escape routes.
- BS EN 14846 - Locks and latches. Electromechanically operated locks and striking plates.

- BS EN 15684 - Mechatronic cylinders. Requirements and test methods.
- prEN 15887 - Uncontrolled door closing devices.
- BS EN 16864 - Mechatronic padlocks. Requirements and test methods.
- BS EN 16867 - Building hardware. Mechatronic door furniture. Requirements and test methods.
- Provide suitable CE marked (non-visible when installed), compatible and complete ironmongery suites of consistent quality and finish.

## Door hanging devices - Not Used

## Window hanging devices - Not Used

## Door operating devices - Not Used

## Door securing devices - Not Used

## Window securing devices - Not Used

## Door furniture - Not Used

## Window furniture - Not Used

Ω End of Section

# Q25
# Slab/ Brick/ Sett/ Cobble Pavings

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 150 PAV-131 Precast Concrete Slab Paving

1. Description: Precast concrete slab paving, as the Design Drawings.
2. Substrate: Refer to the Design Drawings
3. Support pedestals: Height adjustable metal support pedestals adhesive bonded and/ or mechanically fixed to substrates.
4. Paving units:
    4.1. Manufacturer: Marshalls, or acceptable equivalent
        4.1.1. Product reference: To be agreed with the Architect.
    4.2. Type: Precast concrete, hydraulically pressed.
    4.3. Size: 600mm x 600mm.
    4.4. Thickness: 50mm.
    4.5. Edges: Chamfered.
    4.6. Finish: Natural.
    4.7. Colour: Natural.
5. Joints: Closed

### 190 Hard landscaping materials specification

1. Minimum BRE 'Green Guide to Specification' (online) rating: Refer to the Sustainability Report

## System performance

### 210 Design – natural stone slab paving system

1. Design: In accordance with BS 7533-101.
    1.1. Site category: To BS 7533-4
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

### 220 Design – concrete flag paving system

1. Design: In accordance with BS 7533-101.
    1.1. Site category: To BS 7533-4
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed



make

## 230 Design – rigid brick paving system

1. Design: Complete the design of rigid brick paving system in accordance with Brick Development Association 'Design Note 8'.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 240 Design – natural stone sett paving system

1. Design: Complete the design of the natural stone sett paving system in accordance with BS 7533-101.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 250 Design – concrete sett paving system

1. Design: Complete the design of the concrete sett paving system in accordance with manufacturer's instructions.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 260 Design – natural stone cobble paving system

1. Design: In accordance with BS 7533-101.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 270 Design – precast concrete and grass paving system

1. Design: Complete the design of the precast concrete and grass paving system in accordance with manufacturer's instructions.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 275 Design – precast concrete and gravel paving system

1. Design: Complete the design of the precast concrete and gravel paving system in accordance with manufacturer's instructions.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 280 Design – plastics grass reinforcing paving system

1. Design: Complete the design of the plastics grass reinforcing paving system in accordance with manufacturer's instructions.
2. Ground conditions: To be confirmed
3. Performance criteria: To be confirmed

## 285 Design – gravel-filled plastics paving system

1. Design: Complete the design of the gravel-filled plastics paving system in accordance with manufacturer's instructions.
2. Ground conditions: To be confirmed

3. Performance criteria: To be confirmed

## Products

### 370 Cement for site mixed mortar

1. Standard: As section Z21.

### 375 Sand/ fine aggregate for site-mixed mortar

1. Standard: To BS EN 12620

### 380 Lime for site-mixed mortar

1. Standard: Non-hydraulic to BS EN 459-1, BS EN 459-2 and BS EN 459-3, Type CL 90 or, if ready-mixed lime:sand, to BS EN 998-2.

### 390 Sand/ fine aggregate for unbound laying course and jointing of concrete flag paving

1. Standard: To BS 7533-4, unbound construction laying course and jointing material.
2. Purity: Free from deleterious salts, contaminants, lime and cement.
3. Procurement: Obtain from one source and ensure consistent grading.

### 400 Sand/ fine aggregate for laying course of grassed or gravel-filled paving

1. Standard: To BS EN 13242.

### 420 Sand/ fine aggregate for laying course of concrete sett paving

1. Material: Sand/ fine aggregate for laying courses to BS 7533-4

## Execution

### 610 Material samples

1. Samples representative of colour and appearance of designated materials: Submit before placing orders.
    1.1. Designated materials: All pavings

### 615 Control samples

1. Sample areas: Complete as part of the finished work.
    1.1. Types of paving: All paving
    1.2. Size (minimum): One full paving panel
2. Approval of appearance and surface: Obtain before proceeding.

### 620 Adverse weather

1. General
    1.1. Temperature: Do not lay or joint paving if the temperature is below 3°C on a falling thermometer or below 1°C on a rising thermometer.
    1.2. Frozen materials: Do not use. Do not lay bedding on frozen or frost covered bases.
2. Paving with mortar joints and/ or bedding
    2.1. Protect from frost damage, rapid drying out and saturation until mortar has hardened.

3. Paving laid and jointed in sand/ fine aggregate
   3.1. Stockpiled laying course sand/ fine aggregate: Protect from saturation.
   3.2. Exposed areas of unbound laying course and uncompacted areas of unbound paving: Protect from heavy rainfall.
   3.3. Saturated unbound laying course: Remove and replace, or allow to dry before proceeding.
   3.4. Laying dry sand/ fine aggregate jointed paving in damp conditions: Brush in as much jointing sand as possible. Minimize site traffic over paving. As soon as paving is dry, top up joints and complete compaction.

## 625 Laying pavings – general

1. Appearance: Smooth and even with regular joints and accurate to line, level and profile.
2. Falls: To prevent ponding.
3. Bedding of paving units: Firm so that rocking or subsidence does not occur or develop.
   3.1. Bedding/ Laying course: Consistently and accurately graded, spread and compacted to produce uniform thickness and support for paving units.
4. Slopes: Lay paving units upwards from the bottom of slopes.
5. Paving units: Free of mortar and sand stains.
6. Cutting: Cut units cleanly and accurately, without spalling, to give neat junctions with edgings and adjoining finishes.

## 630 Levels of paving

1. Permissible deviation from specified levels
   1.1. Generally: +/-6 mm.
2. Height of finished paving above features
   2.1. At gullies: +6 to +10 mm.
   2.2. At drainage channels and kerbs: +3 to +6 mm.

## 635 Regularity of paved surfaces

1. Maximum variation in gap under a 3 m straight edge placed anywhere on the surface (where appropriate in relation to the geometry of the surface)
   1.1. Precast concrete paving blocks and clay pavers for flexible pavements: 10 mm.
   1.2. Precast concrete flags or natural stone slabs: 3 mm.
2. Difference in level between adjacent paving units (maximum): 2 mm.
3. Sudden irregularities: Not permitted.

## 637 Regularity of paved surfaces

1. Maximum undulations in the surface of pavings (except tactile paving surfaces) under a 1 m straight edge placed anywhere on the surface (where appropriate in relation to the geometry of the surface): 3 mm.
2. Joints between paving units or utility access covers
   2.1. Joints flush with the surface: Difference in level between adjacent units to be no more than twice the joint width (with a 5 mm maximum difference in level).
   2.2. Recessed, filled joints: Difference in level between adjacent units to be no greater than 2 mm; the recess to be no deeper than 5 mm.

2.3. Unfilled joints: Difference in level between adjacent units to be no greater than 2 mm.

3. Sudden irregularities: Not permitted.

## 640 Colour banding

1. General: Unless premixed by manufacturer, select from at least 5 separate packs in rotation to avoid colour banding.

## 645 Protection

1. Cleanliness: Keep paving clean and free from mortar droppings, oil and other materials likely to cause staining.
2. Materials storage: Do not overload pavings with stacks of materials.
3. Handling: Do not damage paving unit corners, arrises, or previously laid paving.
4. Mortar-bedded pavings: Keep free from traffic after laying:
   4.1. Pedestrian traffic (minimum): Four days
   4.2. Vehicular traffic (minimum): Seven days
5. Access: Restrict access to paved areas to prevent damage from site traffic and plant.

## 650 Cementitious bases and sub-bases

1. General: Protect from moisture loss, if not covered by another pavement course within 2 hours of completion.

## 655 Condition of sub-bases/ bases before spreading laying course

1. Trenches and excavation of soft or loose spots in subgrade: Fill and thoroughly compact.
2. Granular surfaces: Lay and compact so as to be sound, clean, smooth and close-textured enough to prevent migration of bedding/ laying course materials into the sub-base during compaction and use, free from movement under compaction plant and free from compaction ridges, cracks and loose material.
3. Prepared existing and new bound bases (roadbases): Sound, clean, free from rutting or major cracking. Remove sharp stones, projections and debris.
4. Sub-base/ Roadbase level tolerances: To BS 7533-101, annex A.
5. Levels and falls: Accurate and within the specified tolerances.
6. Drainage outlets: Within 0-10 mm of the required finished level.
7. Features in unbound paving (including mortar-bedded restraints and drainage ironwork): Complete to required levels; adequately bed and haunch in mortar.
8. Sub-bases containing cement/ hydraulic binder: Cure for minimum times specified in BS 7533-4.

## 660 Drainage holes in existing bases

1. Location: Impervious layers of existing road/ paving.
2. Drainage: Form regular grid of holes, through base and any additional build up, down to sub-base:
   2.1. Spacing in both directions: 750 mm
   2.2. Minimum clear opening: 30 mm
   2.3. Do not weaken or excessively disturb road/ paving.

3. Completion: Remove jagged or protruding edges. Fill voids with washed sharp sand Ram down to form flush smooth surface.

## 665 Planing and repairs to existing bases

1. Existing macadam/ asphalt surfaces: Plane to required levels.
2. Repairs: Cut out to remove ruts and cracks over 25 mm wide
3. Building up existing surfaces to required levels: Regulate using asphalt concrete to BS EN 13108-1

## 670 Laying granular layer over existing bases

1. Thickness To be confirmed.
2. Compaction: Laying granular sub-bases for vehicular and pedestrian areas

## 675 Laying geotextile sheet edging strips

1. Location: Immediately below the laying course, abutting features which interrupt the laying course, including:
    1.1. Perimeters/ Edge restraints/ Kerbs.
    1.2. Other types of paving.
    1.3. Drainage fittings, e.g. channels and manholes.
2. Edge detail: Turn sheet up to a height not less than thickness of the laying course to form an upstand fitted neatly against features.
3. Width: As the manufacturers recommendations
4. Jointing: Lap by As the manufacturers recommendations.

## 680 Laying geotextile sheet patches over drainage holes

1. Laying: Lay geotextile patches on the base, centred over each hole.
    1.1. Size of patch: As the manufacturers recommendations

## 685 Laying geotextile sheet overlays

1. Location: Immediately below the laying course.
2. Laying: Fit neatly at edge restraints and other features that interrupt the laying course, e.g. drainage fittings, channels, manholes and kerbs.
3. Edge detail: Turn sheet up to form an upstand against features, height not less than thickness of the laying course.
4. Width: As the manufacturers recommendations
5. Jointing: Lap by As the manufacturers recommendations.

## 690 Reclaimed natural stone

1. Lifting/ Storage/ Protection: Stack neatly on pallets and shrink wrap in plastics sheeting
2. Preparation: Clean using mild detergent solution

## 695 Site-mixed mortar

1. Mix: 1:3 cement:sand
2. Consistency: As the manufacturers recommendations to suit the conditions of service
3. Admixtures: None

## 700 Site mixed fine concrete

1. Standard: In accordance with BS 7533-101.

## 710 Laying flag and slab paving – sand/ fine aggregate laying course and jointing

1. Standard: In accordance with BS 7533-4.
2. Flag installation and cutting: To Interpave publication Concrete flag paving.
3. Laying course
    3.1. Nominal thickness after compaction: 25 mm
4. Joint width: 2–5 mm.

## 715 Laying flag and slab paving – mortar laying course and jointing

1. Standard generally: In accordance with BS 7533-4.
2. Flag installation and cutting: To Interpave publication Concrete flag paving.
3. Laying course
    3.1. Nominal thickness: 25 mm after compaction
4. Laying and jointing: To BS 7533-4
5. Joint width (nominal): 5–10 mm

## 725 Laying rigid brick paving

1. Standard generally: In accordance with Brick Development Association (BDA) Design Note 8.
2. Bedding and jointing method: Simultaneous bedding and jointing with stiff plastic mortar.
3. Cement slurry: Apply thin slurry (1–3 mm) of neat cement or 1:1 cement:soft sand over the freshly laid mortar bed immediately prior to laying bricks.
4. Laying: Wet bricks as necessary (but do not soak), butter joint faces and press down firmly to give a level surface with 10 mm regular joints.
5. Nominal thickness of bed: As per the manufacturers recommendations
6. Joint profile: Bucket handle

## 730 Laying natural stone sett paving

1. Standard generally: In accordance with BS 7533-101.
2. Laying type: Rigid.
    2.1. Laying and jointing method: Contractor's choice
3. Laying course
    3.1. Target thickness after compaction: 35 mm
4. Joint width (nominal): 5 mm

## 735 Laying concrete sett paving – mortar-bedded

1. Laying: Spread and level a bed of mortar. Individually lay and hammer down the setts so that tops are level, leaving joints open.
2. Laying course
    2.1. Nominal thickness after compaction: 20 mm
3. Jointing: Fill joints and finish neatly. Clean mortar from face of setts before it sets.

4. Joint width (nominal): 5 mm
    4.1. Recess mortar joints below sett surface by: 2 mm

## 750 Laying natural stone cobble paving

1. Bedding, laying, jointing and completion: In accordance with BS 7533-101.

## 755 Laying precast concrete and grass paving

1. Laying: Tamp down into lightly compacted laying course.
    1.1. Nominal thickness of laying course after compaction: 50 mm
2. Filling: Allow to settle and refill level with surface.

## 760 Laying precast concrete and gravel paving

1. Laying: Tamp down into lightly compacted laying course.
    1.1. Nominal thickness of laying course after compaction: 50 mm
2. Filling: Allow to settle and refill level with surface.

## 765 Laying plastics grass reinforcing pavers

1. Laying: Tamp down into lightly compacted laying course Consolidate with vibrating plate compactor.
    1.1. Nominal thickness of laying course after compaction: 50 mm
2. Securing on slopes: Not required
3. Filling: Allow to settle and refill level with surface.

## 770 Laying gravel-filled plastics pavers

1. Laying: Tamp down into lightly compacted laying course Consolidate with vibrating plate compactor.
    1.1. Nominal thickness of laying course after compaction: 50 mm
2. Securing on slopes: Not required
3. Filling: Allow to settle and refill level with surface.

## 780 Laying mortar-bedded accessories

1. Foundation: Concrete
    1.1. Size: As the manufacturers recommendations
2. Bedding
    2.1. Mortar: Ready-mixed
    2.2. Thickness: 10 mm minimum to 40 mm maximum.
3. Laying: Bed units on foundation, and secure with continuous mortar haunching.
    3.1. Keep exposed faces clean and free from mortar.
4. Jointing: Dry, closely butted

## 785 Tooled joints in mortar-bedded units

1. Joints: Completely filled with bedding mortar as work proceeds.
    1.1. Joint width: 6–10 mm
    1.2. Finish: Neat flush profile.

## 790 Tooled coloured joints in mortar-bedded units

1. Joints: Completely filled with bedding mortar as work proceeds.
    1.1. Joint width: 6–10 mm
2. Pointing: 1:3 cement:sand mortar with pigment, colour to match block colour.
    2.1. Depth: 10 mm.

## 795 Sealant movement joints in mortar-bedded units

1. Spacing: 4.5 m
2. Extent of joints: Through edge units, haunching and foundation
3. Joint filler: Compressible cellular rubber or plastics compatible with specified sealant. Build in as work proceeds.
    3.1. Joint width: 10 mm
4. Barrier (joint breaker): As recommended by sealant manufacturer. Position filler and barrier accurately to fully support sealant at recommended distance from exposed faces of units.
5. Sealant
    5.1. Application: As section Z22.

## 800 Applying sealer/ stabilizer to blocks and setts

1. Surface preparation: Ensure sand joints are completely dry and free from contamination
    1.1. Application: To dry paving.
        1.1.1. Method: To manufacturer's recommendations
        1.1.2. Number of coats: Two
        1.1.3. Coverage: As sealer manufacturer's recommendations

## 805 Cleaning existing blocks and setts

1. Area: Refer to the Design Drawings
2. Application
    2.1. Method: High-pressure washer
    2.2. Dosage: As cleaning solution manufacturer's recommendations
3. Sand/ fine aggregate jointed paving: Do not displace jointing sand/ fine aggregate during cleaning process.

## 810 Pedestal installation

1. Surface to accept pedestals: Clean and free of debris.
2. Setting out: Mark centre-point of pedestal on substrate surface, with perpendicular guidelines to ensure square layout.
    2.1. Orientation: Align parallel with adjacent features.
    2.2. Spacing: To suit paving material and dimensions
3. Movement tolerance at perimeter of paver system (maximum): 5 mm.

# Completion

## 915 Completion of paving with dry sand or fine aggregate-filled joints

1. Sand dressing: Leave a thin layer of dry jointing sand/ fine aggregate over the paving, sweep clean before practical completion
2. Final compaction of the surface course: In accordance with BS 7533-3.
3. Vacuum cleaning machines: Not allowed.

## 920 Completion of grassed pavings

1. Protection: Protect from traffic for 6-8 weeks or until grass can tolerate traffic.

## 930 Slip resistance testing

1. Surfaces to be tested:
    1.1. Surface condition: Dry and wet
2. Timing: Two weeks prior to handover, but after initial cleaning
3. Period of notice (minimum): Three working days.
4. Test standard: To BS EN 16165
    4.1. Testing authority: A UKAS-accredited laboratory
    4.2. Witnessing/ Certification: Arrange for tests to be witnessed/ certified by: Contract administrator.
    4.3. Report: Submit.
        4.3.1. Format: As required under BS EN 16165

Ω End of Section



make

# Q50
# Site/ Street Furniture/ Equipment

## Gates, barriers and parking controls

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 110 FFE-801 Turning Circle

1. Description:  Vehicle turntable, as shown on the Design Drawings.
2. Refer to the Transport Consultant's information

## Site and street furniture - Not Used

## Installation - Not Used

Ω End of Section



make

# X12
# Lift Fit-Out/ Platform Lifts

## General

### 100 Design Compliance

1. Section to be read in conjunction with the A and Z Sections which form part of the Specification and Contract Conditions.
2. Design: Refer to and comply with A10.120 and A21.140.

### 330 LFT-101 Passenger Lift Fit-out

1. Description: Fit-out to passenger lifts.
2. Size and configuration: Refer to the Design Drawings
3. Structure:
    3.1. Type/ material: Concealed metal framework/ lift structure as the Structural Engineer's documentation.
4. Lift car finishes:
    4.1. Ceiling: Painted metal ceiling with integrated access panels.
    4.2. Walls:
        4.2.1. Side walls:
            4.2.1.1. Metal panels: Stainless steel
                4.2.1.1.1. Finish: PVD Coated
                4.2.1.1.2. Colour: Bronze
            4.2.1.2. Timber panels/ accents: Solid timber
                4.2.1.2.1. Species: Oak
                4.2.1.2.2. Profile: Refer to the Design Drawings
                4.2.1.2.3. Finish: Matt lacquered.
                4.2.1.2.4. Accent features: Bronze metal panels.
        4.2.2. Back wall: Fabric wall with bronze metal frame positioned in front. Perimeter LED strip lighting integrated around the wall.
    4.3. Skirting: Recessed stainless steel grade 1.4301 skirting.
    4.4. Flooring: Refer to FLS-201 in M40
5. Handrails/protection rails:
    5.1. Type/ material: Mild steel.
        5.1.1. Welded connections at corners with matching flat end caps.
    5.2. Finish: Leather.
    5.3. Colour: Brown.
    5.4. Fixing: Concealed.
6. Lift car doors/ frames/ architraves:
    6.1. Height: Full height internally.
    6.2. Type/ material: Suitable core finished with brushed stainless steel grade 1.4301.
    6.3. Architraves: Standard depth architraves to landing doors.



make

6.4. Integrated fire curtains: Refer to FPS-301 in L20

7. Car operating panel and controls/ functionality:
   7.1. Call buttons: To be selected from the manufacturer's standard range within a non-standard panel.
   7.2. Panel:
      7.2.1. Type/ material: Brushed stainless steel grade 1.4301 panel with 2.5mm thick stainless steel grade 1.4301 surround rings forming the push button surrounds.
      7.2.2. Fixing: Panels to be secured flush/ recessed, using concealed methods and with anti-tamper properties to prevent unauthorised access.
   7.3. Push buttons:
      7.3.1. Type/ material: Brushed stainless steel grade 1.4301 vandal-resisting illuminating push buttons.
         7.3.1.1. Indication numbers and letters to be raised tactile digits on push buttons in standard range typeface.
   7.4. Floor indicator: Light emitting diode (LED) dot matrix digital floor indicator display recessed into the panel and finished flush.
   7.5. Manual door openers: Landing door manual openers to be positioned within door frame.
8. Lighting: Recessed pin lighting and recessed LED lighting. Refer to the Services Engineer's Information
9. Services/ ventilation: Refer to the Services Engineer's Information
10. Accessories: Refer to the Service Engineers documentation and as required to complete the installation
11. Systems shall include fixings, framing, bracketry, support framing/ backing materials, grids, seals and all other components/ accessories necessary to complete the works.

## 336 LFT-110 EOT Lift Fit-out

1. Description: Fit-out to EOT lifts.
2. Size and configuration: Refer to the Design Drawings
3. Structure:
   3.1. Type/ material: Concealed metal framework/ lift structure as the Structural Engineer's documentation.
4. Lift car finishes:
   4.1. Ceiling: Painted metal ceiling with integrated access panels.
   4.2. Walls:
      4.2.1. Side walls:
         4.2.1.1. Metal panels: Stainless steel
            4.2.1.1.1. Finish: PVD Coated
            4.2.1.1.2. Colour: Bronze.
      4.2.2. Back wall: Brushed stainless steel.
   4.3. Skirting: Recessed stainless steel grade 1.4301 skirting.
   4.4. Flooring: Refer to FLS-201 in M40.
5. Handrails/protection rails:
   5.1. Type/ material: Mild steel
      5.1.1. Welded connections at corners with matching flat end caps.

5.2. Finish: Polyester powder coated

5.3. Colour: Black.

5.4. Fixing: Concealed

6. Lift car doors/ frames/ architraves:

6.1. Height: Full height internally.

6.2. Type/ material: Suitable core finished with brushed stainless steel grade 1.4301.

6.3. Architraves: Standard depth architraves to landing doors.

6.4. Integrated fire curtains: Refer to FPS-301 in L20

7. Car operating panel and controls/ functionality:

7.1. Call buttons: To be selected from the manufacturer's standard range within a non-standard panel.

7.2. Panel:

7.2.1. Type/ material: Brushed stainless steel grade 1.4301 panel with 2.5mm thick stainless steel grade 1.4301 surround rings forming the push button surrounds.

7.2.2. Fixing: Panels to be secured flush/ recessed, using concealed methods and with anti-tamper properties to prevent unauthorised access.

7.3. Push buttons:

7.3.1. Type/ material: Brushed stainless steel grade 1.4301 vandal-resisting illuminating push buttons.

7.3.1.1. Indication numbers and letters to be raised tactile digits on push buttons in standard range typeface.

7.4. Floor indicator: Light emitting diode (LED) dot matrix digital floor indicator display recessed into the panel and finished flush.

7.5. Manual door openers: Landing door manual openers to be positioned within door frame.

8. Lighting: Recessed pin lighting and recessed LED lighting. Refer to the Services Engineer's Information

9. Accessories: Refer to the Service Engineers documentation and as required to complete the installation

10. Services/ ventilation: Refer to the Services Engineer's Information

11. Systems shall include fixings, framing, bracketry, support framing/ backing materials, grids, seals and all other components/ accessories necessary to complete the works.

## 341 LFT-121 Goods Lift Fit-out

1. Description: Fit-out to goods lifts.

2. Size and configuration: Refer to the Design Drawings

3. Structure:

3.1. Type/ material: Concealed metal framework/ lift structure as the Structural Engineer's documentation.

4. Lift car finishes:

4.1. Ceiling: Brushed stainless steel panels with integrated access panels.

4.2. Walls:

4.2.1. Side walls: Brushed stainless steel

4.2.2. Back wall: Brushed stainless steel

4.3. Skirting: Recessed stainless steel grade 1.4301 skirting.

4.4. Flooring: Refer to FLS-202 in M40.

5. Handrails/protection rails:
   5.1. Type/ material: Brushed finish stainless steel grade 1.4301 tubular profiles with machined stainless steel grade 1.4301 brackets.
      5.1.1. Welded connections at corners with matching flat end caps.
      5.1.2. Handrail terminates with a smooth curved return to the wall
   5.2. Fixing: Concealed
6. Lift car doors/ frames/ architraves:
   6.1. Height: Full height internally.
   6.2. Type/ material: Suitable core finished with brushed stainless steel grade 1.4301.
   6.3. Architraves: Standard depth architraves to landing doors.
   6.4. Integrated fire curtains: Refer to FPS-301 in L20
   6.5. Door opening cladding: Brushed stainless steel grade 1.4301 cladding to continue from the lift door surround into the full extent of the formed wall opening
7. Car operating panel and controls/ functionality:
   7.1. Call buttons: To be selected from the manufacturer's standard range within a non-standard panel.
   7.2. Panel:
      7.2.1. Type/ material: Brushed stainless steel grade 1.4301 panel with 2.5mm thick stainless steel grade 1.4301 surround rings forming the push button surrounds.
      7.2.2. Fixing: Panels to be secured flush/ recessed, using concealed methods and with anti-tamper properties to prevent unauthorised access.
   7.3. Push buttons:
      7.3.1. Type/ material: Brushed stainless steel grade 1.4301 vandal-resisting illuminating push buttons.
         7.3.1.1. Indication numbers and letters to be raised tactile digits on push buttons in standard range typeface.
   7.4. Floor indicator: Light emitting diode (LED) dot matrix digital floor indicator display recessed into the panel and finished flush.
   7.5. Manual door openers: Landing door manual openers to be positioned within door frame.
8. Services/ lighting/ ventilation: Refer to the Services Engineer's Information
9. Accessories: Refer to the Service Engineers documentation and as required to complete the installation
10. Systems shall include fixings, framing, bracketry, support framing/ backing materials, grids, seals and all other components/ accessories necessary to complete the works.

## 346 LFT-126 Waste Lift Fit-out

1. Description: Fit-out to waste lifts.
2. Size and configuration: Refer to the Design Drawings
3. Structure:
   3.1. Type/ material: Concealed metal framework/ lift structure as the Structural Engineer's documentation.
4. Lift car finishes:
   4.1. Ceiling: Metal ceiling with integrated access panels.

4.2. Walls:

4.2.1. Side walls: Brushed stainless steel

4.2.2. Back wall: Brushed stainless steel

4.3. Skirting: Recessed stainless steel grade 1.4301 skirting.

4.4. Flooring: Refer to FLS-601 in M50.

5. Handrails/protection rails:

5.1. Type/ material: Brushed finish stainless steel grade 1.4301 tubular profiles with machined stainless steel grade 1.4301 brackets.

5.1.1. Welded connections at corners with matching flat end caps.

5.1.2. Handrail terminates with a smooth curved return to the wall.

5.2. Fixing: Concealed

6. Lift car doors/ frames/ architraves:

6.1. Height: Full height internally.

6.2. Type/ material: Suitable core finished with brushed stainless steel grade 1.4301.

6.3. Architraves: Standard depth architraves to landing doors.

6.4. Integrated fire curtains: Refer to FPS-301 in L20

6.5. Door opening cladding: Brushed stainless steel grade 1.4301 cladding to continue from the lift door surround into the full extent of the formed wall opening

7. Car operating panel and controls/ functionality:

7.1. Call buttons: To be selected from the manufacturer's standard range within a non-standard panel.

7.2. Panel:

7.2.1. Type/ material: Brushed stainless steel grade 1.4301 panel with 2.5mm thick stainless steel grade 1.4301 surround rings forming the push button surrounds.

7.2.2. Fixing: Panels to be secured flush/ recessed, using concealed methods and with anti-tamper properties to prevent unauthorised access.

7.3. Push buttons:

7.3.1. Type/ material: Brushed stainless steel grade 1.4301 vandal-resisting illuminating push buttons.

7.3.1.1. Indication numbers and letters to be raised tactile digits on push buttons in standard range typeface.

7.4. Floor indicator: Light emitting diode (LED) dot matrix digital floor indicator display recessed into the panel and finished flush.

7.5. Manual door openers: Landing door manual openers to be positioned within door frame.

8. Services/ lighting/ ventilation: Refer to the Services Engineer's Information

9. Accessories: Refer to the Service Engineers documentation and as required to complete the installation

10. Systems shall include fixings, framing, bracketry, support framing/ backing materials, grids, seals and all other components/ accessories necessary to complete the works.

## 351 LFT-131 Firefighting Lift Fit-out

1. Description: Fit-out to firefighting lifts.

2. Size and configuration: Refer to the Design Drawings

3. Structure:

3.1. Type/ material: Concealed metal framework/ lift structure as the Structural Engineer's documentation.

4. Lift car finishes:
   4.1. Ceiling: Metal ceiling with integrated access panels.
   4.2. Walls:
      4.2.1. Side walls: Brushed stainless steel
      4.2.2. Back wall: Brushed stainless steel
   4.3. Skirting: Refer to FLS-601 in M50.
   4.4. Flooring: To be agreed with the Architect.
5. Handrails/protection rails:
   5.1. Type/ material: Brushed finish stainless steel grade 1.4301 tubular profiles with machined stainless steel grade 1.4301 brackets.
      5.1.1. Welded connections at corners with matching flat end caps.
   5.2. Fixing: Concealed
6. Lift car doors/ frames/ architraves:
   6.1. Height: Full height internally.
   6.2. Type/ material: Suitable core finished with brushed stainless steel grade 1.4301.
   6.3. Architraves: Standard depth architraves to landing doors.
   6.4. Integrated fire curtains: Refer to FPS-301 in L20
   6.5. Door opening cladding: Brushed stainless steel grade 1.4301 cladding to continue from the lift door surround into the full extent of the formed wall opening
7. Car operating panel and controls/ functionality:
   7.1. Call buttons: To be selected from the manufacturer's standard range within a non-standard panel.
   7.2. Panel:
      7.2.1. Type/ material: Brushed stainless steel grade 1.4301 panel with 2.5mm thick stainless steel grade 1.4301 surround rings forming the push button surrounds.
      7.2.2. Fixing: Panels to be secured flush/ recessed, using concealed methods and with anti-tamper properties to prevent unauthorised access.
   7.3. Push buttons:
      7.3.1. Type/ material: Brushed stainless steel grade 1.4301 vandal-resisting illuminating push buttons.
         7.3.1.1. Indication numbers and letters to be raised tactile digits on push buttons in standard range typeface.
   7.4. Floor indicator: Light emitting diode (LED) dot matrix digital floor indicator display recessed into the panel and finished flush.
   7.5. Manual door openers: Landing door manual openers to be positioned within door frame.
8. Services/ lighting/ ventilation: Refer to the Services Engineer's Information
9. Accessories: Refer to the Service Engineers documentation and as required to complete the installation
10. Systems shall include fixings, framing, bracketry, support framing/ backing materials, grids, seals and all other components/ accessories necessary to complete the works.

## 356 LFT-501 Accessible Platform Lifts

1. Description: Standard free-standing platform lift for accessible access and use.
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Standard: To BS EN 81-41.
4. Size and configuration: Refer to the Design Drawings
5. Structure:
    5.1. Type/ material: Concealed metal framework/ lift structure, Refer to the Structural Engineer's Information
        5.1.1. Finish: Framing, accessories and ironmongery to be stainless steel grade 1.4301.
6. Car finish: To be selected from the manufacturer's standard range.
7. Integrated door: As supplied
8. Power supply: Suitable for single-phase 230 V a.c. 50 Hz operation
9. Battery backup: To return the platform to the lowest landing whilst under maximum working load
10. Accessories: As required to complete the installation

## 357 LFT-141 Vertical Lifting Platform Systems

1. Description: Standard free-standing platform lift for vertical lifting.
2. Manufacturer: To be agreed with the Architect.
    2.1. Product reference: To be agreed with the Architect.
3. Standard: To BS EN 81-41.
4. Size and configuration: Refer to the Design Drawings
5. Structure:
    5.1. Type/ material: Concealed metal framework/ lift structure, Refer to the Structural Engineer's Information
        5.1.1. Finish: Framing, accessories and ironmongery to be stainless steel grade 1.4301.
6. Car finish: To be selected from the manufacturer's standard range.
7. Integrated door: As supplied
8. Power supply: Suitable for single-phase 230 V a.c. 50 Hz operation
9. Battery backup: To return the platform to the lowest landing whilst under maximum working load
10. Accessories: As required to complete the installation

## 358 LFT-142 Vertical Lifting Platform Systems

1. Description: Open platform lift for vertical lifting.
2. Manufacturer: Guldmann, or acceptable equivalent
    2.1. Product reference: Stepless SLP model A.
3. Standard: To BS EN 81-41.
4. Size and configuration: Refer to the Design Drawings
5. Structure:
    5.1. Type/ material: Concealed metal framework/ lift structure, Refer to the Structural Engineer's Information

5.1.1. Finish: Bespoke bronze finishes. Floor finish to match adjacent areas.

6. Car finish: To be selected from the manufacturer's standard range.
7. Integrated door: As supplied
8. Power supply: Suitable for single-phase 230 V a.c. 50 Hz operation
9. Battery backup: To return the platform to the lowest landing whilst under maximum working load
10. Accessories: As required to complete the installation

### 358 LFT-151 Lift Access Point

1. Description: Touchscreen operating panel
2. Manufacturer: KONE, or acceptable equivalent
3. Size and configuration: Refer to the Design Drawings
4. Type/ material: To be selected from the manufacturer's standard range.
5. Finish: To be selected from the manufacturer's standard range.
6. Accessories: As required to complete the installation

## System performance

### 200 Samples

1. Provide the following:
   1.1. Ceiling.
   1.2. Wall panels.
   1.3. Skirting.
   1.4. Floor finishes.
   1.5. Handrail.
   1.6. Mirror.
   1.7. Control panel.
   1.8. Door cladding.

### 205 Performance Generally

1. Achieve a slip resistance value of 36PTV in both dry and wet conditions when carrying out tests using a TRL Pendulum Test.
2. Achieve a minima value of 20 µm RZ when carrying out tests with a surface roughness meter.
3. Lift entrances (car and landing doors) to be fire rated.
4. Lift finishes and components to remain stable and in place with no distortion, reduction in performance, deformity or change in appearance during operation or use.
5. Upon completion and when in use there shall be no rattling, creaking or shaking of any component or finishes system within the lift car.
6. Individual elements of the lift car such as handrail, ceiling panels, control boards and panelling are to be removable when in place for maintenance and replacement purposes.

### 210 Design of vertical lifting platform and homelift systems

1. Design: Complete the design of the vertical lifting platform and homelift systems.
2. Standard: To BS 5900 and BS 6440

3. Environment:
4. Duty cycle (minimum): Ten return journeys per hour
5. Number of levels served (including lowest): Refer to the Design Drawings
6. Rated speed (maximum): 0.15 m/s
7. Level access at lowest level: Via fixed ramp
8. Dwell time: 5.0 s
9. Lighting to landings and enclosed liftways (minimum): 50 lux
10. Proposals: Submit dimensioned drawings, plans, elevations and sections, building loadings, builders' work requirements, controls and wiring diagrams, manufacturers' literature, mounting and fixing details, and schedule of labels.
11. User group: Private dwelling occupant or public

## Products

### 395 Landing controls

1. Manufacturer: Submit proposals
    1.1. Product reference: Submit proposals
2. Functions: Submit proposals

## Execution

### 610 Removing vertical lifting platform and homelift systems

1. Scope: Complete installation

### 620 Workmanship generally

1. Lifts to be manufactured at a specialist factory in full compliance with QA/ QC requirements and within tolerances shown on the Working Drawings.
2. Carry out and survey of lift shafts on-Site prior to commencing manufacture or ensure that lift shafts are built to the required size within stated tolerances.
3. Ensure adequate space is allowed for controls and services in so far as they affect the lift finishes.
4. All components to be flawless and accurately manufactured such that when installed they are a perfect fit, are square, align and operate efficiently.
5. Preform all joints, mitres, etc. Joints to align between horizontal and vertical linings as shown on the Working Drawings.
6. Internal linings to be 1:1000 flatness criteria unless agreed otherwise.
7. All fixings to be hidden (non-visible).
8. Panels to be manufactured ±2mm in all directions with maximum ±1mm twist.
9. Shadow gaps/ plane offset not to exceed ±1mm.

### 630 Layout and location

1. General: Refer to the Design Drawings

# Completion

## 910 Testing and commissioning

1. As recommended/ published by the manufacturer.
2. Operational tests: Undertake.
3. Test certificate: Submit.
    3.1. Number of copies: Two

## 920 Electrical inspection and testing

1. Electrical inspection and testing: In accordance with BS 7671.

## 935 Equipment labelling

1. Switches, controls, enclosures and terminations: Clearly and indelibly label describing their purpose. Identify the off position.

## 940 Spares and consumables

1. Supply the following spares
    1.1. Deliberately operated devices: Two of each type
    1.2. Detectors: Two of each type
    1.3. Protective switches: Two of each type
    1.4. Lamps for enclosure or lift car lighting: Four of each type

## 970 Documentation

1. Operation and maintenance instructions: Submit.
2. Record drawings: Submit.
3. Certificates: Submit.
    3.1. Number of copies: Two
4. Instruction manual: Submit.
    4.1. Number of copies: Two
5. Log book: Individual for each lift.
    5.1. Type: Hardback cover embossed with the lift name and unique lift identification reference with A4 lined paper, minimum 100 pages.

## 975 Training

1. Timing: Before completion.
2. Scope to include
    2.1. Daily lift operation.
    2.2. Routine and general maintenance.
    2.3. Emergency passenger release procedure.

## 980 Maintenance

1. Servicing and maintenance: Undertake.
    1.1. Duration: As recommended by the Manufacturer.

Ω End of Section



make

# Z01
# Various Materials

## To be read with preliminaries/ general conditions.

## 100 Cement/ Aggregate/ Water/ Sand

1. Standards: Comply with
    1.1. BS EN 197-6 Cement. Cement with recycled building materials.
    1.2. BS EN 1008 Mixing water for concrete - Specification for sampling, testing and assessing the suitability of water, including water recovered from processes in the concrete industry, as mixing water for concrete.
    1.3. BS EN 12620 Aggregates for concrete.
    1.4. BS EN 13055 Lightweight aggregates.
    1.5. BS EN 13139 Aggregates for mortar.
    1.6. PD 6682-3 Aggregates. Aggregates for mortar - guidance on the use of BS EN 13139.
    1.7. PD 6682-4 Aggregates. Lightweight aggregates for concrete, mortar and grout - Guidance on the use of BS EN 13055-1.
    1.8. Glassfibre Reinforced Concrete Association (GRCA) recommendations.
2. Materials
    2.1. Sand:
        2.1.1. Minimise organic impurities.
        2.1.2. Restrict the use of deleterious reactive materials containing alkalis in cement.
        2.1.3. Use coarse aggregate or dust/ salt/ silt free natural sand.
        2.1.4. Less than 10% of material to be able to pass through a 150 micron sieve.
        2.1.5. Maintain +/- 20% variation in fineness modulus for facing mix design.
        2.1.6. Remove all soluble material.
        2.1.7. Carefully control mix variables.
    2.2. Water: To remain clean, fresh, potable and impurity free from a known and reliable source.

## 200 Plasterboard

1. Standards: Comply with:
    1.1. BS 8000-8 Workmanship on construction sites. Design and installation of dry lining systems - Code of practice.
    1.2. BS EN 520 Gypsum plasterboards - Definitions, requirements and test methods.
    1.3. GPDA (Gypsum Products Development Association) recommendations.
    1.4. FIS (Finishes and Interiors Sector) recommendations.
2. Materials:
    2.1. Quality:
        2.1.1. To be high performance type of 1500kg/$m^2$ density when used in high humidity/ damp areas and/ or when walls are to receive tiling, linings and the like.
        2.1.2. In areas likely to be wet or very damp use moisture resistant plasterboard.
        2.1.3. Use Type A/ H/ E and P where standard sheets are required.



make

2.1.4. Use Type I in high density areas such as cores where glass fibre is required as reinforcement.

2.1.5. Metal foil backed sheets are not acceptable as a water/ vapour barrier.

2.1.6. Before applying wet render of plaster to moisture resistant boards without suitable pre-treatment with an appropriate bonding agent. Such treatment to be agreed in advance.

2.1.7. All boards to be new.

3. Installation:

3.1. Workmanship:

3.1.1. Multiple layer boarding to be staggered by 600mm.

3.1.2. Boards to be undamaged, clean and dry with no sign of having been wet prior to installation.

3.1.3. Fix to suitably prepared and stable framework/ substrate.

3.1.4. Only install in dry interior areas.

3.1.5. Joints to be tight butts without any lipping, imperfect edges or indentations.

3.1.6. Cuts to be straight and neat.

3.1.7. Cut edges to be lightly sanded, sealed and smooth forming a tight continuous fit.

3.1.8. Avoid using cut boards at perimeter conditions or at edges of penetrations.

3.1.9. Use only recommended and appropriate fixings installed strictly in accordance with the manufacturer's recommendations.

3.1.10. Apply fixings at appropriate centres commencing in the middle of each board.

3.1.11. Final surfaces to be smooth and flat without indentations caused by fixings.

3.1.12. Taper edge joints and butt each sheet with a minimal gap of no more than 2mm.

3.1.13. Stagger joints either side of the partition.

3.1.14. Install appropriate flush beading/ reinforcement at edges, openings, perimeters, interfaces, stress points, etc. where necessary.

3.1.15. Tape and fill all joints/ gaps to provide a smooth, flat and continuous surface finish using 100% secured tape and apply plaster skim coat of appropriate thickness.

## 300 Rigid Sheet Boarding

1. Standards: Comply with:

1.1. BS 476-6 Fire tests on building materials and structures. Method of test for fire propagation for products.

1.2. BS 476-7 Fire tests on building materials and structures. Method of test to determine the classification of the surface spread of flame of products.

1.3. BS 1088 Marine plywood - requirements.

1.4. BS 3757 Specification for Rigid PVC sheet.

1.5. BS EN 300 Oriented Strand Boards (OSB) - definitions, classification and specifications.

1.6. BS EN 309 Particleboards. Definition and classification.

1.7. BS EN 310 Wood-based panels - determination of modulus of elasticity in bending and of bending strength.

1.8. BS EN 311 Wood-based panels - Surface soundness - Test method.

1.9. BS EN 312 Particleboards - specifications.

1.10. BS EN 313-1 Plywood - classification and terminology. Classification.

1.11. BS EN 314-1 Plywood - bonding quality. Test methods.
1.12. BS EN 314-2 Plywood - bonding quality. Requirements.
1.13. BS EN 315 Plywood - tolerances for dimensions.
1.14. BS EN 316 Wood fibreboards - definition, classification and symbols.
1.15. BS EN 322 Wood-based panels - Determination of moisture content.
1.16. BS EN 323 Wood-based panels - Determination of density.
1.17. BS EN 335 Durability of wood and wood-based products - Use classes: definitions, application to solid wood and wood-based products - Durability - Class 3.
1.18. BS EN 438-3 High-pressure decorative laminates (HPL) - sheets based on thermosetting resins (usually called laminates). Classification and specifications for laminates less than 2 mm thick intended for bonding to supporting substrates.
1.19. BS EN 438-4 High-pressure decorative laminates (HPL) - sheets based on thermosetting resins (usually called laminates). Classification and specifications for compact laminates of thickness 2 mm and greater.
1.20. BS EN 438-7 High-pressure decorative laminates (HPL) - sheets based on thermosetting resins (usually called laminates). Compact laminate and HPL composite panels for internal and external wall and ceiling finishes.
1.21. BS EN 622-1 Fibreboards - specifications. General requirements.
1.22. BS EN 622-2 Fibreboards - specifications. Requirements for hardboards.
1.23. BS EN 622-3 Fibreboards - specifications. Requirements for medium boards.
1.24. BS EN 622-4 Fibreboards - Specifications. Requirements for softboards.
1.25. BS EN 622-5 Fibreboards - specifications. Requirements for dry process boards (MDF).
1.26. BS EN 633 Cement-bonded particleboards. Definition and classification.
1.27. BS EN 634-1 Cement-bonded particle boards - specification. General requirements.
1.28. BS EN 634-2 Cement-bonded particleboards - specifications. Requirements for OPC bonded particleboards for use in dry, humid and exterior conditions.
1.29. BS EN 636 Plywood - Specifications.
1.30. BS EN 942 Timber in joinery - general requirements.
1.31. BS EN 12467 Fibre-cement flat sheets - product specification and test methods.
1.32. BS EN 12871 Wood-based panels - determination of performance characteristics for load bearing panels for use in floors, roofs and walls.
1.33. BS EN 14322 Wood-based panels - melamine faced board for interior uses - definition, requirements and classification.
1.34. BRE Digest 477 Part 1 Wood-based panels: oriented strand board (OSB).
1.35. BRE Digest 477 Part 2 Wood-based panels: particleboard (chipboard).
1.36. BRE Digest 477 Part 3 Wood-based panels: cement bonded particleboard.
1.37. BRE Digest 477 Part 4 Wood-based panels: plywood.
1.38. BRE Digest 477 Part 5 Wood-based panels: medium density fibreboard (MDF).
1.39. BRE Digest 477 Part 6 Wood-based panels: hardboard, medium board and softboard.
1.40. BRE Digest 477 Part 7 Wood-based panels: selection.
1.41. CEN/TS 1099 Plywood - Biological durability - Guidance for the assessment of plywood for use in different use classes - grade H or better.
1.42. PD CEN/TR 12872:2014 Wood-based panels - guidance on the use of load-bearing boards in floors, walls and roofs.

1.43. Approved Document Part B - Building Regulations - Bitumen impregnated fibreboard.

1.44. BFP (British Plastic Federation) - recommendations for laminated sheets.

1.45. BLFA (British Laminate Fabricators Association) - recommendations for laminated sheets.

2. Materials:

2.1. General: Board grade to be sufficiently and suitably robust, sound, of suitable class/ thickness and moisture resistant to suit prevailing conditions and use.

2.2. Fibreboard: Only use bitumen impregnated fibreboard with cavity barriers.

2.3. Laminated faced panels:

2.3.1. Core substrate to be of suitable type and quality of either:

- Plywood.
- Grade P5 particleboard.
- MDF.
- Blockboard.
- Laminboard.

2.3.2. Control moisture content to suit prevailing environmental conditions and core material.

2.3.3. Bond sheets in presses.

2.3.4. Fully condition sheets prior to lamination.

2.3.5. Oversize holes to compensate for shrinkage of the laminate.

2.3.6. Final board thickness (including laminate) to provide:

- Sufficient structural strength.
- Impact resistance.
- Stability.

2.3.7. Laminated panels to be balanced to stop warping, bowing, etc.

2.3.8. Allow a minimum of 36 hours atmospheric conditioning prior to commencement of high pressure laminate bonding.

2.3.9. All edges to be smooth cut and match adjoining panels/ surfaces.

2.3.10. Apply laminate to edges first followed by main surfaces.

2.3.11. Edge and top joints to be chamfered at 45°.

2.3.12. Black lines at edges are not acceptable.

2.3.13. Final finish to be from from bowing/ twisting/ chips/ pimpling/ scratches/ cracks/ surface contamination, debris and marks/ depressions/ stains/ colour, texture and pattern irregularities.

2.3.14. Edge conditions to be agreed as either:

- Hardwood lipping.
- Post formed edges.
- 3mm thick PVC in matching or agreed colour.
- Decorative laminate in matching or agreed colour, using radio frequency edgebanders.
- No edging.

3. Workmanship:

3.1. Board fixing:

3.1.1. Run edges along supports.

3.1.2. At 150mm c/c maximum to intermediate supports.

3.1.3. At 100mm c/c maximum to edge boards.

3.1.4. Inset 12mm from edge.

3.2. Plywood:

3.2.1. Take care storing on site to ensure protection from adverse environmental conditions with free airflow maintained at all times.

3.2.2. Fix plywood:

- Long edges running at 90° to supports.
- End joints to be staggered in alignment with supports.
- 150mm c/c at perimeters/ edges.
- 300mm c/c at intermediate supports.
- Use 50mm x 3mm annular ring shank nails.
- Bond floor joints using PVA adhesive.

3.3. Chipboard:

3.3.1. Long edges running at 90° to supports.

3.3.2. End joints to be staggered in alignment with supports.

3.3.3. 300mm c/c at perimeters/ edges/ interim supports.

3.3.4. Use 50mm x 3mm annular ring shank nails.

3.3.5. Bond floor joints using PVA adhesive.

3.3.6. Leave a 10mm gap around all perimeter edges for expansion.

3.4. Laminated panels: No lipping with flush joints and flat surfaces panel to panel in all directions.

## 400 Glass / Coatings

1. Standards: Comply with:

1.1. BS 476-20 Fire tests on building materials and structures. Method for determination of the fire resistance of elements of construction (general principles).

1.2. BS 476-22 Fire tests on building materials and structures. Methods for determination of the fire resistance of non-loadbearing elements of construction.

1.3. BS 952-1 Glass for glazing. Classification.

1.4. BS 952-2 Glass for glazing. Terminology for work on glass.

1.5. BS 5516-1 Patent glazing and sloping glazing for buildings. Code of practice for design and installation of sloping and vertical patent glazing.

1.6. BS 5516-2 Patent glazing and sloping glazing for buildings. Code of practice for sloping glazing.

1.7. BS 6206 Specification for impact performance requirement for flat safety glass and safety plastics for use in buildings (Partially superseded but remains current).

1.8. BS 6262-1 Glazing for buildings. General methodology for the selection of glazing.

1.9. BS 6262-2 Glazing for buildings - Code of practice for energy, light and sound.

1.10. BS 6262-3 Glazing for buildings - Code of practice for fire, security and wind loading.

1.11. BS 6262-4 Glazing for buildings. Code of practice for safety related to human impact.

1.12. BS 6262-6 Glazing for buildings - Code of practice for special applications.

1.13. BS 6262-7 Glazing for buildings. Code of practice for the provision of information.

1.14. BS EN 572-1 Glass in building - basic soda-lime silicate glass products. Definitions and general physical and mechanical properties.

1.15. BS EN 572-2 Glass in building - basic soda lime silicate glass products. Float glass.
1.16. BS EN 572-3 Glass in building - Basic soda lime silicate glass products. Polished wired glass.
1.17. BS EN 572-4 Glass in building - Basic soda lime silicate glass products. Drawn sheet glass.
1.18. BS EN 572-5 Glass in building - Basic soda lime silicate glass products. Patterned glass.
1.19. BS EN 572-6 Glass in building - Basic soda lime silicate glass products. Wired patterned glass.
1.20. BS EN 572-7 Glass in building - Basic soda lime silicate glass products. Wired or unwired channel shaped glass.
1.21. BS EN 572-8 Glass in building - basic soda-lime silicate glass products. Supplied and final cut sizes.
1.22. BS EN 572-9 Glass in building - basic soda lime silicate glass products. Evaluation of conformity/product standard.
1.23. BS EN 1036-1 Glass in building. Mirrors from silver-coated float glass for internal use. Definitions, requirements and test methods.
1.24. BS EN 1036-2 Glass in building - mirrors from silver-coated float glass for internal use - evaluation of conformity; product standard.
1.25. BS EN 1096-1 Glass in building - coated glass. Definitions and classification.
1.26. BS EN 1096-2 Glass in building - coated glass. Requirements and test methods for class A, B and S coatings.
1.27. BS EN 1096-3 Glass in building - coated glass. Requirements and test methods for class C and D coatings.
1.28. BS EN 1096-4 Glass in building - coated glass. Product standard.
1.29. BS EN 1279-1 Glass in building - insulating glass units. Generalities, system description, rules for substitution, tolerances and visual quality.
1.30. BS EN 1279-2 Glass in building - Insulating glass units. Long term test method and requirements for moisture penetration.
1.31. BS EN 1279-3 Glass in building - Insulating glass units. Long term test method and requirements for gas leakage rate and gas concentration tolerances.
1.32. BS EN 1279-4 Glass in building - insulating glass units. Methods of test for the physical attributes of edge seal components and inserts.
1.33. BS EN 1279-5 Glass in building - Insulating glass units. Product standard.
1.34. BS EN 1279-6 Glass in building - insulating glass units. Factory production control and periodic tests.
1.35. BS EN 1288-2 Glass in building - Determination of the bending strength of glass. Coaxial double ring test on flat specimens with large test surface areas.
1.36. BS EN 1288-3 Glass in building - determination of the bending strength of glass. Test with specimen supported at two points (four point bending).
1.37. BS EN 1748-1-1 Glass in building - Special basic products - Borosilicate glasses. Definition and general physical and mechanical properties.
1.38. BS EN 1748-1-2 Glass in building - Special basic products - Borosilicate glasses. Evaluation of conformity/product standard.
1.39. BS EN 1748-2-1 Glass in building. Special basic products. Glass ceramics. Definitions and general physical and mechanical properties.
1.40. BS EN 1748-2-2 Glass in building. Special basic products. Glass ceramics. Evaluation of conformity/Product standard.

1.41. BS EN 1863-1 Glass in buildings - heat strengthened soda lime silicate glass. Definition and description.

1.42. BS EN 1863-2 Glass in building - heat strengthened soda lime silicate glass. Evaluation of conformity/product standard.

1.43. BS EN 1873 Prefabricated accessories for roofing - individual rooflights of plastics - product specification and test methods.

1.44. BS EN 1991-1-1 Eurocode 1: Actions on structures. General actions - Densities, self-weight, imposed loads for buildings.

1.45. BS EN 1991-1-2 Eurocode 1 - Actions on structures. Actions on structures exposed to fire.

1.46. BS EN 1991-1-3 Eurocode 1 - actions on structures. General actions - Snow loads.

1.47. BS EN 1991-1-4 Eurocode 1: Actions on structures. General actions - Wind actions.

1.48. BS EN 1991-1-5 Eurocode 1 - Actions on structures. General actions - Thermal actions.

1.49. BS EN 1991-1-6:2005 Eurocode 1: Actions on structures. General actions - Actions during execution.

1.50. BS EN 1991-1-7 Eurocode 1 - Actions on structures. General actions - Accidental actions.

1.51. BS EN 12150-1 Glass in building - thermally toughened soda lime silicate safety glass. Definition and description.

1.52. BS EN 12150-2 Glass in building - Thermally toughened soda lime silicate safety glass. Evaluation of conformity/Product standard.

1.53. BS EN 12600 Glass in building - Pendulum test - Impact test method and classification for flat glass.

1.54. BS EN 12758 Glass in building - Glazing and airborne sound insulation - Product descriptions, determination of properties and extension rules.

1.55. BS EN 13024-1 Glass in building - Thermally toughened borosilicate safety glass. Definition and description.

1.56. BS EN 13024-2 Glass in building - Thermally toughened borosilicate safety glass. Evaluation of conformity/Product standard.

1.57. BS EN 13541 Glass in building - security glazing - testing and classification of resistance against explosion pressure.

1.58. BS EN 14179-1 Glass in building - heat soaked thermally toughened soda lime silicate safety glass. Definition and description.

1.59. BS EN 14179-2 Glass in building - heat soaked thermally toughened soda lime silicate safety glass. Evaluation of conformity/product standard.

1.60. BS EN 14449 Glass in building - Laminated glass and laminated safety glass. Evaluation of conformity/product standard.

1.61. BS EN 15682-2 Glass in building - heat soaked thermally toughened alkaline earth silicate safety glass. Evaluation of conformity/product standard.

1.62. BS EN 16477-1 Glass in building — Painted glass for internal use Part 1: Requirements.

1.63. BS ISO 11485-1 Glass in building - curved glass. Terminology and definitions.

1.64. BS ISO 11485-2 Glass in building - curved glass. Quality requirements.

1.65. BS ISO 11485-3 Glass in building - Curved glass. Requirements for curved tempered and curved laminated safety glass..

1.66. BS EN ISO 12543-1 Glass in building - laminated glass and laminated safety glass. Vocabulary and description of component parts.

1.67. BS EN ISO 12543-2 Glass in building - laminated glass and laminated safety glass. Laminated safety glass.

1.68. BS EN ISO 12543-3 Glass in building - laminated glass and laminated safety glass. Laminated glass.

1.69. BS EN ISO 12543-4 Glass in building - laminated glass and laminated safety glass. Test methods for durability.

1.70. BS EN ISO 12543-5 Glass in building - laminated glass and laminated safety glass. Dimensions and edge finishing.

1.71. BS EN ISO 12543-6 Glass in building - laminated glass and laminated safety glass. Appearance.

1.72. BS ISO 18543 Glass in building. Electrochromic glazings. Accelerated ageing test and requirements.

1.73. CWCT Technical Note 66 Safety and fragility of glazed roofing: guidance on specification.

1.74. Building Regulation Approved Document Part B - Fire safety.

1.75. Building Regulation Approved Document Part K - Protection.

1.76. Glass and Glazing Federation Safety and Security Glazing - Good Practice Guide.

2. Materials

2.1. General:

2.1.1. Use only float process to manufacture annealed glass.

2.1.2. Glass to be clear, undistorted and free from reflection.

2.1.3. Glass packaging to show place of manufacture.

2.1.4. Glass to be sourced from a single source for each element of the Works.

2.1.5. Only use kitemarked safety glass.

2.1.6. Glass to be free from impurities.

2.1.7. Glass/ glazing systems to be fully warranted and guaranteed to suit design and service life requirements.

2.1.8. Glass/ glazing systems to be suitable for intended use, of an appropriate type/ quality, of adequate thickness and to an acceptable finish.

2.1.9. Glass/ glazing systems to comply with all performance and visual intent requirements.

2.1.10. Mirror glass is not acceptable unless specifically requested.

2.1.11. Glazing installations to be visually consistent and acceptable throughout, maintaining the appearance and quality of all accepted samples, prototypes and benchmarks.

2.1.12. Visual acceptance to be determined by observation/ inspection from a distance of 1000mm at multiple angles with both blue sky and cloudy backgrounds.

2.1.13. The following are unacceptable:

- Visual distortion.
- Air bubbles.
- Imperfections of any kind.
- Smoke vanes.

2.1.14. Glazing systems to be manufactured in specialist facilities using appropriate equipment to ensure pristine accuracy in terms of cutting, finishing, assembly, etc.

2.1.15. Edges, joints, to be smooth, even, ground and arrised to provide a visually acceptable product.

2.1.16. Assemblies to be installed safely into frames within the recommended/ agreed tolerances as shown on the Working Drawings.

2.1.17. Panels to be designed to facilitate individual removal for maintenance/ removal purposes.

2.1.18. Glazing systems are not to exert stress or external force on any component/ assembly in any condition that reduces its performance, visual appearance, shape or safety.

2.1.19. Coatings and finishes to remain pristine.

2.1.20. Carry out full stress analysis to confirm compliance.

2.1.21. Wired glass to be minimum 3mm thick, using clear annealed sheet glass with 1.3mm gauge georgian wire embedded during glass manufacture.

2.1.22. Fire resistant safety glass to be Category B/ Class 2B2 minimum being clear and unwired incorporating fire rated beading.

2.2. Laminated glass:

2.2.1. To be carried out at a specialist facility.

2.2.2. Bond a minimum of two layers of flat glass with a suitable clear interlayer between each panel.

2.2.3. Accurately align each sheet to prevent overlapping at edges.

2.2.4. No distortions are acceptable.

2.2.5. Support bottom edges such that loads are supported evenly along the full length of each unit.

2.2.6. Point loading is not acceptable.

2.2.7. Fully seal all edges.

2.3. Heat strengthening glass:

2.3.1. To be carried out at a specialist facility.

2.3.2. Fully control strengthening process and anisotropy to avoid reduction in performance, quality or visual intent.

2.3.3. Edge dip not to exceed 0.25mm/300mm.

2.3.4. Glass edge bowing to be less than 0.12mm/300mm.

2.3.5. Flatness criteria to be 1mm in 1000mm in any direction and non accumulative over 3000mm.

2.3.6. Minimise roller wave distortion such that it is not visible/ detectable in any direction/ orientation.

2.3.7. Small visible ripples may be accepted provided that they are horizontally consistent and evenly spaced over the whole panel.

2.4. Toughened glass:

2.4.1. Generally unacceptable unless agreed in advance in writing.

2.4.2. Must be heat soaked, tested with full audit traceability and audit trail.

2.4.3. Provide full technical details and guarantee of performance, suitability and adherence to visual intent if toughening is proposed.

2.5. Curved glass:

2.5.1. To be carried out at a specialist facility.

2.5.2. Maintain a consistent radius.

2.5.3. Use specially prepared and appropriate types of glass. Demonstrate compliance through calculations and samples/ prototypes.

2.5.4. Maintain performance requirements, visual intent and quality.

2.5.5. Maintain tight tolerances to guarantee a consistent curve when multiple panels and connected.

2.5.6. Maintain perfect alignment at vertical and horizontal edges which are to be continuous with consistent joint widths.

2.5.7. Maximum allowable deviation in curvature of a complete installation to be a maximum of ±4mm from the theoretical position.

2.6. Mirror glass: To be:

2.6.1. New.

2.6.2. Clean.

2.6.3. Flat.

2.6.4. Clear.

2.6.5. Smooth.

2.6.6. Free of defects, marks or blemishes.

2.6.7. Completely undistorted.

2.7. Horizontal/ inclined glazing:

2.7.1. Use only suitably strengthened glass.

2.7.2. Fix into a suitable and adequate support framing system which is structurally sound.

2.7.3. Fixing/ restraining methods to guarantee a safe installation.

2.7.4. Performance requirements and visual intent to be maintained.

2.7.5. Glass to remain in place (i.e. no pieces of glass or support system will fall to the ground) should damage or breakage occur.

2.7.6. System to be sufficiently sound to resist all live and dead loads to which it is subjected including accidental damage, maintenance, cleaning, etc.

2.7.7. Carry out impact tests to an energy level of 1200J minimum.

## 500 Insulation

1. Standards: Comply with:

1.1. BS EN 13162 Thermal insulation products for buildings - Factory made mineral wool (MW) products - Specification.

1.2. BS EN 13163 Thermal insulation products for buildings - factory made expanded polystyrene (EPS) products - specification.

1.3. BS EN 13164 Thermal insulation products for buildings - factory made extruded polystyrene foam (XPS) products - specification.

1.4. BS EN 13165 Thermal insulation products for buildings - factory made rigid polyurethane foam (PU) products - specification.

1.5. BS EN 13166 Thermal insulation products for buildings - Factory made phenolic foam (PF) products - Specification.

1.6. BS EN 13167 Thermal insulation products for buildings - Factory made cellular glass (CG) products - Specification.

1.7. BS EN 13168 Thermal insulation products for buildings - Factory made wood wool (WW) products - Specification.

1.8. BS EN 13169 Thermal insulation products for buildings - Factory made expanded perlite board (EPB) products - Specification.

1.9. BS EN 13170 Thermal insulation products for buildings - Factory made products of expanded cork (ICB) - Specification.

1.10. BS EN 13171 Thermal insulation products for buildings - Factory made wood fibre (WF) products - Specification.

1.11. BS EN 13501-1 Fire classification of construction products and building elements. Classification using data from reaction to fire tests.

1.12. BS EN 13501-2 Fire classification of construction products and building elements. Classification using data from fire resistance and/or smoke control tests, excluding ventilation services.

1.13. BS EN 13501-3 Fire classification of construction products and building elements. Classification using data from fire resistance tests on products and elements used in building service installations: fire resisting ducts and fire dampers.

1.14. BS EN 13501-4 Fire classification of construction products and building elements. Classification using data from fire resistance tests on components of smoke control systems.

1.15. BS EN 13501-5 Fire classification of construction products and building elements. Classification using data from external fire exposure to roof tests.

1.16. BS EN 13501-6 Fire classification of construction products and building elements. Classification using data from reaction to fire tests on power, control and communication cables.

1.17. BS EN ISO 844 Rigid cellular plastics - Determination of compression properties.

1.18. BS EN ISO 4590 Rigid cellular plastics - determination of the volume percentage of open cells and of closed cells and closed cell percentages.

1.19. BS ISO 1922 Rigid cellular plastics - determination of shear properties.

1.20. BS ISO 4898 Rigid cellular plastics - thermal insulation products for buildings - specifications.

1.21. BS ISO 20310 Thermal insulation for building equipment and industrial installations - aluminosilicate wool products - specification.

1.22. BRE Green Guide to Specification - Sustainability.

2. Materials:

2.1. General: Insulation to be:

2.1.1. Durable, Inert, vermin and rot proof.

2.1.2. Euroclass A2 except in zones between sub-base and ground floor slab or ground floor slab and screed or areas with zero risk of combustion where Euroclass E is acceptable. Agree the use of E class material in advance.

2.1.3. Have sufficient compressive strength to support products, materials, systems, maintenance/ use traffic, temporary loads, etc. to which it is subjected from above without sagging, delamination or bulging.

2.1.4. Provide insulation material that is of a suitable type/ thickness to sustain anticipated use without reduction in performance or stability.

2.1.5. Suitable to receive necessary overlay and finish surface.

2.1.6. Resistant to, water, water vapour/ moisture and extreme temperature ranges.

2.2. EPS: Do not use expanded polystyrene unless agreed in advance.

2.3. Sustainability:

2.3.1. Rating A/ A+ (Green Guide).

2.3.2. Material to be free from CFC (chlorofluorocarbon) and HFC (hydrochlorofluorocarbon).

2.3.3. Material to be free from ODP (Ozone Depleting Potential).

2.3.4. Material to have a GWP (Global Warming Potential) of □5.

2.3.5. In accordance with BREEAM requirements, material to have a minimal environmental impact.

3. Workmanship:

3.1. Lay flat and even with surface contact.

3.2. Provide a solid and stable underlayer at all times and in all conditions.

3.3. Joints to be tight with no lipping or gaps.

## 600 Membranes

1. Standards: Comply with:

1.1. BS 743 Specification for materials for damp proof courses (Partially superseded but remains current).

1.2. BS 1521 Specification for waterproof building papers.

1.3. BS 6014 Ethylene propylene rubber compounds - Specification.

1.4. BS 6398 Specification for bitumen damp-proof courses for masonry.

1.5. BS 6515 Specification for polyethylene damp-proof courses for masonry.

1.6. BS 8000-0 Workmanship on construction sites. Introduction and general principles.

1.7. BS 8000-3 Workmanship on construction sites. Masonry - code of practice.

1.8. BS 8102 Protection of below ground structures against water ingress - code of practice.

1.9. BS 8215 Code of practice for design and installation of damp-proof courses in masonry construction.

1.10. BS 8747 Reinforced bitumen membranes (RBMs) for roofing - guide to selection and specification.

1.11. BS EN 13984 Flexible sheets for waterproofing - plastic and rubber vapour control layers - definitions and characteristics.

2. Materials:

2.1. General: Membranes to be:

2.1.1. Durable, inert, vermin and rot proof.

2.1.2. Non combustible, with Class 0 surface spread of flame.

2.1.3. Airtight.

2.1.4. Water resistant/ impermeable.

2.1.5. Vapour permeable.

2.1.6. Sufficiently flexible to accommodate structural/ environmental movement without any reduction in performance.

2.1.7. Resistant to chemical attack.

2.1.8. Non degradable.

2.1.9. Of sufficient type and gauge to resist puncturing.

2.1.10. Compatible with all materials which it is in direct contact without degradation or reduction in performance.

2.1.11. Agrément certified.

2.1.12. Used as part of a complete waterproofing system.

2.1.13. Compatible with all fixing methods, accessories and components recommended by the manufacturer.

2.2. DPM (Damp Proof Membrane): To be EPDM (Ethylene Propylene Diene Monomer) unless deemed inappropriate by the manufacturer/ Contractor.

2.3. Breather membrane: To be a non woven, superbonded, high density polypropylene unless deemed inappropriate by the manufacturer/ Contractor.

2.4. Vapour control barriers: To be:

2.4.1. High performance reinforced membrane of suitable strength and thickness.

2.4.2. Membrane to be adequately protected on both sides as recommended by the manufacturer.

2.4.3. Moisture vapour transmission (g/□/hour) to be sufficient to satisfy performance requirements.

2.4.4. Moisture vapour resistance (MNs/ g) to be sufficient to satisfy performance requirements.

2.4.5. Bituminous vapour barriers to be a self adhesive or fully bonded polyester based material of suitable type and grade, with an expandable aluminium core and SBS (Styrene Butadiene Styrene) coating.

2.5. Waterproof sheet membranes: To be:

2.5.1. PVC (Polyvinylchloride) or FPO (Flexible Polyolefin).

2.5.2. PVC to be coated on both sides with liquid plastisol.

2.5.3. FPO to be coated on both sides with compounded flexible plasticised polyolefins.

2.5.4. Membrane to be fleece backed.

2.5.5. Suitable for use in conditions of exposure to Ultra Violet light and direct sunlight.

3. Workmanship:

3.1. Installed membranes to provide a watertight barrier at all conditions and interfaces.

3.2. Lay on adequately and suitably prepared base which:

- Sufficiently flat, smooth and without voids or sharp protrusions.
- When laid on hardcore shall be blinded with 12mm sift sand or other suitable material and a 50mm standard mix concrete screed (GEN 1 or ST1/ST2).

3.3. Lay membrane flat, smooth and fully bonded/ mechanically using suitable fixings and flat smooth galvanised steel bars/ washers to substrate without air bubbles, ridges, ripples, etc.

3.4. Membrane to be suitably lapped and sealed at all joints.

3.5. Laps to be staggered with a minimum 75mm overlap at edges and a minimum 150mm at ends.

3.6. Watertight seals to be provided at all interfaces including drains, services and structural penetrations.

3.7. Use adequate protection boards where necessary.

3.8. Tears or punctures to be repaired immediately or replaced to guarantee future performance and watertightness. Report all such instances.

3.9. Use primers as appropriate and necessary in accordance with the manufacturer's recommendations.

3.10. Installation to accommodate without reduction in performance all movement to which it is exposed including those resulting from thermal expansion, moisture, structure, environmental conditions, etc.

3.11. When jointing polyethylene use a continuous mastic strip, appropriate tape and folded welts at upstands.

3.12. Flexible sheet bitumen to be fixed using bitumen lap cement.

3.13. Gas retardant bitumen to be fixed using fully bonded hot bitumen.

3.14. Geocomposite drainage to be sealed with non-combustible fabric at all exposed edges.

3.15. Vapour control layers to be installed to provide an unbroken and continuous barrier. Fix to warm side of thermal insulation material where required.

3.16. DPC and cavity trays to:

- Cover the full wall width including surface finishes.
- Be fully sealed and laid on a bed of fresh pliable mortar.
- Sealed laps to comprise adhesive, mastic or by torching.
- Be installed with special care at any change of direction or level. Measures such as preformed cloaks and welding where appropriate are acceptable provided continuity of performance is maintained at all conditions and details.

## 700 Stone

1. Standards: Comply with:

1.1. BS 8298-2 Design and installation of natural stone cladding and lining. Traditional handset external cladding - code of practice.

1.2. BS 8298-3 Design and installation of natural stone cladding and lining. Stone-faced precast concrete cladding systems - code of practice.

1.3. BS 8298-4 Design and installation of natural stone cladding and lining. Stone cladding on rainscreen support systems - code of practice.

1.4. BS 8298-5 Design and installation of natural stone cladding and lining. Internal cladding and lining - Code of practice.

1.5. BS EN 1341 Slabs of natural stone for external paving - Requirements and test methods.

1.6. BS EN 1468 Natural stone - rough slabs - requirements.

1.7. BS EN 1469 Natural stone products - Slabs for cladding - Requirements.

1.8. BS EN 1925 Natural stone test methods - Determination of water absorption coefficient by capillarity.

1.9. BS EN 1926 Natural stone test methods - determination of uniaxial compressive strength.

1.10. BS EN 1936 Natural stone test methods - determination of real density and apparent density, and of total and open porosity.

1.11. BS EN 12057 Natural stone products - modular tiles - requirements.

1.12. BS EN 12058 Natural stone products - slabs for floors and stairs - requirements.

1.13. BS EN 12059 Natural stone products - dimensional stone work - requirements.

1.14. BS EN 12370 Natural stone test methods - Determination of resistance to salt crystallisation.

1.15. BS EN 12371 Natural stone test methods - Determination of frost resistance.

1.16. BS EN 12372 Natural stone test methods - determination of flexural strength under concentrated load.

1.17. BS EN 12407 Natural stone test methods - petrographic examination.

1.18. BS EN 12440 Natural stone - Denomination criteria.

1.19. BS EN 12670 Natural stone - terminology.

1.20. BS EN 13161 Natural stone test methods - Determination of flexural strength under constant moment.

1.21. BS EN 13755 Natural stone test methods - Determination of water absorption at atmospheric pressure.

1.22. BS EN 14066 Natural stone test methods - determination of resistance to ageing by thermal shock.

1.23. BS EN 14157 Natural stone test methods - determination of the abrasion resistance.

1.24. BS EN ISO 10456 Building materials and products - Hygrothermal properties - Tabulated design values and procedures for determining declared and design thermal values.

1.25. PD CEN/TR 17024 Natural stones - guidance for use of natural stones.

2. Materials:

2.1. Testing:

2.1.1. All tests to be carried out or certified by an independent testing specialist/ laboratory.

2.1.2. Test or demonstrate testing of a minimum of three slabs for each stone type.

2.1.3. Check compliance with samples and technical data provided.

2.1.4. Material to be certified and CE marked to demonstrate compliance with performance and strength requirements.

2.2. Sourcing:

2.2.1. Identify quarry or supplier and facilitate inspection.

2.2.2. Provide samples from identified potential sources.

2.2.3. Confirm adequate quantities available from a single source.

2.2.4. Confirm quality and performance compliance.

2.2.5. Provide sufficient samples within colour and finish range to be selected.

2.2.6. All material used in the Works to be within the agreed range of colour and texture.

2.2.7. Confirm location of preparation, including cutting, finishing and packing.

2.3. Quality: Slabs to be:

2.3.1. Visually acceptable within agreed range.

2.3.2. Of sufficient strength.

2.3.3. Free from imperfections, stains, marking, cracks, holes, chips, etc.

2.3.4. Free from contaminants or harmful constituents.

2.3.5. Grout to be visually acceptable and technically compatible.

2.3.6. Sealer to be visually acceptable and technically compatible.

3. Workmanship

3.1. Install to line and level to provide a flat and horizontal/ vertical surface.

3.2. Alignment of slabs to be precise with no lipping, inconsistent joints or change of level.

3.3. Cuts to be straight and accurate.

3.4. Edges to be straight and smooth with no imperfections.

3.5. No surface marking or visual imperfections.

3.6. Apply cured sealer coats as required to dry, clean and dust free surfaces.

3.7. Sealer to be completely dry before application of further coats.

3.8. Control areas to be sealed so as to maintain consistency of finish and appearance.

3.9. Replace areas not accepted in terms of workmanship quality, finish or visual intent.

3.10. Before polishing, clean and prepare the surfaces so that they can be treated immediately.

3.11. Allow for visual inspection of material at all times.

3.12. Replace any slabs deemed to be:

- Damaged.
- Of insufficient quality.
- Stained.
- Outside of the agreed sample range.
- Unacceptable for failure to comply with the design intent.

## 800 Homogeneous Solid Surfaces

1. Standards: Comply with:
   1.1. ASTM D638/ D638M - Tensile properties of plastics.
   1.2. BS EN ISO 178 Plastics - Determination of flexural properties.
   1.3. BS EN ISO 438-2 High-pressure decorative laminates (HPL) - Sheets based on thermosetting resins (usually called laminates) Part 2: Determination of properties.
   1.4. BS EN ISO 527-1 Plastics - determination of tensile properties. General principles.
   1.5. BS EN ISO 527-2 Plastics - determination of tensile properties. Test conditions for moulding and extrusion plastics.
   1.6. BS EN ISO 527-3 Plastics - determination of tensile properties. Test conditions for films and sheets.
   1.7. BS EN ISO 527-4 Plastics - Determination of tensile properties. Test conditions for isotropic and orthotropic fibre-reinforced plastic composites.
   1.8. BS EN ISO 527-5 Plastics. Determination of tensile properties. Test conditions for unidirectional fibre-reinforced plastic composites.
   1.9. BS EN ISO 1183-1 Plastics - methods for determining the density of non-cellular plastics. Immersion method, liquid pycnometer method and titration method.
   1.10. BS EN ISO 1183-2 Plastics - methods for determining the density of non-cellular plastics. Density gradient column method.
   1.11. BS EN ISO 1183-3 Plastics. Methods for determining the density of non-cellular plastics. Gas pyknometer method.
   1.12. BS EN ISO 4892-2 Plastics - Methods of exposure to laboratory light sources - Part 2: Xenon-arc lamps.
   1.13. DIN 68861-4 Furniture surfaces - Part 4: Behaviour under scratch stress.
   1.14. DIN 68861-7 Furniture surfaces - Part 7: Behaviour subjection to dry heat.
   1.15. DIN 68861-8 Furniture surfaces - Part 8: Behaviour on subjection to wet heat.
   1.16. ISO 4586-1 High-pressure decorative laminates (HPL, HPDL) — Sheets based on thermosetting resins (usually called laminates) - Dimensional stability @ 20°C.
2. Certification: Provide LGA Hygiene Certificate for each material.

## 900 Gaskets

1. Standards: Comply with:
   1.1. BS 4255-1 Rubber used in preformed gaskets for weather exclusion from buildings. Specification for non-cellular gaskets.
   1.2. BS EN 12365-1 Building hardware - gasket and weatherstripping for doors, windows, shutters and curtain walling. Performance requirements and classification.

2. Materials:
   2.1. Provide certification from the manufacturer confirming that products are appropriate and suitable for their intended purpose.
   2.2. Manufacture from suitable material of appropriate grade and hardness characteristics.
   2.3. Accommodate maximum movement when fixed in position without any distortion or reduction in performance.
   2.4. Provide required acoustic, air-tightness and weather tightness performance characteristics.
   2.5. Heat seal butt joints to gasket joints where necessary.
   2.6. Frame corners to be pre-formed and factory vulcanised in ladders.
   2.7. Colour to be black unless agreed or instructed otherwise.
   2.8. Extrusion to be designed to best suit the condition using extruded and moulded elements, all to comply with the visual intent.
   2.9. Appear and perform as an unbroken and unjointed single element.
   2.10. Maintain elasticity throughout Service Life of the product.
   2.11. Be compatible with all substrates, interface materials and components.
   2.12. Avoid contact with materials/ products which may cause staining.
   2.13. Be resistant to attach from chemical, insect or any other form of attack
   2.14. Do not shrink, warp, distort, deteriorate or disfigure under any circumstance during Service Life.
   2.15. Retain glazed elements under loading and use.

Ω End of Section

# Z10
# Purpose Made Joinery

## To be read with preliminaries/ general conditions.

## 100 Standards

1. Comply with:
   1.1. BS 1186-2 Timber for and workmanship in joinery. Specification for workmanship.
   1.2. BS 4978 Visual strength grading of softwood - specification.
   1.3. BS 5756 Visual strength grading of temperate hardwood - Specification.
   1.4. BS EN 204 Classification of thermoplastic wood adhesives for non-structural applications.
   1.5. BS EN 338 Structural timber - strength classes.
   1.6. BS EN 350 Durability of wood and wood-based products - testing and classification of the durability to biological agents of wood and wood-based materials.
   1.7. BS EN 942 Timber in joinery - general requirements.
   1.8. BS EN 1313-1 Round and sawn timber - permitted deviations and preferred sizes. Softwood sawn timber.
   1.9. BS EN 1313-2 Round and sawn timber - permitted deviations and preferred sizes. Hardwood sawn timber.
   1.10. BS EN 1912 Structural timber - Strength classes - Assignment of visual grades and species.
   1.11. BS EN 1995 series of standards - Timber structure design.
   1.12. BS EN 13556 Round and sawn timber - Nomenclature of timbers used in Europe.
   1.13. BS EN 13986 Wood-based panels for use in construction - characteristics, evaluation of conformity and marking.
   1.14. BS EN 14081-1 Timber structures - strength graded structural timber with rectangular cross section. General requirements.
   1.15. BS EN 14081-2 Timber structures - strength graded structural timber with rectangular cross section. Machine grading; additional requirements for type testing.
   1.16. BS EN 14081-3 Timber structures - strength graded structural timber with rectangular cross section. Machine grading; additional requirements for factory production control.
   1.17. BS EN 14220 Timber and wood-based materials in external windows, external door leaves and external doorframes - Requirements and specifications.
   1.18. BS EN 14221 Timber and wood-based materials in internal windows, internal door leaves and internal doorframes - Requirements and specifications.
   1.19. BS EN 16737 Structural timber - visual strength grading of tropical hardwood.

## 105 Timber

1. Suitability: Use only suitable dried, marked and milled wood.
2. Quality: Straight, defect free, knot free, crack free with no discolouration, insect holes or distortion.
3. Class: E1 - zero formaldehyde release.
4. Hardwood: Kiln dried, high grade, Class J2 with controlled moisture content of between 8% and 12%. Mill from quarter sawn stock.

## 110 Fabrication

1. Standard: To BS 1186-2 Timber for and workmanship in joinery. Specification for workmanship.
2. Sections: Accurate in profile and length, and free from twist and bowing. Formed out of solid unless shown otherwise.
    2.1. Machined surfaces: Smooth and free from tearing, wooliness, chip bruising and other machining defects.
3. Joints: Tight and close fitting.
4. Assembled components: Rigid. Free from distortion.
5. Screws: Provide pilot holes.
    5.1. Screws of 8 gauge (4 mm diameter) or more and screws into hardwood: Provide clearance holes.
    5.2. Countersink screws: Heads sunk at least 2 mm below surfaces visible in completed work.
6. Adhesives: Compatible with wood preservatives applied and end uses of timber.
7. Repairs: Replace and do not repair visible/ semi visible Class J2/ J5/ CSH timber.

## 120 Cross section dimensions of timber

1. General: Dimensions on Working Drawings are finished sizes.
2. Maximum permitted deviations from finished sizes
    2.1. Softwood sections: To BS EN 1313-1 Round and sawn timber - permitted deviations and preferred sizes. Softwood sawn timber:
        2.1.1. Clause 6 for sawn sections.
    2.2. Hardwood sections: To BS EN 1313-2 Round and sawn timber - permitted deviations and preferred sizes. Hardwood sawn timber:
        2.2.1. Clause 6 for sawn sections.
        2.2.2. Clause NA.3 for further processed sections.

## 130 Preservative treated wood

1. Cutting and machining: Completed as far as possible before treatment.
2. Extensively processed timber: Retreat timber sawn lengthways, thicknessed, planed, ploughed, etc.
3. Surfaces exposed by minor cutting and/ or drilling: Treat as recommended by main treatment solution manufacturer.
4. General:
    4.1. Apply before cutting, notching, concealing, ploughing, sawing, planning or processing.
    4.2. Retreat areas again after machining, cutting, notching, concealing, ploughing, sawing, planning or processing.
    4.3. Pay special attention to cut edges, including drilled holes.
    4.4. Apply a minimum of two coats.

## 140 Moisture content

1. Wood and wood based products: Maintained within range specified for the component during manufacture and storage.
2. Control: Maintain within an acceptable range during all stages of manufacture and installation to maintain stability and minimise movement/ expansion/ contraction/ shrinkage.

3. Temperature/ humidity: Maintain consistent and acceptable temperature/ humidity conditions through all stages of delivery, manufacture, storage and installation. Take regular meter readings and report.

## 210 Laminated plastics veneered boards/ panels

1. Fabrication: To British Laminated Plastics Fabricators Association (BLFA) fabricating standards.
2. Balancing veneer: From decorative veneer manufacturer and of similar composition. Applied to reverse side of core material.
3. Finished components: Free from defects, including bow, twist, scratches, chipping, cracks, pimpling, indentations, glue marks, staining and variations in colour and pattern.
4. Joints visible in completed work: Tight butted, true and flush.
5. Preperation: Suitable and fully prepared surfaces to receive fully bonded veneer.

## 220 Wood veneered boards/ panels

1. Core material and veneers: Conditioned before bonding.
2. Setting out: Veneer features and grain pattern aligned regularly and symmetrically unless instructed otherwise.
3. Balancing veneer: Applied to reverse side of core material.
    3.1. Moisture and temperature movement characteristics: As facing veneer.
4. Veneer edges: Tight butted and flush, with no gaps.
5. Tolerance of veneer thickness (maximum): ± 0.5 mm.
6. Finished components: Free from defects, including bow, twist, scratches, chipping, splits, blebs, indentations, glue marks and staining.
7. Surface finish: Fine, smooth, free from sanding marks, no visible defects (including telegraphing). Carefully sand where agreed in advance without damaging or changing appearance.
8. Veneer Selection requirements: Must include grain orientation requirements:. Where not indicated on the Design Drawings, obtain confirmation of grain orientation from the Architect.

## 250 Finishing

1. Surfaces: Smooth, even and suitable to receive finishes.
    1.1. Arrises: Eased unless shown otherwise on Working Drawings.
2. End grain in external components: Sealed with primer or sealer as section M60 and allowed to dry before assembly.
3. Final treatment: To be smooth, flat and colour consistent.

Ω End of Section

# Z11
# Purpose Made Metalwork

## To be read with preliminaries/ general conditions.

### 310 Materials generally

1. Grades of metals, section dimensions and properties: To appropriate British Standards. When not specified, select grades and sections appropriate for the purpose.
2. Prefinished metal: May be used if methods of fabrication do not damage or alter appearance of finish, and finish is adequately protected.
3. Fasteners: To appropriate British Standards and, unless specified otherwise, of same metal as component being fastened, with matching coating or finish.
4. Design: To BS EN 1993 series of standards.
5. Quality/ workmanship:
    5.1. Vent holes in hollow sections to be completely sealed.
    5.2. All permanent connections to be fitted accurately and neatly.
    5.3. No distortion is acceptable, including that caused by welding.
    5.4. All welds to be shown on the Working Drawings.
    5.5. Visible lines/ depressions on external surfaces resulting from internal welds to hollow sections are not acceptable.
6. Surface preparation:
    6.1. Remove all traces of rust, scale and surface contamination.
    6.2. Surface finish quality to be Sa 2.5 as BS EN ISO 8501 series of standards.
    6.3. Do not Acid Pickle to remove contaminants.
7. Lettering: Raised lettering, hard stamping and the like will not be acceptable where visible in the finished Works.

### 320 Steel long and flat products

1. Hot rolled structural steels (excluding structural hollow sections and tubes): To BS EN 10025-1 Hot rolled products of structural steels. General technical delivery conditions.
2. Fine grain steels, including special steels:
    2.1. To BS EN 10025-3 Hot rolled products of structural steels. Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels.
    2.2. To BS EN 10025-4 Hot rolled products of structural steels. Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels..
3. Steels with improved atmospheric corrosion resistance: To BS EN 10025-5:2019 Hot rolled products of structural steels. Technical delivery conditions for structural steels with improved atmospheric corrosion resistance.

### 330 Steel plate, sheet and strip

1. Plates and wide flats, high yield strength steel: To BS EN 10025-6 Hot rolled products of structural steels. Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition.

## 340 Hot rolled steel plate, sheet and strip

1. Flat products, high yield strength for cold forming:
    1.1. To BS EN 10149-1 Hot rolled flat products made of high yield strength steels for cold forming. General technical delivery conditions.
    1.2. To BS EN 10149-2 Hot rolled flat products made of high yield strength steels for cold forming. Technical delivery conditions for thermomechanically rolled steels.
    1.3. To BS EN 10149-3 Hot rolled flat products made of high yield strength steels for cold forming. Technical delivery conditions for normalized or normalized rolled steels.
2. Carbon steel sheet and strip for cold forming: To BS EN 10111 Continuously hot rolled low carbon steel sheet and strip for cold forming - technical delivery conditions.
3. Narrow strip, formable steel and steel for general engineering purposes:
    3.1. To BS 1449-1-8 Steel plate, sheet and strip. Carbon and carbon-manganese plate, sheet and strip. Specification for hot rolled narrow strip based on formability.
    3.2. To BS 1449-1-14 Steel plate, sheet and strip. Carbon and carbon-manganese plate, sheet and strip. Specification for hot rolled narrow strip supplied in a range of conditions for heat treatment and general engineering purposes.

## 350 Cold rolled steel plate, sheet and strip

1. Steel sections: To BS EN 10162 Cold rolled steel sections - Technical delivery conditions - Dimensions and cross-sectional tolerances.
2. Flat products, high yield strength micro-alloyed steels for cold forming: To BS EN 10268 Cold rolled steel flat products with high yield strength for cold forming - technical delivery conditions.
3. Carbon steel flat products for cold forming:
    3.1. To BS EN 10130 Cold rolled low carbon steel flat products for cold forming - Technical delivery conditions.
    3.2. To BS EN 10131 Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming - Tolerances on dimensions and shape.
4. Uncoated carbon steel narrow strip for cold forming:
    4.1. To BS EN 10139 Cold rolled uncoated low carbon steel narrow strip for cold forming - technical delivery conditions.
    4.2. To BS EN 10140 Cold rolled narrow steel strip. Tolerances on dimensions and shape.
5. Narrow strip steel for general engineering purposes: To BS EN 10132 Cold rolled narrow steel strip for heat treatment - technical delivery conditions.
6. Carbon steel flat products for vitreous enamelling: To BS EN 10209 Cold-rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions.

## 360 Coated steel flat products

1. Hot dip zinc coated carbon steel sheet and strip for cold forming:
    1.1. To BS EN 10143 Continuously hot-dip metal coated steel sheet and strip - tolerances on dimensions and shape.
    1.2. To BS EN 10346 Continuously hot-dip coated steel flat products for cold forming - technical delivery conditions.
2. Hot dip zinc coated structural steel sheet and strip:

2.1. To BS EN 10143 Continuously hot-dip metal coated steel sheet and strip - tolerances on dimensions and shape.

2.2. To BS EN 10346 Continuously hot-dip coated steel flat products for cold forming - technical delivery conditions.

3. Hot dip zinc-aluminium (za) coated sheet and strip: To BS EN 10346 Continuously hot-dip coated steel flat products for cold forming - technical delivery conditions.
4. Hot dip aluminium-zinc (az) coated sheet and strip: To BS EN 10346 Continuously hot-dip coated steel flat products for cold forming - technical delivery conditions.
5. Organic coated flat products: To BS EN 10169 Continuously organic coated (coil coated) steel flat products - technical delivery conditions.

## 370 Steel structural hollow sections (SHS)

1. Non alloy and fine grain steels, hot finished:
   1.1. To BS EN 10210-1 Hot finished structural hollow sections of non-alloy and fine grain steels. Technical delivery conditions.
   1.2. To BS EN 10210-2 Hot finished structural hollow sections. Tolerances, dimensions and sectional properties.
2. Non-alloy and fine grain steels, cold formed welded: To BS EN 10219-2 Cold formed welded steel structural hollow sections. Tolerances, dimensions and sectional properties.
3. Weather resistant steels, hot finished: To BS 7668 Weldable structural steels - hot finished structural hollow sections in weather resistant steels - specification.

## 380 Other steel sections

1. Equal flange tees: To BS EN 10055 Hot rolled steel equal flange tees with radiused root and toes - Dimensions and tolerances on shape and dimensions.
2. Equal and unequal angles:
   2.1. To BS EN 10056-1 Structural steel equal and unequal leg angles. Dimensions.
   2.2. To BS EN 10056-2 Specification for structural steel equal and unequal leg angles. Tolerances on shape and dimensions.
3. Wire, carbon steel for general engineering purposes: To BS 1052 Specification for mild steel wire for general engineering purposes.
4. Wire and wire products, general: To BS EN 10218-2 Steel wire and wire products. General. Wire dimensions and tolerances.
5. Tubes
   5.1. Seamless circular: To BS EN 10297-1 Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions. Non-alloy and alloy steel tubes.
   5.2. Seamless cold drawn: To BS EN 10305-1 Steel tubes for precision applications - technical delivery conditions. Seamless cold drawn tubes.
   5.3. Welded and cold sized square and rectangular: To BS EN 10305-3 Steel tubes for precision applications - Technical delivery conditions. Welded cold sized tubes.
   5.4. Welded circular: To BS EN 10296-1 Welded circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions. Non-alloy and alloy steel tubes.
   5.5. Welded cold drawn: To BS EN 10305-2 Steel tubes for precision applications - technical delivery conditions. Welded cold drawn tubes.

5.6. Welded cold sized: To BS EN 10305-3 Steel tubes for precision applications - Technical delivery conditions. Welded cold sized tubes.

## 400 Stainless steel products

1. Chemical composition and physical properties: To BS EN 10088-1 Stainless steels. List of stainless steels.
2. Sheet, strip and plate: To BS EN 10088-2 Stainless steels. Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes.
3. Semi-finished products bars, rods and sections: To BS EN 10088-3 Stainless steels. Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes.
4. Wire: To BS EN 10088-3 Stainless steels. Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes.
5. Tubes
   5.1. Welded circular: To BS EN 10296-2 Welded circular steel tubes for mechanical and general engineering purposes - technical delivery conditions. Stainless steel.
   5.2. Seamless circular: To BS EN 10297-2 Seamless circular steel tubes for mechanical and general engineering purposes - technical delivery conditions. Stainless steel.
6. Fasteners:
   6.1. To BS EN ISO 3506-1 Fasteners - mechanical properties of corrosion-resistant stainless steel fasteners. Bolts, screws and studs with specified grades and property classes.
   6.2. To BS EN ISO 3506-2 Fasteners - mechanical properties of corrosion-resistant stainless steel fasteners. Nuts with specified grades and property classes.
   6.3. To BS EN 10263-5 Steel rod, bars and wire for cold heading and cold extrusion - Part 5: Technical delivery conditions for stainless steels - wire and cold forged.
7. Tolerances:
   7.1. To BS EN ISO 9445-1 Continuously cold-rolled stainless steel - tolerances on dimensions and form. Narrow strip and cut lengths.
   7.2. To BS EN ISO 9445-2 Continuously cold-rolled stainless steel - tolerances on dimensions and form. Wide strip and plate/sheet.
   7.3. To BS EN 10029 Hot-rolled steel plates 3mm thick or above - Tolerances on dimensions and shape.
   7.4. To BS EN 10048 Hot rolled narrow steel strip - tolerances on dimensions and shape.
   7.5. To BS EN 10051 Continuously hot-rolled strip and plate/sheet cut from wide strip of non-alloy and alloy steels - Tolerances on dimensions and shape.
   7.6. To BS EN 10095 Heat resisting steels and nickel alloys.
8. Electrodes: To BS EN ISO 3581 Welding consumables - Covered electrodes for manual metal arc welding of stainless and heat-resisting steels - Classification.
9. Forgings: To BS EN 10250-4 Open die steel forgings for general engineering purposes. Stainless steels.
10. Masonry wall ties: To BS EN 845-1:2013 Specification for ancillary components for masonry. Wall ties, tension straps, hangers and brackets.
11. Castings:
    11.1. To BS EN 10283 Corrosion resistant steel castings.

11.2. To BS 3146-2 Specification for Investment Castings in Metal - Part 2: Corrosion and Heat Resisting Steels, Nickel and Cobalt Base Alloys.

12. Surface finish:
    12.1. Avoid cracking, tensile stress, corrosion, residual stress, etc.
    12.2. Utilise lost wax manufacturing process or acceptable equivalent, avoiding exposed feeder portland die lines.
    12.3. Acceptable finishing processes to include at least two of the following:
        12.3.1. Bead blasting.
        12.3.2. Electropolishing.
        12.3.3. Blasting.
        12.3.4. Acid pickling.
    12.4. Treatment:
        12.4.1. Brushed 240 grit at internal locations.
        12.4.2. Brushed 2K with less than 0.5 microns Ra at external locations.
        12.4.3. Do not use aluminium oxide abrasives.
13. Surface Roughness: To ASTM A802 Standard Practice for Steel Castings, Surface Acceptance Standards, Visual Examination - SCRATA A2.
14. Material grades (minimum):
    14.1. Grade 1.4401 - Use in visible external areas (including fixings).
    14.2. Grade 1.4301 - Use in non-visible internal areas.
    14.3. Grade 1.4408 - Corrosion resistant castings.
15. Cleaning: Comply with BSSA (British Stainless Steel Association) recommendations. In particular those relating to:
    15.1. Characteristic resistivity.
    15.2. Preventing ferritic contamination.
16. Terne coating:
    16.1. To BS EN 502 Roofing products from metal sheet - specification for fully supported roofing products of stainless steel sheet.
    16.2. To BS EN 508-3 Roofing and cladding products from metal sheet - Specification for self-supporting products of steel, aluminium or stainless steel sheet. Stainless steel.
    16.3. Continuously hot dip in lead tin alloy or a continuous coating of tin using the electrodeposition method to a uniform finish.
17. Weathering steel:
    17.1. To ASTM A242/A242M Standard specification for high-strength low-alloy structural steel.
    17.2. To ASTM A588/A588M Standard specification for high-strength low-alloy structural steel, up to 50 ksi [345 Mpa] minimum yield point, with atmospheric corrosion resistance.
    17.3. To ASTM A606/A606M Standard specification for steel, sheet and strip, high-strength, low-alloy, hot-rolled and cold-rolled, with improved atmospheric corrosion resistance.
    17.4. To BS EN 10025 series of standards.
    17.5. Achieve:
        17.5.1. Low alloy.
        17.5.2. High strength.
        17.5.3. Resistance to atmospheric corrosion.

17.6. Blast clean (or pickle)surfaces to remove mill scale.

17.7. Avoid contamination from grease, oil or other shop marking process.

17.8. Provide oxide finish to components, fixings, welds and perforations.

## 410 Aluminium alloy products

1. Designations
   1.1. Designation system, chemical composition and forms:
      1.1.1. To BS EN 573-1 Aluminium and aluminium alloys - Chemical composition and form of wrought products. Numerical designation system.
      1.1.2. To BS EN 573-2 Aluminium and aluminium alloys - chemical composition and form of wrought products. Chemical symbol based designation system.
      1.1.3. To BS EN 573-3 Aluminium and aluminium alloys - Chemical composition and form of wrought products. Chemical composition and form of products.
      1.1.4. To BS EN 573-5 Aluminium and aluminium alloys - chemical composition and form of wrought products. Codification of standardized wrought products.
   1.2. Temper designations: To BS EN 515 Aluminium and aluminium alloys - wrought products - temper designations.
2. Sheet, strip and plate:
   2.1. To BS EN 485-1 Aluminium and aluminium alloys - sheet, strip and plate. Technical conditions for inspection and delivery.
   2.2. To BS EN 485-2 Aluminium and aluminium alloys - sheet, strip and plate. Mechanical properties.
   2.3. To BS EN 485-3 Aluminium and aluminium alloys - sheet, strip and plate. Tolerances on dimensions and form for hot-rolled products.
   2.4. To BS EN 485-4 Aluminium and aluminium alloys - Sheet, strip and plate. Tolerances on shape and dimensions for cold-rolled products.
3. Cold drawn rods, bars and tubes:
   3.1. To BS EN 754-1 Aluminium and aluminium alloys - cold drawn rod/bar and tube. Technical conditions for inspection and delivery.
   3.2. To BS EN 754-2 Aluminium and aluminium alloys - cold drawn rod/bar and tube. Mechanical properties.
4. Extruded rods, bars, tubes and profiles:
   4.1. To BS EN 755-1 Aluminium and aluminium alloys - extruded rod/bar, tube and profiles. Technical conditions for inspection and delivery.
   4.2. To BS EN 755-2 Aluminium and aluminium alloys - extruded rod/bar, tube and profiles. Mechanical properties.
5. Drawn wire:
   5.1. To BS EN 1301-1 Aluminium and aluminium alloys. Drawn wire. Technical conditions for inspection and delivery.
   5.2. To BS EN 1301-2 Aluminium and aluminium alloys - drawn wire. Mechanical properties.
   5.3. To BS EN 1301-3 Aluminium and aluminium alloys. Drawn wire. Tolerances on dimensions.
6. Rivet, bolt and screw stock: To BS 1473 Wrought aluminium and aluminium alloys - For general engineering purposes - Rivet, bolt and screw stock.
7. Structural sections: To BS 1161 Specification for aluminium alloy sections for structural purposes.

8. Alloy profiles:
    8.1. To BS EN 12020-1 Aluminium and aluminium alloys - Extruded precision profiles in alloys EN AW-6060 and EN AW-6063. Technical conditions for inspection and delivery.
    8.2. To BS EN 12020-2 Aluminium and aluminium alloys - Extruded precision profiles in alloys EN AW-6060 and EN AW-6063. Tolerances on dimensions and form.
9. Structural design: To BS EN 1999 series of standards and PD 6702-1 Structural use of aluminium. Recommendations for the design of aluminium structures to BS EN 1999.
10. Sheet thickness: 3mm unless specified or agreed otherwise.
11. Alloys: Use relevant grade with profiles EN AW-6060 and/ or EN AW-6063. Use J57S for flat panels to be anodised.
12. Die drawings: Submit as Working Drawings.
13. Protection: Low tack adhesive film.
14. Grain in panels: To run in same direction.
15. Bracketry/ cleats, etc: Use alloy of suitable grade to visually match attached and surrounding panels and frames.
16. Unacceptable items: No warping, sagging, dimpling, bowing, pillowing, rippling, oil canning, visible irregularities/ discolouration, lipping/ irregular transitions at joints, etc., or any other visible irregularities.

## 420 Copper alloy products

1. Sheet, strip, plate and circles for general purposes: To BS EN 1652 Copper and copper alloys - plate, sheet, strip and circles for general purposes.
2. Sheet and strip for building purposes: To BS EN 1172 Copper and copper alloys - sheet and strip for building purposes.
3. Rods: To BS EN 12163 Copper and copper alloys - rod for general purposes.
4. Profiles and rectangular bars: To BS EN 12167 Copper and copper alloys - profiles and bars for general purposes.
5. Wire: To BS EN 12166 Copper and copper alloys - wire for general purposes.
6. Tubes: To BS EN 12449 Copper and copper alloys - seamless, round tubes for general purposes.
7. Springs/ connectors: To BS EN 1654 Copper and copper alloys - Strip for springs and connectors.
8. Forging stock (wrought/ unwrought): To BS EN 12165 Copper and copper alloys - Wrought and unwrought forging stock.
9. Forgings: To BS EN 12420 Copper and Copper Alloys - Forgings.
10. Copper sheet roofs/ wall coverings: To CP 143-12 Sheet roof and wall coverings. Copper. Metric units.
11. Trade Association: To recommendations of CDA (Copper Development Association).
12. Appearance: Colour, appearance and texture consistent with naturally aged copper composition. Visible pinholes, inclusions, laminations, etc. are not acceptable.

## 425 Zinc products

1. Standards:
    1.1. BS EN 501 Roofing products from metal sheet - specification for fully supported roofing products of zinc sheet.
    1.2. BS EN 988 Zinc and zinc alloys - specification for rolled flat products for building.

1.3. CP 143-5:1964 Sheet roof and wall coverings. Zinc.

1.4. Recommendations of ZIC (Zinc Information Centre).

2. Quality: Pinholes, laminations and inclusions are not acceptable.

# Fabrication

## 515 Fabrication generally

1. Contact between dissimilar metals in components: Avoid.
2. Finished components: Rigid and free from distortion, cracks, burrs and sharp arrises.
   2.1. Moving parts: Free moving without binding.
3. Corner junctions of identical sections: Mitre.
4. Aluminium/ metal sheets: To be ridged fully welded/ sealed, tempered, and visually flat (1:1000 over 3000mm in all directions), without distortion.
5. Corners: Corners to comprise purpose made components and not cut unless shown otherwise on the Working Drawings. Colour and finish to match adjoining elements.
6. Protection: Provide adequate protection at all times.
7. Insulation: Provide, where necessary, to negate drumming and other noise.
8. Joints: To be concealed, continuous and sealed.
9. Ends: Aluminium ends to be mill finished. Metal ends to be hot dipped galvanised.

## 520 Cold formed work

1. Profiles: Accurate, with straight arrises.

## 525 Adhesive bonding

1. Preparation of surfaces of metals to receive adhesives
   1.1. Degrease.
   1.2. Abrade mechanically or chemically etch.
   1.3. Prime: To suit adhesive.
2. Adhesive bond: Form under pressure.

## 527 Welding

1. Description: Continuous and suitable.
2. Welding procedures
   2.1. Method and standard: Full penetration welds, flush ground where visible. Use jigs/ clamps, as necessary, to ensure a precise fit.
   2.2. Welding Procedure Specification (WPS): Provide written procedure statement.
3. Preparation
   3.1. Joint preparation: Clean thoroughly.
   3.2. Surfaces of materials that will be self-finished and visible in the completed work: protect from weld splatter, residue build up and splatter.
4. Jointing
   4.1. Joints: Fully bond parent and filler metal throughout with no inclusions, holes, porosity or cracks.
   4.2. Dissimilar metals: Use appropriate material and techniques.

4.3. Strength requirements: Welds to achieve design loads.

4.4. Heat straightening: Only when agreed in advance.

4.5. Complex assemblies: Agree priority for welding members to minimize distortion caused by subsequent welds.

4.6. Tack welds: Use only for temporary attachment.

4.7. Jigs: Provide to support and restrain members during welding.

4.8. Filler plates: Obtain approval.

4.9. Lap joints: Minimum 5 x metal thickness or 25 mm, whichever is greater.

4.10. Weld terminations: Clean and sound.

4.11. Workmanship: Visible holes, inclusions, ridges, cracks, undercuts or porosity are not acceptable.

4.12. Stainless steel welds:

4.12.1. Grind to leave smooth and seamless surface.

4.12.2. Remove heat tints using light abrasives, wire brushing, pickling paste or other similar and acceptable method.

4.12.3. Comply with BS 499 series of standards and BS EN 1011 series of standards.

4.12.4. Minimise distortion during welding.

4.12.5. All welds, methods and consumables to achieve, as a minimum, the same level of corrosion resistance as the base material.

## 530 Stainless steel fabrication

1. Guillotining or punching: Do not use for metal thicknesses greater that 10 mm.
2. Thermal cutting
   2.1. Carbonation in the heat affected zone: Remove, after cutting.
3. Bending
   3.1. Plates or bars: Cold bending radius not less than material thickness.
   3.2. Tubes: Cold bending radius not less than 2 x tube diameter.
4. Welding: In addition to general welding requirements:
   4.1. Protect adjacent surfaces from weld spatter.
   4.2. Pickle all welds before post fabrication treatments.
5. Protection: Provide protection to fabricated components during transit and on site.
6. Corrosion: Not acceptable.

## 555 Brazing

1. Standard: To BS EN 14324 Brazing. Guidance on the application of brazed joints.
2. Testing
   2.1. Destructive testing: To BS EN 12797 Brazing - Destructive tests of brazed joints.
   2.2. Nondestructive testing: To BS EN 12799 Brazing. Non-destructive examination of brazed joints.

## 620 Workmanship general

1. Standards:

1.1. To BS EN 1011-1 Welding - Recommendations for welding of metallic materials. General guidance for arc welding and BS EN 1011-2 Welding - Recommendations for welding of metallic materials. Arc welding of ferritic steels.

1.2. To BS EN ISO 15614-1 Specification and qualification of welding procedures for metallic materials - welding procedure test. Arc and gas welding of steels and arc welding of nickel and nickel alloys.

1.3. To BS EN 14324 Brazing. Guidance on the application of brazed joints.

1.4. To BS EN ISO 17672 Brazing - Filler metals.

1.5. To BS EN ISO 9606-1 Qualification testing of welders - Fusion welding. Steels.

1.6. To BS EN ISO 9606-2 Qualification test of welders - Fusion welding. Aluminium and aluminium alloys.

1.7. To the NFRC (National Federation of Roofing Contractors) Profiled sheet roofing and cladding. The guide to design and best practice.

1.8. To MCRMA (Metal Cladding and Roofing Manufacturers Association) Technical Paper 6 Profiled metal roofing design guide.

2. Dissimilar metals:
   2.1. Keep apart where moisture is present or likely to occur.
   2.2. Prevent bimetallic corrosion between dissimilar materials.
   2.3. Retain connectivity to lightning protection system as necessary.
3. Coatings:
   3.1. Apply post fabrication and drilling.
   3.2. Thoroughly clean to remove all contaminants and blemishes prior to application.
   3.3. Repair defects prior to application.
   3.4. Apply only to smooth and thoroughly prepared surfaces ready to receive coating.
   3.5. Final finish to be smooth, blemish free and acceptable.
4. Finished components: Remain rigid and free from cracking, imprints, distortion, burrs, rolling marks, sharp arrises.

## Finishing

### 710 Finishing welded and brazed joints visible in complete work

1. Standard: To BS EN ISO 8501-3 Preparation of steel substrates before application of paints and related products. Visual assessment of surface cleanliness. Preparation grades of welds, edges and other areas with surface imperfections.
   1.1. Preparation grade: To be proposed by Contractor and agreed.
2. Butt joints: Smooth, and flush with adjacent surfaces.
3. Fillet joints: Neat.
4. Grinding: Grind smooth where indicated on drawings.

### 745 Preparation for application of coatings

1. General: Complete fabrication, and drill fixing holes before applying coatings.
2. Paint, grease, flux, rust, burrs and sharp arrises: Remove.

## 750 Liquid organic coating for aluminium alloy components

1. Standard: To BS 4842 Liquid organic coatings for application to aluminium alloy extrusions, sheet and preformed sections for external architectural purposes, and for the finish on aluminium alloy extrusions etc.

## 760 Zinc and cadmium plating of iron and steel surfaces

1. Zinc plating: To BS EN ISO 2081 Metallic and other inorganic coatings - electroplated coatings of zinc with supplementary treatments on iron or steel.
2. Cadmium plating: To BS EN ISO 2082 Metallic and other inorganic coatings - electroplated coatings of cadmium with supplementary treatments on iron or steel.

## 770 Chromium plating

1. Standard: To BS EN ISO 1456 Metallic and other inorganic coatings. Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium.

## 780 Galvanizing

1. Standard: To BS EN ISO 1461 Hot dip galvanized coatings on fabricated iron and steel articles - specifications and test methods.
2. Preparation
    - 2.1. Vent and drain holes: Provide in accordance with BS EN ISO 14713-1 and BS EN ISO 14713-2. Seal after sections have been drained and cooled.
    - 2.2. Components subjected to cold working stresses: Heat treat to relieve stresses before galvanizing.
    - 2.3. Welding slag: Remove.
    - 2.4. Component cleaning: To BS EN ISO 8501-3 Preparation of steel substrates before application of paints and related products. Visual assessment of surface cleanliness. Preparation grades of welds, edges and other areas with surface imperfections.
    - 2.5. Grade: To be proposed by Contractor and agreed.

## 790 Vitreous enamelling

1. Standard: To BS EN ISO 28722 Vitreous and porcelain enamels - Characteristics of enamel coatings applied to steel panels intended for architecture.
2. Substrate metal: Steel to BS EN 10209 Cold-rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions.

# Completion

## 910 Documentation

1. Submit
    - 1.1. Manufacturer's maintenance instructions.
    - 1.2. Guarantees, warranties, test certificates, record schedules and log books.

## 920 Completion

1. Protection: Remove.
2. Cleaning and maintenance: Carry out in accordance with procedures detailed in fabricators' guarantees.

Ω End of Section

# Z12
# Preservative/ Flame-retardant Treatment

## To be read with preliminaries/ general conditions.

## 105 Standards

1. BS 8417 Preservation of wood - code of practice.
2. BS EN 351-1 Durability of wood and wood-based products - Preservative-treated solid wood. Classification of preservative penetration and retention.
3. BS EN 351-2 Durability of wood and wood-based products - Preservative-treated solid wood. Guidance on sampling for the analysis of preservative-treated wood.
4. BS EN 335 Durability of wood and wood-based products - Use classes: definitions, application to solid wood and wood-based products.
5. BS EN 599-1 Durability of wood and wood-based products - efficacy of preventive wood preservatives as determined by biological tests. Specification according to use class.
6. BS EN 599-2 Durability of wood and wood-based products - efficacy of preventive wood preservatives as determined by biological tests. Labelling.

## 110 Treatment application

1. Timing: After cutting and machining timber, and before assembling components.
2. Cutting and machining: Carry out as much as possible before treatment.
3. Extensively processed timber: Retreat edges that have been sawn, planed, etc.
4. Surfaces exposed by minor cutting or drilling: Treat with two flood coats of a solution recommended by main treatment solution manufacturer
5. Site-cut timber surfaces: Liberally brush with a compatible preservative treatment
6. Processor: WPA benchmark-accredited for the specified treated components

## 120 Preservative treatment performance

1. Standard: In accordance with Table 5 of the WPA Code of practice.
2. Desired service life: Refer to the A Sections.

## 130 Preservative treatment solution strengths/ treatment cycles

1. General: Select to achieve specified service life and to suit treatability of specified wood species

## 140 Copper-organic preservative treatment

1. Solution
    1.1. Manufacturer: To be as stated in the relevant work section or an acceptable firm proposed by the Contractor.
        1.1.1. Product reference: To be as stated in the relevant work section or an acceptable firm proposed by the Contractor.
    1.2. Colour: To be as stated in the relevant work section as per samples provided and accepted.
    1.3. Application: High-pressure impregnation.
2. Moisture content of wood



make

2.1. At time of treatment: As specified for the timber/ component at time of fixing.

2.2. After treatment: Timber to be surface dry before use.

## 150 Water-based organic preservative treatment

1. Solution
   1.1. Manufacturer: To be as stated in the relevant work section as per samples provided and accepted.
      1.1.1. Product reference: To be as stated in the relevant work section as per samples provided and accepted.
   1.2. Application: High-pressure impregnation.
2. Moisture content of wood
   2.1. At time of treatment: As specified for the timber/ component at time of fixing.
   2.2. After treatment: Timber to be surface dry before use.

## 160 Organic solvent preservative treatment

1. Solution
   1.1. Manufacturer: Submit proposals
      1.1.1. Product reference: Submit proposals
   1.2. Application: Double-vacuum and low-pressure impregnation, or immersion.
2. Moisture content of wood
   2.1. At time of treatment: As specified for the timber/ component at time of fixing.
   2.2. After treatment: Timber to be surface dry before use.

## 165 Water-based microemulsion preservative treatment

1. Solution
   1.1. Manufacturer: Submit proposals
      1.1.1. Product reference: Submit proposals
   1.2. Application: Double-vacuum and low-pressure impregnation.
2. Moisture content of wood
   2.1. At time of treatment: As specified for the timber/ component at time of fixing.
   2.2. After treatment: Timber to be surface dry before use.

## 167 Boron compound preservative treatment

1. Solution
   1.1. Manufacturer: To be as stated in the relevant work section as per samples provided and accepted.
      1.1.1. Product reference: To be as stated in the relevant work section as per samples provided and accepted.
   1.2. Application: High-pressure impregnation.
2. Moisture content of wood
   2.1. At time of treatment: As specified for the timber/ component at time of fixing.
   2.2. After treatment: Timber to be surface dry before use.



make

## 170 Leach resistant fire retardant treatment

1. Solution
   1.1. Manufacturer: To be as stated in the relevant work section as per samples provided and accepted.
      1.1.1. Product reference: To be as stated in the relevant work section as per samples provided and accepted.
   1.2. Application: High-pressure impregnation.
2. Moisture content of wood
   2.1. At time of treatment: As specified for the timber/ component at time of fixing.
   2.2. After treatment: Timber to be surface dry before use.

## 210 Flame-retardant treatment

1. Standard: In accordance with the WPA publication Flame Retardant Treatments - An Overview and supporting guidance notes
2. Solution type: Submit proposals.
   2.1. Manufacturer: To be as stated in the relevant Work Section or an acceptable firm proposed by the Contractor.
      2.1.1. Product reference: To be as stated in the relevant Work Section or an acceptable firm proposed by the Contractor.
   2.2. Application: Vacuum pressure treatment.
3. Reaction to fire performance: To BS EN 13501-1, refer to the Fire Strategy Report.
4. Moisture content of wood
   4.1. At time of treatment: As specified for the timber/ component at time of fixing
   4.2. After treatment: Timber to be re-dried slowly at temperatures not exceeding 60°C to minimize distortion and degradation

## 610 Making good to preservative treatment on site

1. Preservative solution: Compatible with off-site treatment.
2. Application: In accordance with preservative manufacturer's recommendations.

## 620 Making good to flame-retardant treatment on site

1. Flame-retardant: Compatible with off-site treatment.
2. Application: In accordance with flame-retardant manufacturer's recommendations.

Ω End of Section



make

# Z20
# Fixings and Adhesives

## Products

### 310 Fasteners generally

1. Materials need to have:
   1.1. Bimetallic corrosion resistance appropriate to items being fixed.
   1.2. Atmospheric corrosion resistance appropriate to fixing location.
   1.3. Fire resistance appropriate to fixing location and application.
   1.4. Material safety and data sheets (MSDS).
   1.5. Suitable strength and properties to safely accommodate attachments and movements.
   1.6. To be appropriate, secure and firmly fixed in place.
   1.7. Bowing, distortion, vibration and other movement is not acceptable.
   1.8. Unless shown otherwise on the Working Drawings, fixings are not to be visible.
   1.9. Fixings to aluminium frames and cappings to match material to which they are attached.
   1.10. Removable items to have invisible secret fixings unless shown otherwise on the Working Drawings.
   1.11. Fixings to comply with all relevant Standards and Codes of Practice, being suitable for their intended purpose in terms of strength and type.
   1.12. Maximise use of Allen headed bolts on proprietary systems.
   1.13. Fixing systems to be complete and include all accessories necessary such as bearings, brackets, flashings, closures etc.
   1.14. Galvanise steel subframe assemblies.
   1.15. When fixing to concrete utilise cast-in channels as appropriate.
   1.16. When fixing to structural steel use direct fixings as appropriate.
2. Appearance: Submit samples on request.
   2.1. To PD 6484 Commentary on corrosion at bimetallic contacts and its alleviation.
   2.2. To BS 6037-1 Planning, design, installation and use of permanently installed access equipment - code of practice. Suspended access equipment - Inspection of structural fixings.
   2.3. To BS 4190 ISO metric black hexagon bolts, screws and nuts - specification.
   2.4. To BS 1473 Wrought aluminium and aluminium alloys - For general engineering purposes - Rivet, bolt and screw stock.
   2.5. To BS 4078-2 Powder actuated fixing systems. Specification for tools.
   2.6. To BS EN ISO 4016 Hexagon head bolts - product grade C.
   2.7. To BS EN ISO 4034 Hexagon regular nuts (style 1) - Product grade C.
   2.8. To BS EN 20898-7 Mechanical properties of fasteners. Torsional test and minimum torques for bolts and screws with nominal diameters 1 mm to 10 mm.
   2.9. To BS EN ISO 898-1 Mechanical properties of fasteners made of carbon steel and alloy steel. Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread.
   2.10. To BS EN ISO 898-2 Fasteners - Mechanical properties of fasteners made of carbon steel and alloy steel. Nuts with specified property classes.

2.11. To BS EN ISO 898-5 Mechanical properties of fasteners made of carbon steel and alloy steel. Set screws and similar threaded fasteners with specified hardness classes - coarse thread and fine pitch thread.

2.12. To BS EN ISO 3506 series of standards - Mechanical properties of corrosion resistant stainless steel fasteners.

## 320 Packings

1. Materials: Noncompressible, corrosion proof.
2. Area of packings: Sufficient to transfer loads.

## 330 Nailed timber fasteners

1. Standard:
   1.1. Steel:
   To BS 1202-1 Specification for nails. Steel nails or To BS EN 10230-1 Steel wire nails. Loose nails for general applications.
   1.2. Copper: To BS 1202-2 Nails. Copper nails.
   1.3. Aluminium: To BS 1202-3 Nails. Aluminium nails.

## 350 Plugs

1. Type: Proprietary types to suit substrate, loads to be supported and conditions expected in use.
2. Quality: Locate plugs accurately in appropriate size holes.

## 360 Anchors

1. Type:
   1.1. Expansion: For use in substrate strong enough to resist forces generated by expansion of anchor.
   1.2. Adhesive or chemical:
      1.2.1. For use in substrate where expansion of anchor would fracture substrate.
      1.2.2. For use in irregular substrate where expansion anchors cannot transfer load on anchor.
   1.3. Cavity: For use where the anchor is retained by toggles of the plug locking onto the inside face of the cavity.
2. Environmental
   2.1. Embodied carbon content:
   2.2. Recycled content:
   2.3. Recyclability:
3. Verification
   3.1. Submittals:
   3.2. Timing:

## 370 Wood screws

1. Type
   1.1. Wood screws (traditional pattern).
      1.1.1. Standard: To BS 1210 Wood screws.

1.2. Wood screws.

1.2.1. Pattern: Parallel, fully threaded shank or twin thread types.

2. Washers and screw cups: Where required are to be of same material as screw.

## 380 Screws

1. Screw details
   1.1. Type: To suit the fixing requirement of the components and substrate.
      1.1.1. Pattern: Self-tapping, metallic drive screws, or power driven screws.
   1.2. Washers: Where required to be of same material as screw.
2. Environmental
   2.1. Embodied carbon content:
   2.2. Recycled content:
   2.3. Recyclability:
3. Verification
   3.1. Submittals:
   3.2. Timing:

## 390 Adhesives

1. Standards:
   1.1. Hot-setting phenolic and aminoplastic: To BS 1203 Hot-setting phenolic and aminoplastic wood adhesives - Classification and test methods.
   1.2. Thermosetting wood adhesives: To BS EN 12765 Classification of thermosetting wood adhesives for non-structural applications.
   1.3. Thermoplastic adhesives: To BS EN 204 Classification of thermoplastic wood adhesives for non-structural applications.
   1.4. Non-structural wood adhesives: To BS EN 205 Adhesives - Wood adhesives for non-structural applications - Determination of tensile shear strength of lap joints.
   1.5. Loadbearing timber structure adhesive, phenolic and aminoplastic: To BS EN 301 Adhesives, phenolic and aminoplastic, for load-bearing timber structures - Classification and performance requirements.
   1.6. Test methods for loadbearing timber structure adhesives: To BS EN 302 series of standards - Adhesives for load-bearing timber structures.
   1.7. Timber structure design: To BS EN 1995-1-1 Eurocode 5: Design of timber structures. General - Common rules and rules for buildings.
   1.8. Animal glues: To BS EN ISO 9665 Adhesives - Animal Glues - Methods for Sampling and Testing.
2. Compatability: Compatible with VOC requirements, finished surfaces and preservative/ fire retardant treatments.
3. Environmental
   3.1. Volatile organic compounds (VOC) emissions level (maximum):
   3.2. Embodied carbon content:
   3.3. Recycled content:
4. Verification
   4.1. Submittals:

4.2. Timing:

## 410 Powder actuated fixing systems

1. Types of fastener, accessories and consumables: As recommended by tool manufacturer.

# Execution

## 610 Fixing generally

1. Integrity of supported components: Select types, sizes, quantities and spacings of fixings, fasteners and packings to retain supported components without distortion or loss of support.
2. Components, substrates, fixings and fasteners of dissimilar metals: Isolate with washers/ sleeves to avoid bimetallic corrosion.
3. Fixing to structure: Record all primary fixings to include size, type, spacing, embedment, precise location, edge dimensions, shimming, isolation techniques for dissimilar metals, locking washers, etc. and test results.
4. Bolts: After tightening, leave a minimum of one clear thread above each nut.
5. Appearance: Fixings to be in straight lines at regular centres.
6. Compliance: To Approved Document Part A of the Building Regulations.
7. Tools: Use purpose made tooling systems where appropriate and necessary to ensure consistent and adequate depths and perpendicular installation.
8. Screws: Countersunk screw heads visible in the finished works require countersinks to be machined in order to receive them.

## 625 Fixing using clips

1. Fastener spacing: In accordance with system manufacturers details/ recommendations and the requirements of the testing/ certification criteria applicable to each system to achieve the specific performance requirements and to suit the service conditions of each individual instance.
2. Accessories : As necessary to complete the system design and installation, achieve the performance requirements and to suit the service conditions

## 630 Fixing packings

1. Function: To take up tolerances and prevent distortion of materials and components.
2. Limits: Do not use packings beyond thicknesses recommended by fixings and fasteners manufacturer.
3. Locations: Not within zones to be filled with sealant.
4. Quality:
   4.1. Use only rot-proof, non-corrodible and non-compressible materials next to fixing points.
   4.2. Packings not to be visible.
   4.3. In floors, allow 10mm gap at each edge, parallel to the lie of the boards.

## 640 Fixing cramps

1. Cramp positions: Maximum 150 mm from each end of frame sections and at 600 mm maximum centres.
2. Fasteners: Fix cramps to frames with screws of same material as cramps.
3. Fixings in masonry work: Fully bed in mortar.

## 650 Fixing with nails

1. Fastener spacing: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
    1.1. Marking : Mark locations of the nails on the substrate surface.
2. Pilot holes: Drill using a hammer drill.
    2.1. Diameter: 0.2 mm larger than diameter of nail.
    2.2. Depth: 20mm.
    2.3. Cleaning: Blow out holes with compressed air of a minimum 80 psi for four seconds to remove all debris and dust or brush out holes with a nylon brush for a minimum of four cycles to remove all debris and dust.
3. Installation: Drive nail in without splitting or crushing timber.
4. Finished level of nail head
    4.1. Exposed: Flush with surface
    4.2. Punched (holes filled or stopped): Sink minimum 2 mm below surface.

## 660 Screw fixing

1. Fastener spacing : In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
    1.1. Marking: Mark locations of the screws on the substrate surface.
2. Pilot holes : Drill using a hammer drill.
    2.1. Diameter: 0.2 mm larger than diameter of the screw.
    2.2. Depth: 20mm.
    2.3. Cleaning: Blow out holes with compressed air of a minimum 80 psi for four seconds to remove all debris and dust or brush out holes with a nylon brush for a minimum of four cycles to remove all debris and dust.
3. Installation: Dive screw into pilot hole using a drill. Ensure not to split or damage the timber.
4. Finished level of countersunk screw heads
    4.1. Exposed: Flush with surface.
    4.2. Concealed (holes filled or stopped): Sink minimum 2 mm below surface.
    4.3. Hardwood: Drill pilot holes to half the diameter of the shank when using 8 gauge (or more) screws.
5. Testing:

## 670 Pelleted countersunk screw fixing

1. Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
2. Pellets: Cut from matching timber, match grain and glue in to full depth of hole.
3. Finished level of pellets: Flush with surface.

## 680 Plugged countersunk screw fixing

1. Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
2. Plugs: Glue in to full depth of hole.
3. Finished level of plugs: Projecting above surface.

## 690 Using powder actuated fixing systems

1. Powder actuated fixing tools: To BS 4078-2 Powder actuated fixing systems. Specification for tools and Kitemark certified.
2. Operatives: Trained and certified as competent by tool manufacturer.
3. Primer: Apply zinc rich primer to external fastener heads or locations subject to possible dampness.
4. Isolation: Isolate stainless steel components from external cartridge fired nails using top hat section plastic washers.

## 700 Applying adhesives

1. Surface preparation: Dry, clean, and free from oil, grease and other contaminants. Adjust regularity and texture to suit bonding and gap filling characteristics of adhesive.
2. Temporary support and clamping: Provide as necessary. Do not mark surfaces of (or distort) components being fixed.
3. Finished joints: Fully bonded and free of surplus adhesive.
4. Conditions: Do not use during unsuitable climatic/ environmental conditions, after recommended “use by” date/ open pot times.
5. Other requirements: In accordance with the manufacturers details and recommendations to achieve the performance requirements and to suit the service conditions.
6. Application
    6.1. Use the appropriate primer which is left to dry before adhesive is applied.
    6.2. Use recommended spreaders/ applicators/ tools to ensure adequate and even coverage.
    6.3. Remove surplus adhesive before it dries.

Ω End of Section



make

# Z21
# Mortars

## Cement gauged mortars

### 110 Cement gauged mortar mixes

1. Specification: Proportions and additional requirements for mortar materials are specified elsewhere.
2. Standards:
   2.1. To BS EN 1996-1-1 Eurocode 6 - Design of masonry structures. General rules for reinforced and unreinforced masonry structures. Reference Table NA2.
   2.2. To BS EN 998-1 Specification for mortar for masonry. Rendering and plastering mortar.
   2.3. To BS EN 998-2 Specification for mortar for masonry. Masonry mortar.
3. Appearance: No colour difference above and below DPC level or between different daily installations.
4. Performance: Mix to meet performance requirements in all conditions and locations.
5. Workmanship:
   5.1. Mix thoroughly in a suitable forced action mechanical mixer to produce uniform/ consistent colour and workable consistency.
   5.2. Do not use a free fall drum mixer.
   5.3. Minimise water content whilst maintaining required consistency/ workability.
   5.4. Batch by volume using dry sand.
   5.5. Maintain a mix ratio of 1:3.5/ 4 cement to sand when mixing semi-dry mortar and ensure that it does not crumble or deform when squeezed or form a film of water on the surface.
   5.6. Maintain a mix ratio of no less than 1:3.5/ 4 cement to sand when mixing a floating mortar bed and ensure that it does not crumble or deform when squeezed or form a film of water on the surface. Lay on a smooth and thoroughly cleaned base material with a minimum 500 gauge polythene separating lay over insulation with minimum 100mm laps at joints. Mix to be semi-dry incorporating steel mesh as necessary.
6. Proprietary mortars: Ensure that allowance is made for selection/addition of project-specific aggregate such as may be required to obtain the visual acceptance of the Architect.

### 120 Sand for site made cement gauged masonry mortars

1. Standard: To BS EN 13139:2002 Aggregates for mortar.
2. Grading: 0/2 (FP or MP).
   2.1. Fines content where the proportion of sand in a mortar mix is specified as a range (e.g. 1:1:5-6):
      2.1.1. Lower proportion of sand: Use category 3 fines.
      2.1.2. Higher proportion of sand: Use category 2 fines.
3. Sand for facework mortar: Maintain consistent colour and texture. Obtain from one source.
4. Marine sand: Do not use with sulphate resisting or super sulphated cement.

### 131 Ready-Mixed lime:sand for cement gauged masonry mortars

1. Standard: To BS EN 998-2 Specification for mortar for masonry. Masonry mortar.

2. Lime: Nonhydraulic to BS EN 459-1 Building lime. Definitions, specifications and conformity criteria.
   2.1. Type: CL 90S.
3. Pigments for coloured mortars: To BS EN 12878 Pigments for the colouring of building materials based on cement and/or lime - specifications and methods of test.

## 135 Site made lime:sand for cement gauged masonry mortars

1. Permitted use: Where a special colour is not required and in lieu of factory made ready-mixed material.
2. Lime: Nonhydraulic to BS EN 459-1 Building lime. Definitions, specifications and conformity criteria.
   2.1. Type: CL 90S.
3. Mixing: Thoroughly mix lime with sand, in the dry state. Add water and mix again. Allow to stand, without drying out, for at least 16 hours before using.

## 160 Cements for mortars

1. Cement: To BS EN 197-1 Cement. Composition, specifications and conformity criteria for common cements and UKCA marked or CE marked.
   1.1. Types: Portland cement, CEM I.
      1.1.1. Portland limestone cement, CEM II/A-L or CEM II/A-LL.
2. Portland slag cement, CEM II/B-S.
3. Portland fly ash cement, CEM II/B-V.
   3.1. Strength class: 32.5, 42.5 or 52.5.
4. White cement: To BS EN 197-1 Cement. Composition, specifications and conformity criteria for common cements and UKCA marked or CE marked.
   4.1. Type: Portland cement, CEM I.
   4.2. Strength class: 52.5.
5. Sulfate resisting Portland cement
   5.1. Type: To BS EN 197-1 Sulfate resisting Portland cement, CEM I/SR and UKCA marked or CE marked.
6. To BS EN 197-1 fly ash cement, CEM II/B-V and CE marked.
   6.1. Strength class: 32.5, 42.5 or 52.5.
7. Masonry cement: To BS EN 413-1 Masonry cement. Composition, specifications and conformity criteria and UKCA marked or CE marked.
   7.1. Class: MC 12.5.

## 180 Admixtures for site made cement gauged mortars

1. Air entraining (plasticizing) admixtures: To BS EN 934-3 and compatible with other mortar constituents.
2. Other admixtures: Submit proposals.
3. Prohibited admixtures: Calcium chloride, ethylene glycol and any admixture containing calcium chloride.
4. Pigments: To BS EN 12878 - Colouring. Submit product data, characteristics and testing criteria for review and acceptance before use.

## 190 Retarded ready to use cement gauged mortar

1. Standard: To BS EN 998-2 Specification for mortar for masonry. Masonry mortar.
2. Lime for cement:lime:sand mortars: Nonhydraulic to BS EN 459-1 Building lime. Definitions, specifications and conformity criteria.
    2.1. Type: CL 90S.
3. Pigments for coloured mortars: To BS EN 12878 Pigments for the colouring of building materials based on cement and/or lime - specifications and methods of test.
4. Time and temperature limitations: Use within limits prescribed by mortar manufacturer.
    4.1. Retempering: Restore workability with water only within prescribed time limits.

## 200 Storage of cement gauged mortar materials

1. Sands and aggregates: Keep different types/ grades in separate stockpiles on hard, clean, free-draining bases.
2. Factory made ready-mixed lime:sand/ ready to use retarded mortars: Keep in covered containers to prevent drying out or wetting.
3. Bagged cement/ hydrated lime: Store off the ground in dry conditions.

## 210 Making cement gauged mortars

1. Batching: By volume. Use clean and accurate gauge boxes or buckets.
    1.1. Mix proportions: Based on dry sand. Allow for bulking of damp sand.
2. Mixing: Mix materials thoroughly to uniform consistency, free from lumps.
    2.1. Mortars containing air entraining admixtures: Mix mechanically. Do not overmix.
3. Working time (maximum): Two hours at normal temperatures.
4. Contamination: Prevent intermixing with other materials.

## 250 Grout

1. Batching: By volume. Use clean and accurate gauge boxes or buckets.
2. General:
    2.1. To be latex modified, pigmented and colour matched.
    2.2. To be impervious, resistant to chemical attack/ mould, non-staining and of a type that is a proprietary product manufactured by a specialist manufacturer.
3. Sealer: To be an impervious, uncontaminated, clear water based proprietary product which does not have any detrimental effect on the grout (performance or colour).
4. Staining: Not acceptable.

# Lime:sand mortars

## 310 Lime:sand mortar mixes

1. Specification: Proportions and additional requirements for mortar materials are specified elsewhere.

## 320 Sand for lime:sand masonry mortars

1. Type: Sharp, well graded.
    1.1. Quality, sampling and testing: To BS EN 13139 Aggregates for mortar.

1.2. Grading/ Source: As specified elsewhere in relevant mortar mix items.

## 330 Ready prepared lime putty

1. Type: Slaked directly from CL 90 quicklime to BS EN 459-1, using an excess of water.
   1.1. Maturation: In pits/ containers that allow excess water to drain away.
   1.2. Density of matured lime putty: 1.3 - 1.4 kg/litre.
2. Maturation period before use (minimum): As recommended by the manufacturer/ supplier.

## 335 Ready prepared lime putty

1. Manufacturer: To be recommended by the Contractor and accepted.
   1.1. Product reference: To be recommended by the Contractor and accepted.
2. Maturation period before use (minimum): As recommended by the manufacturer/ supplier.

## 340 Pozzolanic additives for nonhydraulic lime:sand mortars

1. Manufacturer/ Supplier: To be recommended by the Contractor and accepted.
   1.1. Product reference: To be recommended by the Contractor and accepted.
2. Mixing: Mix thoroughly into mortar during knocking up.

## 345 Admixtures for hydraulic lime:sand mortars

1. General: Only use those which have been demonstrated as being essential and are acceptable.
2. Air entraining (plasticizing) admixtures: To BS EN 934-3 and compatible with other mortar constituents.
3. Prohibited admixtures: Calcium chloride, ethylene glycol and any admixture containing calcium chloride.

## 350 Storage of lime:sand mortar materials

1. Sands and aggregates: Keep different types/ grades in separate stockpiles on hard, clean, free-draining bases.
2. Ready prepared nonhydraulic lime putty: Prevent drying out and protect from frost.
3. Nonhydraulic lime:sand mortar: Store on clean bases or in clean containers that allow free drainage. Prevent drying out or wetting and protect from frost.
4. Bagged hydrated hydraulic lime: Store off the ground in dry conditions.

## 360 Making lime:sand mortars generally

1. Batching: By volume. Use clean and accurate gauge boxes or buckets.
2. Mixing: Mix materials thoroughly to uniform consistency, free from lumps.
3. Contamination: Prevent intermixing with other materials, including cement.
4. Plasticisers: Only to be used when recommended by the manufacturer and accepted. Use is not to adversely affect performance or appearance.

## 370 Site prepared nonhydraulic lime:sand mortars

1. Mixing: Mix materials thoroughly by compressing, beating and chopping. Do not add water.
   1.1. Equipment: Roller pan mixer or submit proposals.
2. Maturation period before use (maximum): To be agreed.

## 380 Ready to use nonhydraulic lime:sand mortars

1. Manufacturer: To be recommended by the Contractor and accepted.
   1.1. Product reference: To be recommended by the Contractor and accepted.
2. Materials: Select from:
   2.1. Lime putty slaked directly from quicklime to BS EN 459-1 and mixed thoroughly with sand.
   2.2. Quicklime to BS EN 459-1 slaked directly with sand.
3. Maturation period before use (maximum): To be agreed.

## 390 Knocking up nonhydraulic lime:sand mortars

1. Knocking up before and during use: Achieve and maintain a workable consistency by compressing, beating and chopping. Do not add water.
   1.1. Equipment: Roller pan mixer or submit proposals.

## 400 Making hydraulic lime:sand mortars

1. Mixing hydrated hydraulic lime:sand: Follow the lime manufacturer's recommendations for each stage of the mix.
   1.1. Water quantity: Only sufficient to produce a workable mix.
2. Working time: Within limits recommended by the hydraulic lime manufacturer.

## 500 Testing

1. Laboratory: Tests to be carried out by a laboratory/ facility accredited by UKAS (United Kingdom Accreditation Service).
2. Standards:
   2.1. To BS 4551 Mortar - Methods of test for mortar and screed - Chemical analysis and physical testing.
   2.2. To BS EN 1015 series of standards - Masonry mortar.
3. Apparatus: Maintain and retain on site all necessary thermometers to ensure accurate readings.
4. Methodology:
   4.1. Samples for testing to be taken from the mixing point.
   4.2. Repeat tests until performance requirements are satisfied.
   4.3. Prepare test samples/ specimens at least 42 days prior to planned use in the Works.
   4.4. Replace and re test any walling erected using mortar which fails a test.

## 510 Workmanship

1. Storage: Store materials in a dry location, raised above ground and under cover to protect from rain, damp, frost, etc.
2. Equipment: Keep all tools and equipment clean and in good working order.
3. Techniques: Fully trained operatives to employ tried and tested techniques to achieve a workable, lump free and appropriate/ consistent mix. Mix by suitable mechanical means using dry sand and mortar containing air-entraining admixtures where appropriate or recommended by the manufacturer.
4. Weather conditions: Mortar temperature to be a minimum of 4□ at all times. Replace mortar which has been affected by frost or temperatures below 4□.

5. Standards:
    5.1. Butter masonry with mortar prior to laying.
    5.2. Each course to be filled.
    5.3. Maintain a consistent thickness of all joints.
    5.4. Remove excess mortar as the work proceeds and before it hardens.
    5.5. Brush and wash off smears with soft brush and clean water.
    5.6. Avoid staining caused by excessive washing or scrubbing.
    5.7. Do not use cleaning or acid based products to clean.

Ω End of Section



make

# Z22
# Sealants

## Products

### 100 Standards/ certification

1. Materials:
    1.1. To BS EN 681 series of standards - Electromeric seals.
    1.2. To BS EN 14188-1 Joint fillers and sealants. Specifications for hot applied sealants.
    1.3. To BS EN 14188-2 Joint fillers and sealants. Specifications for cold applied sealants.
    1.4. To BS EN 15651 series of standards - Non-structural joints.
    1.5. To BS EN ISO 9000 Quality management systems - fundamentals and vocabulary - Quality Control (manufacture date and batch number recognition).
    1.6. To BS EN ISO 11600 Building construction - Jointing products - Classification and requirements for sealants.
2. Installation:
    2.1. To BS 6093 Design of joints and jointing in building construction - Guide.
    2.2. To BS 6213 Selection of construction sealants. Guide.
    2.3. To BS 8000-16 Workmanship on building sites. Code of practice for sealing joints in buildings using sealants.
    2.4. To BS EN ISO 9001 Quality management systems - requirements.
3. Testing:
    3.1. To BS 3712-1 Building and construction sealants. Methods of test for homogeneity, relative density and penetration.
    3.2. To BS 3712-2 Building and construction sealants. Methods of test for seepage, staining, shrinkage, shelf life and paintability.
    3.3. To BS 3712-3 Building and construction sealants. Methods of test for application life, skinning properties and tack-free time.
    3.4. To BS 3712-4 Building and construction sealants. Methods of test for adhesion in peel, tensile extension and recovery and loss of mass after heat ageing.
4. Compliance:
    4.1. Provide an independent UK performance certificate relating to fire resistance.
    4.2. Comply with the VOC requirements.
    4.3. Provide independent evidence to confirm that neither discolouration, bleeding or staging will occur.
    4.4. Provide evidence to confirm external sealants will be waterproof and weatherproof.
    4.5. Carry out site testing as necessary.

### 110 Sealants

1. Performance:
    1.1. Provide a watertight seal.
    1.2. Provide necessary Service Life guarantee.
    1.3. Discolouration or degradation is not acceptable.
    1.4. To be fuel resistant.

1.5. Accommodate movement to which it is subjected without reduction in performance or visual appearance.

1.6. Resistant to impact/ indentation caused by wear, abrasion, vandalism, etc.

2. Material:

2.1. Types and use to be as recommended by the sealant manufacturer.

2.2. To be of appropriate grade and type to suit its location.

2.3. Must not be a health (or potential) hazard.

2.4. Samples to be provided as necessary.

2.5. Sealants and primers to be fully compatible with the surfaces/ joints to which they are applied and in appropriate locations.

### 310 Joints

1. Description: As shown on the Design Drawings.
2. Primer, backing strip, bond breaker: Types recommended by sealant manufacturer.
3. Fire performance

3.1. Fire resistance: In accordance with the requirements of the Fire Strategy Report.

3.2. Reaction to fire (for rateable products): In accordance with the requirements of the Fire Strategy Report.

## Execution

### 610 Suitability of joints

1. Presealing checks

1.1. Joint dimensions: Within limits specified for the sealant.

1.2. Substrate quality: Surfaces regular, undamaged and stable.

2. Primer: Joints to be carefully primed, without spillage or seeping, to achieve a full bond, without causing damage or change in appearance. Use masking tape where necessary.

### 620 Preparing joints

1. Surfaces to which sealant must adhere

1.1. Remove temporary coatings, tapes, loosely adhering material, dust, oil, grease, surface water and contaminants that may affect bond.

1.2. Clean using materials and methods recommended by sealant manufacturer.

1.3. Non porous surfaces such as metal, to be cleaned with an appropriate chemical agent, which does not damage the substrate or sealant in any way.

2. Vulnerable surfaces adjacent to joints: Mask to prevent staining or smearing with primer or sealant.
3. Backing strip and/ or bond breaker installation: Insert into joint to correct depth, without stretching or twisting, leaving no gaps.
4. Protection: Keep joints clean and protect from damage until sealant is applied.

### 630 Applying sealants

1. Substrate: Dry (unless recommended otherwise) and unaffected by frost, ice or snow.

2. Trade recommendations: In accordance with the "Industry Guide to the Professional Application of Construction Sealants on Site" published by BASA (British Adhesives and Sealants Association).
3. Environmental conditions: Do not dry or raise temperature of joints by heating.
4. Sealant application: Fill joints evenly, completely and neatly, ensuring firm adhesion to substrates, without bubbling using appropriate tools.
5. Sealant profiles
    5.1. Butt and lap joints: Slightly concave.
    5.2. Fillet joints: Flat or slightly convex.
6. Joint fillers:
    6.1. To be entirely compatible and suitable.
    6.2. Gaps in joints to be consistent with depth of sealant.
    6.3. Sealant cross section in joint to be a ratio of 2:1 - width to depth - unless agreed otherwise.
    6.4. Generate fillers from closed cell foam.
    6.5. Fillers are not to adhere to cured sealant - use bond breaker tape if necessary.
7. Protection: Protect finished joints from contamination or damage until sealant has cured.

Ω End of Section

# Z30
# General Metalwork Finishes

## To be read with preliminaries/ general conditions.

## 100 Standards

1. Comply with:
   1.1. BS 1344-7 Methods of Testing Vitreous Enamel Finishes - Part 7: Determination of Resistance to Heat.
   1.2. BS 1344-16 Methods of Testing Vitreous Enamel Finishes - Part 16: Resistance to Thermal Shock of Coatings on Cooking Utensils.
   1.3. BS 1344-21 Methods of Testing Vitreous Enamel Finishes - Part 21: Determination of the Resistance of Vitreous Enamelled Articles to Impact: Pistol Test.
   1.4. BS 1344-22 Methods of Testing Vitreous Enamel Finishes - Part 22: Determination of Resistance to Thermal Shock of Coatings for Fabricated Sheet Steel Components Such as Burners and Pan Supports for Cookers.
   1.5. BS 4842 Liquid organic coatings for application to aluminium alloy extrusions, sheet and preformed sections for external architectural purposes, and for the finish on aluminium alloy extrusions etc.
   1.6. BS 6001-1 Sampling procedures for inspection by attributes. Sampling schemes indexed by acceptance quality limit (AQL) for lot-by-lot inspection - Acceptable Quality Level 2 - Minimum 1% for every finish and colour required.
   1.7. BS ISO 2859-2 Sampling procedures for inspection by attributes. Sampling plans indexed by limiting quality (LQ) for isolated lot inspection - Limited Quality (LQ) to be (Pa=10%)+5% for each element identified as being damaged or without a test certificate.
   1.8. BS EN ISO 1456 Metallic and other inorganic coatings. Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium.
   1.9. BS EN ISO 1461 Hot dip galvanized coatings on fabricated iron and steel articles - specifications and test methods.
   1.10. BS EN 10027-1 Designation systems for steel. Steel names - Steel designation to grade DC 03 ED to receive vitreous enamel finish.
   1.11. BS EN 10209 Cold-rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions.
   1.12. BS EN ISO 2063-1 Thermal spraying - zinc, aluminium and their alloys. Design considerations and quality requirements for corrosion protection systems.
   1.13. BS EN ISO 2063-2 Thermal spraying - zinc, aluminium and their alloys. Execution of corrosion protection systems.
   1.14. BS EN ISO 2081 Metallic and other inorganic coatings - electroplated coatings of zinc with supplementary treatments on iron or steel.
   1.15. BS EN ISO 2082 Metallic and other inorganic coatings - electroplated coatings of cadmium with supplementary treatments on iron or steel.
   1.16. BS EN ISO 2746 Vitreous and porcelain enamels - High voltage test.
   1.17. BS EN ISO 4534 Vitreous and porcelain enamels - Determination of fluidity behaviour - Fusion flow test.
   1.18. BS EN ISO 6370-1 and 2 - Abrasion resistance.

1.19. BS EN ISO 6370-1 Vitreous and porcelain enamels - Determination of the resistance to abrasion Part 1: Abrasion testing apparatus.

1.20. BS EN ISO 6370-2 Vitreous and porcelain enamels - Determination of the resistance to abrasion Part 2: Loss in mass after sub-surface abrasion.

1.21. BS EN ISO 8501-1 Preparation of steel substrates before application of paints and related products. Visual assessment of surface cleanliness. Rust grades and preparation grades of uncoated steel substrates and of steel substrates after overall removal of previous coating - Surface preparation to Sa 2.5.

1.22. BS EN ISO 8503-1 Preparation of steel substrates before application of paints and related products - Surface roughness characteristics of blast-cleaned steel substrates. Specifications and definitions for ISO surface profile comparators for the assessment of abrasive blast-cleaned surfaces.

1.23. BS EN ISO 8503-2 Preparation of steel substrates before application of paints and related products - Surface roughness characteristics of blast-cleaned steel substrates. Method for the grading of surface profile of abrasive blast-cleaned steel - Comparator procedure.

1.24. BS EN ISO 8503-5 Preparation of steel substrates before application of paints and related products - surface roughness characteristics of blast-cleaned steel substrates. Replica tape method for the determination of the surface profile.

1.25. BS EN ISO 12944-2 Paints and varnishes - Corrosion protection of steel structures by protective paint systems. Classification of environments.

1.26. BS EN ISO 14713 - Zinc coatings.

1.27. BS EN ISO 14713-1 Zinc coatings - guidelines and recommendations for the protection against corrosion of iron and steel in structures. General principles of design and corrosion resistance.

1.28. BS EN ISO 14713-2 Zinc coatings - guidelines and recommendations for the protection against corrosion of iron and steel in structures. Hot dip galvanizing.

1.29. BS EN ISO 14713-3 Zinc coatings - guidelines and recommendations for the protection against corrosion of iron and steel in structures. Sherardizing.

1.30. BS EN ISO 17834 Thermal spraying Coatings for protection against corrosion and oxidation at elevated temperatures.

1.31. BS EN ISO 28706-1 Vitreous and porcelain enamels - Determination of resistance to chemical corrosion Part 1: Determination of resistance to chemical corrosion by acids at room temperature.

1.32. BS EN ISO 28706-2 Vitreous and porcelain enamels — Determination of resistance to chemical corrosion Part 2: Determination of resistance to chemical corrosion by boiling acids, boiling neutral liquids, alkaline liquids and/or their vapours.

1.33. BS EN ISO 28722 Vitreous and porcelain enamels - Characteristics of enamel coatings applied to steel panels intended for architecture.

1.34. BS EN ISO 28764 Vitreous and porcelain enamels - Production of specimens for testing enamels on sheet steel, sheet aluminium and cast iron.

2. Other standards:

2.1. Zinc Information Centre recommendations for zinc metal and galvanising applications (sprayed).

2.2. PEI (Porcelain Enamel Institute) S-1001 (ALS-100) Specifications for architectural porcelain enamel - 60 microns local thickness minimum/ 500 microns local thickness maximum.

## 302 Preparation/ quality

1. Surfaces to receive treatment to be:
    1.1. Bare metal.
    1.2. Clean and free from any existing surface finish.
    1.3. Smooth and regular.
    1.4. Uncontaminated.
    1.5. Consistent in terms of colour and texture.
    1.6. Uniform.
    1.7. Unblemished and undistorted.
    1.8. Free of scratches, dents or other imperfections.
2. Finished surfaces to be:
    2.1. Of acceptable quality and finish.
    2.2. Colour and texture consistent/ uniform.
    2.3. Resistant to fading, prevailing environmental conditions.
    2.4. Stable.
    2.5. Unaffected by UV light.
    2.6. Free of surface imperfection.
3. Allow for metallics, including RAL 9006 and RAL 9007. 30% gloss. Allow for bespoke colour matching to samples held at the Architect's office.

## 305 Galvanised Finishes

1. Workmanship/ Quality:
    1.1. Evenly applied to an agreed, uniform and appropriate thickness to achieve the design intent and visual requirements.
    1.2. Smooth, uniform and consistent with no edges, ridges, burrs and the like.
    1.3. Clean with no staining or visible drip marks.
    1.4. Fill pinholes prior to application.
    1.5. Confirm suitability of steel to be coated prior to galvanising, including suitability for acid pickling in diluted hydrochloric acid and passivated if required.
    1.6. Single dip application only.
    1.7. Do not touch up and repair any coated surface without prior written agreement.
    1.8. Do not use Galvafroid or any paint applied finish.
    1.9. Minimise distortion and keep within visual tolerances that are acceptable.

## 306 Vitreous enamel finishes

1. Base material:
    1.1. To be decarbonised.
    1.2. Maximum 0.006% carbon content allowable unless stabilised by addition of appropriate amount of titanium. All to be agreed in advance.
    1.3. Internal stresses resulting from cold rolling of steel are not acceptable.
    1.4. Corrosion free.
    1.5. Clause 302 above applies as relevant.

2. Finish:
    2.1. Colour, texture and finish to be consistent and acceptable.
    2.2. Variance of gloss levels to remain within seven units of that agreed.
    2.3. Mirroring is not acceptable.
3. Quality/ workmanship
    3.1. All surfaces to be fully coated.
    3.2. Welding material to be consistent and ground smooth at all locations including corners and whirls.
    3.3. Maintain full coating/ testing records on site prior to installation for checking when requested.

Ω End of Section



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# Z31
# Powder Coatings

## To be read with preliminaries/ general conditions.

### 120 Powder coating materials

1. Manufacturer: Submit proposals
   1.1. Product reference: Submit proposals
2. Submit the following details of the proposed manufacturer before commencement of powder coating:
   - Name and contact details.
   - Details of accreditation schemes.
   - Technical data of product including current Agrément certificates.
3. Comply with:
   - BBA - British Board of Agrément requirements/ recommendations.
   - GSB International Quality Regulations for aluminium and steel.
   - RAL colour range as selected.
   - BS 950-1 Specification for artificial daylight for the assessment of colour. Illuminant for colour matching and colour appraisal.
   - BS EN 485 series of standards.
   - BS EN 515 Aluminium and aluminium alloys - wrought products - temper designations.
   - BS EN 573 series of standards, Grade 1200/3103 - Sheet materials.
   - BS EN 755 series of standards, Grade EW AW-6063, EN AW-6060 - Alloys.
   - COSHH Regulations 2002 as applicable.
   - Health and Safety Regulations as applicable.
   - Environmental Protection Act 1990 Part 1.
4. Testing requirements:
   4.1. General:
      - BS 476-7 - Class 1 - coatings.
      - BS EN 12206-1.
      - BS EN 13438.
      - BS EN ISO 2064 - Organic and metallic coatings.
      - Florida testing method.
   4.2. Elements to be tested: Each wall type to be tested to be considered as an individual element.
   4.3. Scope of testing:
      - Impact resistance.
      - Permeability/ water ingress.
      - Artificial and natural weathering.
      - Chalking, colour retention and finish.
      - Mortar resistance.
      - Visual inspection from a maximum of 1000mm.
      - Panel size for testing to be a minimum of 200mm x 100mm.
      - Extrusion size for testing to be a minimum of 300mm.
   4.4. Provide detailed reports of each test to include:



make

- Product name.
- Manufacturer/ supplier.
- Colour code/ reference.
- Batch number.
- Place of manufacture.
- Time and location of test.
- Details of test failures and re-tests.
- Independent inspections carried out, including on Site.

4.5. Acceptable quality level to be <1% for every colour and finish.

## 210 Working procedures

1. Comply with the following standards
   1.1. Aluminium components: BS EN 12206-1.
   1.2. Steel components: BS EN 13438.
   1.3. Safety standards: British Coatings Federation publication Code of safe practice: Powder coating. Application of coating powders by electrostatic spraying.
   1.4. Health and safety guidance: Health and Safety Executive publication Reducing risk associated with using coating powders - employers.
   1.5. Adequacy: Ensure aluminium temper and thickness are suitable to receive treatment without reducing performance or colour consistency of the application.
   1.6. Use one, spray applied batch per colour.
   1.7. Use regulated and automated equipment at all times.
   1.8. Dry film thickness between panels not to exceed 20%.
   1.9. Maintain colour consistency across all panels and extrusions.

## 220 Powder coating applicators

1. Applicator requirements
   1.1. Approved by powder-coating manufacturer
   1.2. Currently certified to BS EN ISO 9001
   1.3. Comply with quality procedures, guarantee conditions, standards and tests required by powder-coating manufacturer
   1.4. Selected applicator: Submit details before commencement of powder coating, including:
      - Name and contact details.
      - Details of accreditation schemes.

## 225 Guarantees

1. Powder-coating manufacturer and applicator guarantees
   1.1. Submit sample copies before commencement of powder-coating
   1.2. Submit signed project specific copies on completion of work

## 230 Control samples

1. Prior to ordering materials for the works, obtain approval of appearance for
   1.1. Powder-coated samples: Of various grades and forms of background metal to be used, showing any colour, texture and gloss variation.



make

1.2. Fabrication samples: Showing joint assembly, how powder-coating is affected and how any cut metal edges are finished and protected.

1.3. Where manual application is required, controlled samples should be coated and inspected for colour and gloss stability

2. Samples to include the following information:
    - Product reference.
    - Colour.
    - Reference number.
    - Name.
    - Gloss level.

## 235 Independent inspection at plant

1. Requirement: Contractors/ suppliers of the following designated components must commission an approved independent inspection authority to carry out acceptance inspections to confirm that powder-coating application complies with this specification.
2. Acceptance inspections: Carry out for each variation of colour and finish of each component work package at applicator's plant prior to any fabrication of units, in accordance with the following:
3. Components failing inspection: Reprocess or replace and reinspect.
4. Inspection reports: Independent inspection authority must submit copies.

## 240 Qualicoat quality assurance system

1. Requirement: Powder and coating application to the following designated components is to be tested and approved in accordance with the QUALICOAT specification 'QUALICOAT Specifications 2024 Specifications for a quality label for liquid and powder coatings on aluminium for architectural applications Master version (V01b)'.

## 250 Component design

1. Condition of components to be powder-coated
    1.1. To comply with relevant recommendations of BS 4479-1, BS 4479-3 and BS 4479-4
    1.2. Of suitable size to fit plant capacity
    1.3. Of suitable thickness to withstand oven curing

## 260 General Workmanship

1. Additional Standards requirements:
    1.1. To BS EN ISO 12944 series of standards, C1 to C3 - non aggressive environments.
    1.2. To BS EN ISO 12944 series of standards, C4 and C5 - hostile environments.
2. Quality:
    2.1. Use material from a single source and batch.
    2.2. Do not repair damage on Site.
    2.3. Replace and repair damaged items at the point of manufacture.
    2.4. Replaced and/or repaired elements to colour match and maintain required performance characteristics.
    2.5. Keep all surfaces clean and new up to handover.
    2.6. Maintain 3 month cleaning protocol.

## 310 Pretreatment of aluminium components

1. Condition of components to be pretreated
   1.1. Free from corrosion and damage
   1.2. All welding and jointing completed and finish off as specified
   1.3. Free from impurities including soil, grease and oil
   1.4. Suitable for and compatible with the pretreatment process
2. Conversion coating requirements
   2.1. Chromate system: BS EN 12206-1.
   2.2. Chromate-free system: To BS EN 12206-1. Submit details before using.
3. Rinsing requirements: Use demineralized water. Drain and dry.

## 320 Pretreatment of steel components

1. Condition of components to be pretreated
   1.1. Free from corrosion and damage
   1.2. All welding and jointing completed and finish off as specified
   1.3. Free from impurities including soil, grease and oil
   1.4. Suitable for and compatible with the pretreatment process
2. Conversion coating requirements: To BS EN 13438.
3. Rinsing requirements: Use demineralized water. Drain and dry.

## 330 Pretreatment for protection in aggressive environments

1. Minimum thickness of 60 micrometres across significant and/ or primary surfaces
2. All cut edges, drilled holes and mitres to be fully sealed
3. Cleaning and maintenance: Carried out once every three to 12 months (dependent on proximity to pollutant).

## 430 Extent of powder coatings

1. Application: To visible component surfaces, and concealed surfaces requiring protection. Coated surfaces will be deemed 'significant surfaces' for relevant BS EN 13438 performance requirements.

## 435 Application of powder coatings

1. Surfaces to receive powder coatings: Free from dust or powder deposits.
2. Powder colours: Obtain from one batch of one manufacturer.
3. Commencement of powder-coating: To be continuous from pretreatment.
4. Components to be installed on site in order of application.
5. Jig points: Not visible on coated components.
6. Curing: Controlled to attain metal temperatures and hold periods recommended by powder-coating manufacturer.
7. Stripping and recoating of components: Only acceptable by prior agreement of powder-coating manufacturer. Stripping, pretreatment and powder-coating are to be in accordance with manufacturer's requirements.
8. Overcoating of components: Not acceptable.

## 440 Performance and appearance of powder coatings

1. For aluminium components: BS EN 12206-1
2. For steel components: BS EN 13438
3. Visual inspection after powder-coating: Significant surface viewing distances to be as specified in the relevant Standard, unless specified otherwise.
4. Colour and gloss levels: To conform with approved samples.

## 450 Aluminium alloy fabrications

1. Units may be assembled
   1.1. Before powder-coating
   1.2. From components powder-coated after cutting to size
   1.3. Where approved, from components powder-coated before cutting to size
2. Exposure of uncoated background metal: Not acceptable.
3. Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

## 460 Steel fabrications

1. Unit assembly: Wherever practical, before powder-coating.
2. Exposure of uncoated background metal: Not acceptable.
3. Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

## 470 Fixings

1. Exposed metal fixings: Powder-coat together with components, or coat with matching repair paint system applied in accordance with the powder-coating manufacturer's recommendations.

## 480 Damaged components – repair or replacement

1. Before delivery to site: Check all components for damage to powder coatings. Replace damaged components.
2. Site damage: Submit proposals for repair or replacement.

## 510 Protection

1. Powder-coated surfaces of components: Protect from damage during handling and installation, or by subsequent site operations.
2. Protective coverings must be
   2.1. Resistant to weather conditions
   2.2. Partially removable to suit building in and access to fixing points
3. Protective tapes in contact with powder coatings must be
   3.1. Low tack, self adhesive and light in colour
   3.2. Applied and removed in accordance with tape and powder-coating manufacturers' recommendations. Do not use solvents to remove residues as these are detrimental to the coating
4. Inspection of protection: Carry out monthly. Promptly repair any deterioration or deficiency.

## 520 Protection in hazardous locations

1. Minimum thickness of 60 micrometres across significant and/ or primary surfaces
2. All cut edges, drilled holes and mitres to be fully sealed
3. Cleaning: Carried out once every three to twelve months (dependent on proximity to pollutant).

## 535 Documentation

1. Submit the following information for each batch of powder-coated components
   1.1. Supplier.
   1.2. Trade name.
   1.3. Colour.
   1.4. Type of powder.
   1.5. Method of application.
   1.6. Batch and reference number.
   1.7. Statutory requirements.
   1.8. Test certificates.
   1.9. Maintenance instructions.

## 540 Completion

1. Protection: Remove any protective coverings.
2. Cleaning and maintenance of powder coatings: Carry out in accordance with procedures detailed in powder-coating manufacturer and applicator guarantees.

Ω End of Section

# Z33
# Anodizing

## To be read with preliminaries/ general conditions.

## 105 Standards

1. Comply with:
   1.1. BS 3987 Specification for anodic oxidation coatings on wrought aluminium for external architectural applications.
   1.2. BS 6001-1 Sampling procedures for inspection by attributes. Sampling schemes indexed by acceptance quality limit (AQL) for lot-by-lot inspection - Level 2 general inspection.
   1.3. BS ISO 2859-2 Sampling procedures for inspection by attributes. Sampling plans indexed by limiting quality (LQ) for isolated lot inspection.
   1.4. BS EN ISO 2360 - Coating thickness for non conductive coatings on non magnetic electrically conductive base metal using amplitude sensitive Eddy Current method.
   1.5. BS EN ISO 2931 Anodizing of aluminium and its alloys. Assessment of quality of sealed anodic oxidation coatings by measurement of admittance.
   1.6. BS EN ISO 7599 Anodizing of aluminium and its alloys - Method for specifying decorative and protective anodic oxidation coatings on aluminium.
   1.7. BS EN ISO 8251 Anodizing of aluminium and its alloys. Measurement of abrasion resistance of anodic oxidation coatings - Testing - Wear index <1.4.

## 110 Anodic coating

1. Selected anodizer: Submit details before commencement of anodizing, including:
   1.1. Name and contact details.
   1.2. Details of accreditation schemes.
   1.3. Technical data of product, including current Agrément certificates.
2. Process:
   2.1. Two stage electrolytic using cobalt or nickel colouring agent. (Note - single stage integral process may be used with prior authorisation).
   2.2. Do not use tin electrolytes.
   2.3. Consistency of finish and colour to be the same throughout.
   2.4. Alloy extrusions to be Grade AW-6063.
   2.5. Alloy sheets to be Grade J57S.
   2.6. Provide all necessary certification to demonstrate compliance and suitability.
   2.7. Source from single manufacturer and single batch.

## 210 Working procedures

1. Standard: To BS 3987 for anodic coatings on wrought aluminium.

## 220 Anodizer requirements

1. Processing
   1.1. Approved: By the Aluminium Federation (ALFED).
   1.2. Certified: To BS EN ISO 9001.

1.3. Anodizing plant: Each anodizer to use only one plant.

## 230 Guarantees

1. Anodizer guarantees: Submit sample copies before commencement of anodizing.
2. Project specific guarantees: Submit signed copies on completion of work.
3. Guarantees to cover
   3.1. Life expectancy.
   3.2. Colour: Opacity and consistency.
   3.3. Texture: Gloss, satin or matt.
   3.4. Quality of coating.

## 240 Control samples

1. Sequence: Prior to ordering materials for the works, obtain approval of appearance for:
   1.1. Anodic coated samples: Showing colour and texture variation.
   1.2. Fabrication samples: Showing joint assembly, how anodic coating is affected and how cut metal edges are finished and protected.

## 250 Independent inspection at plant

1. Requirement: Contractors/ suppliers of the following designated components must commission an approved Independent Inspection Authority to carry out acceptance inspections to confirm that anodic coating application complies with this specification.
   1.1. Designated components:
2. Acceptance inspections: Carry out for each variation of colour and finish of each component work package at anodizer's plant prior to any fabrication of units, in accordance with the following:
   2.1. Where three of more production runs are required for application of coatings, not less than three acceptance inspections must be carried out in accordance with BS 6001-1, general inspection level 2, with an acceptance quality limit of 1%.
   2.2. Where less than three production runs are required for application of coatings, one acceptance inspection must be carried out in accordance with BS ISO 2859-2, with a limiting quality of 5% where the probability of acceptance is 10%.
3. Components failing inspection: Reprocess or replace and reinspect.
4. Inspection reports: Independent Inspection Authority must submit copies.

## 255 Quality assurance system

1. Requirement: Powder and coating application to the following designated components is to be tested and approved in accordance with the Qualanod system.
   1.1. Commence coating after all machining and fabrication is complete.
   1.2. All visible elements, including edges to be coated.
   1.3. Uncoated edges, etc. to be non visible in the completed installation.
   1.4. Surfaces to have high quality, consistent and uniform appearance, including colour and texture.
   1.5. Apply appropriate sealants as required.
   1.6. Ensure that anodised finishes are weather resistant, compatible, resistant to abrasion/ impact/ chemical attack. Surface blemishes will not be accepted if visible from 1000mm.

1.7. Ensure that anodised finishes are weather resistant, compatible, resistant to abrasion/ impact/ chemical attack. Impregnated cold sealing is not acceptable.

1.8. Ensure that anodised finishes are weather resistant, compatible, resistant to abrasion/ impact/ chemical attack. Use a sulphuric acid bath process to achieve high quality anodic oxidation.

1.9. Regularly clean extrusion dies.

1.10. Damaged surfaces to be repaired or replaced off site under factory controlled conditions. Repaired elements to match existing in terms of appearance and quality. Use a sulphuric acid bath process to achieve high quality anodic oxidation.

## 270 Component design

1. Condition of components to be anodized:
   1.1. To comply with relevant recommendations of BS 4479-1.
   1.2. Of suitable size to fit plant capacity.

## 310 Pretreatment

1. Condition of components to be anodized
   1.1. Free from corrosion and damage.
   1.2. Suitable for and compatible with the pretreatment and anodizing process.
2. Process: In accordance with the specification requirements for the finish.

## 410 Extent of anodic coatings

1. Application: To visible component surfaces, and concealed surfaces requiring protection. Coated surfaces will be deemed 'significant surfaces' for relevant BS 3987 performance requirements.

## 420 Application of anodic coatings

1. Surfaces to receive anodic coatings: Clean.
2. Commencement of anodic coating: To be continuous from pretreatment.
3. Jig points: To be agreed. Not on visible areas of anodic coated components.
4. Use of touch-up paint: Not acceptable.

## 430 Performance and appearance of anodic coatings

1. Standard: To BS 3987.
2. Visual inspection after anodizing: Significant surfaces to be free from visible coating/ defects when viewed from a distance of not less than 5 m for external and 3 m for internal applications.

## 440 Fabrication

1. Units may be assembled
   1.1. Before anodizing, providing sufficient drainage holes are included in components to fully drain components.
   1.2. From components anodized after cutting to size.
   1.3. Where approved, from components anodized before cutting to size.
   1.4. Exposure of uncoated background metal: Not acceptable.



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1.5. Assembly sealants: Compatible with anodic coatings. Obtain approval of colour if sealants are visible after fabrication.

## 450 Damaged components – repair/ replacement

1. Before delivery to site: Check all components for damage to anodic coatings. Replace damaged components.
2. Site damage: Submit proposals for repair or replacement.

## 510 Protection

1. Anodic coated surfaces of components: Protect from damage during handling and installation, or by subsequent site operations.
2. Protective coverings to be: Must be:
   2.1. Resistant to weather conditions.
   2.2. Partially removable to suit building in and access to fixing points.
3. Protective tapes in contact with anodizing to be
   3.1. Low-tack, self-adhesive and light in colour.
   3.2. Applied and removed in accordance with tape and anodizer's recommendations.
4. Inspection of protection: Carry out weekly. Promptly repair any deterioration or deficiency.

## 530 Documentation

1. Submit the following information for each batch of anodic coated components
   1.1. Supplier.
   1.2. Trade name.
   1.3. Colour (if required).
   1.4. Batch and reference number.
   1.5. Statutory requirements.

## 540 Completion

1. Protection: Remove.
2. Cleaning and maintenance of anodic coatings: Carry out in accordance with procedures detailed in anodizer's guarantees.

Ω End of Section



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Specification created using NBS Chorus